Machine Learning

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UNIVERSITY OF TARTU
Institute of Computer Science

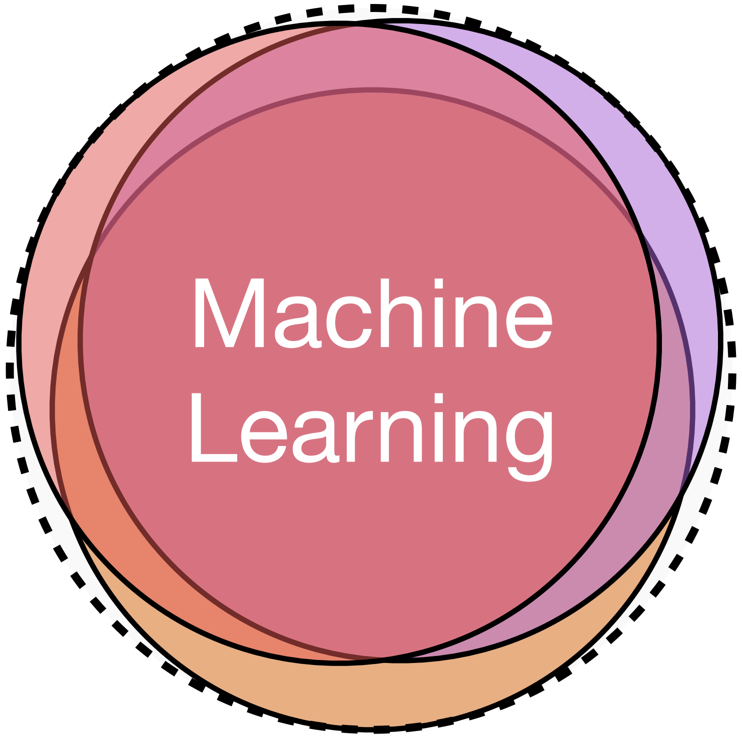
BIIT

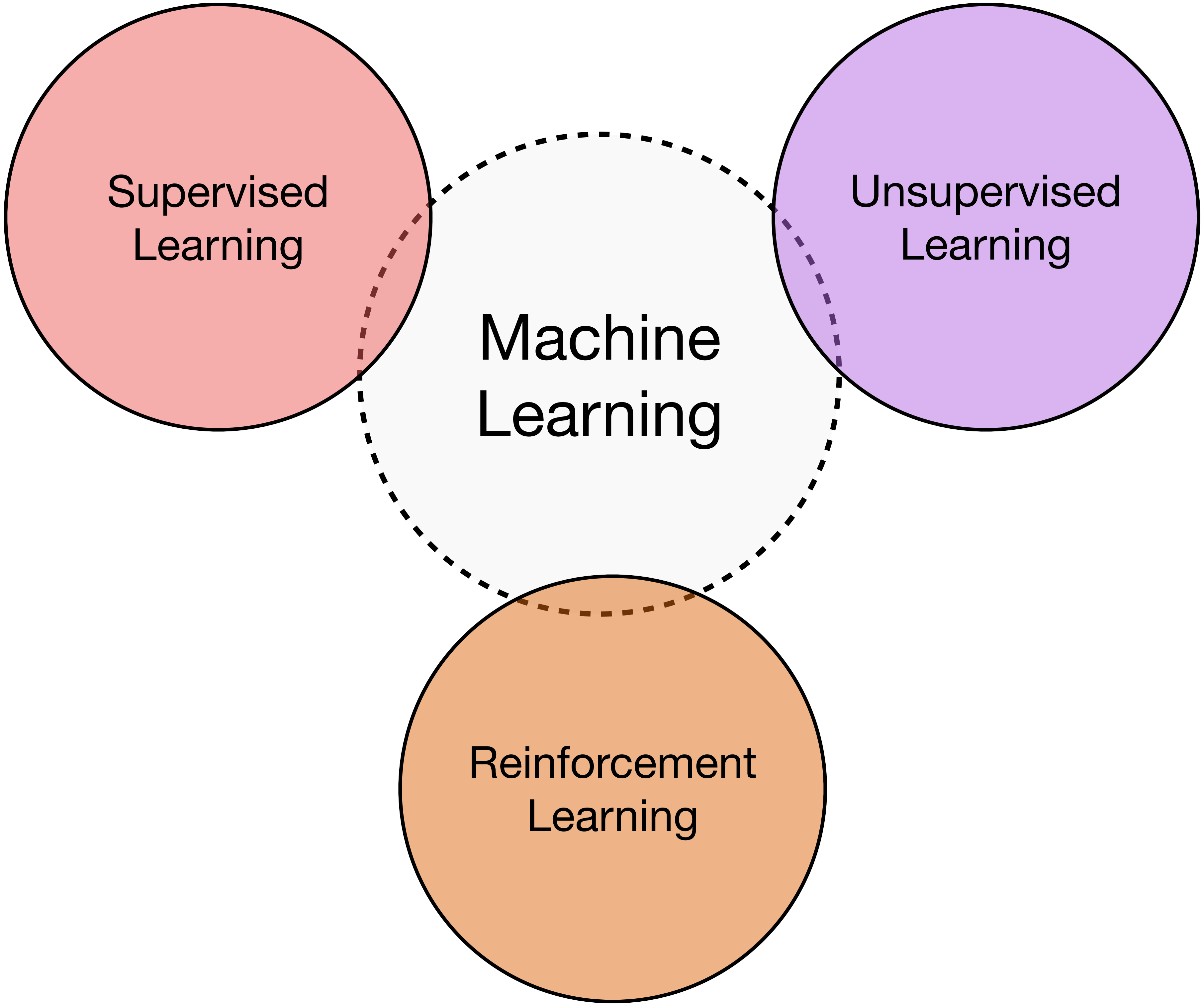
September 13, 2021

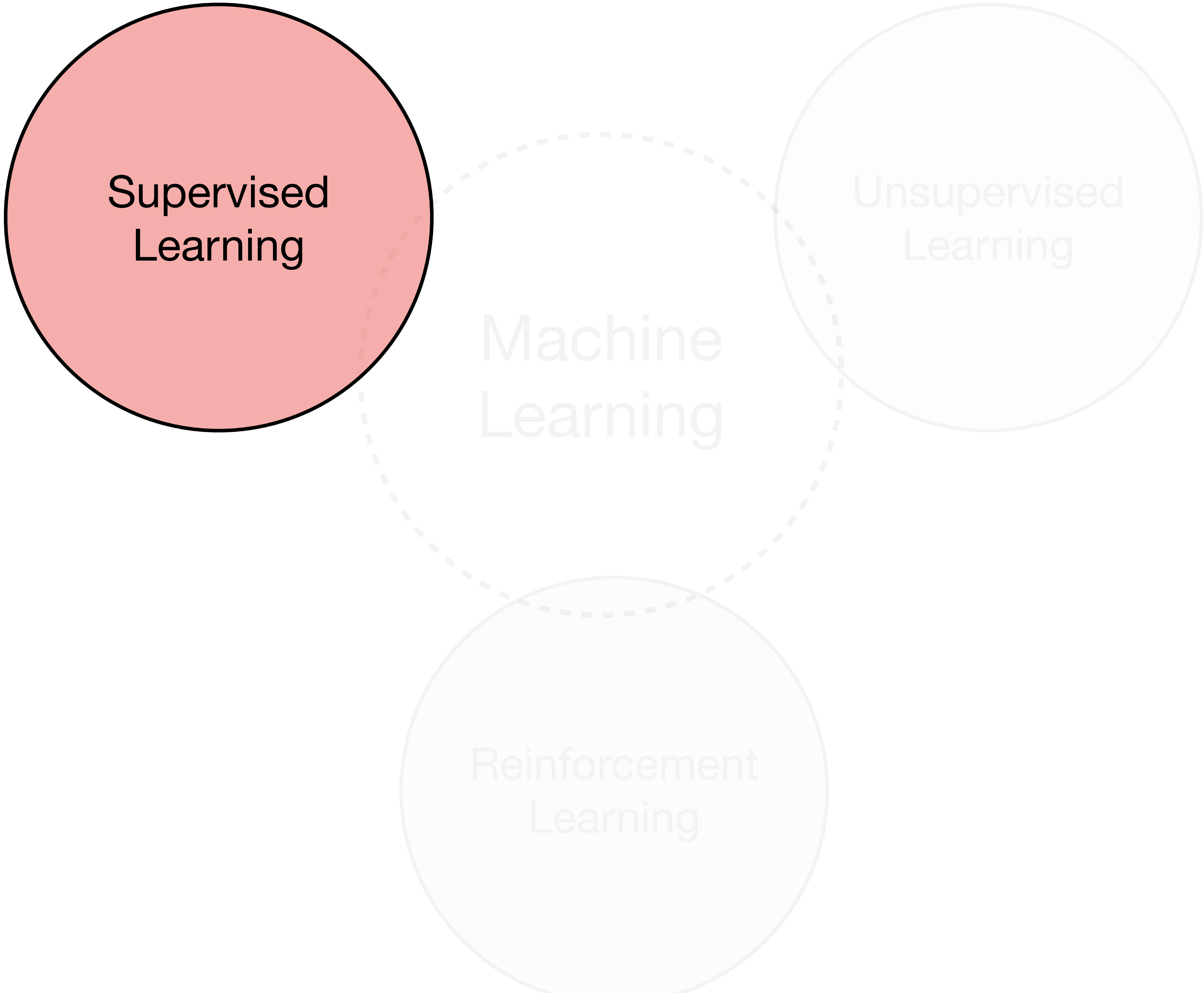
Deadlines

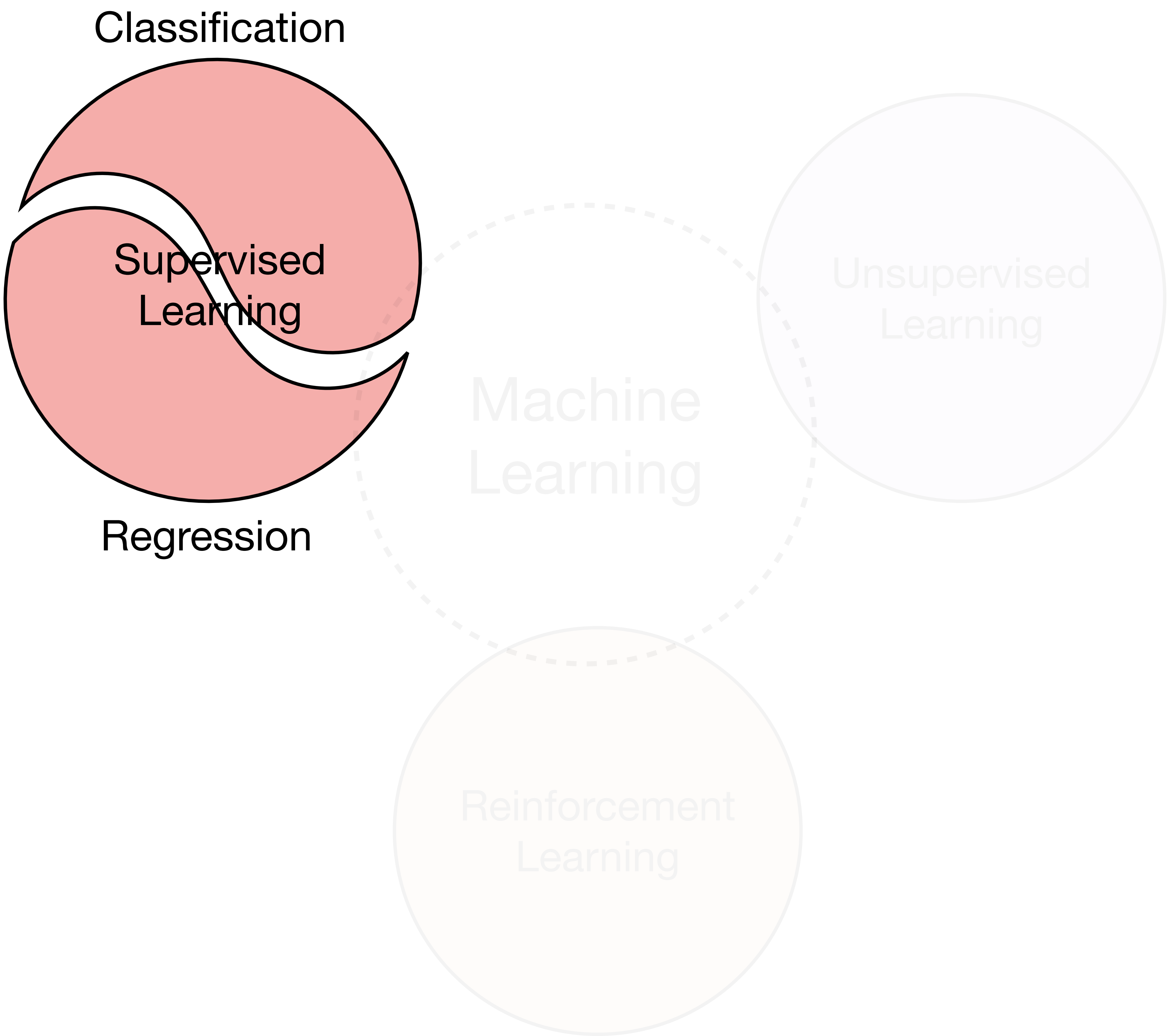
Assignment	Date of assignment	Deadline (**midnight 23:59**)
HW1	**Sep 6**	**Sep 19**
HW2	Sep 20	Oct 3
HW3	Oct 4	Oct 17
Paper summary	Oct 11	Oct 24
HW4	Oct 18	Oct 31
HW5	Nov 1	Nov 14
HW6	Nov 22	Dec 5
Project	**TBA**	Dec 13 - 15

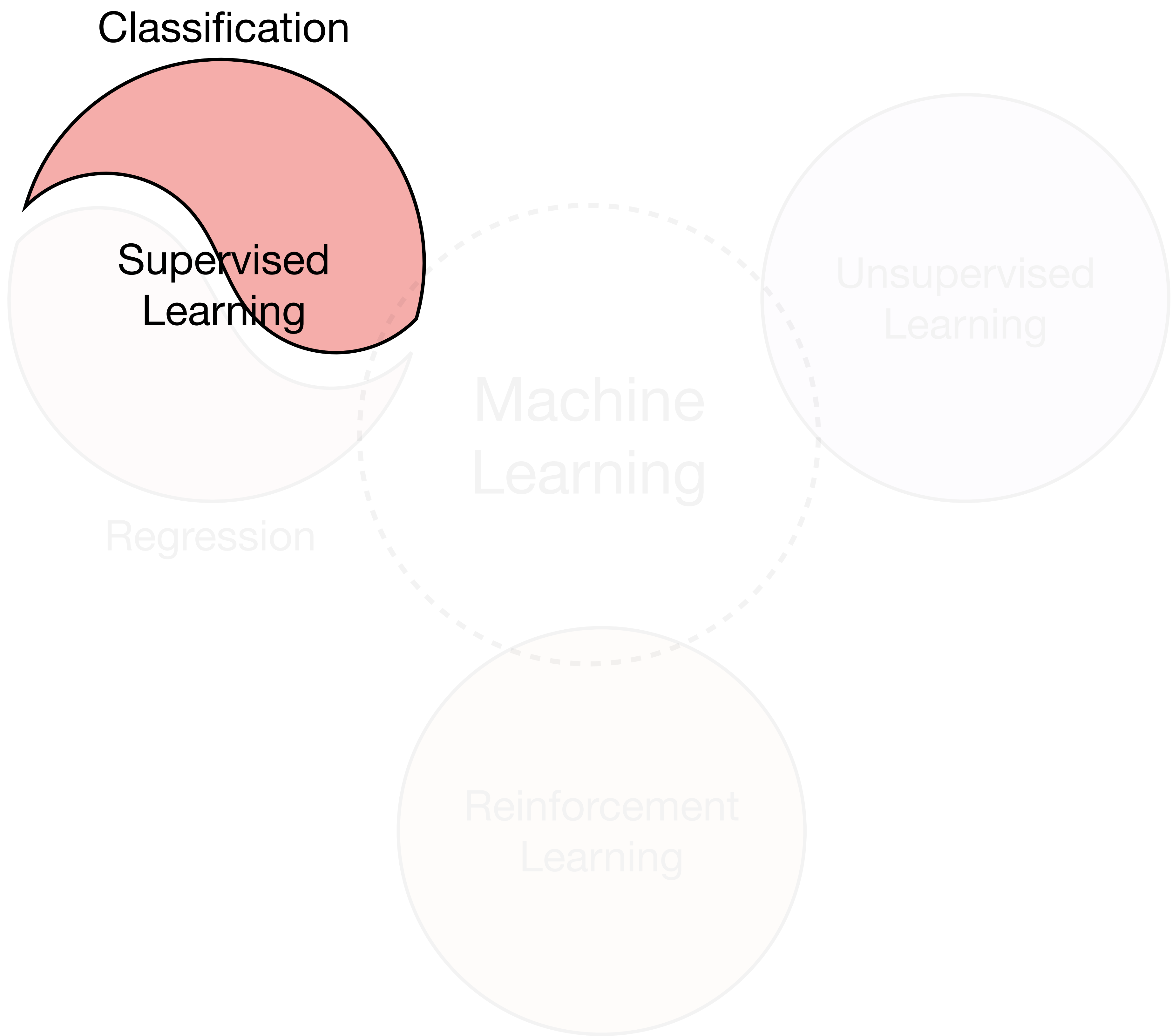
***All deadlines are subject to change, check out CampusWire and website for updates**


Machine
Learning


Supervised Learning
Unsupervised Learning
Machine Learning
Reinforcement Learning


Supervised
Learning
Unsupervised
Learning
Machine
Learning
Reinforcement
Learning


Classification
Supervised Learning
Regression
Machine Learning
Unsupervised Learning
Reinforcement Learning


Classification
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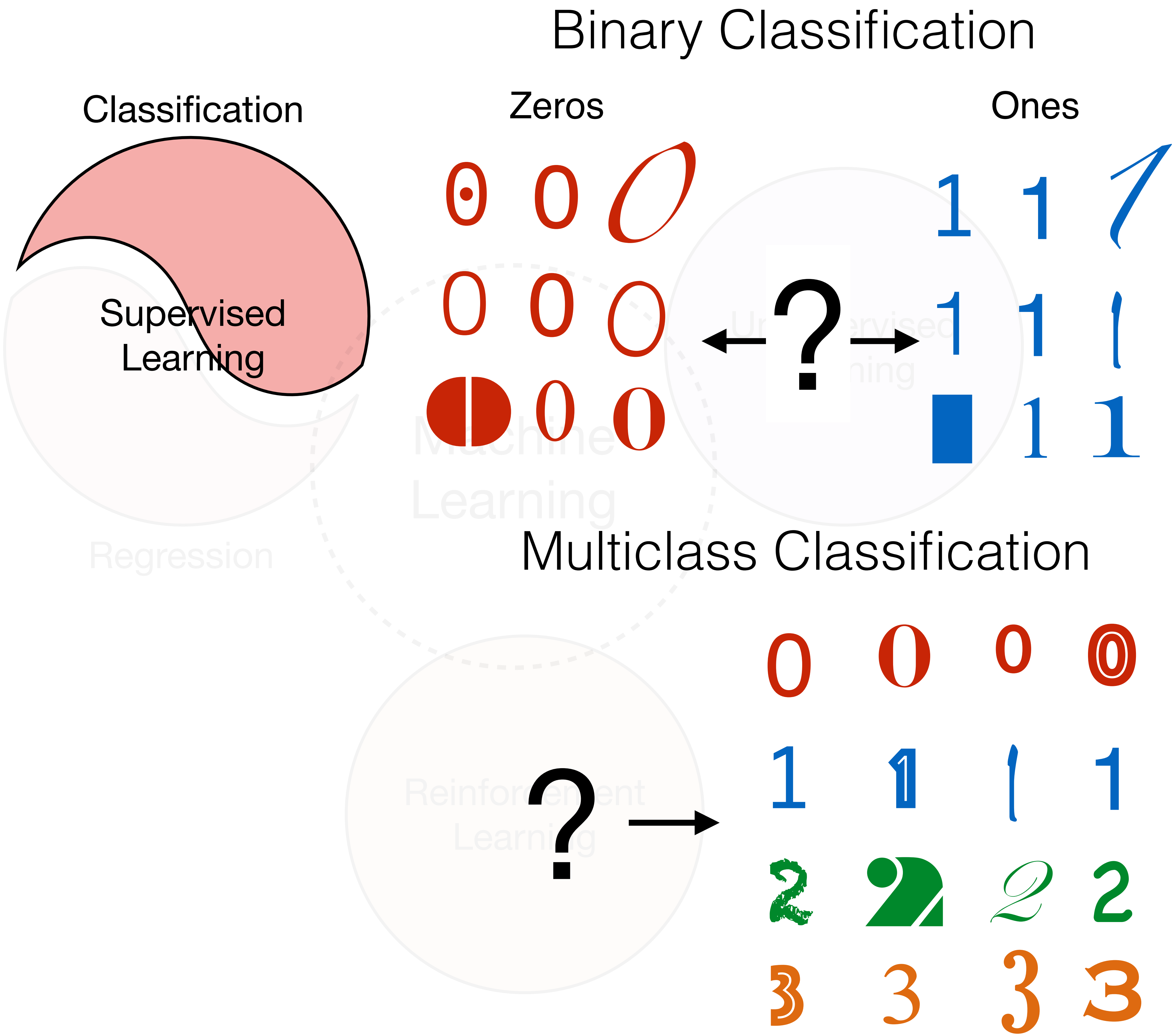
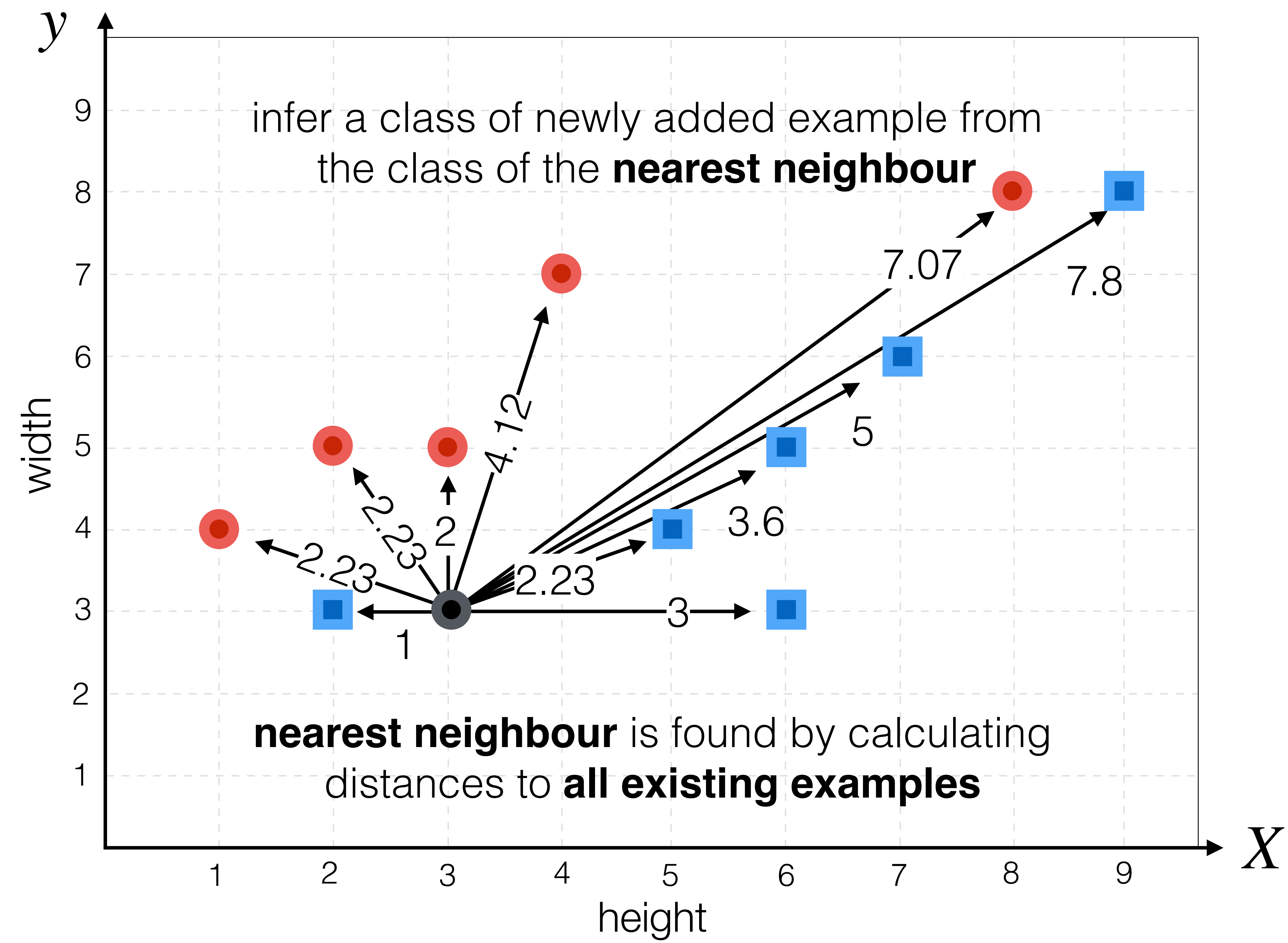
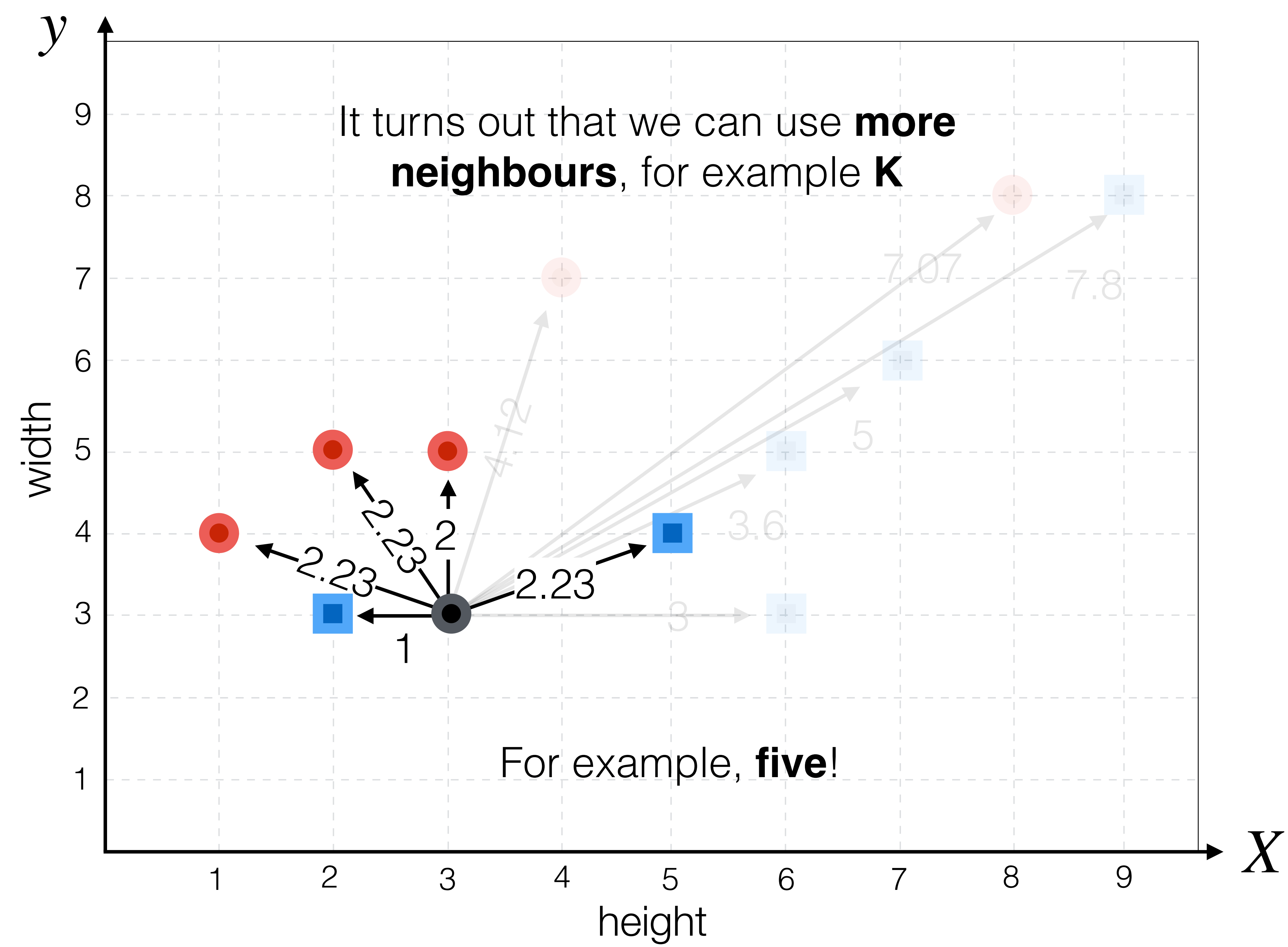
Reinforcement Learning

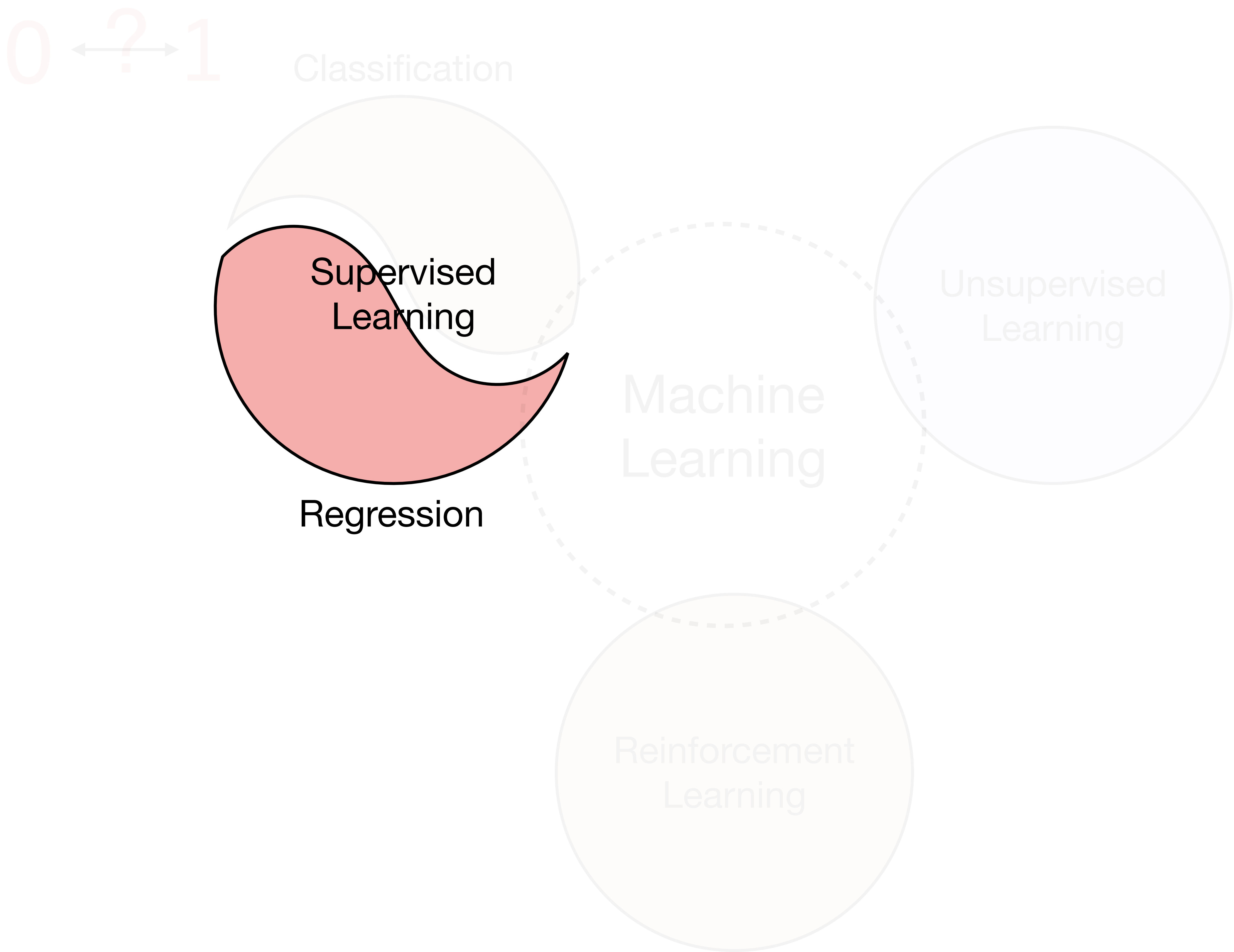

Binary Classification
Classification
Supervised
Learning
Regression
Zeros
Ones
?
Multiclass Classification
?

Nearest Neighbour Classifier



K-Nearest Neighbour Classifier





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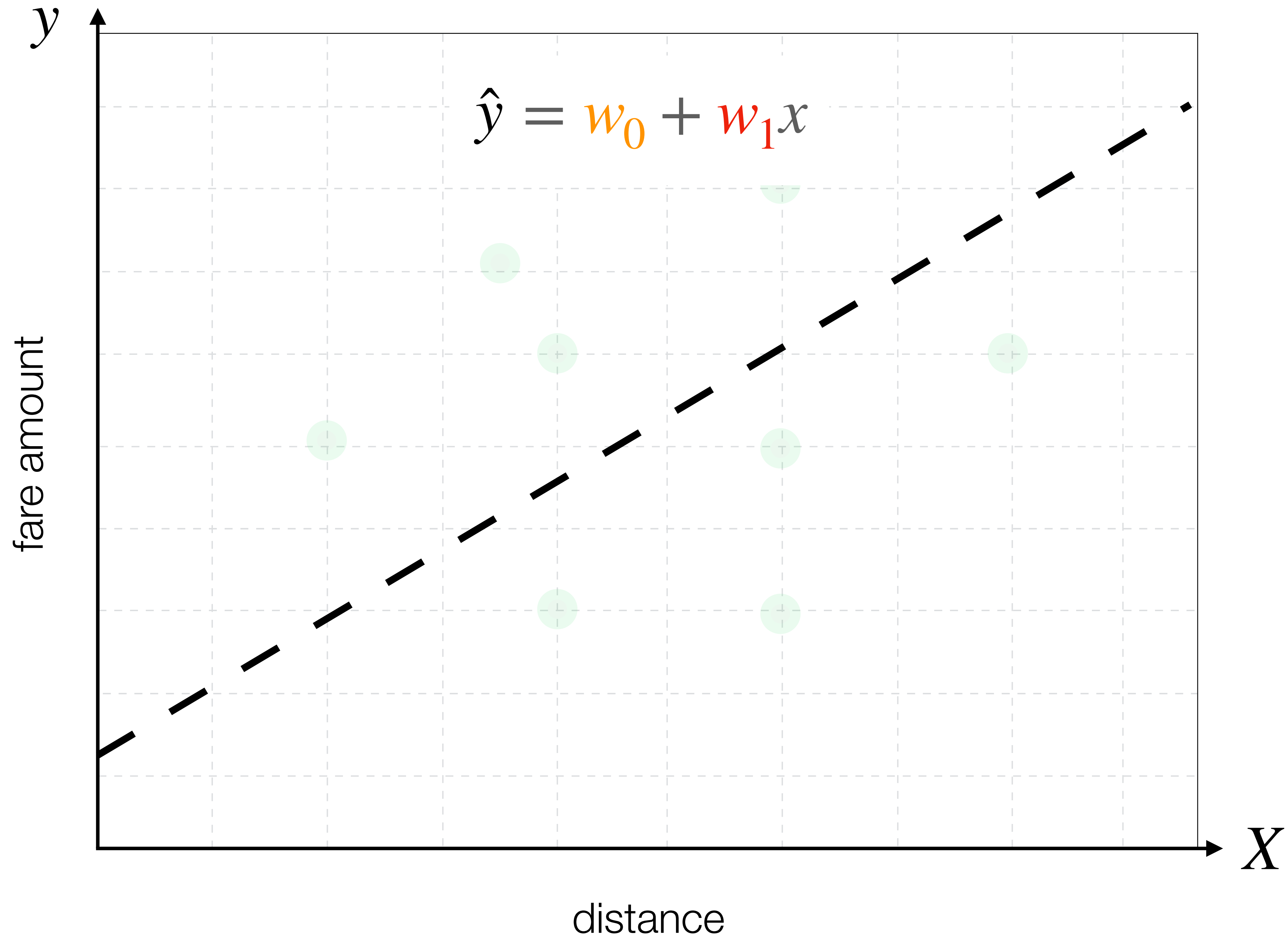
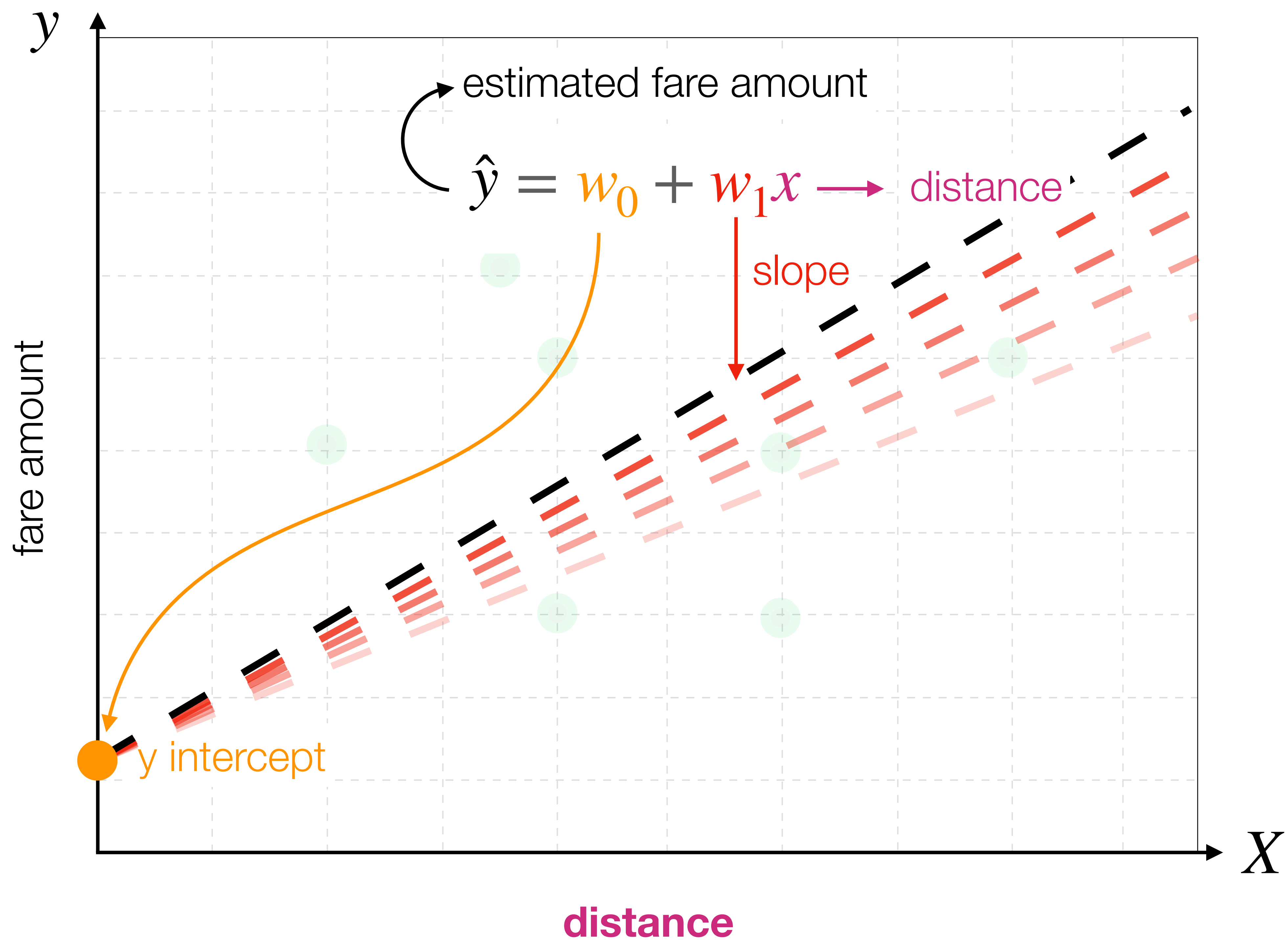
Classification
Supervised
Learning
Regression
Machine
Learning
Unsupervised
Learning
Reinforcement
Learning

Linear Regression

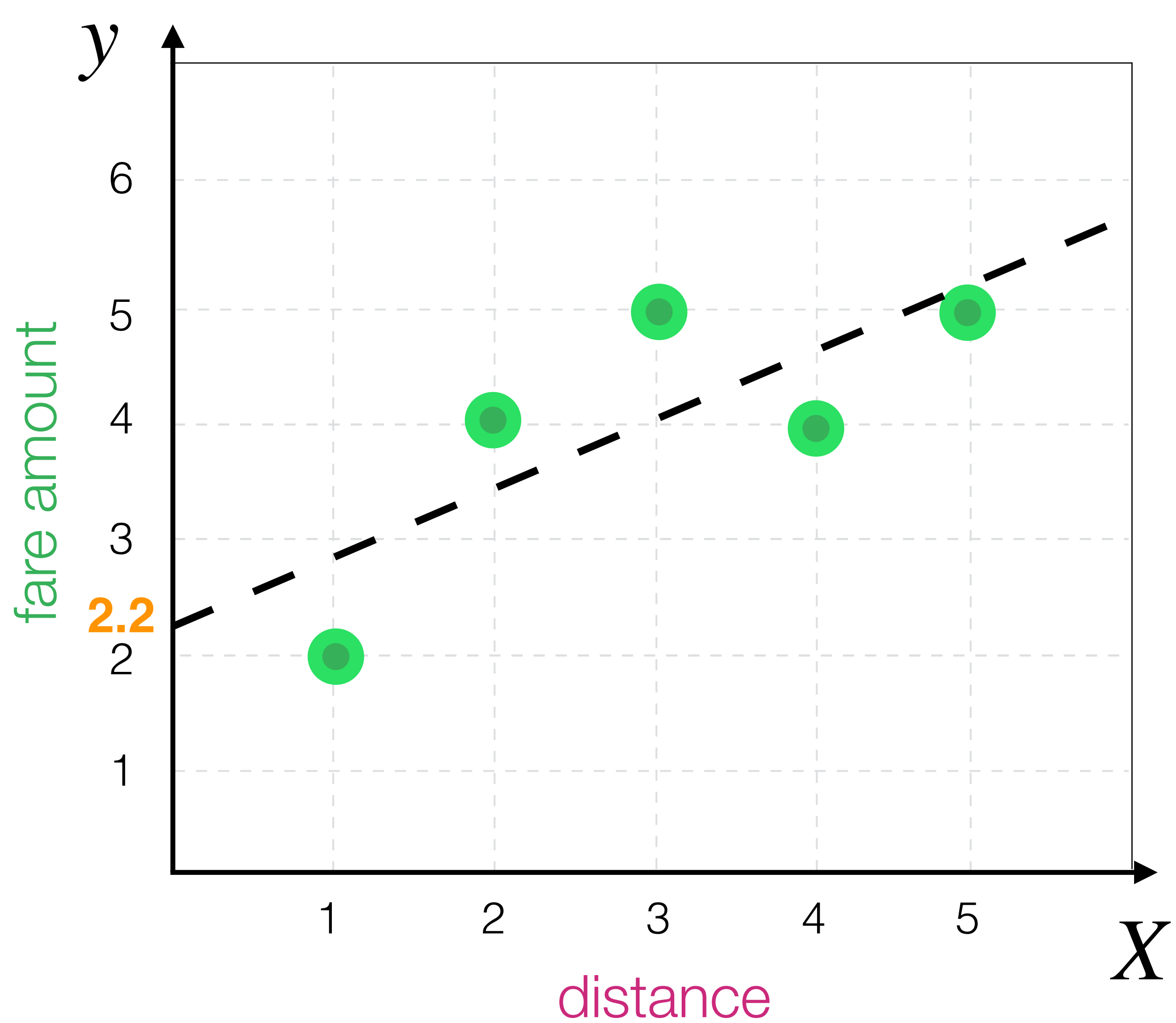


Linear Regression



Linear Regression (example)

$$\hat{y} = 2.2 + .6\ x$$



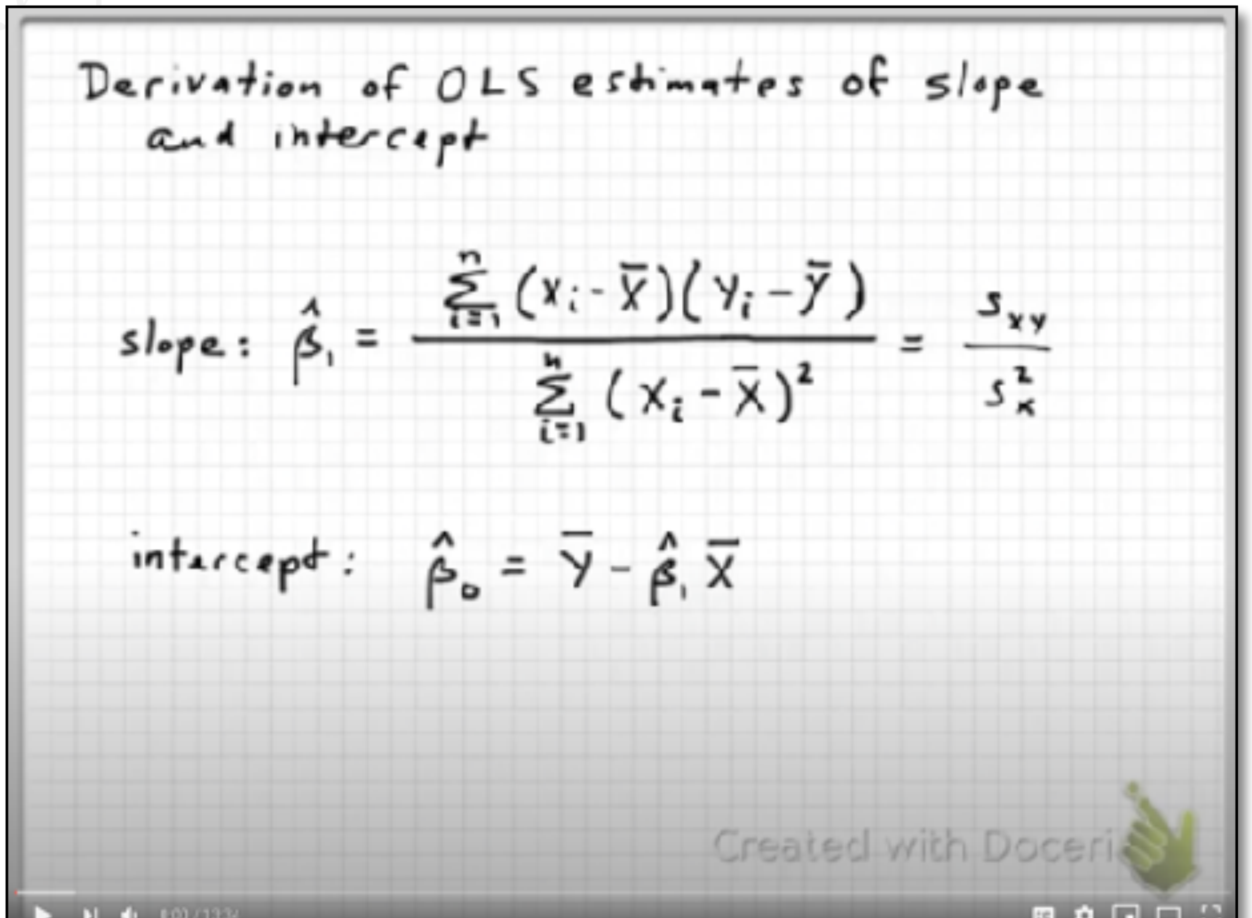
x	y	x - x̄	y - ȳ	(x - x̄)²	(x - x̄)(y - ȳ)
1	2	-2	-2	4	4
2	4	-1	0	1	0
3	5	0	1	0	0
4	4	1	0	1	0
5	5	2	1	4	2
$\bar{x} = 3$	$\bar{y} = 4$			10	6

$$w_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{6}{10} = .6$$

$$4 = w_0 + .6 * 3$$

$$w_0 = 2.2$$

Linear Regression (example)



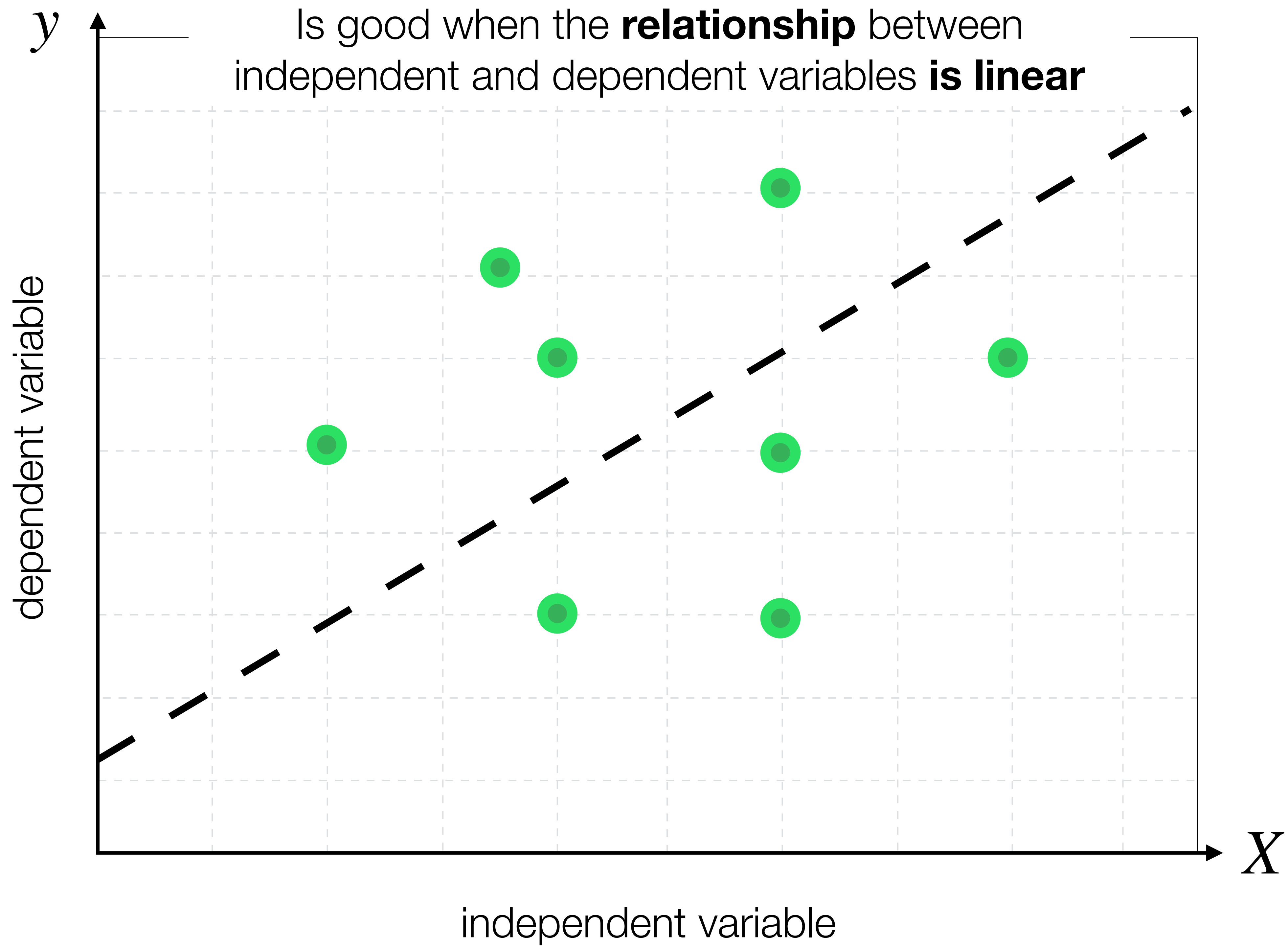
https://youtu.be/**jqoHefilf9U**

Check out the video for how the formulas are derived!

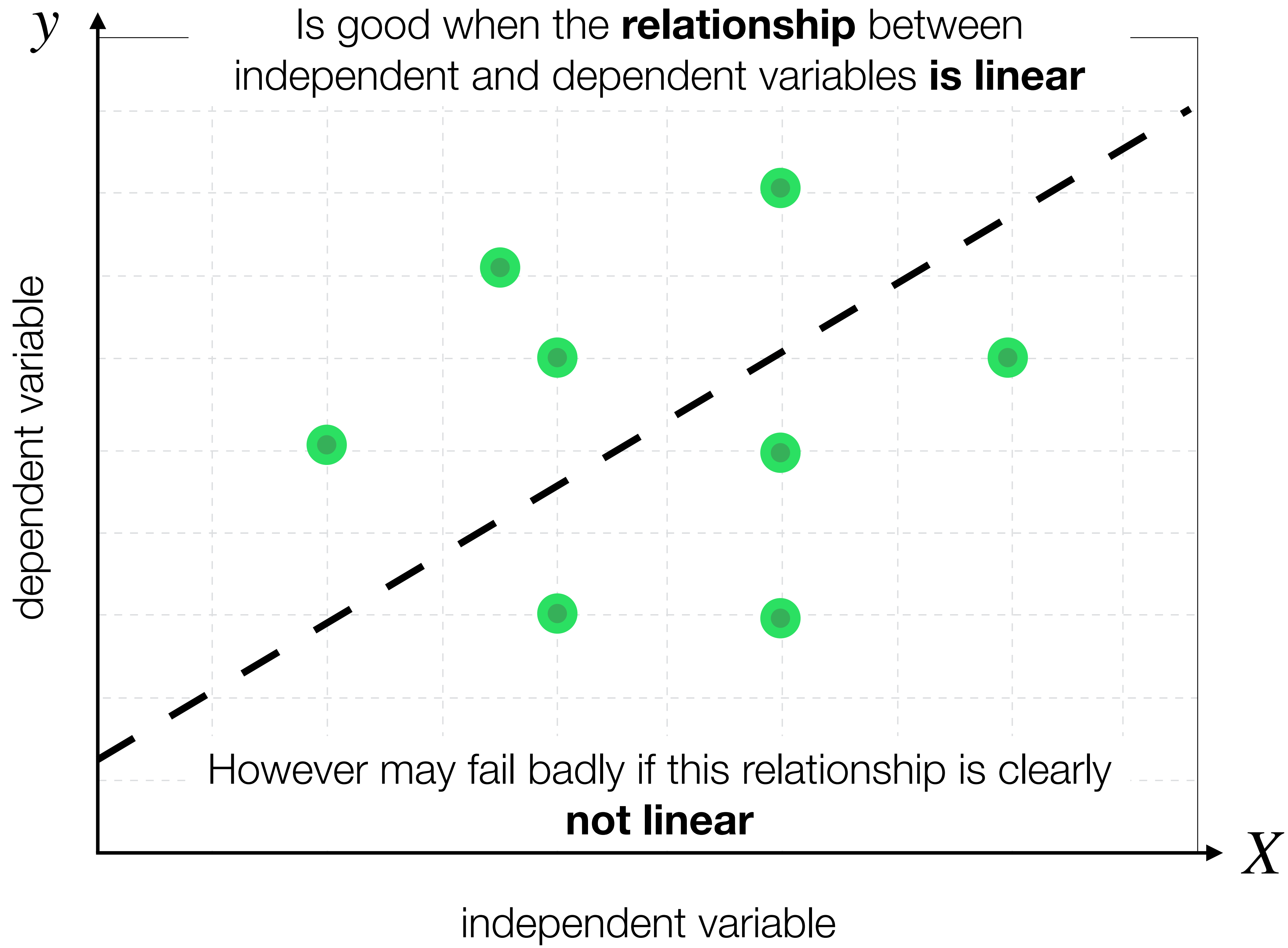
$$w_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$w_0 = \bar{y} - w_1 \bar{x}$$

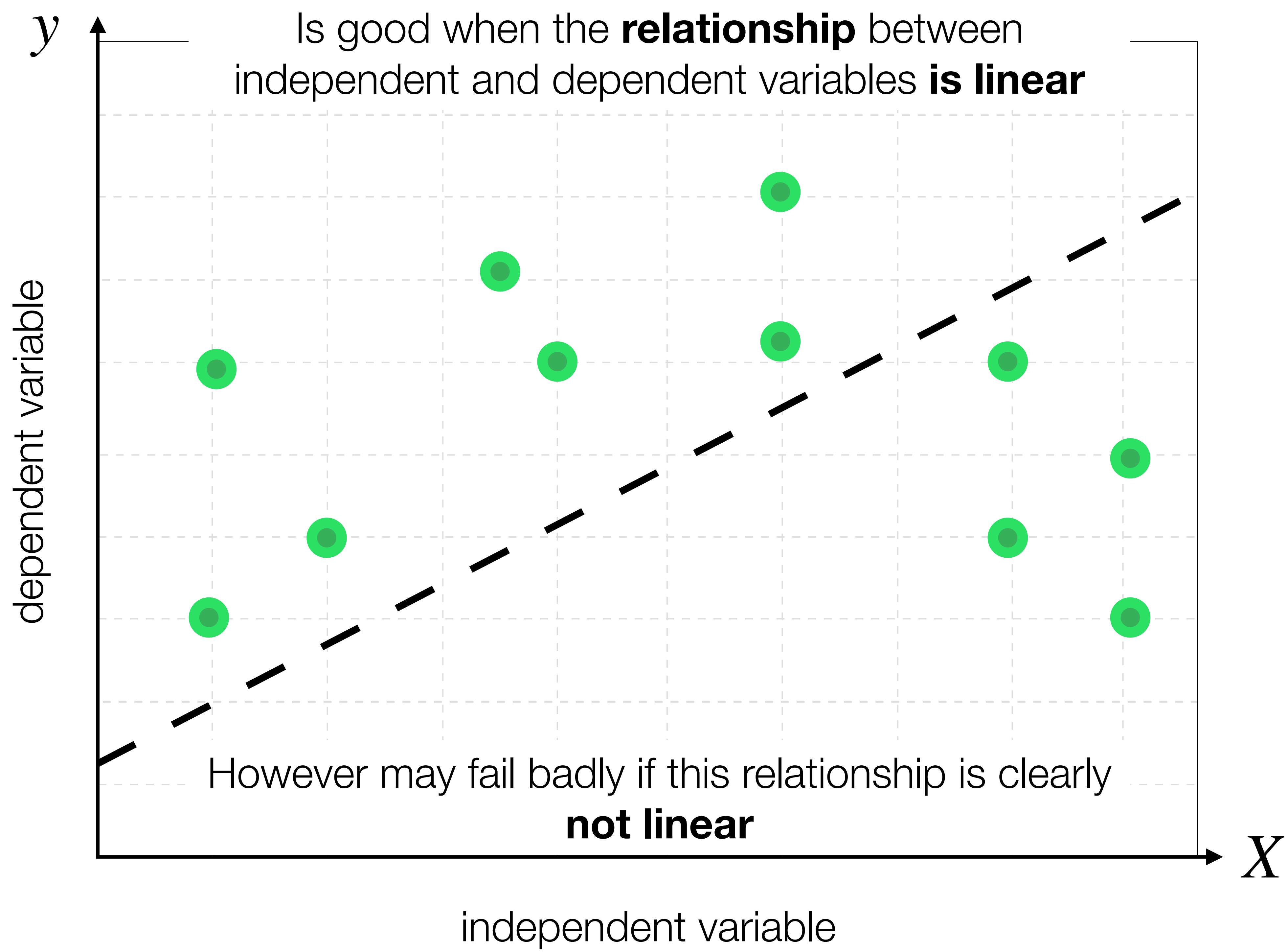
Linear Regression



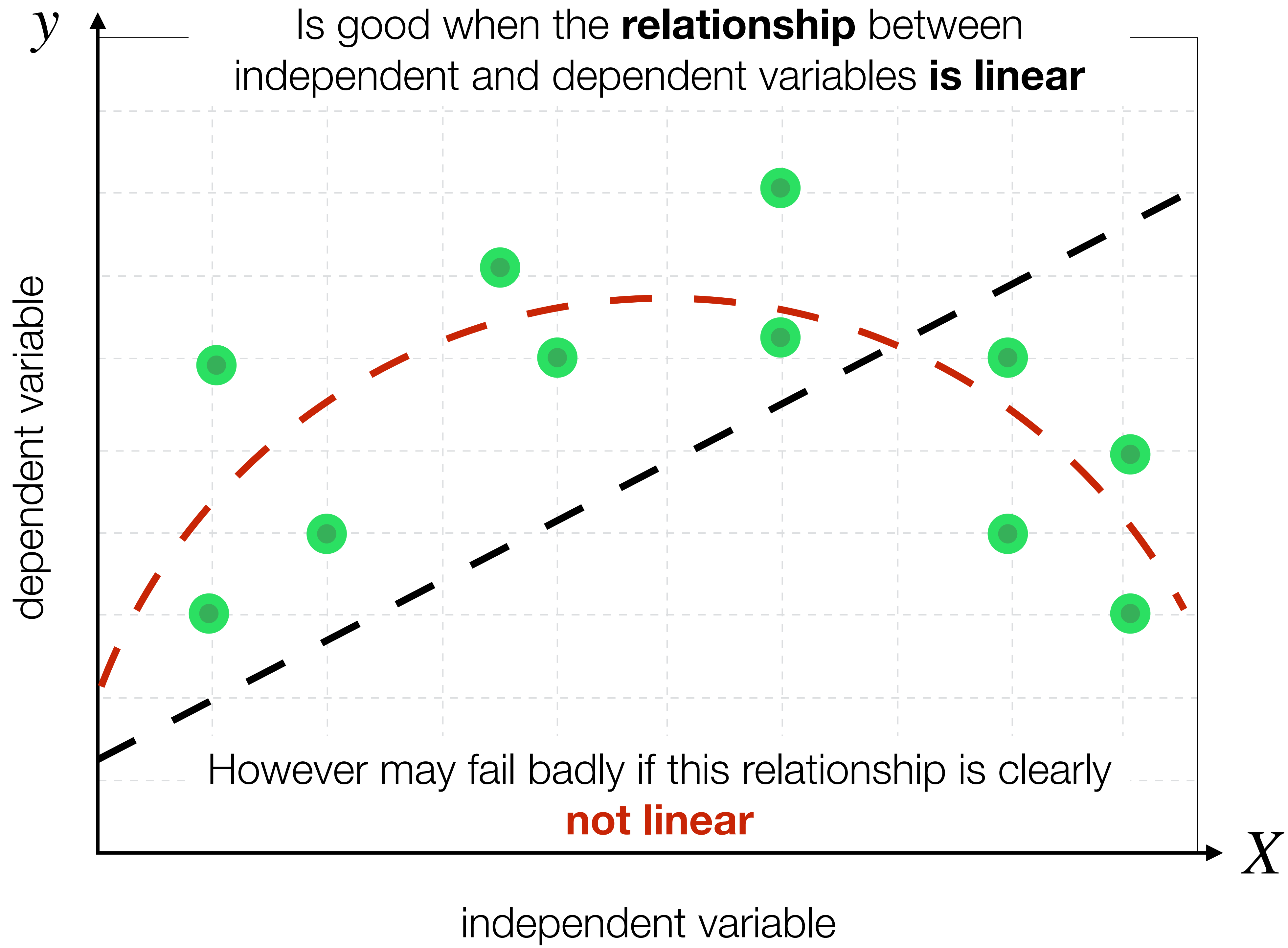
Linear Regression



Linear Regression



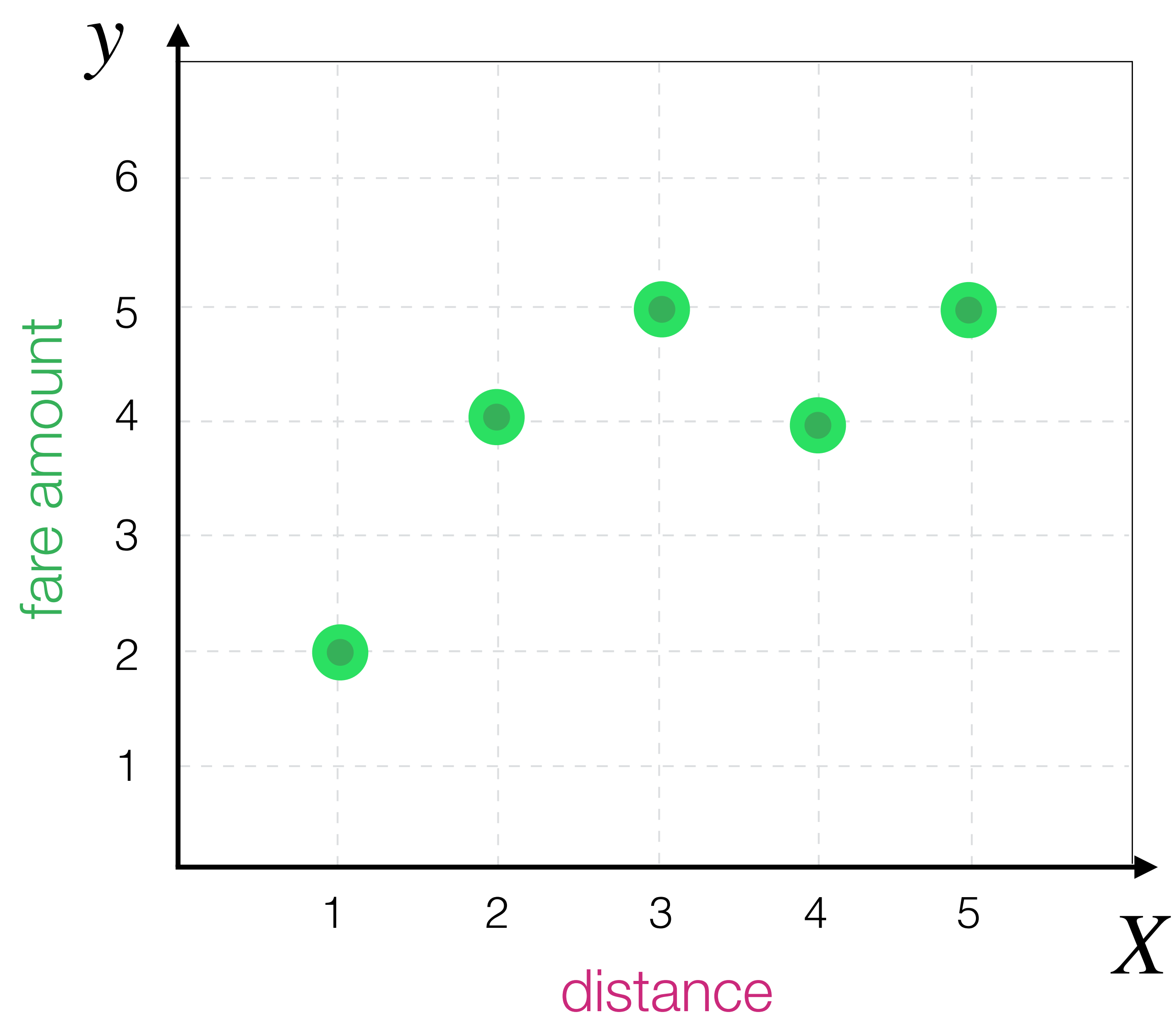
Linear Regression

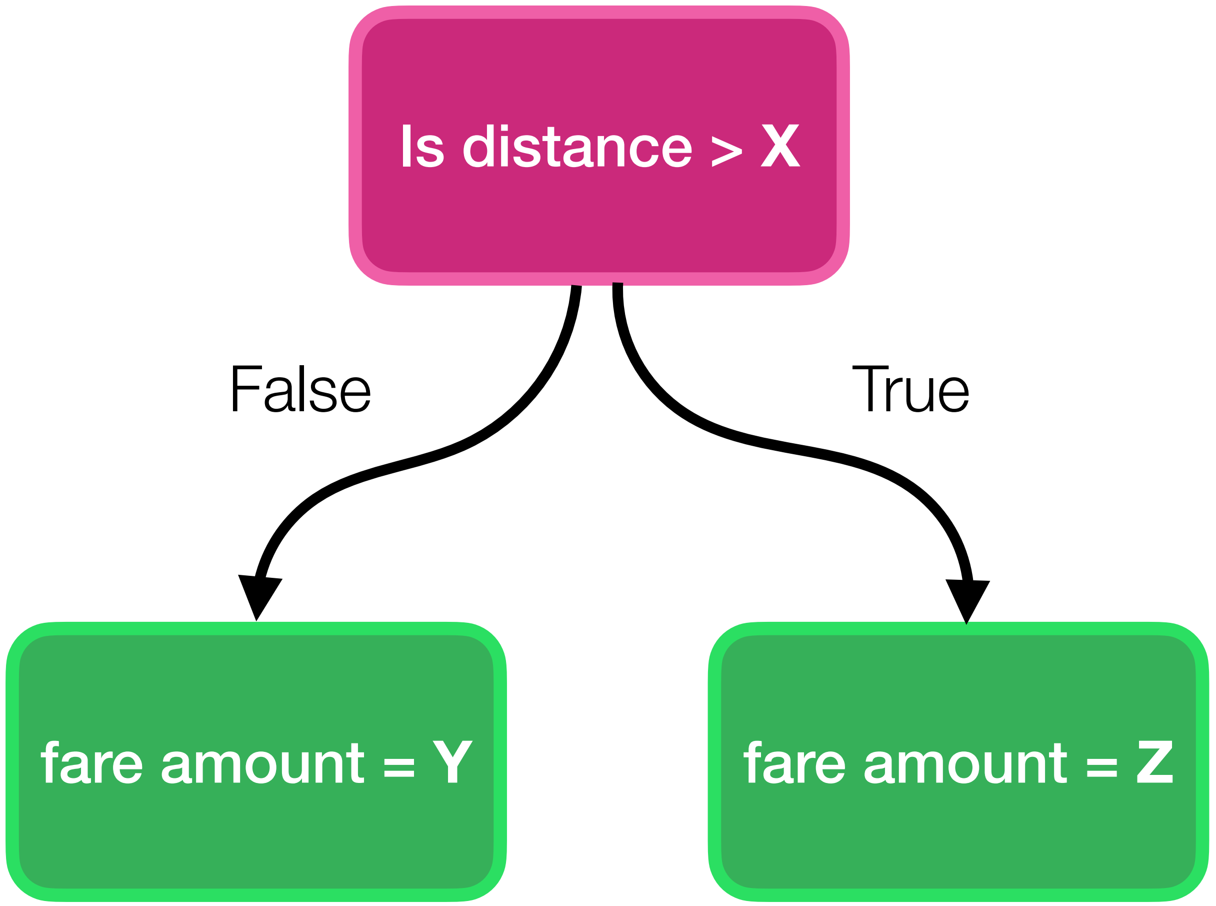


Decision Tree Algorithm

Decision Tree Algorithm

By asking a simple **question** about value of **independent variable** it tries to predict a value of **dependent variable**





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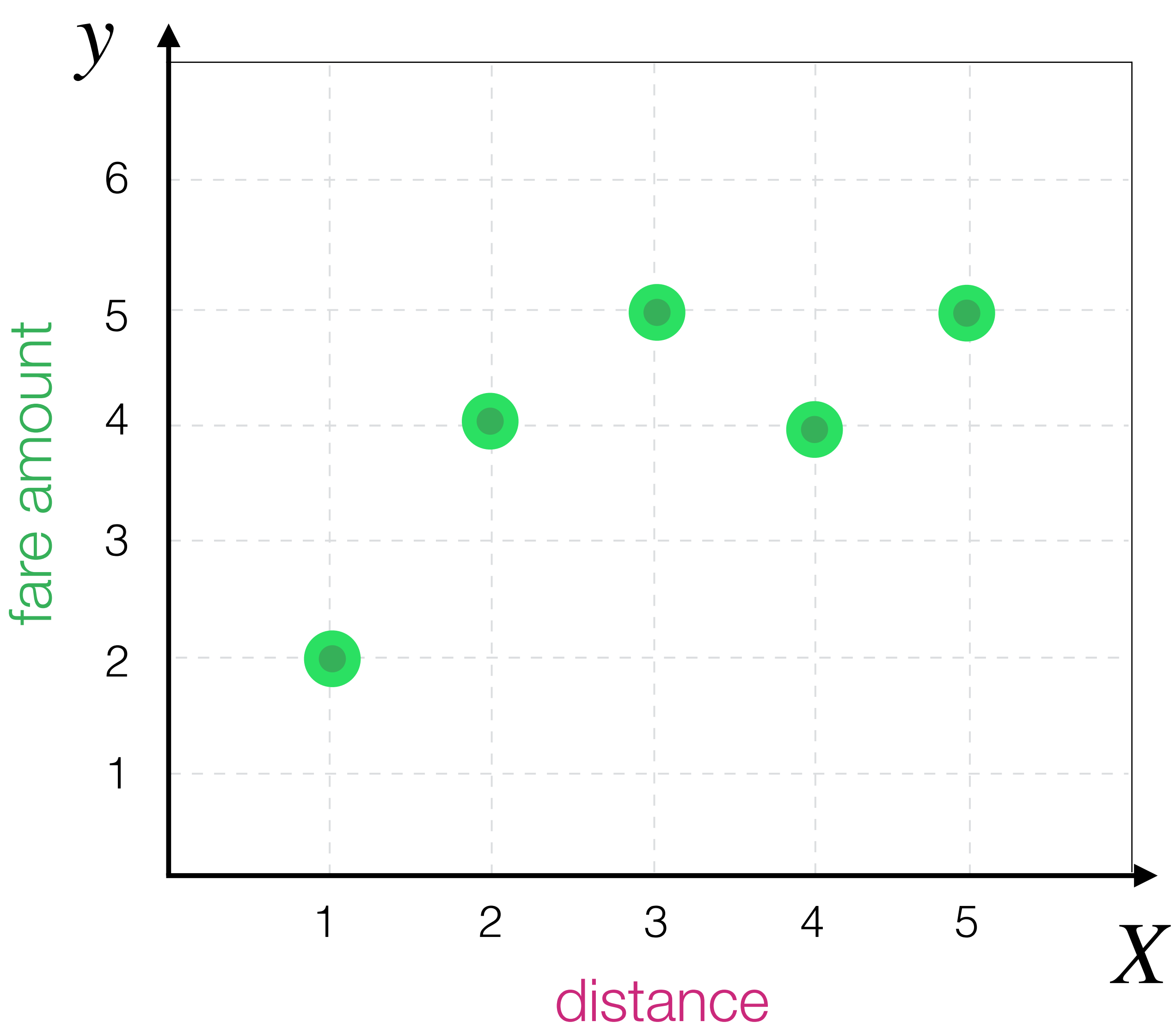

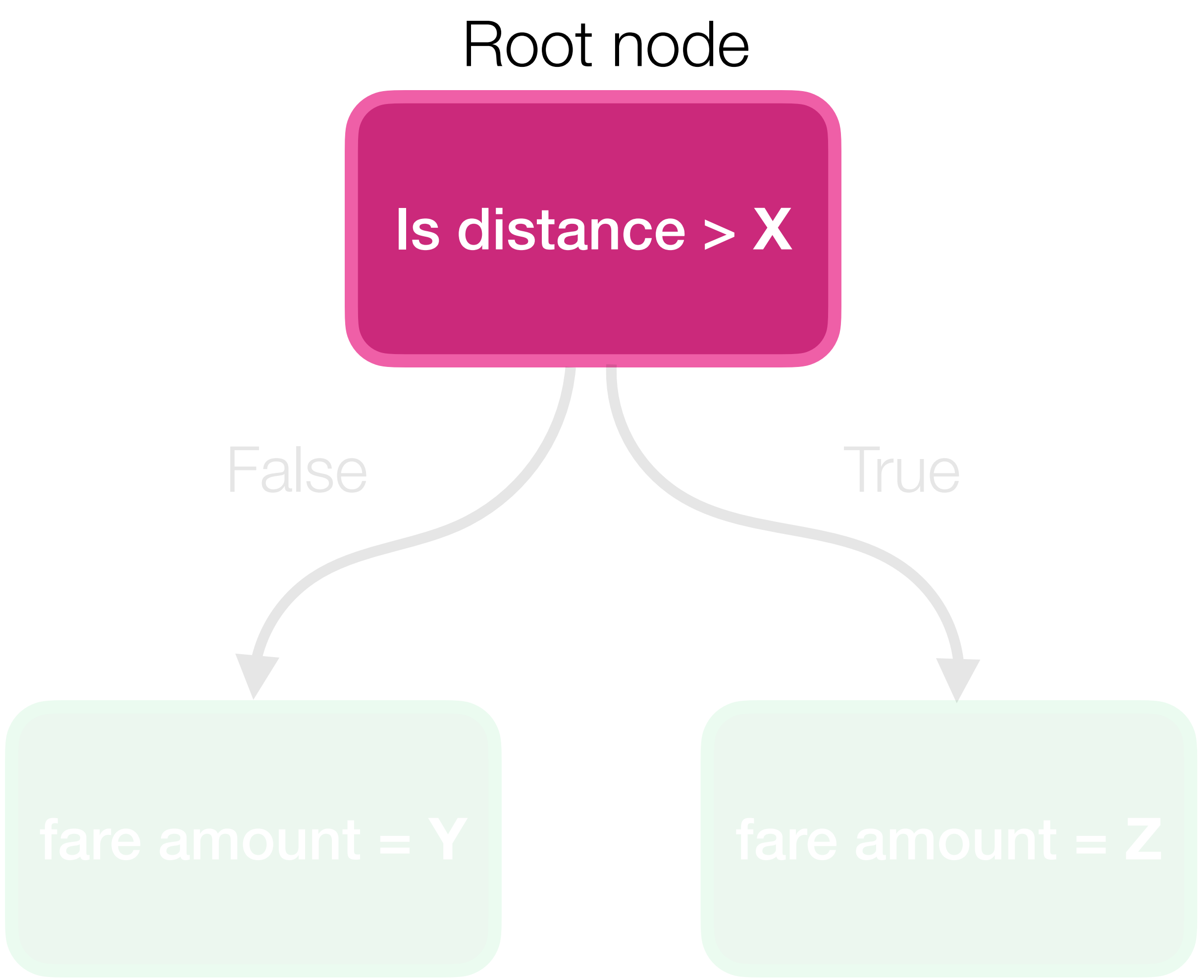

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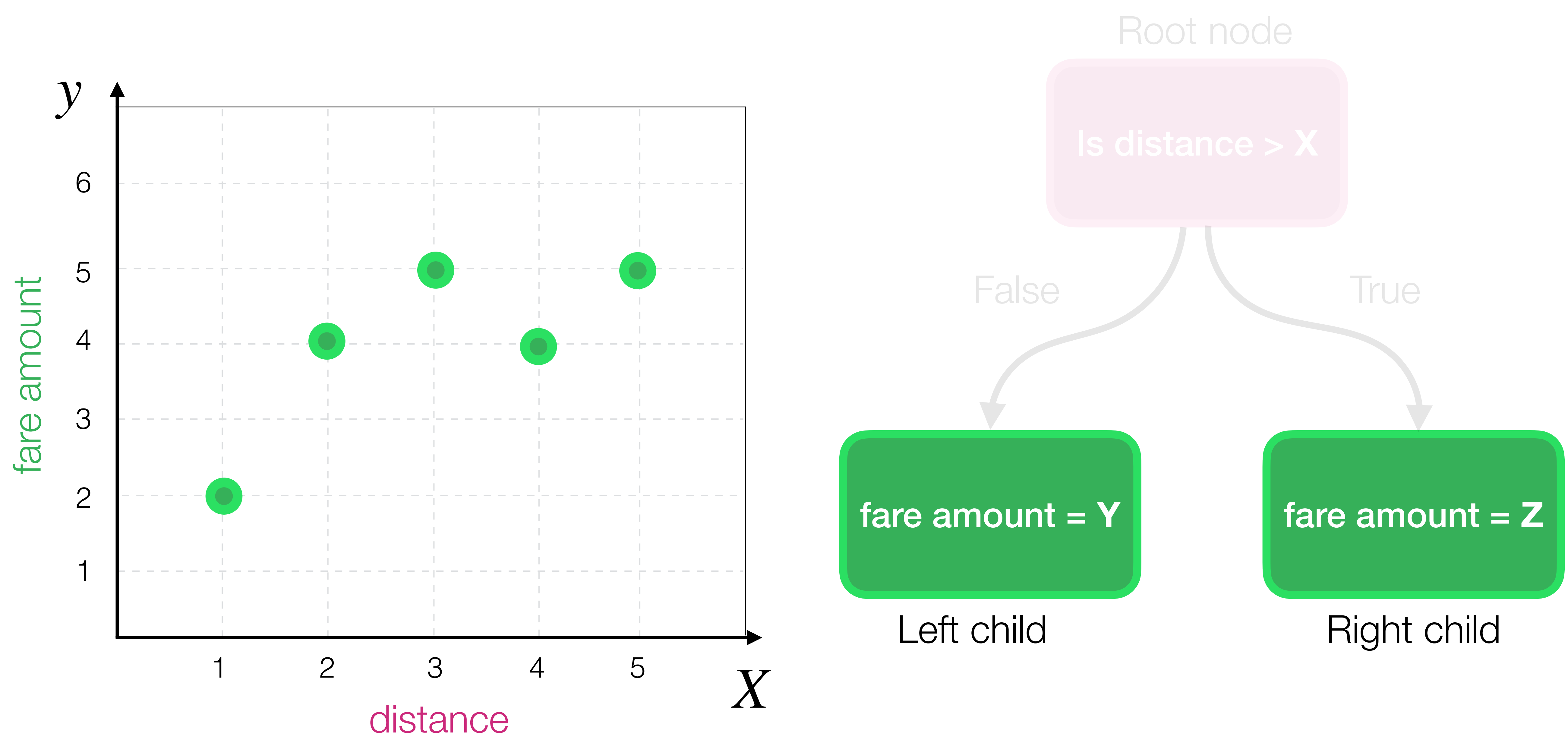

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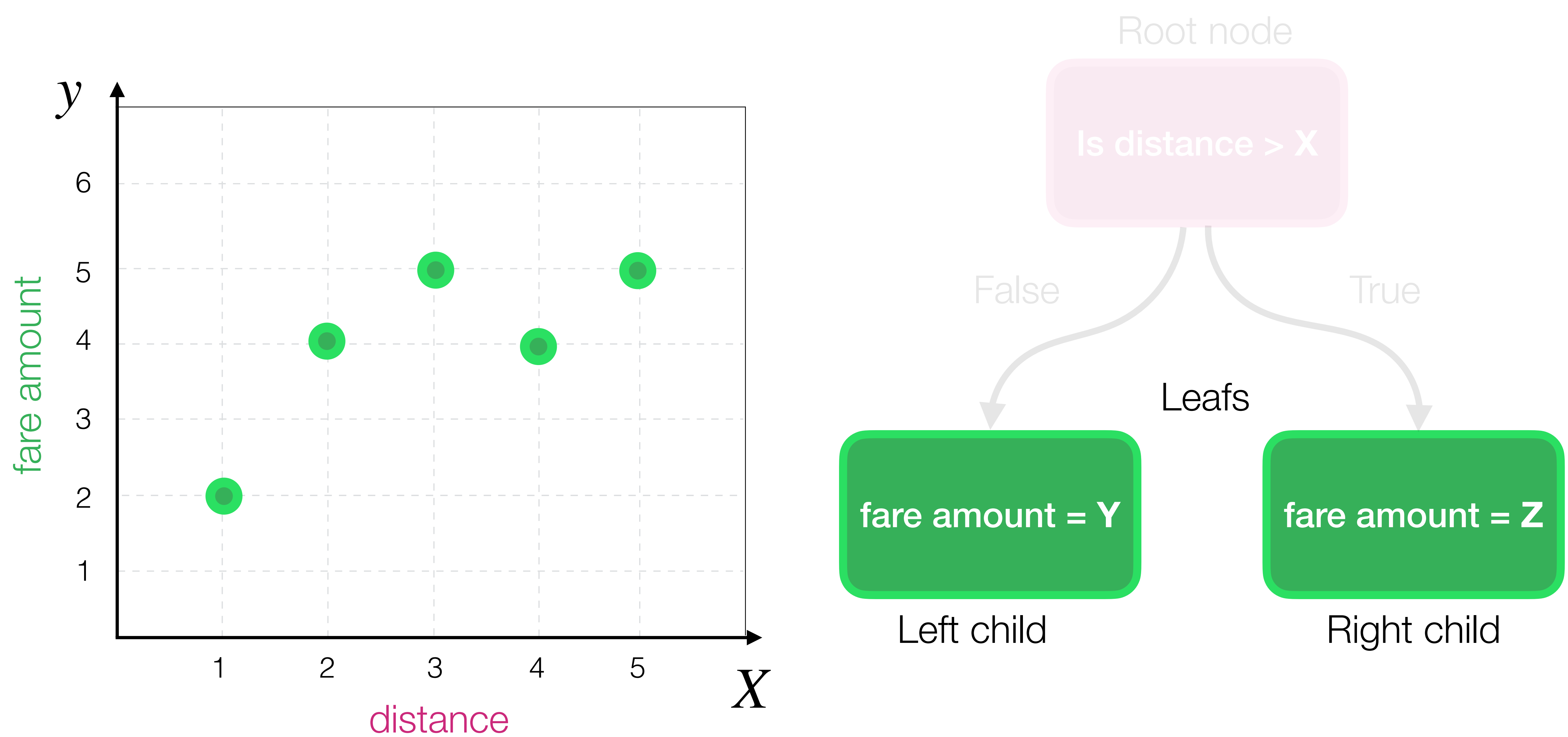

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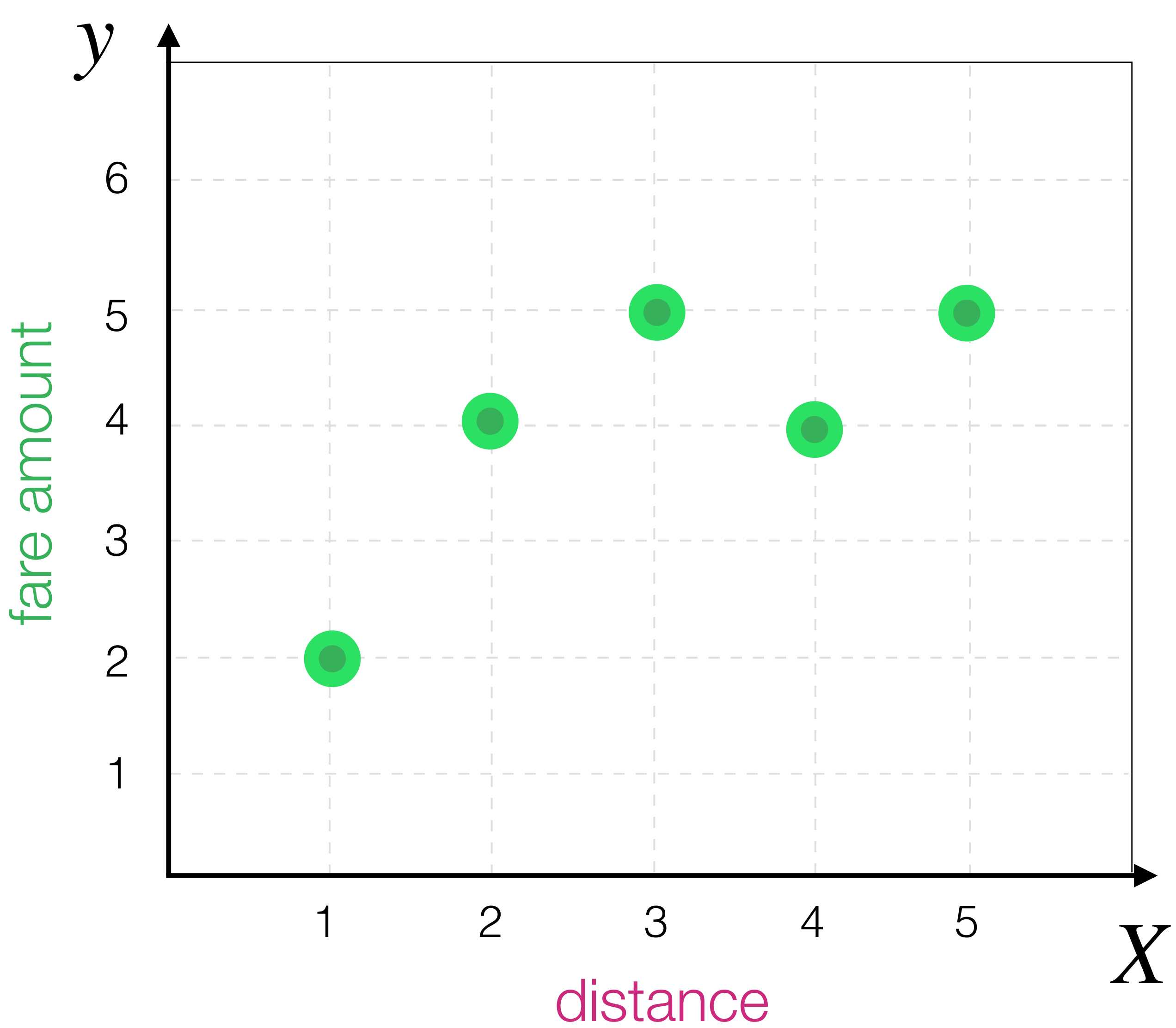
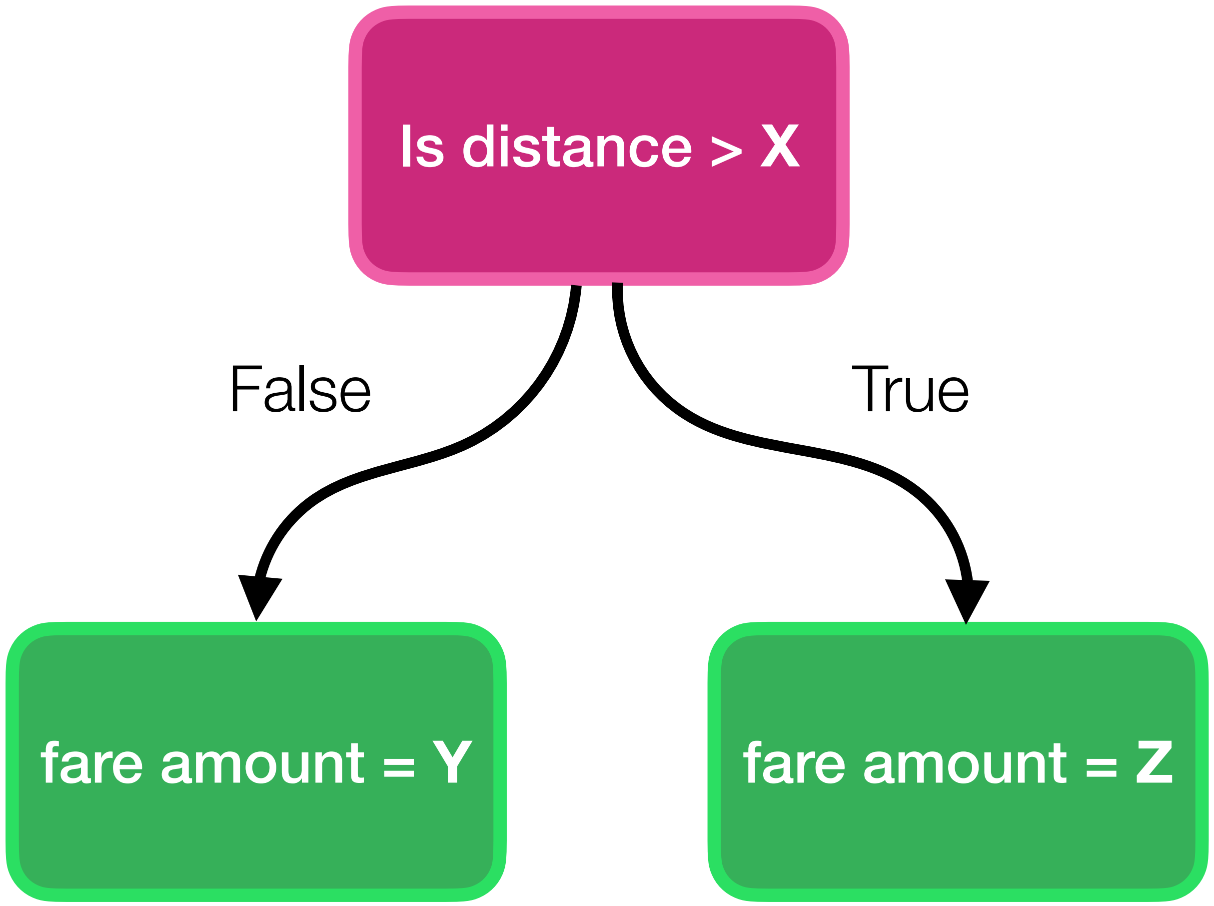
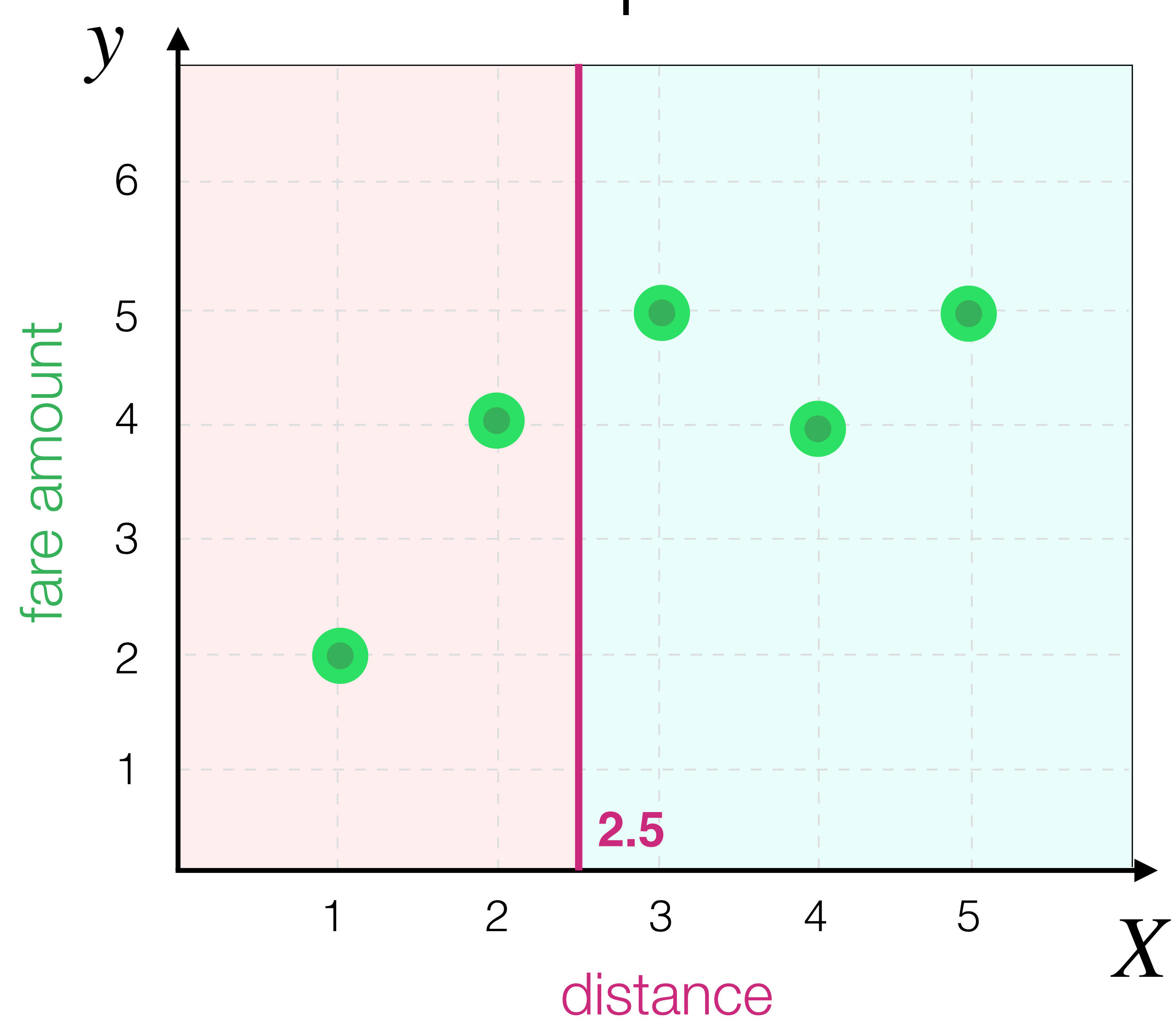


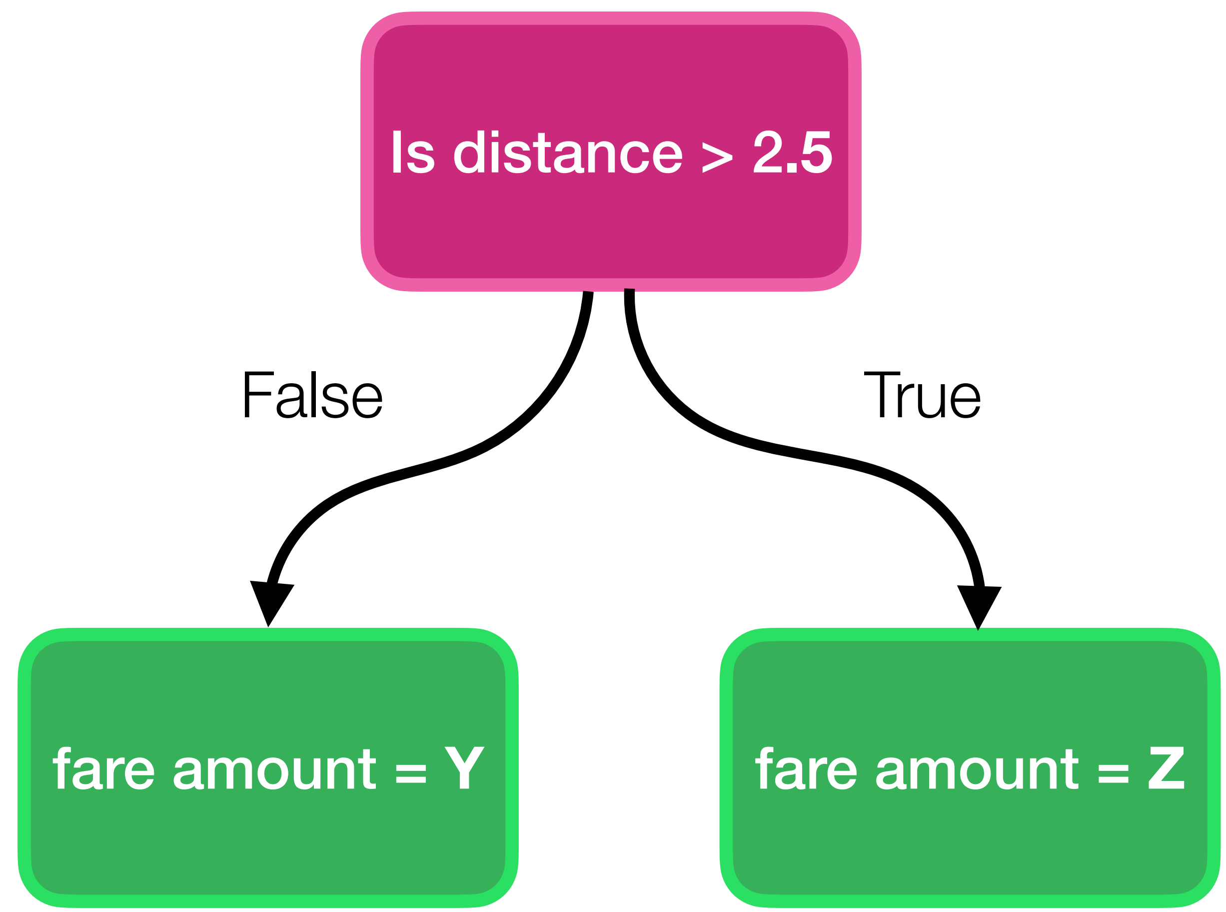


Decision Tree Algorithm

Here, **X** may correspond to any vertical line.

For example if X = 2.5:

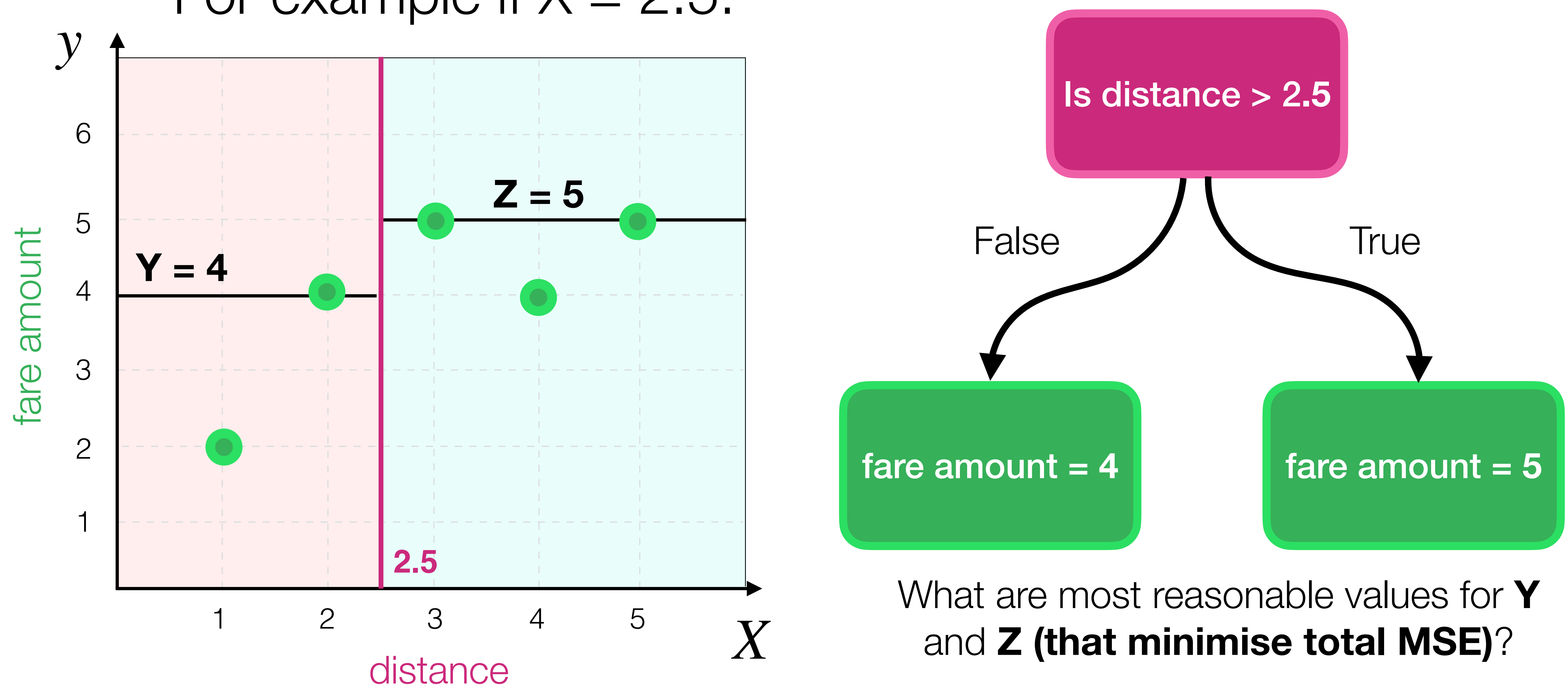




Decision Tree Algorithm

What would be MSE if **Y = 4** and **Z = 5**?

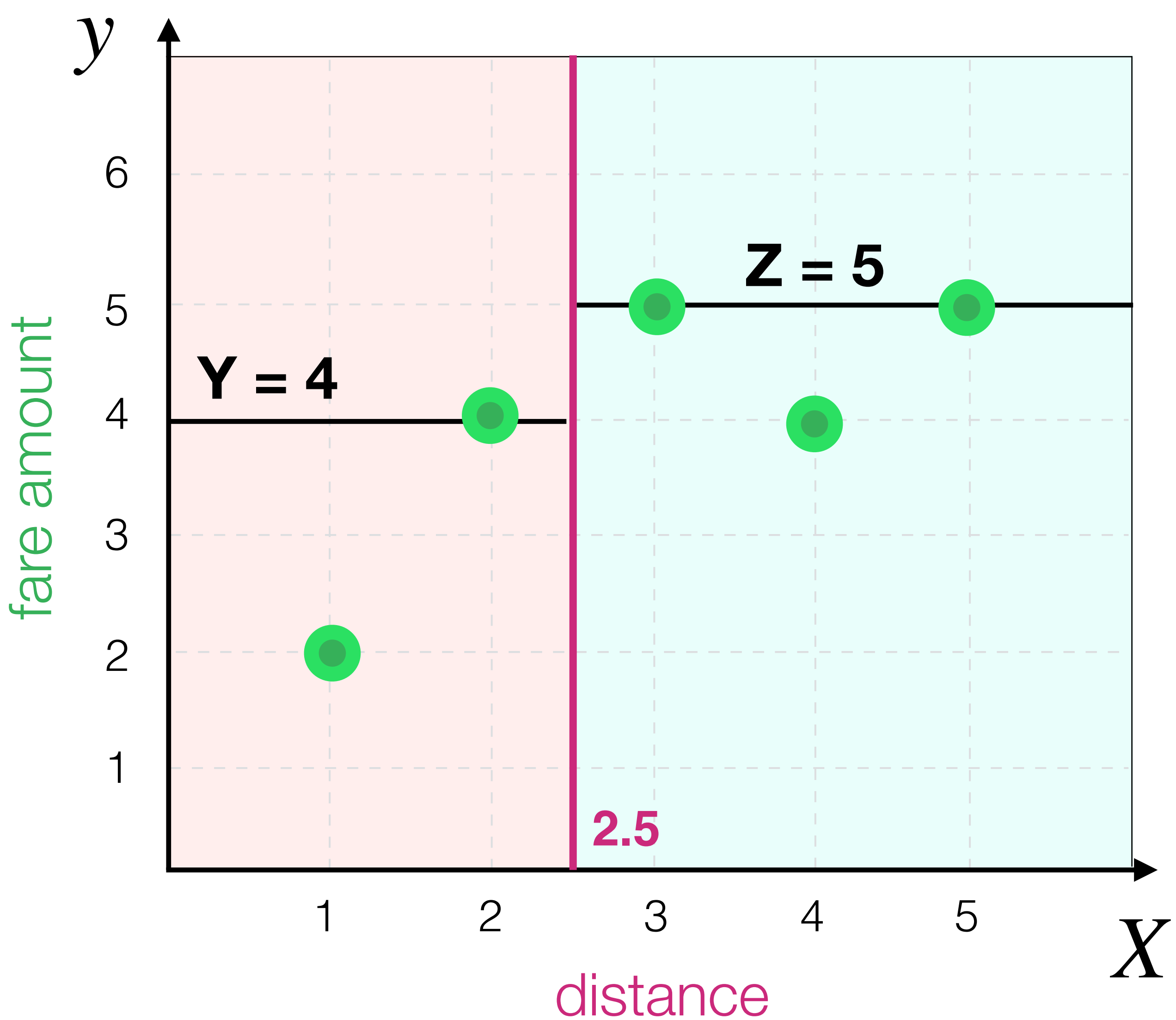
For example if X = 2.5:

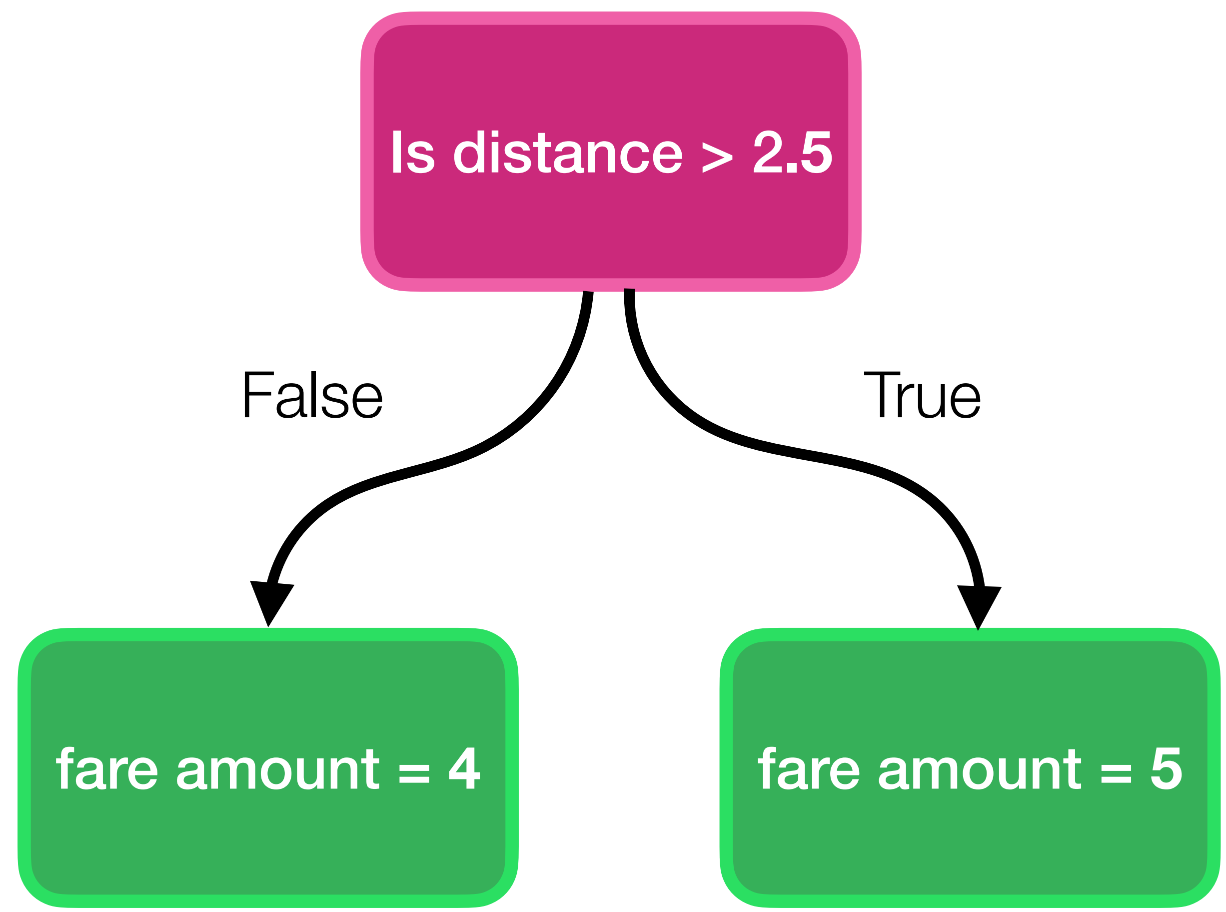


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$





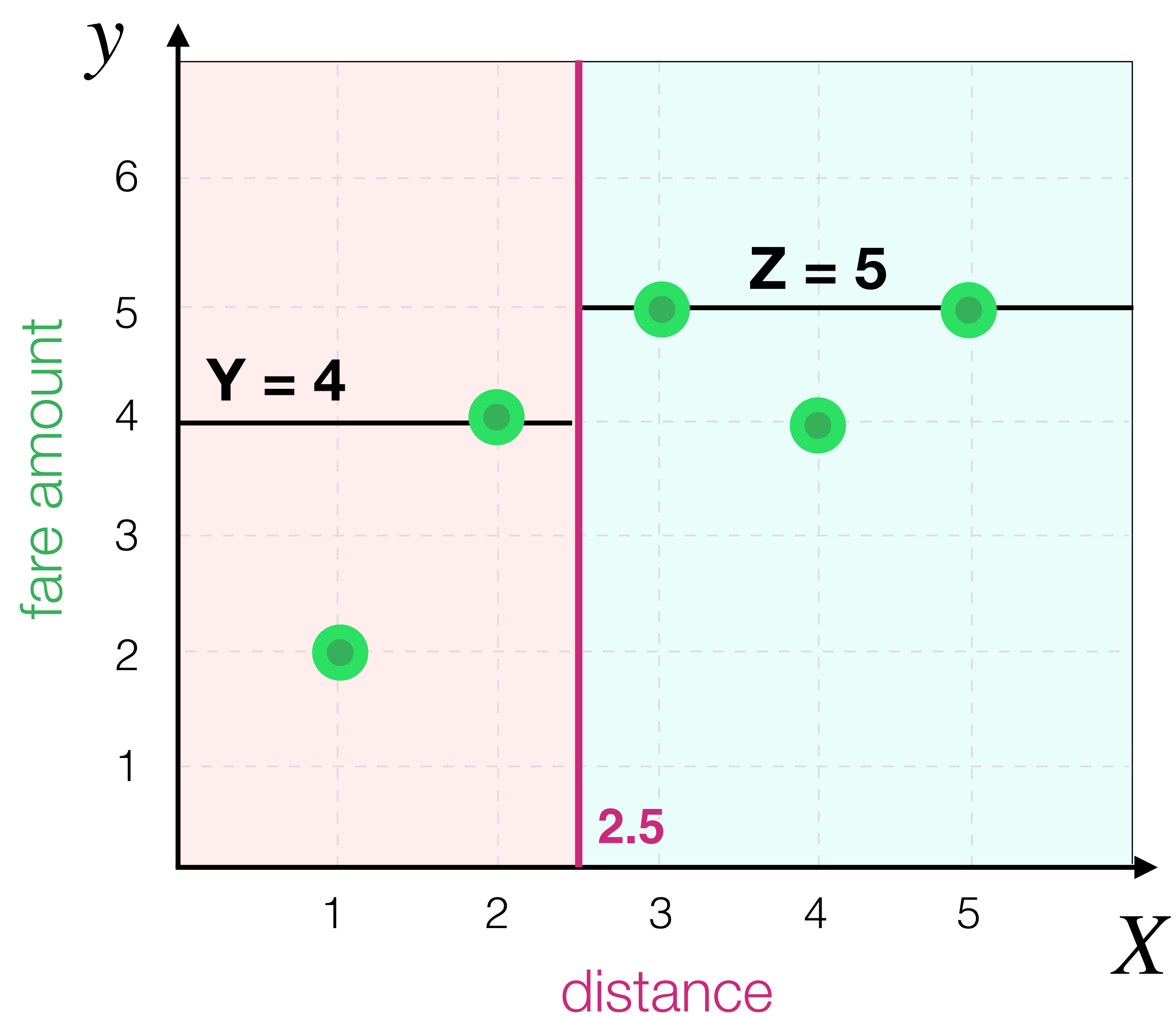
What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

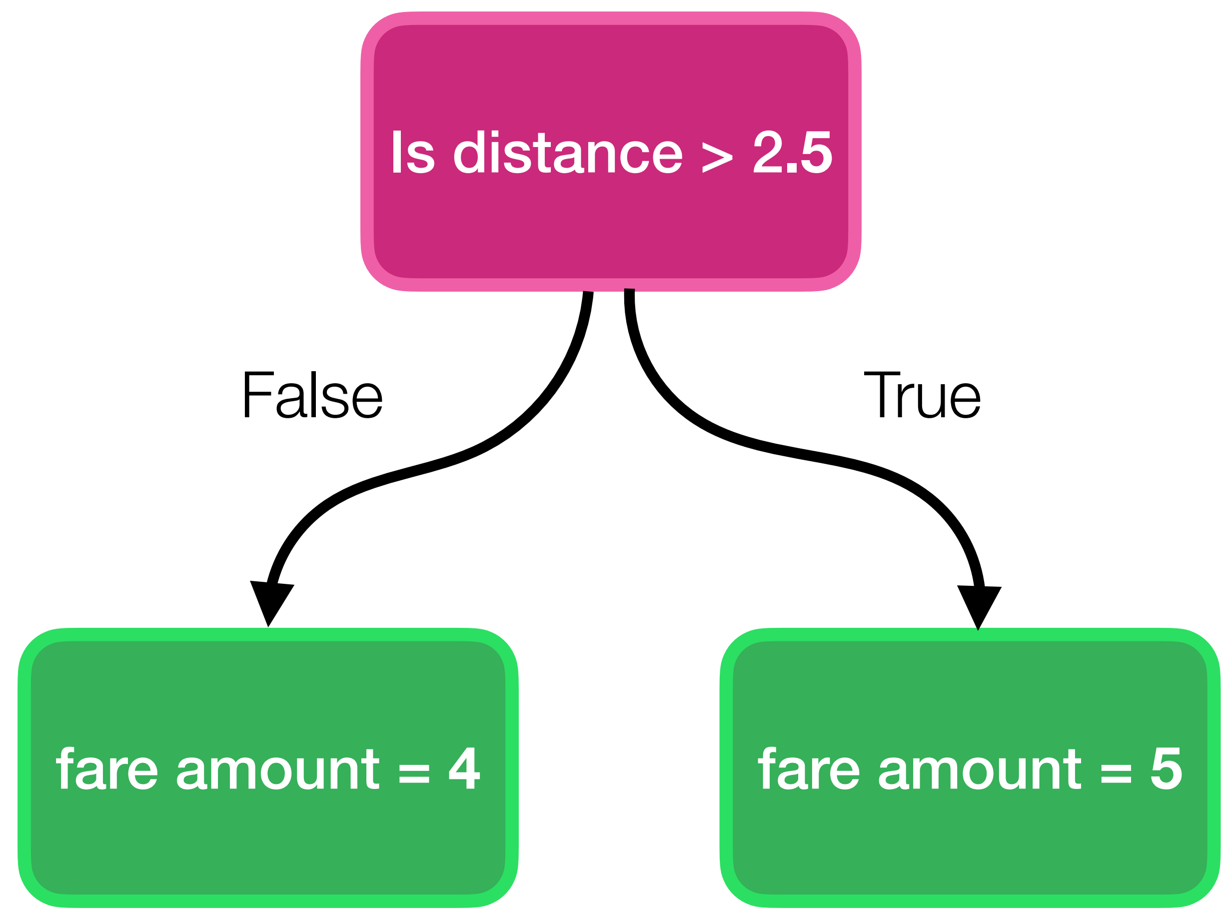
Decision Tree Algorithm

real value

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

predicted value

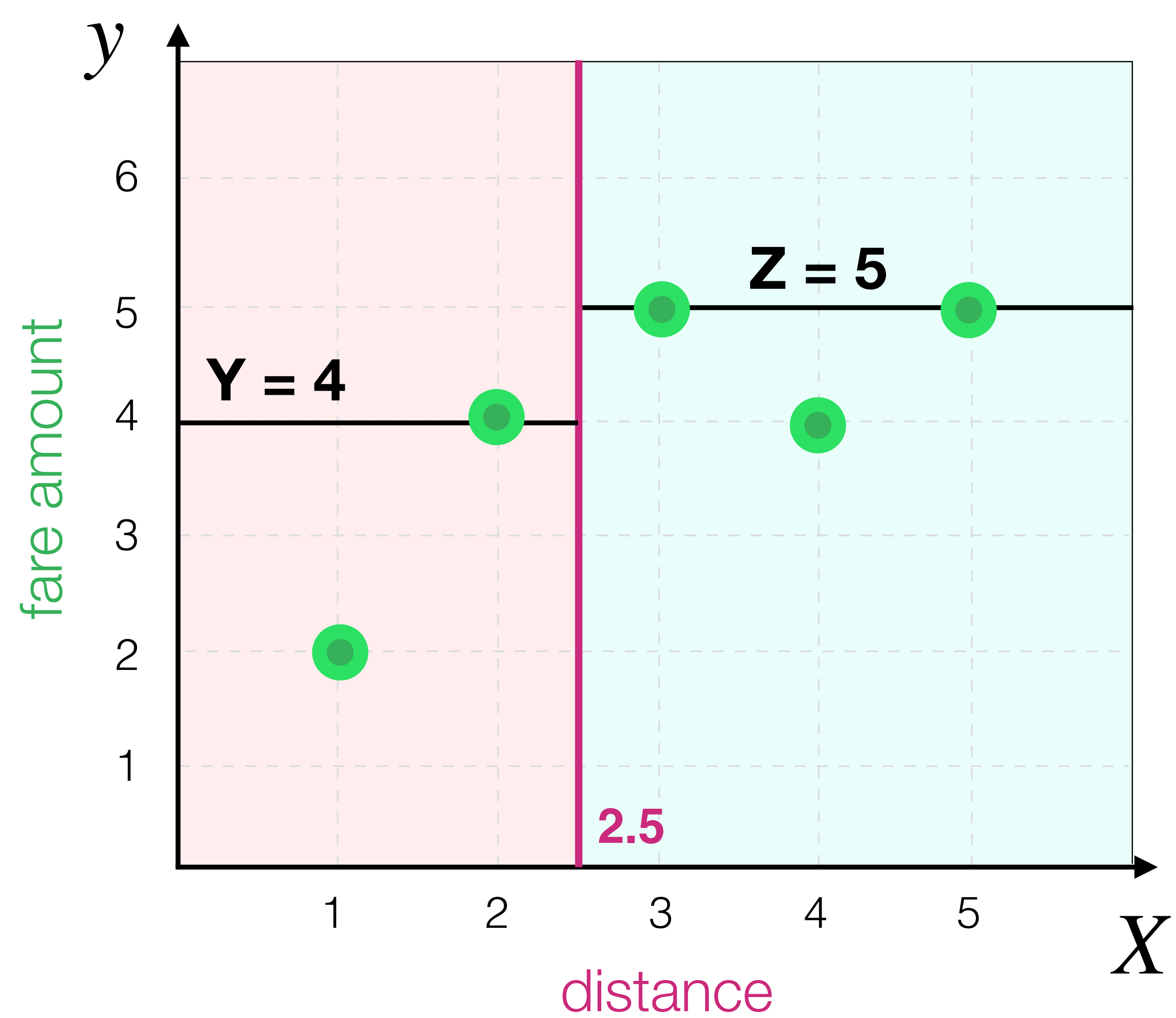


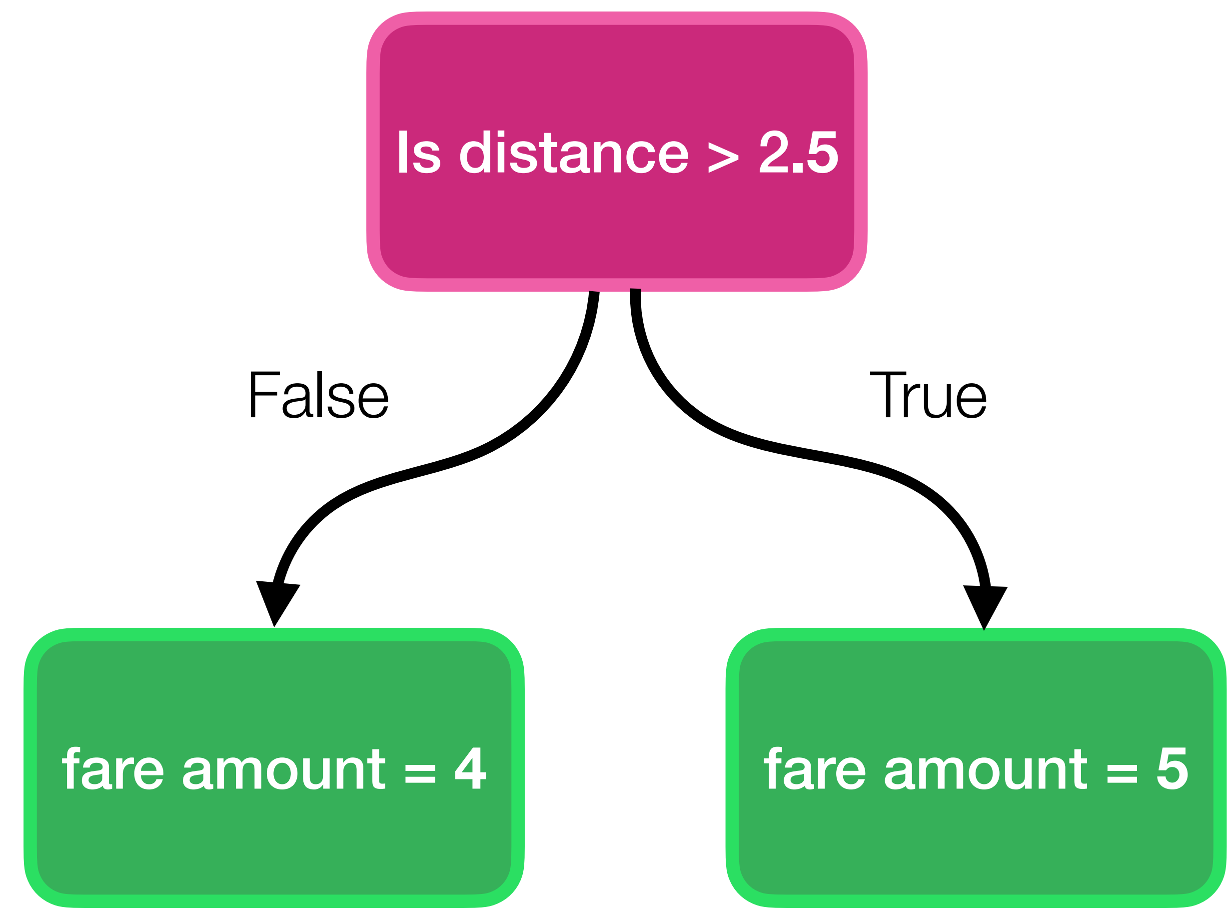


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(y_1-\hat{y}_1)^2+(y_2-\hat{y}_2)^2+(y_3-\hat{y}_3)^2+(y_4-\hat{y}_4)^2+(y_5-\hat{y}_5)^2}{5}$$

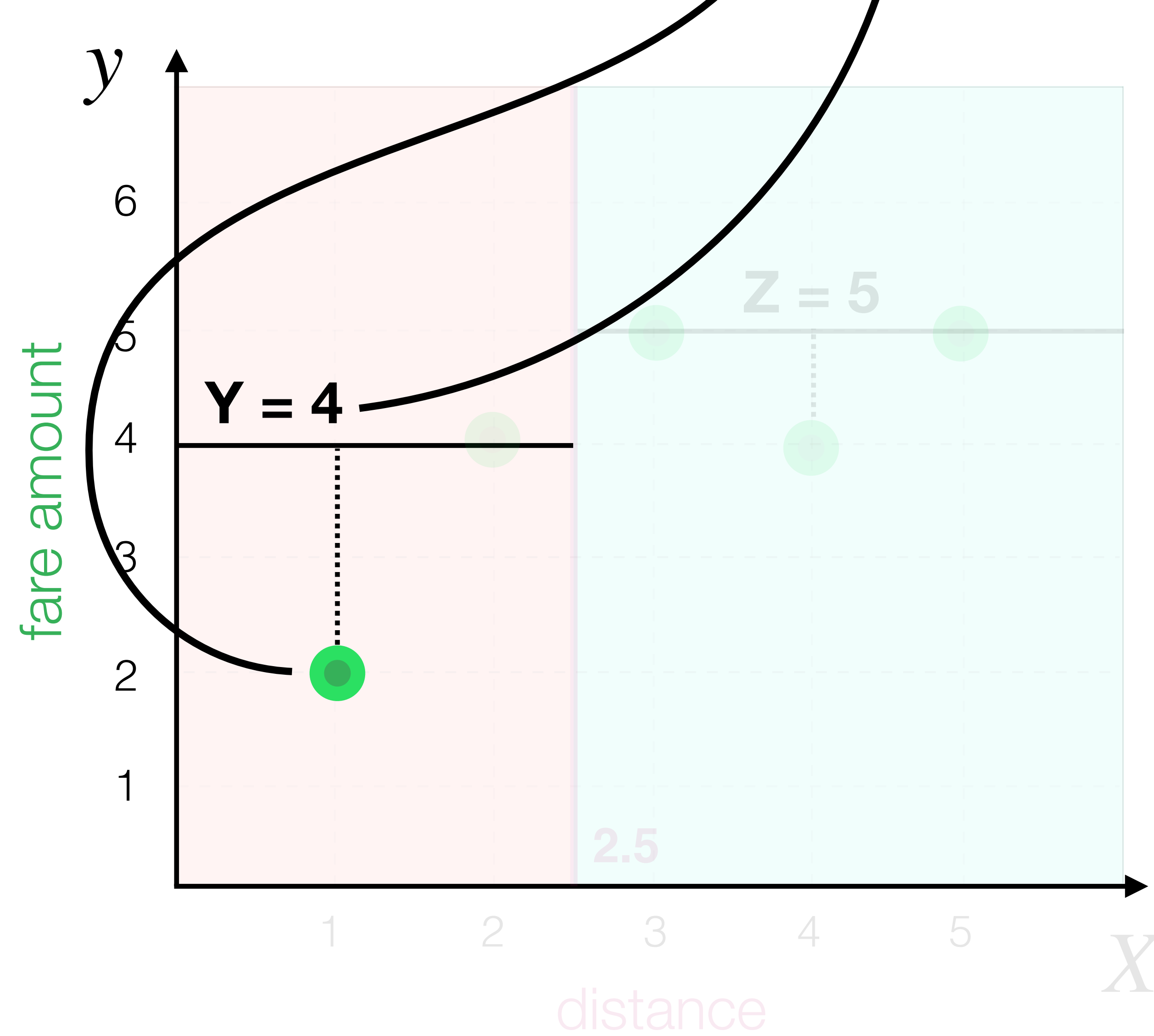


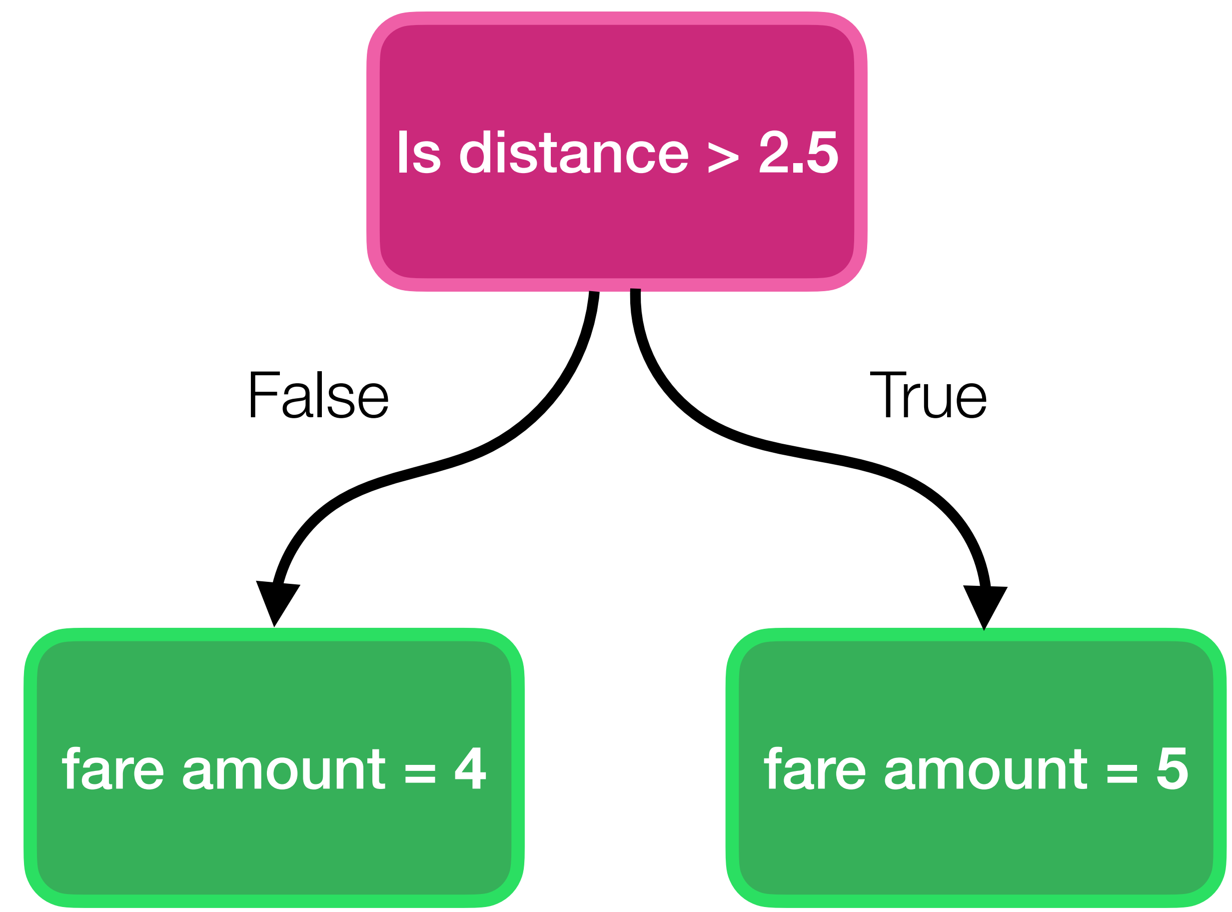


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(2-4)^2 + (y_2-\hat{y}_2)^2 + (y_3-\hat{y}_3)^2 + (y_4-\hat{y}_4)^2 + (y_5-\hat{y}_5)^2}{5}$$





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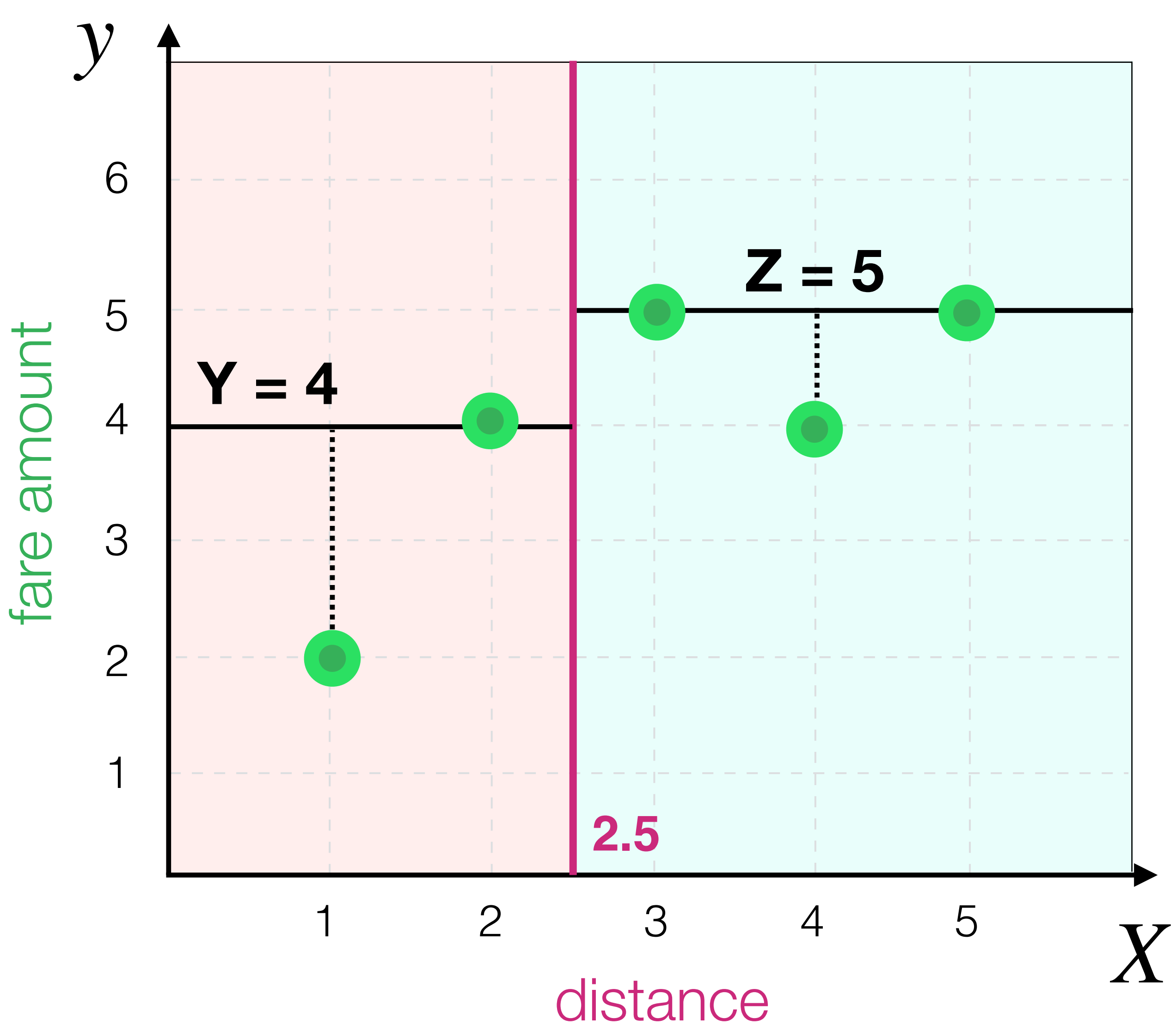
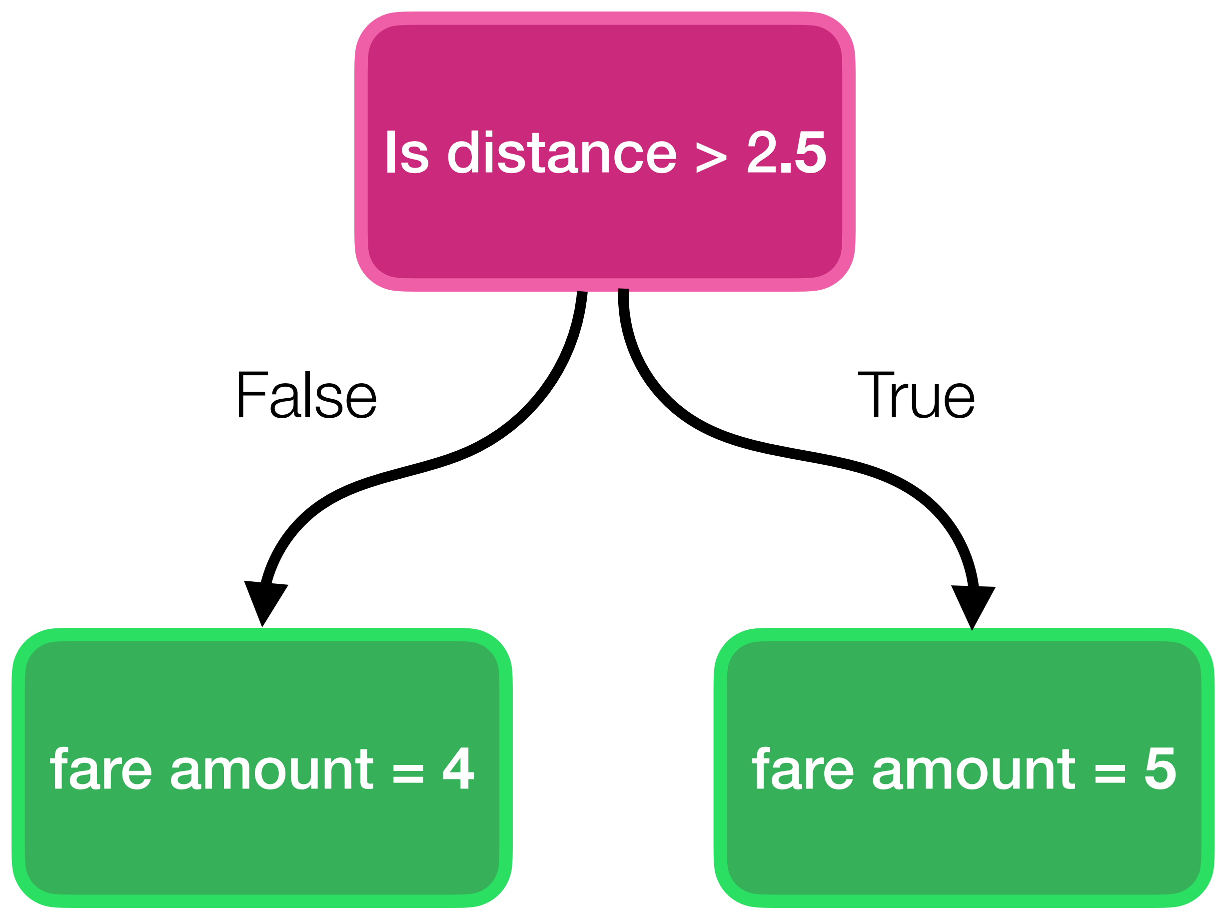
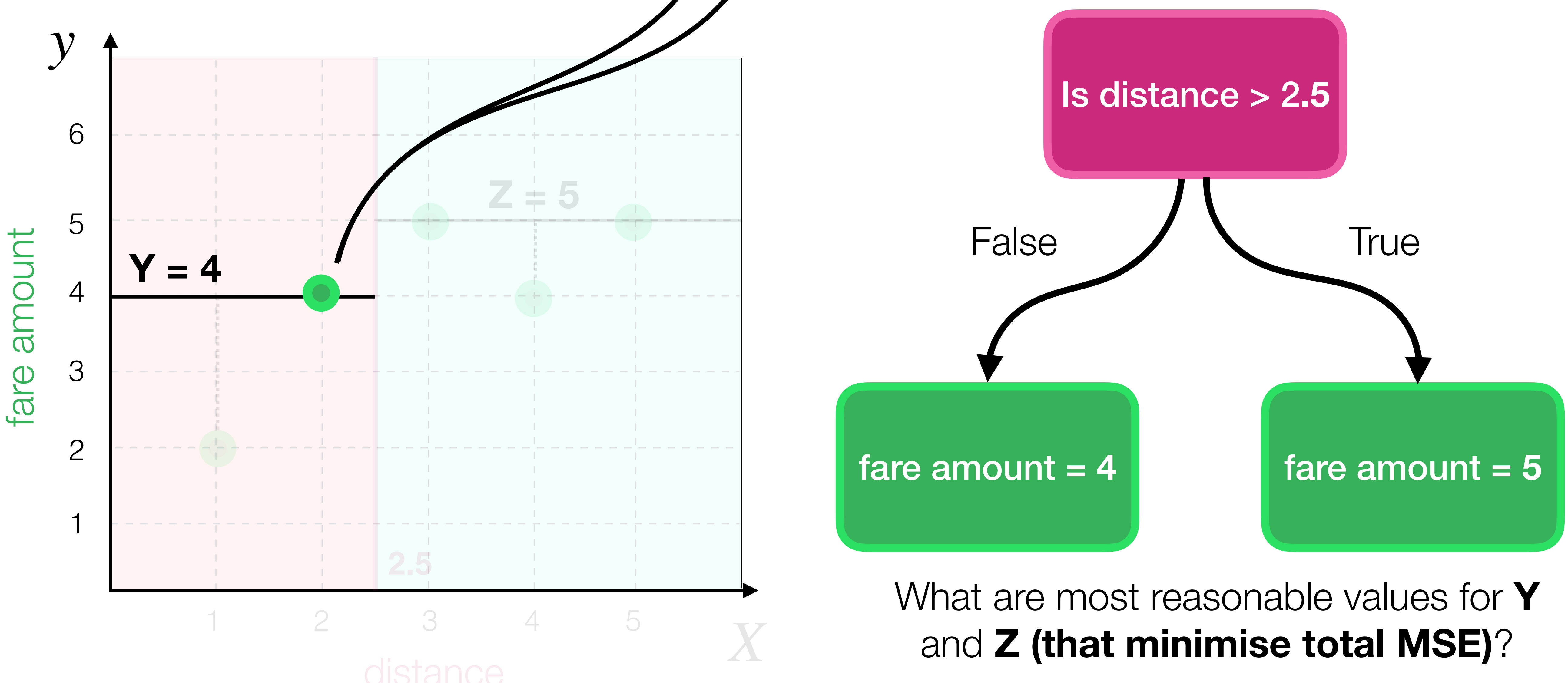




What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(2-4)^2 + (4-4)^2 + (y_3-\hat{y}_3)^2 + (y_4-\hat{y}_4)^2 + (y_5-\hat{y}_5)^2}{5}$$



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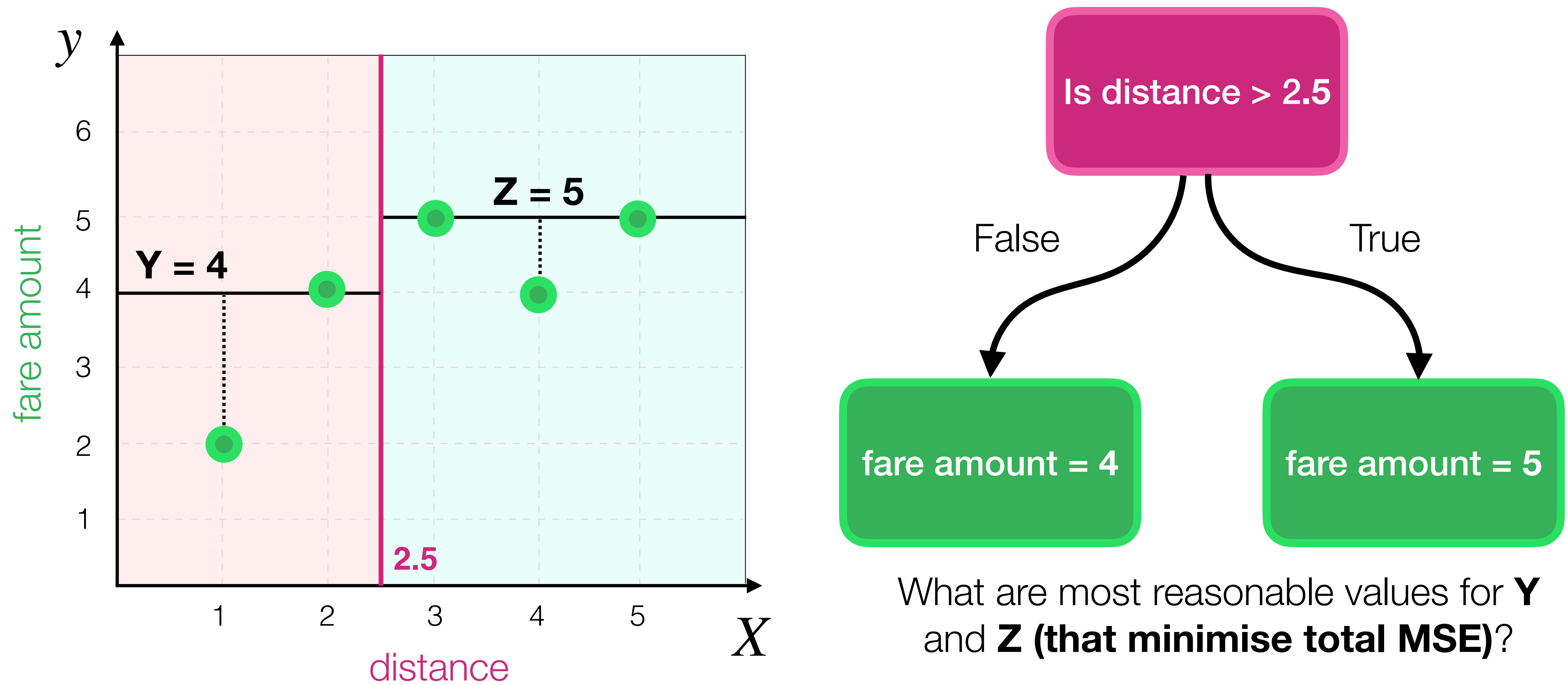

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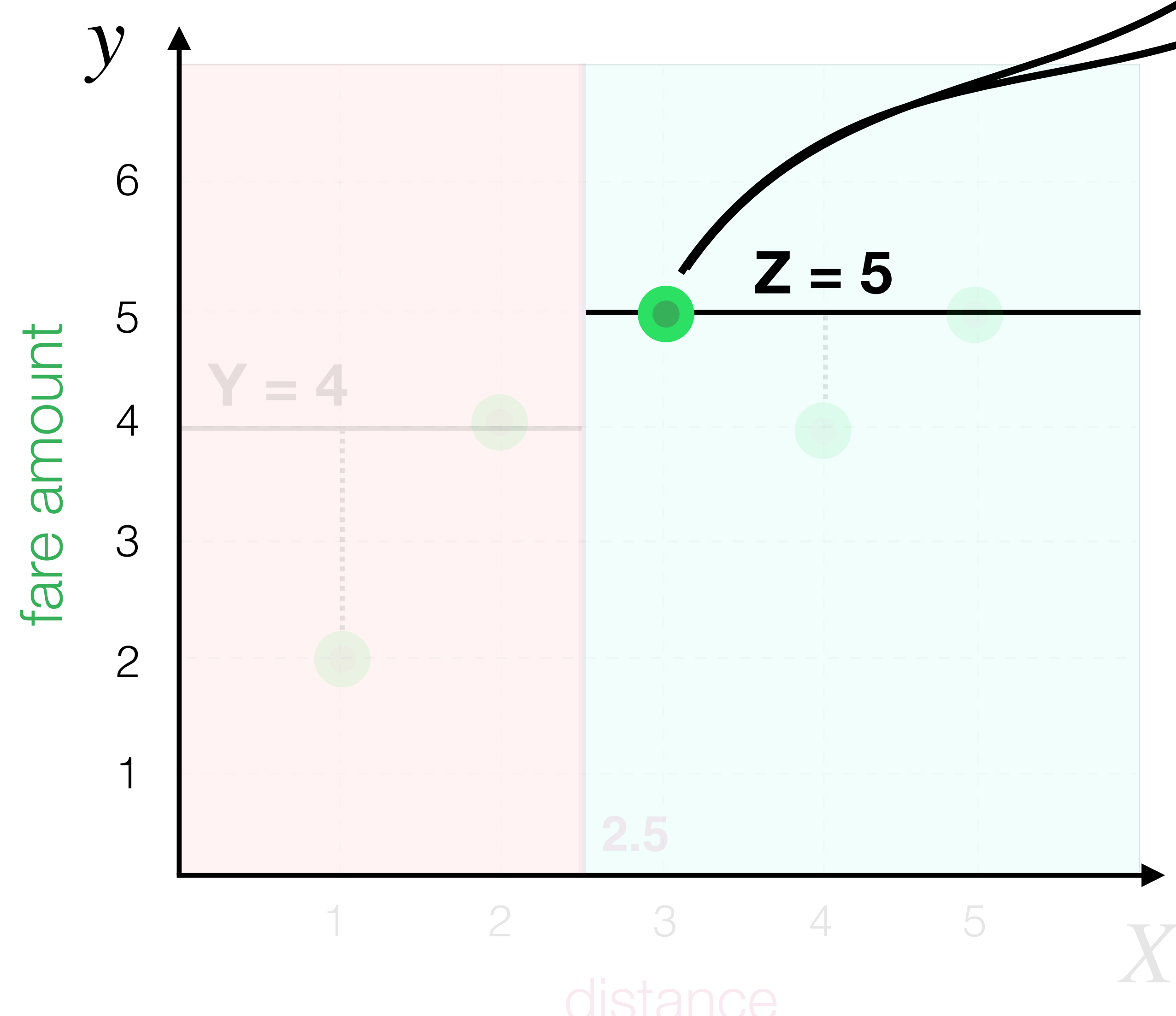
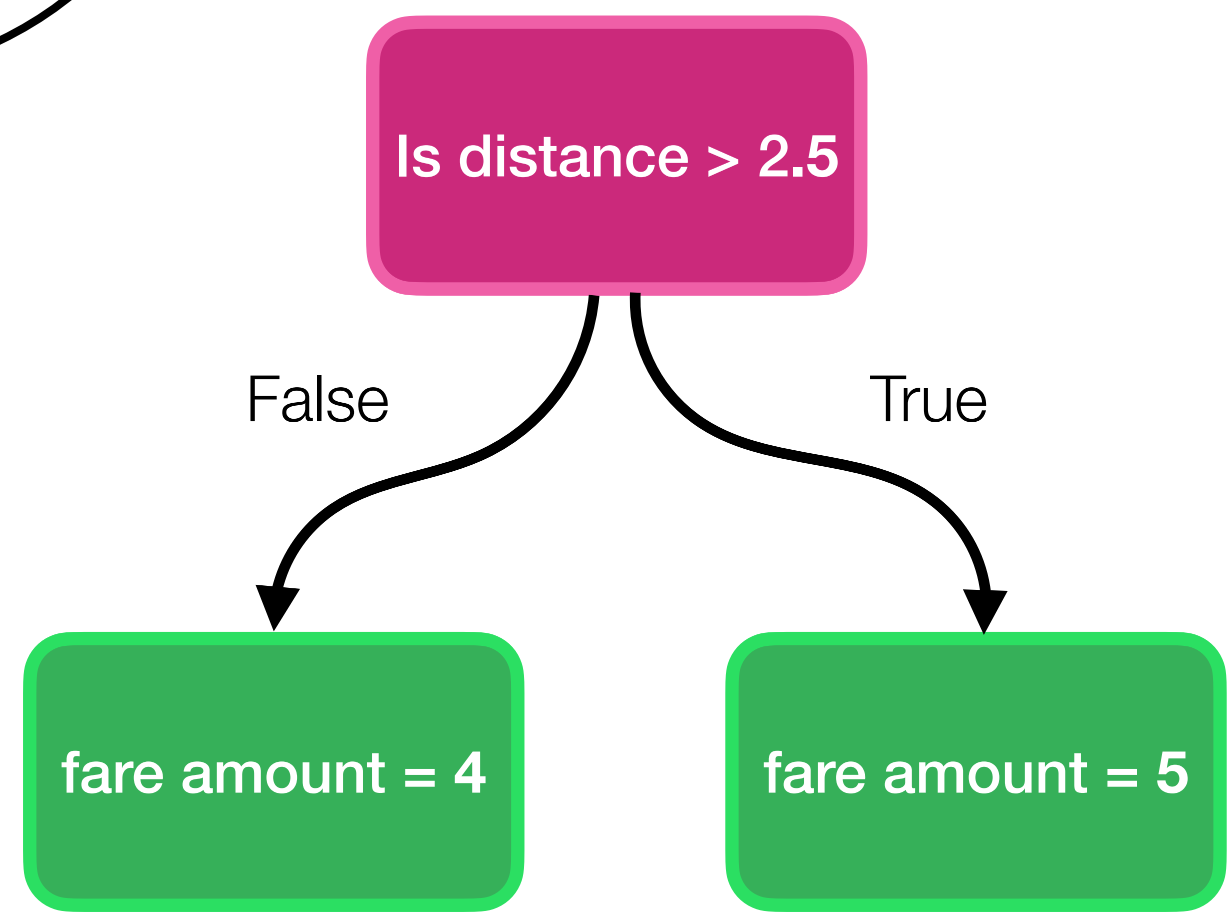
Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(2-4)^2 + (4-4)^2 + (5-5)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$





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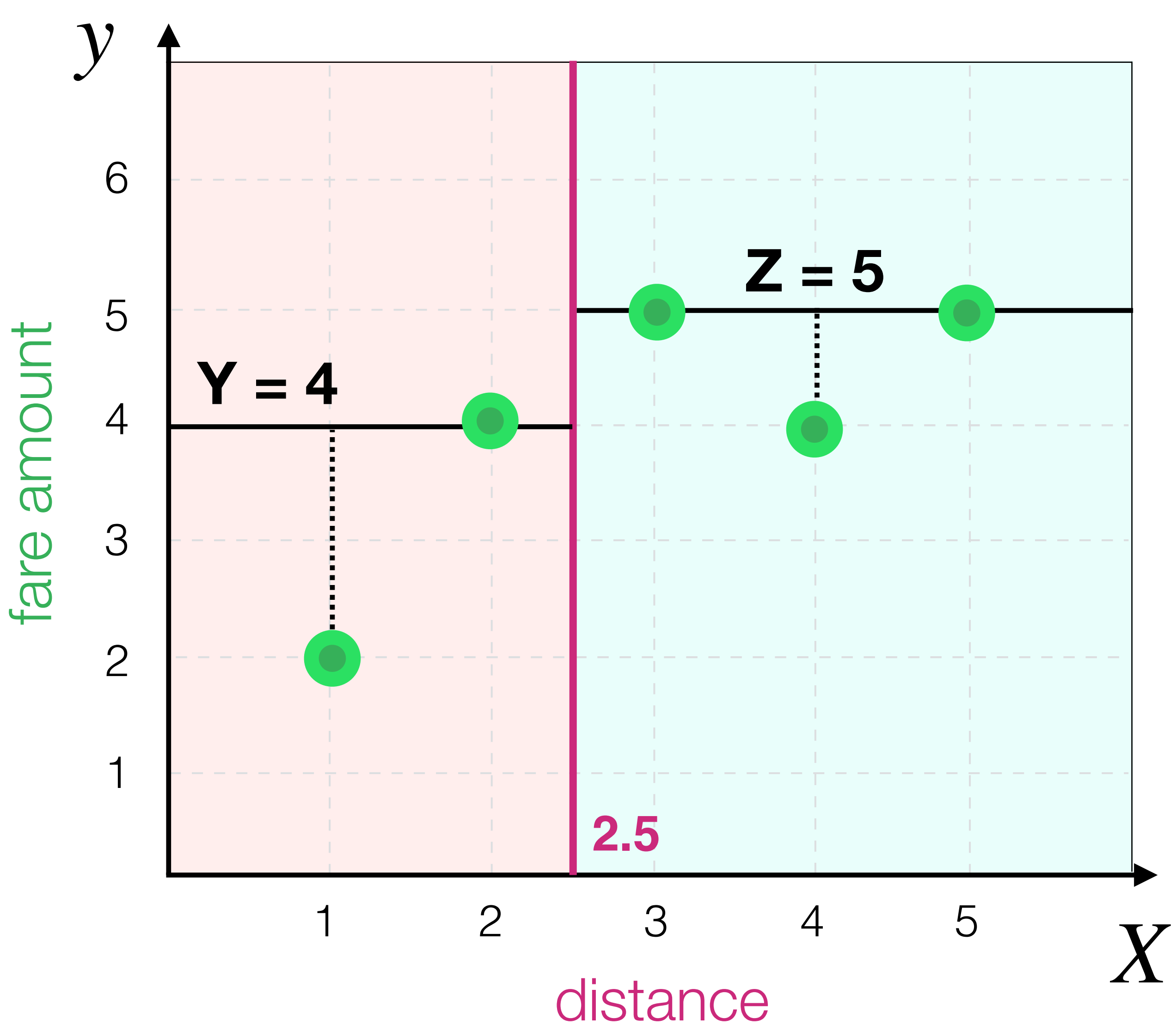

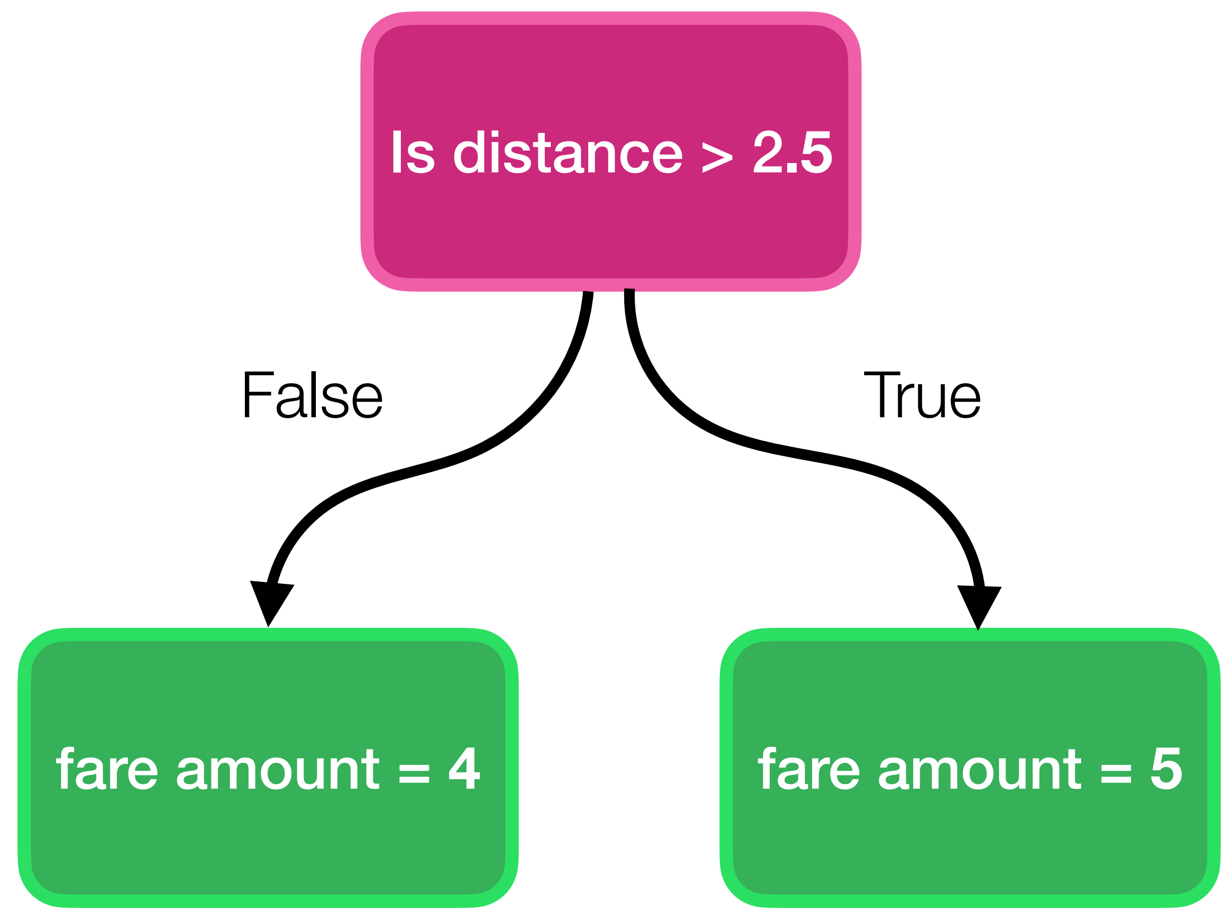

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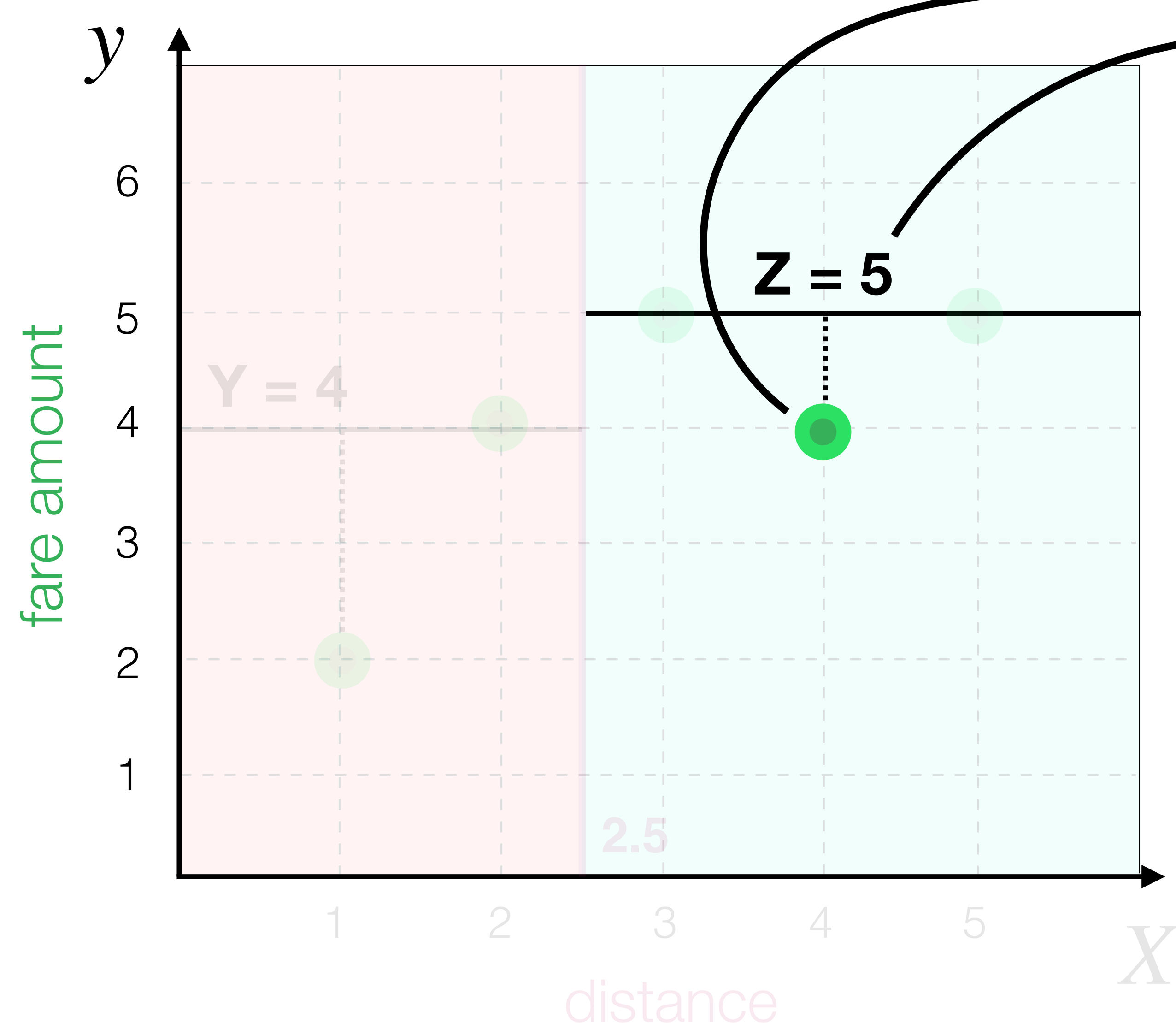

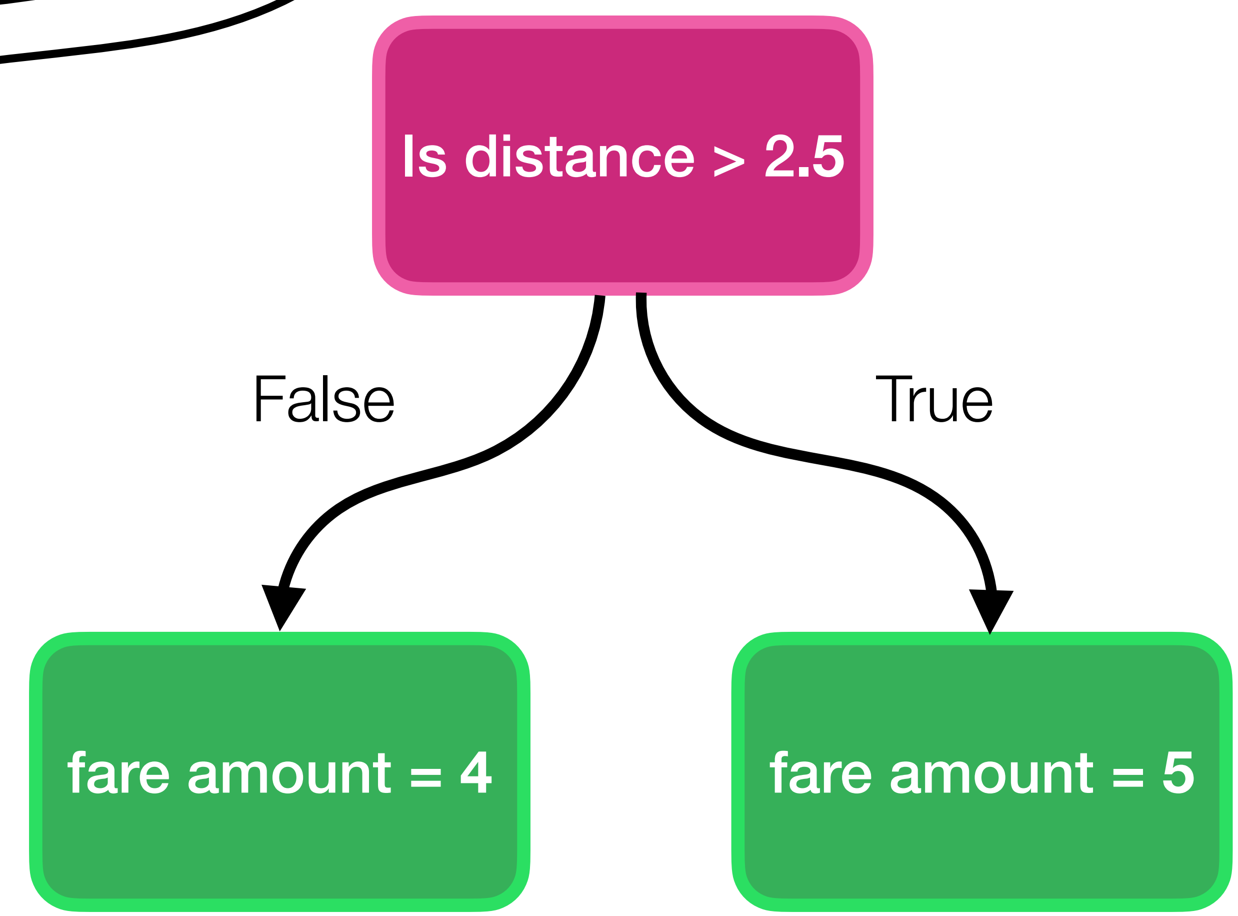

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Decision Tree Algorithm

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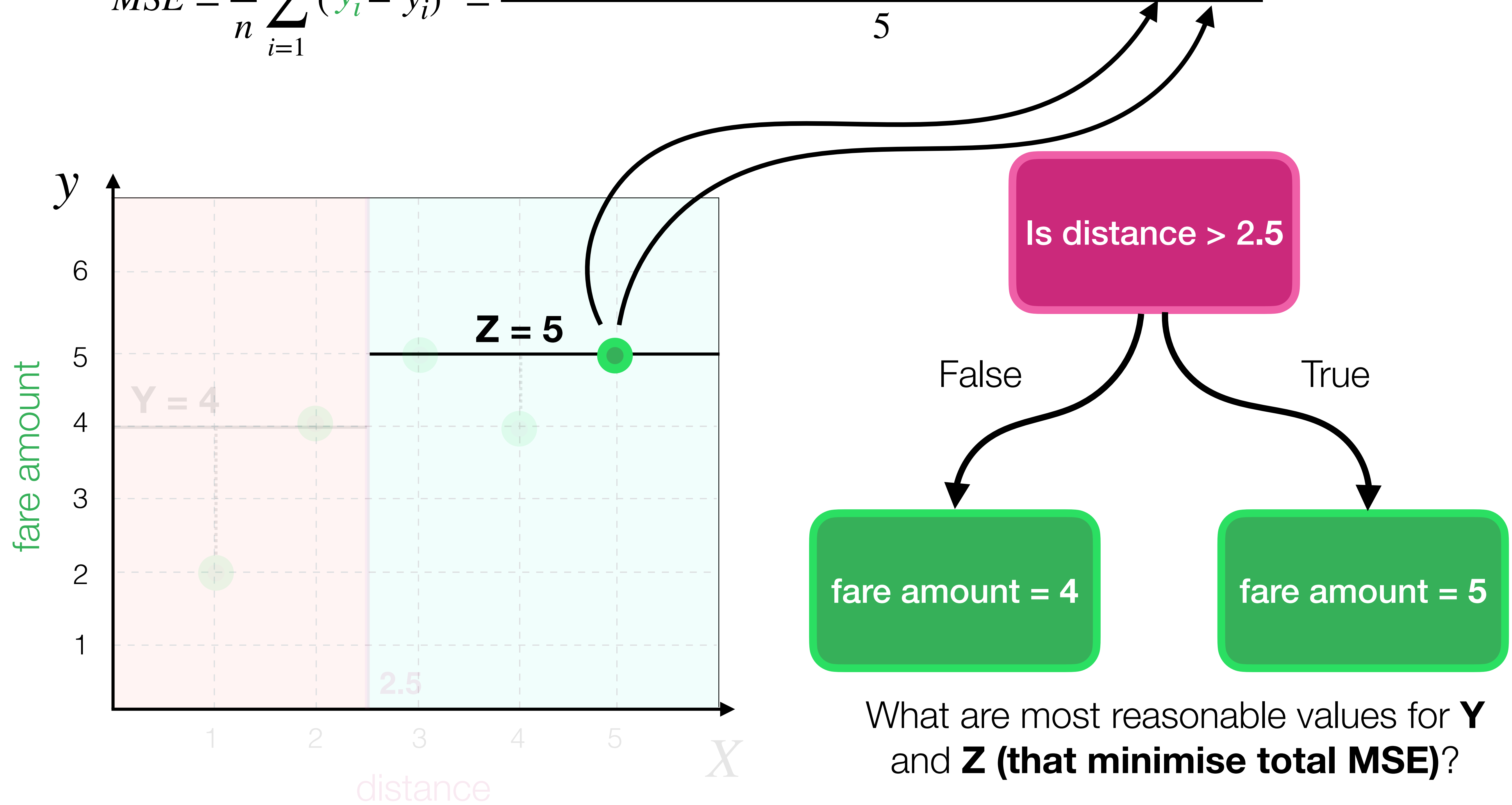

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Decision Tree Algorithm

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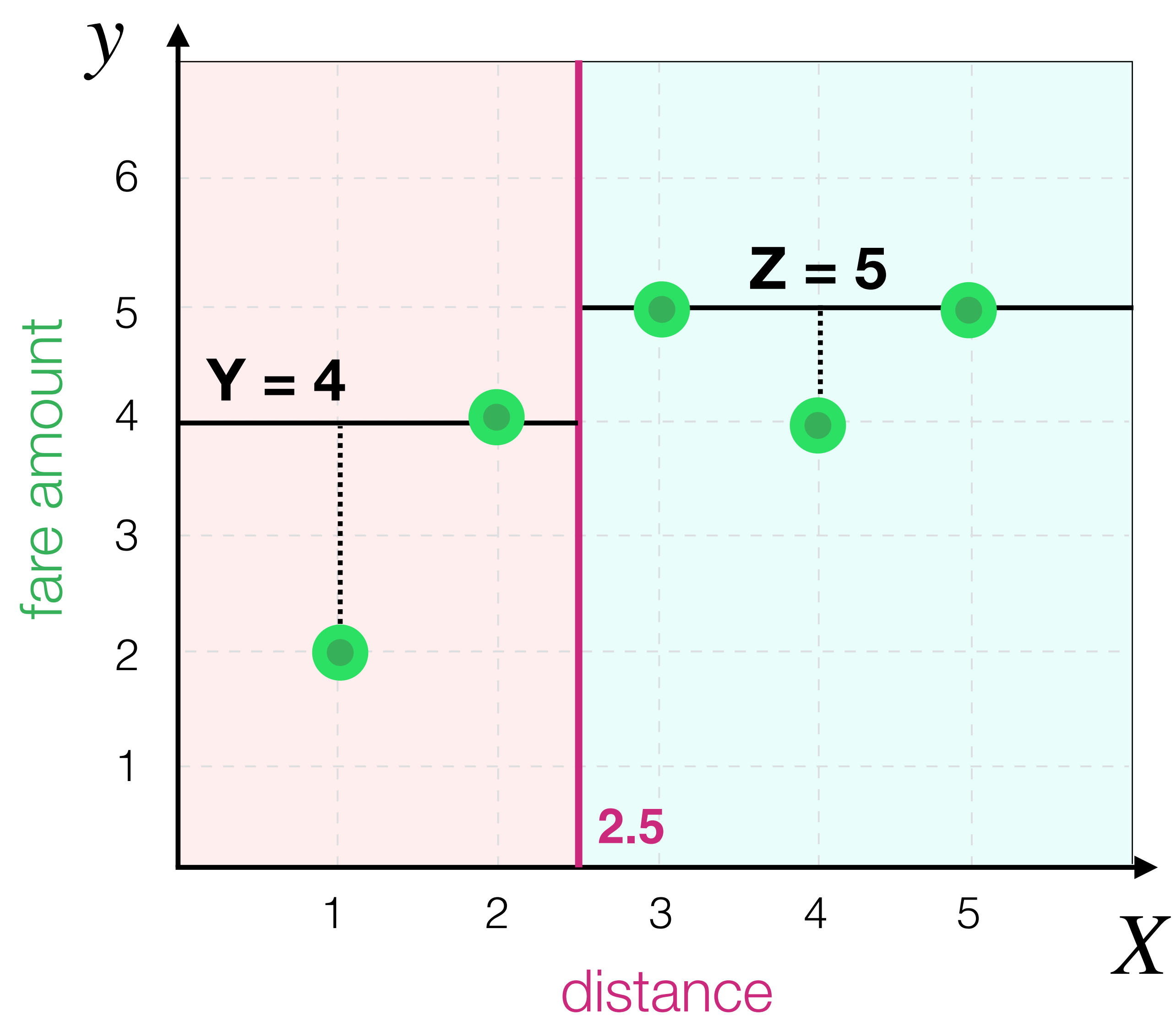
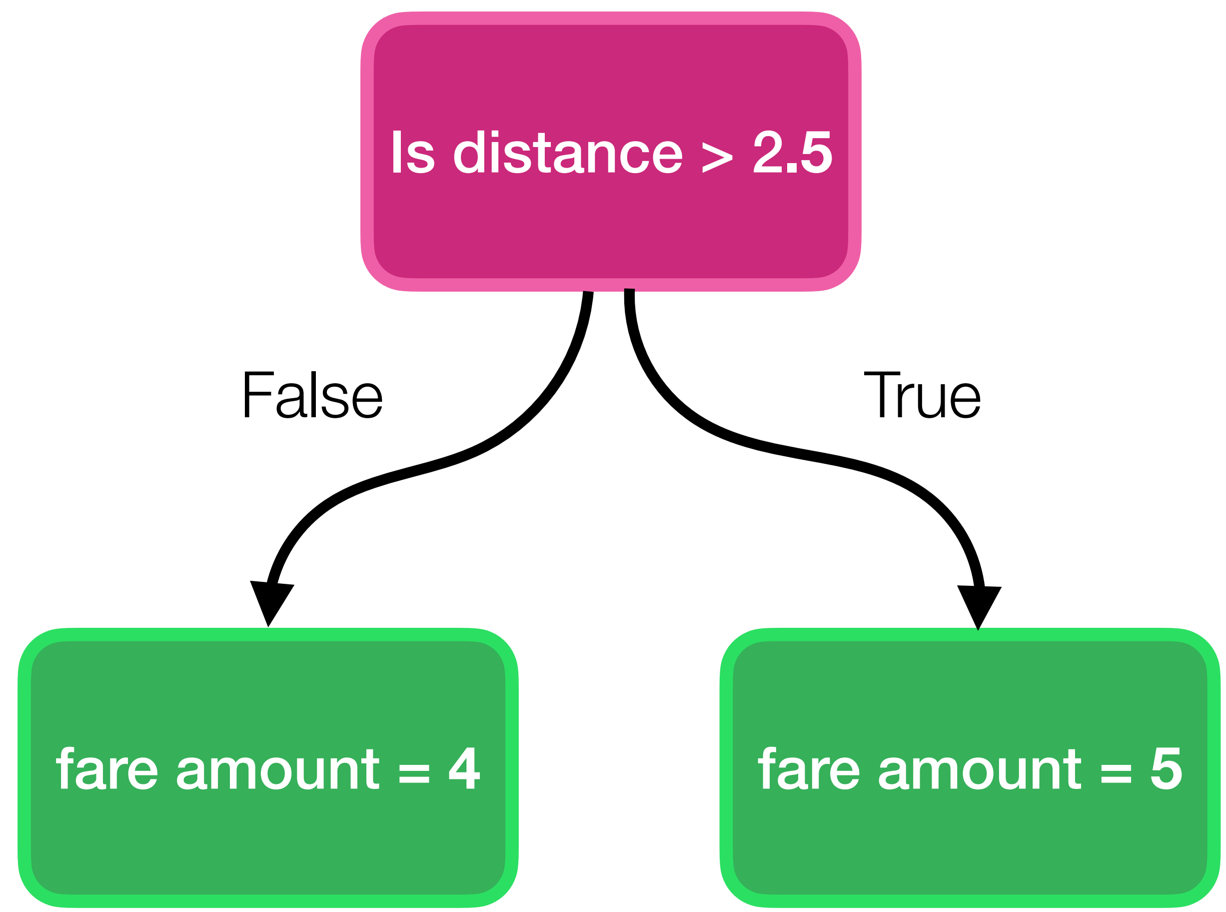
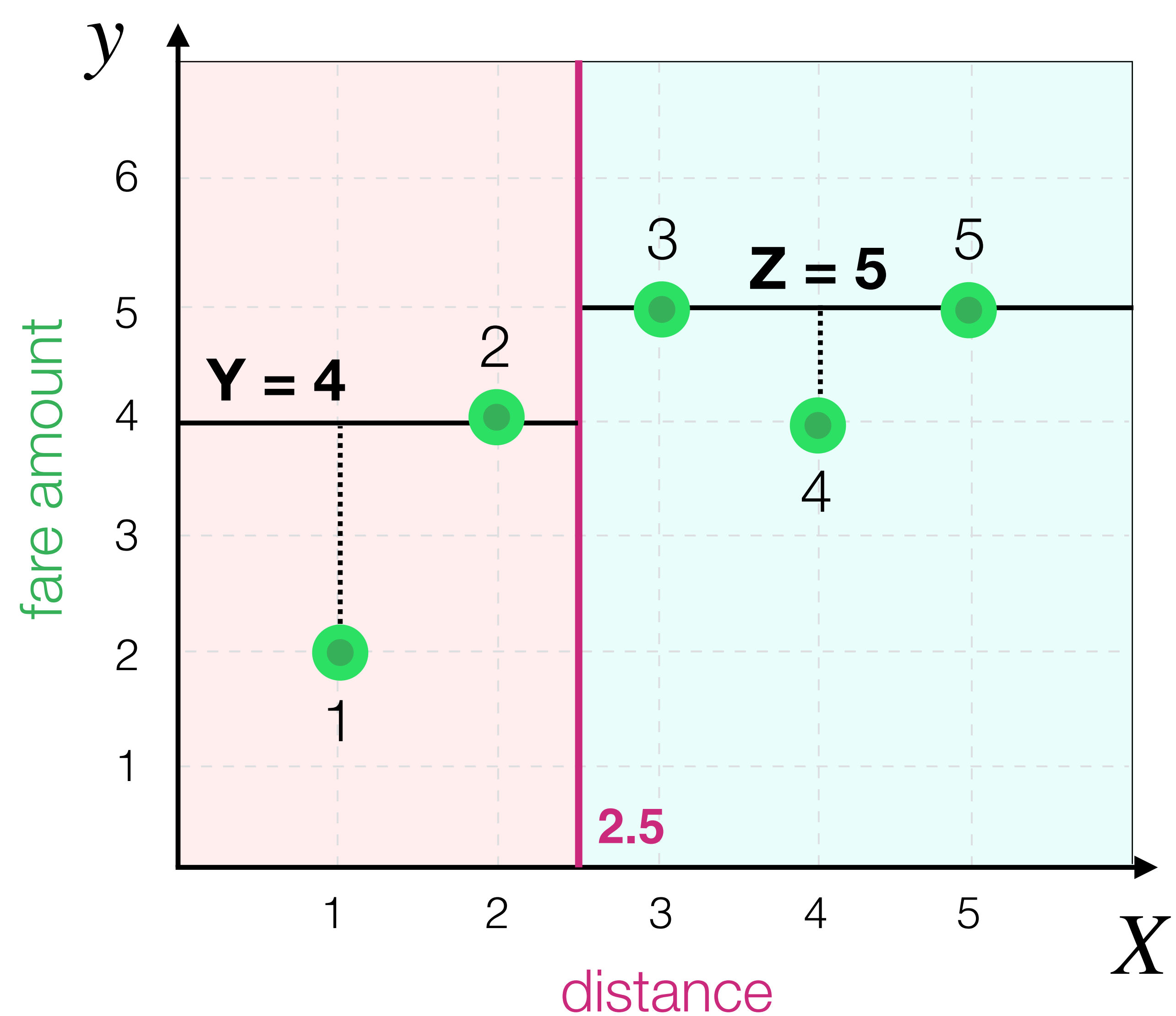


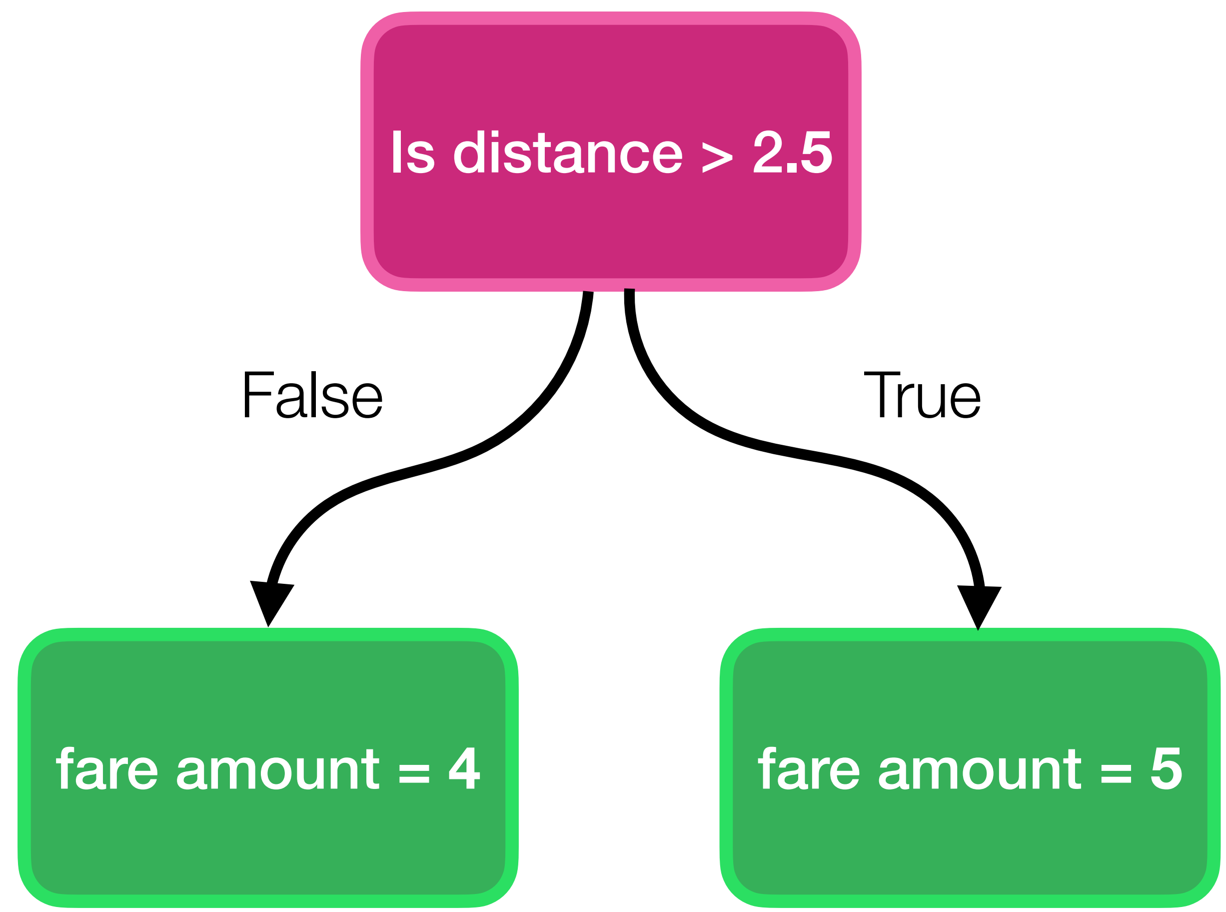


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(-2)^2 + (0)^2 + (0)^2 + (-1)^2 + (0)^2}{5}$$

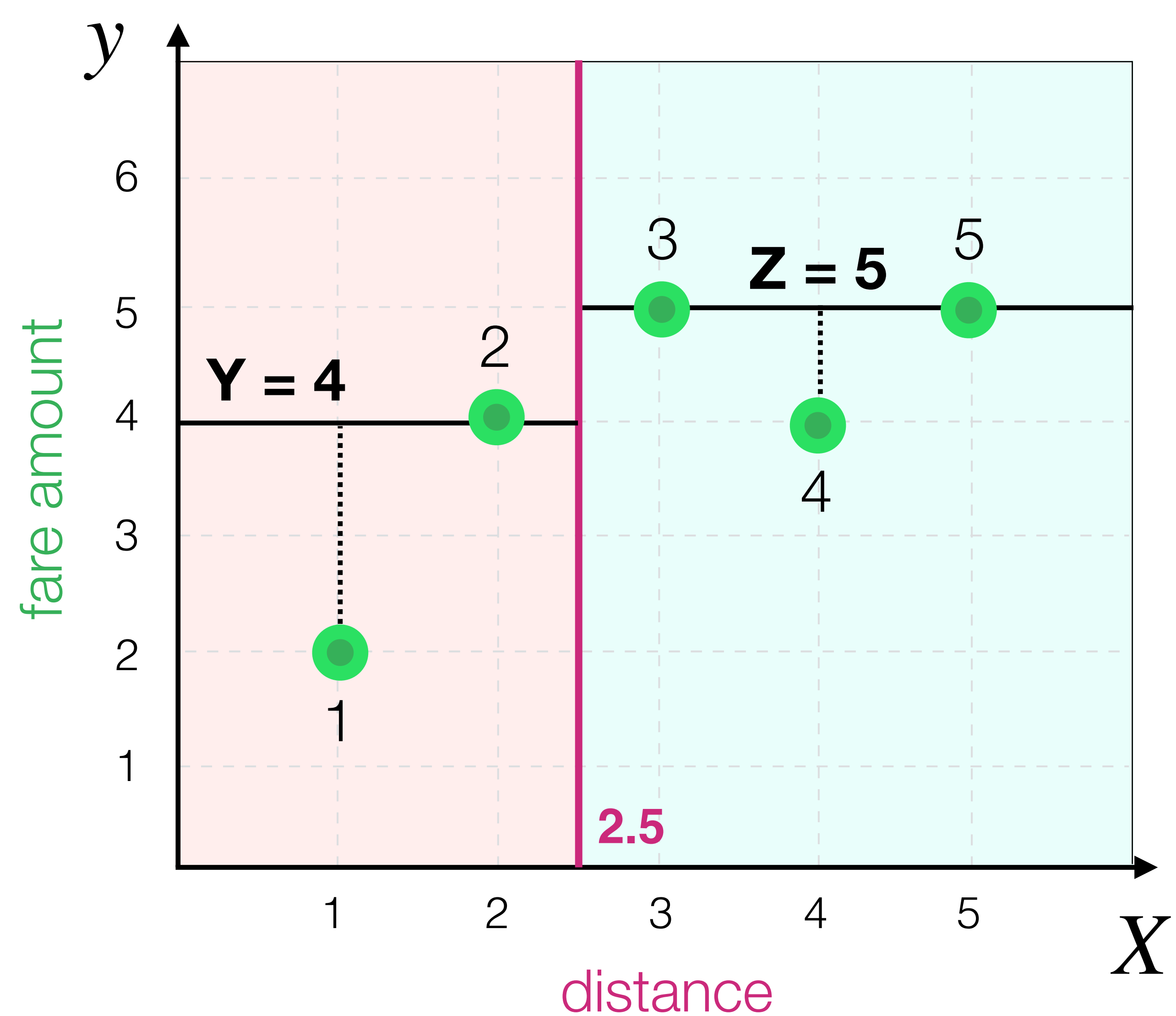


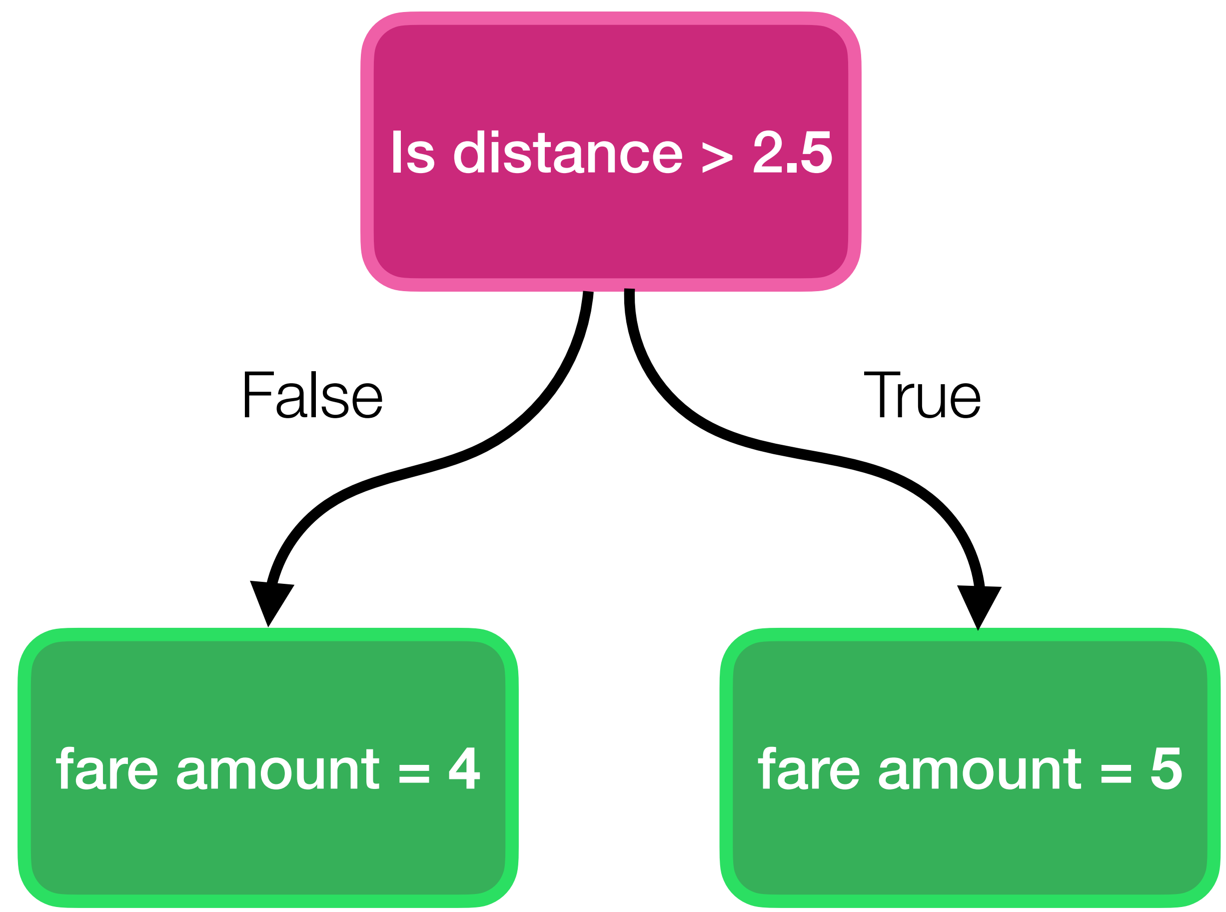


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{4+0+0+1+0}{5}$$

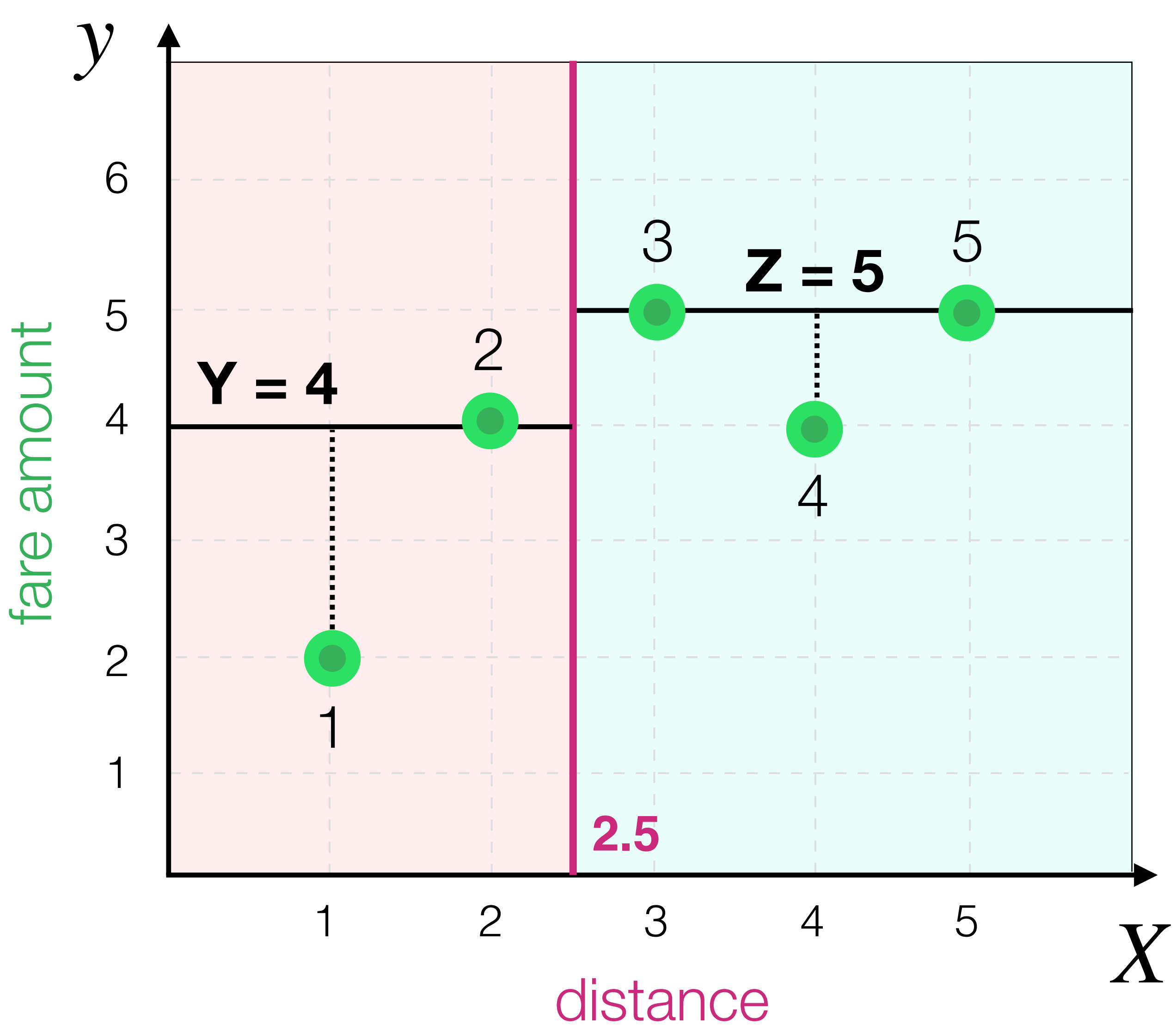


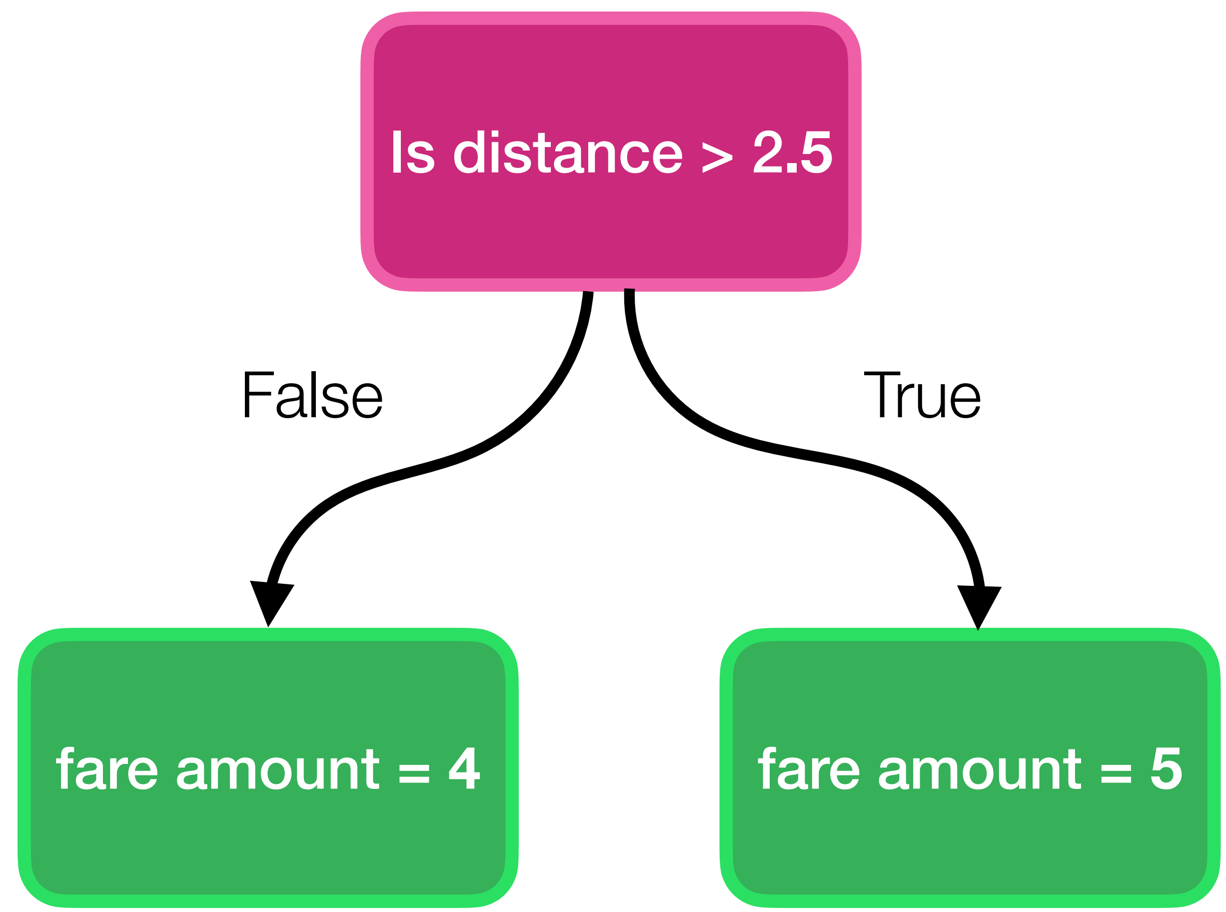


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{5}{5}$$

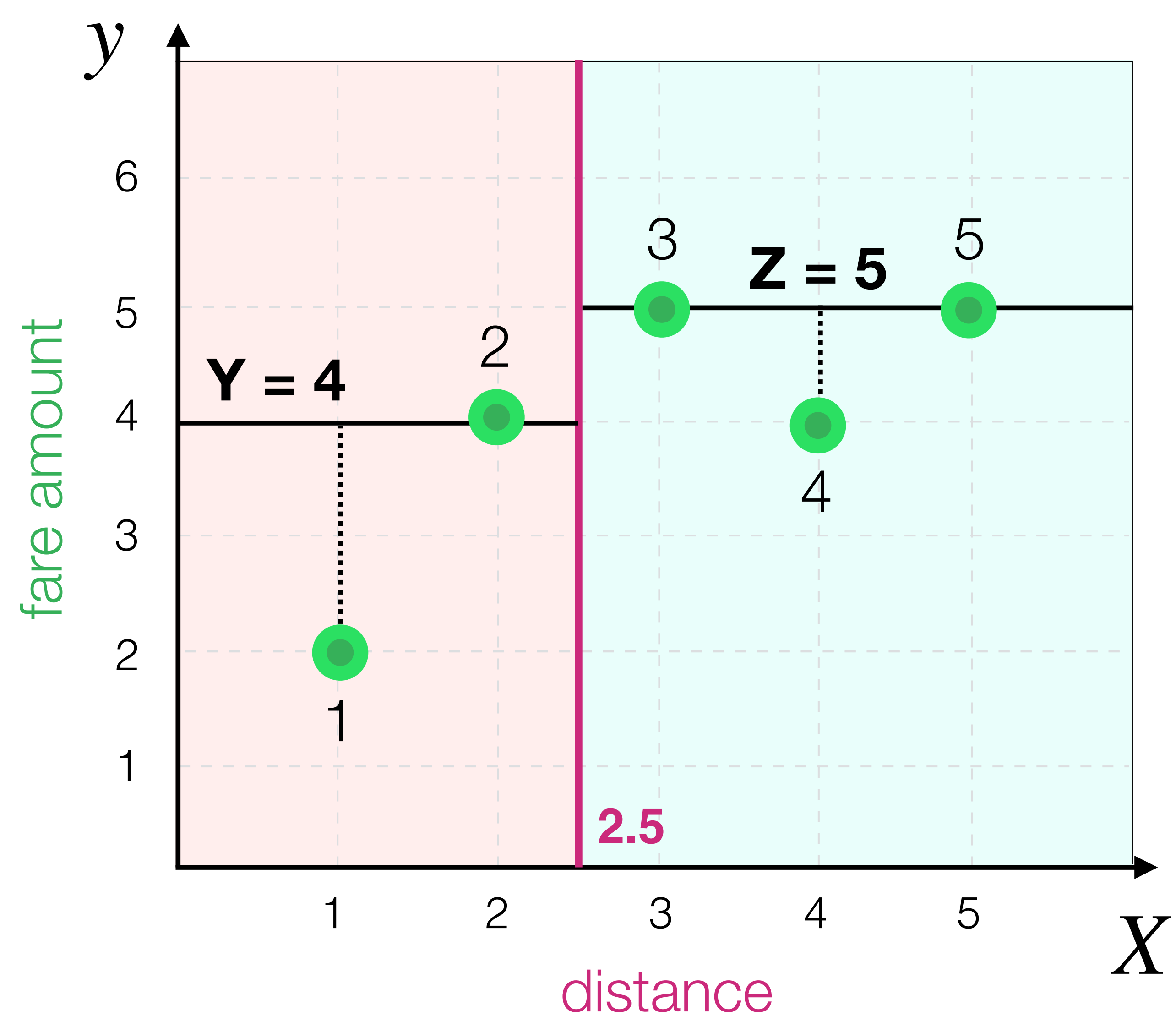


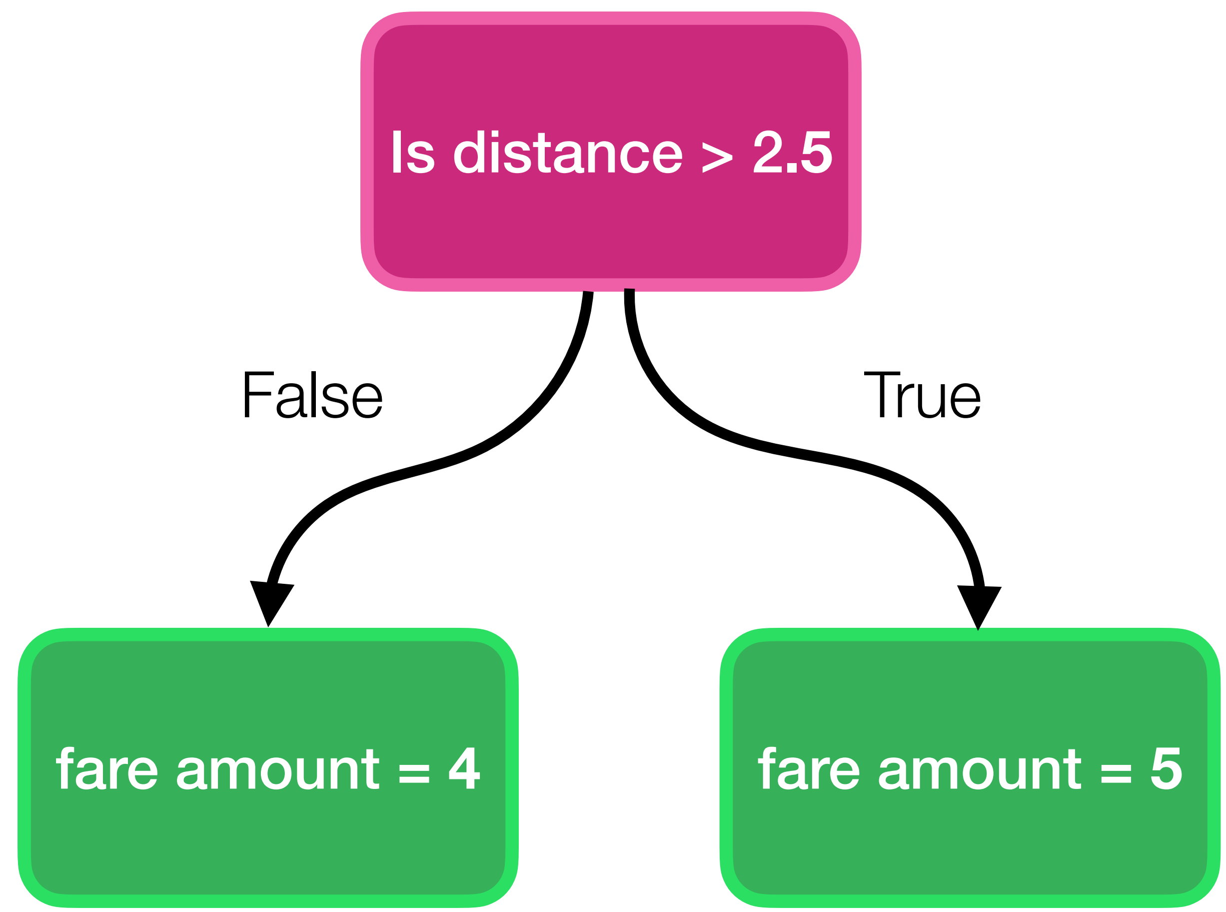


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = 1$$

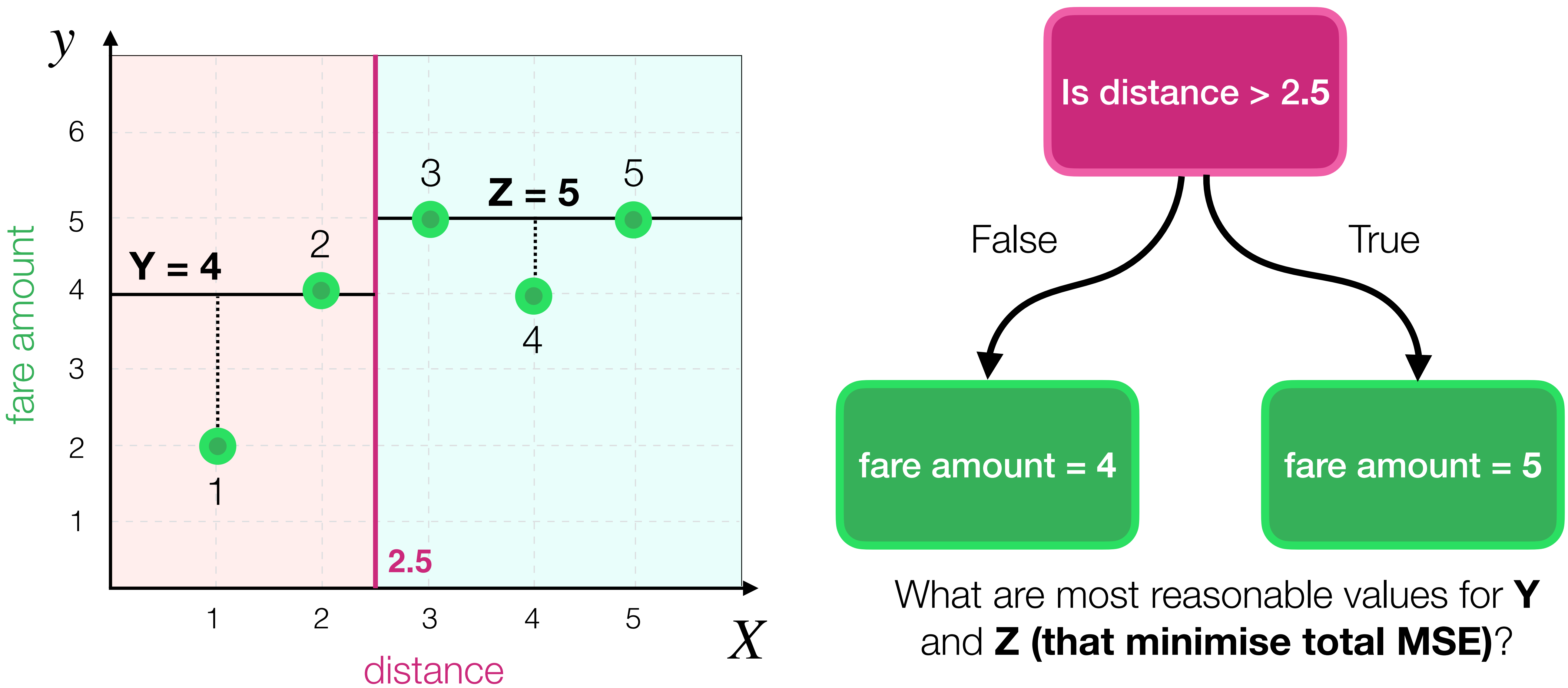




What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = 1$ so, if **X** = **2.5**, **Y** = **4** and **Z** = **5**, MSE is **1**

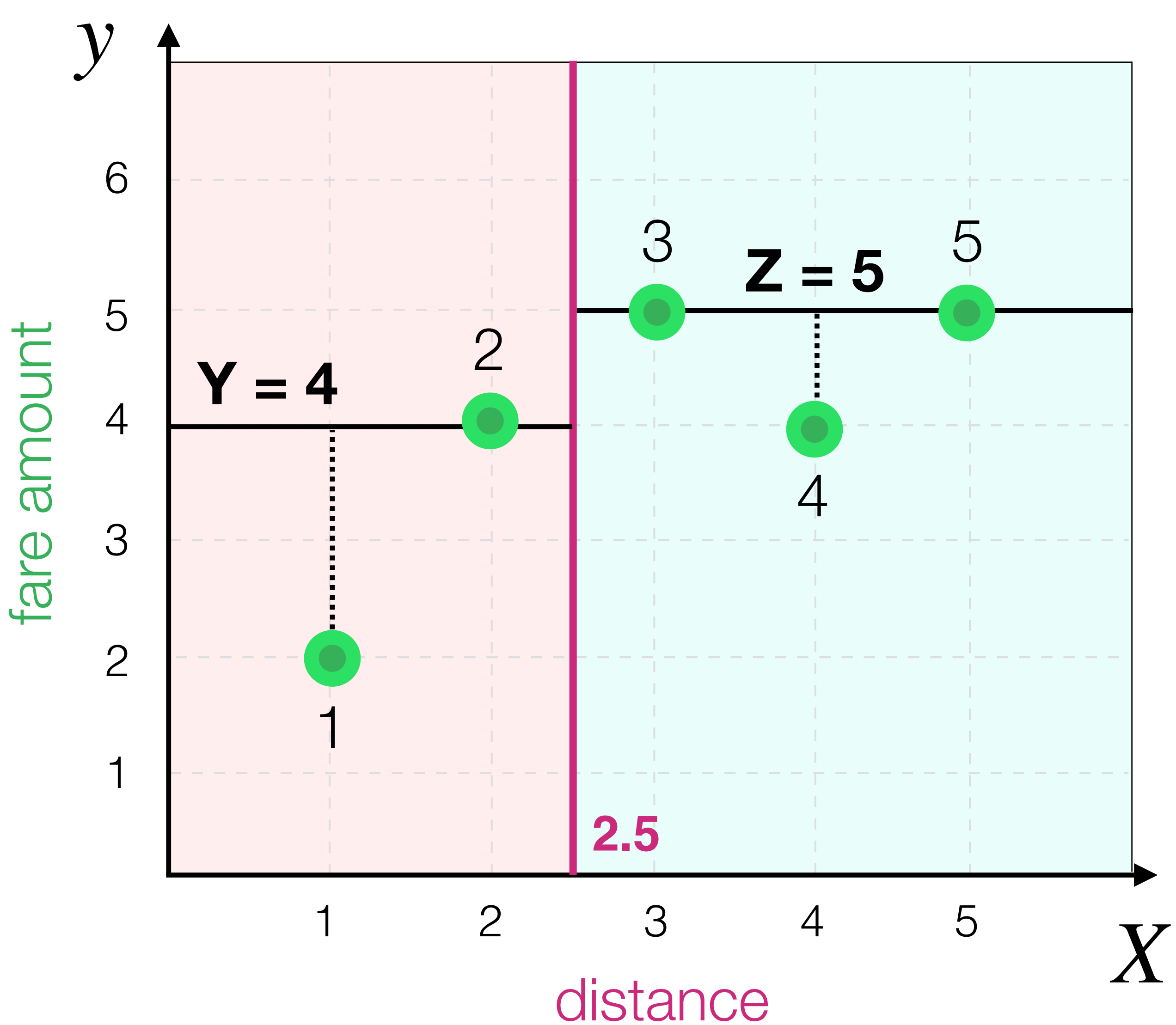


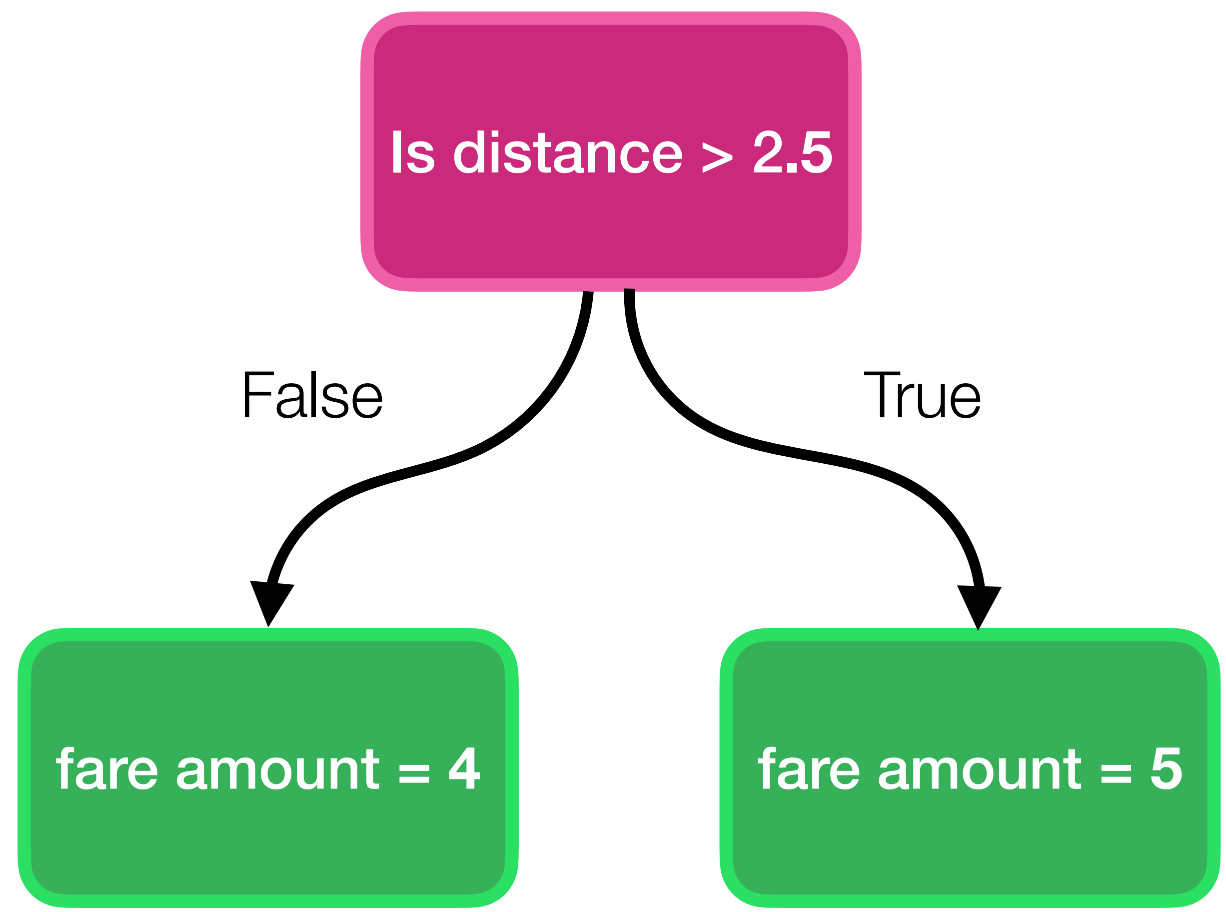
What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = 1$ so, if **X** = **2.5**, **Y** = **4** and **Z** = **5**, MSE is **1**

Can we find better **Y** and **Z**?

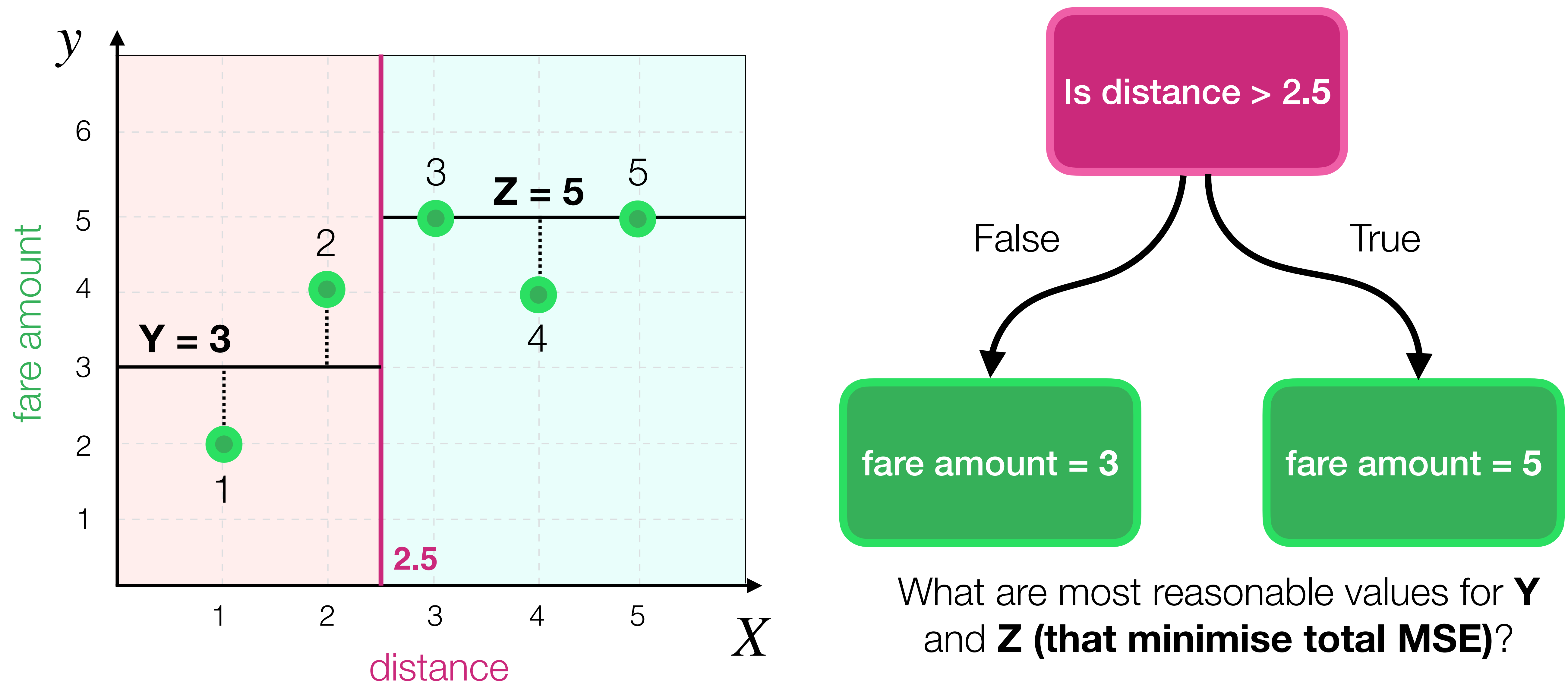




What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

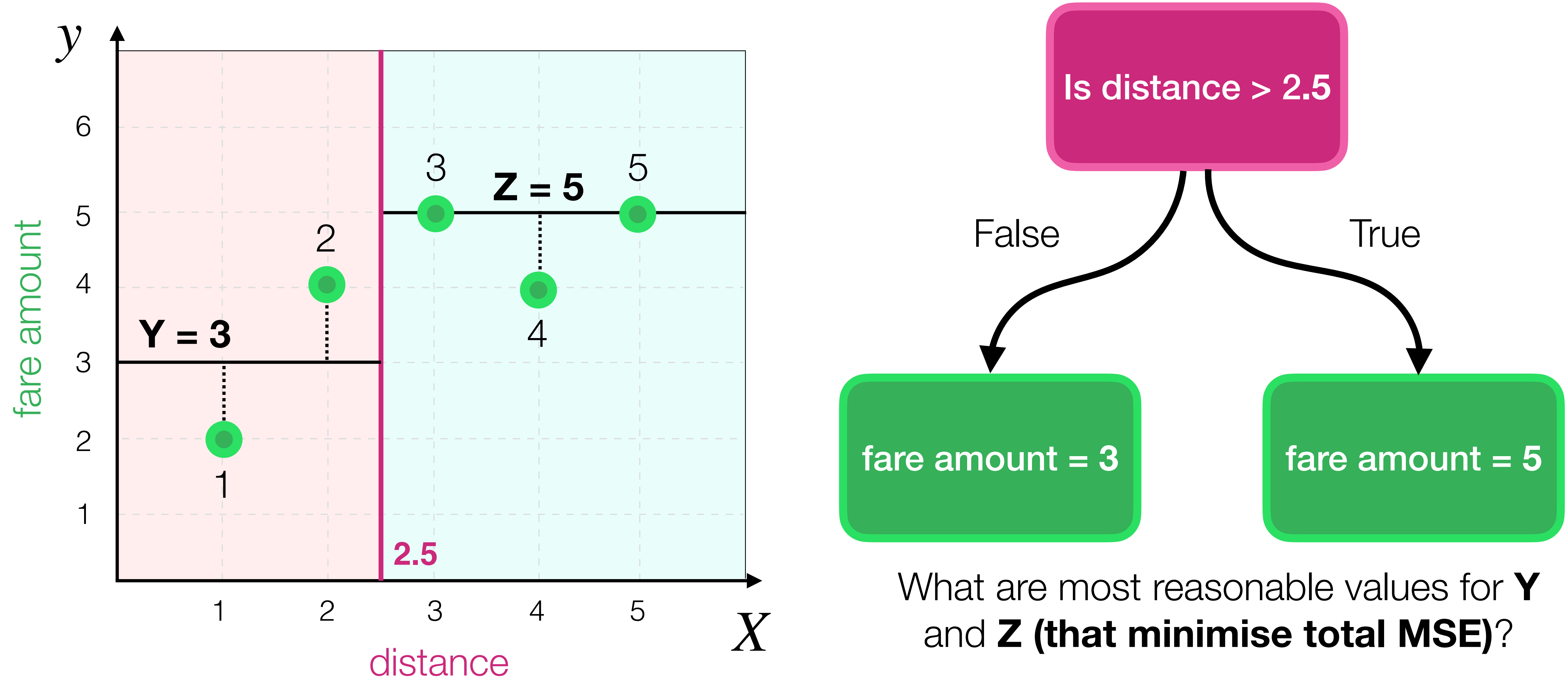
$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(y_1-\hat{y}_1)^2+(y_2-\hat{y}_2)^2+(y_3-\hat{y}_3)^2+(y_4-\hat{y}_4)^2+(y_5-\hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

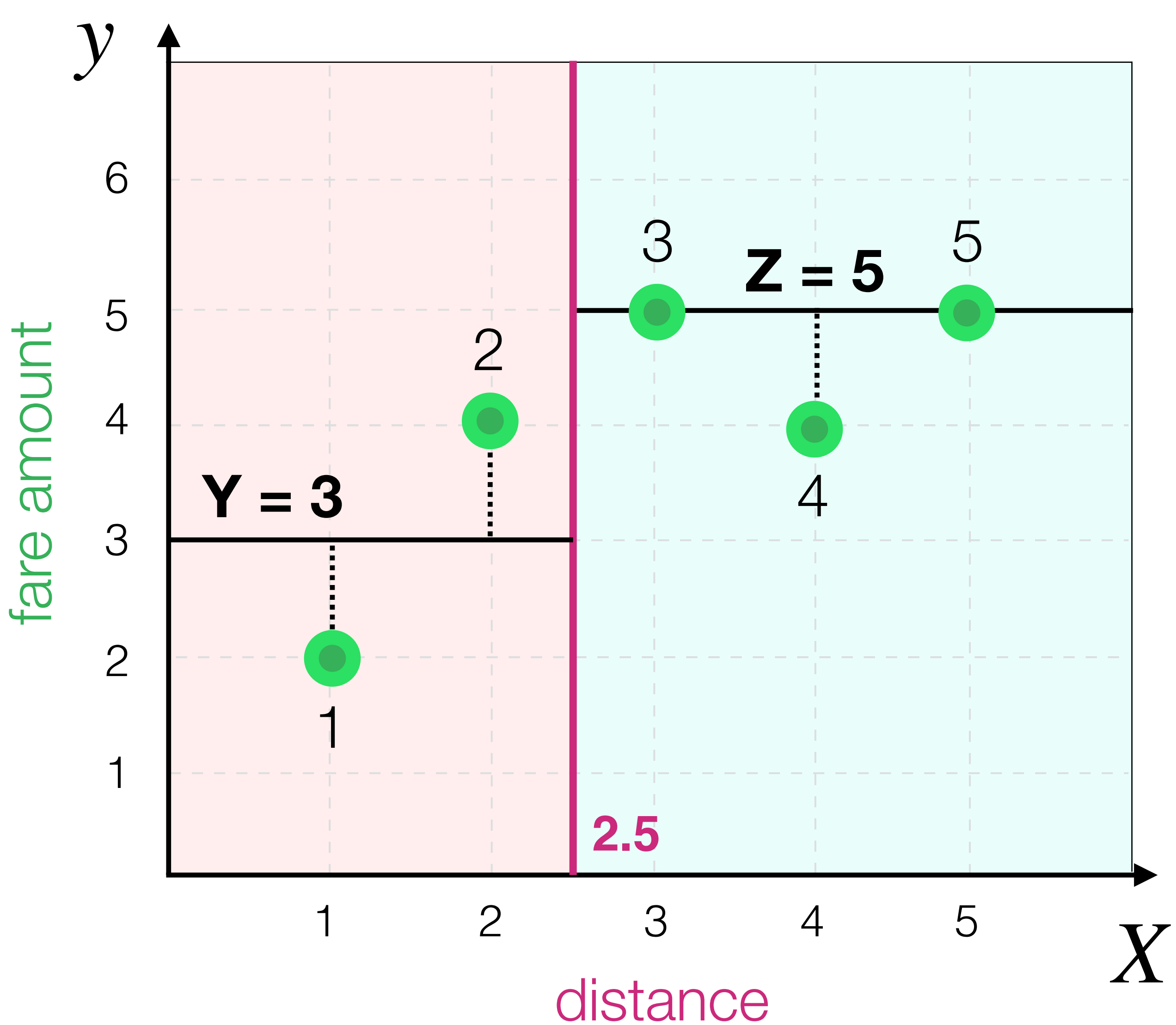
$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(2-3)^2 + (4-3)^2 + (5-5)^2 + (4-5)^2 + (5-5)^2}{5}$$

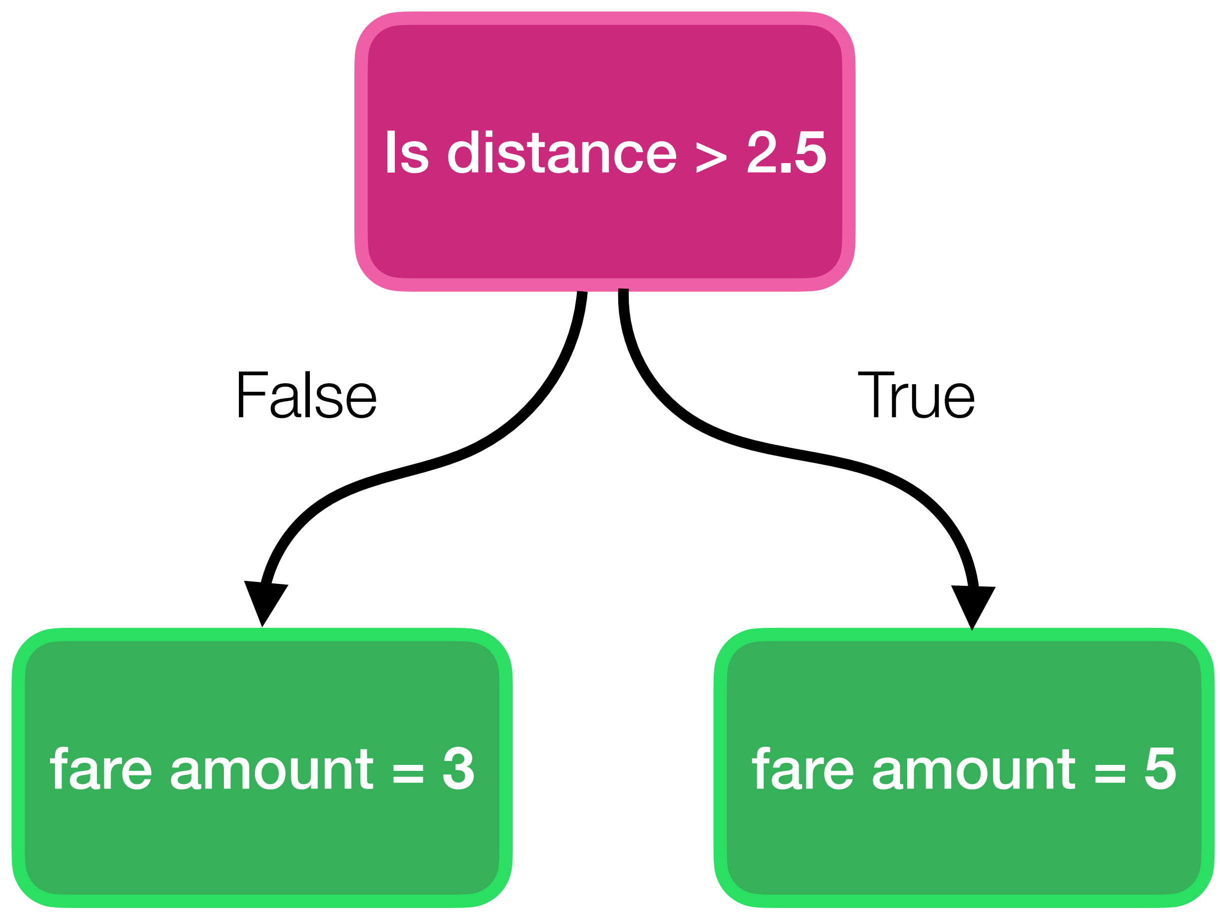


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1+1+0+1+0}{5}$$

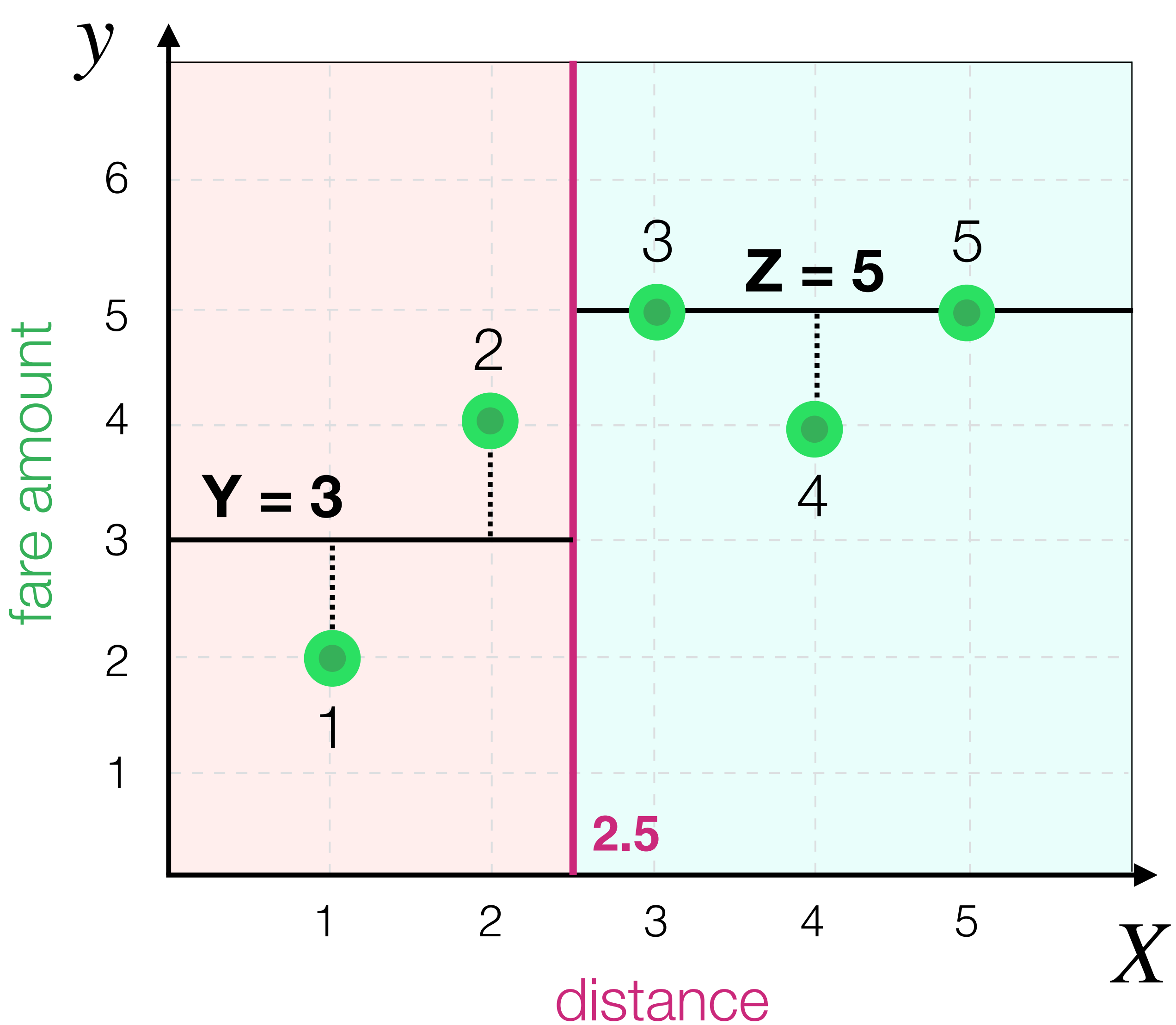


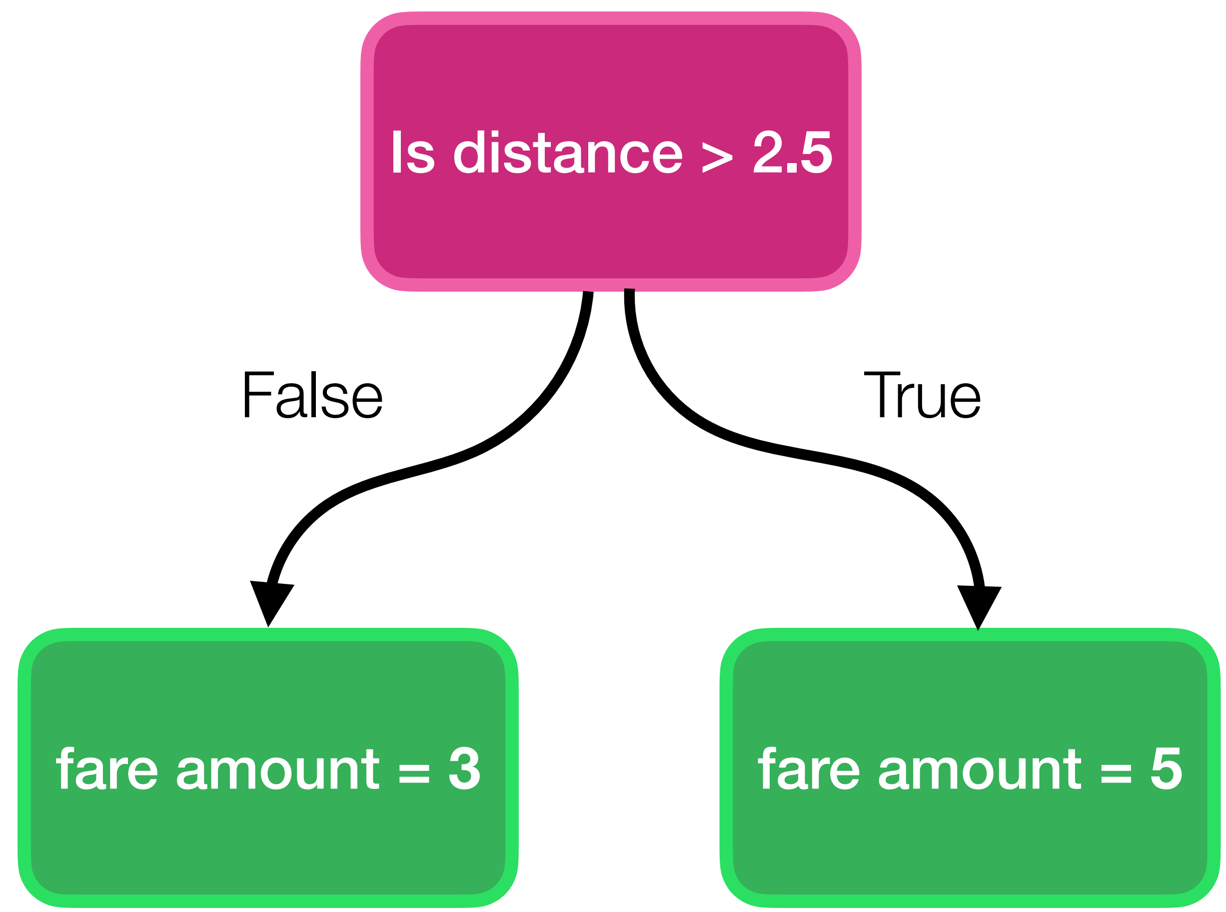


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{3}{5} = 0.6$$



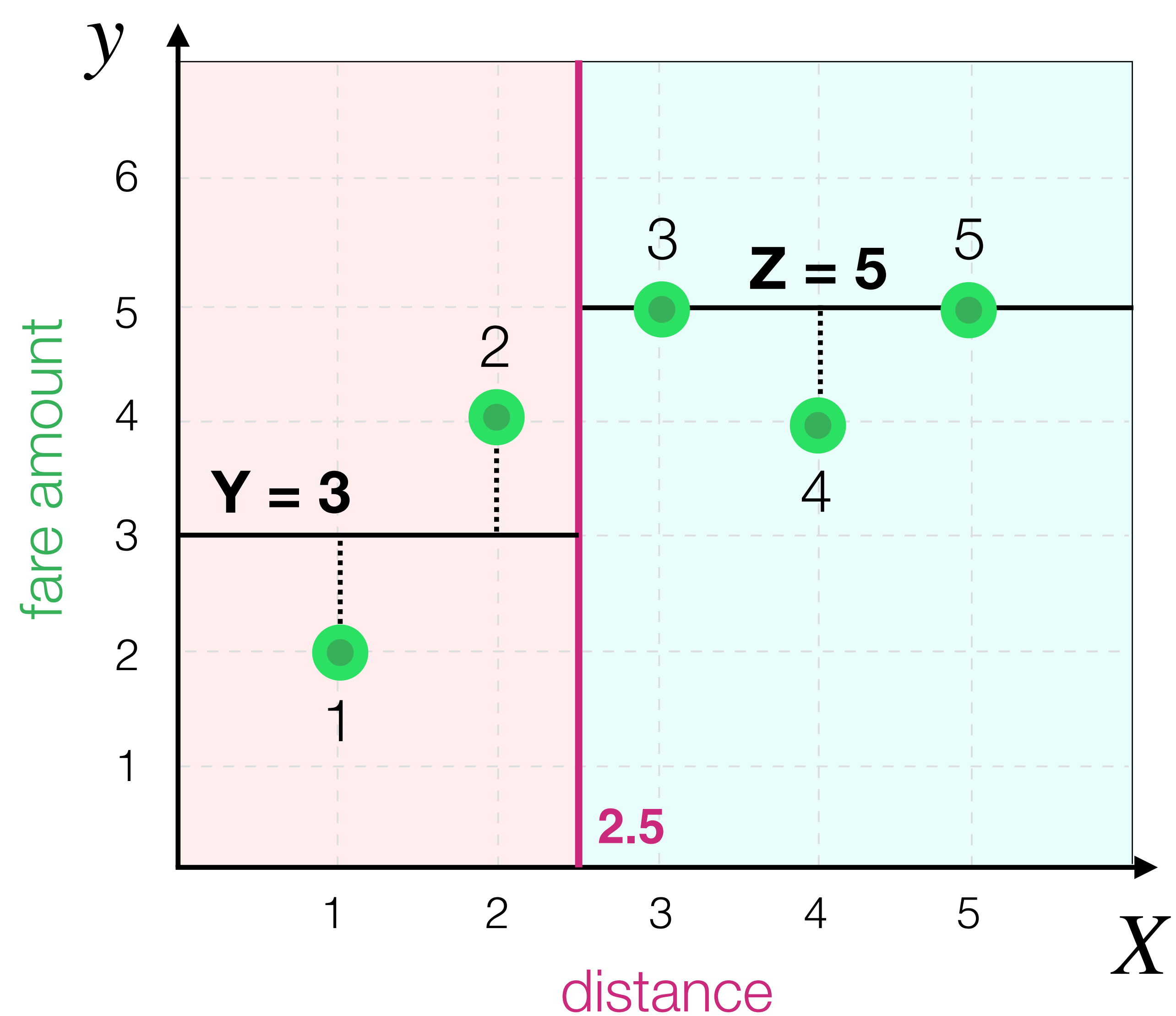


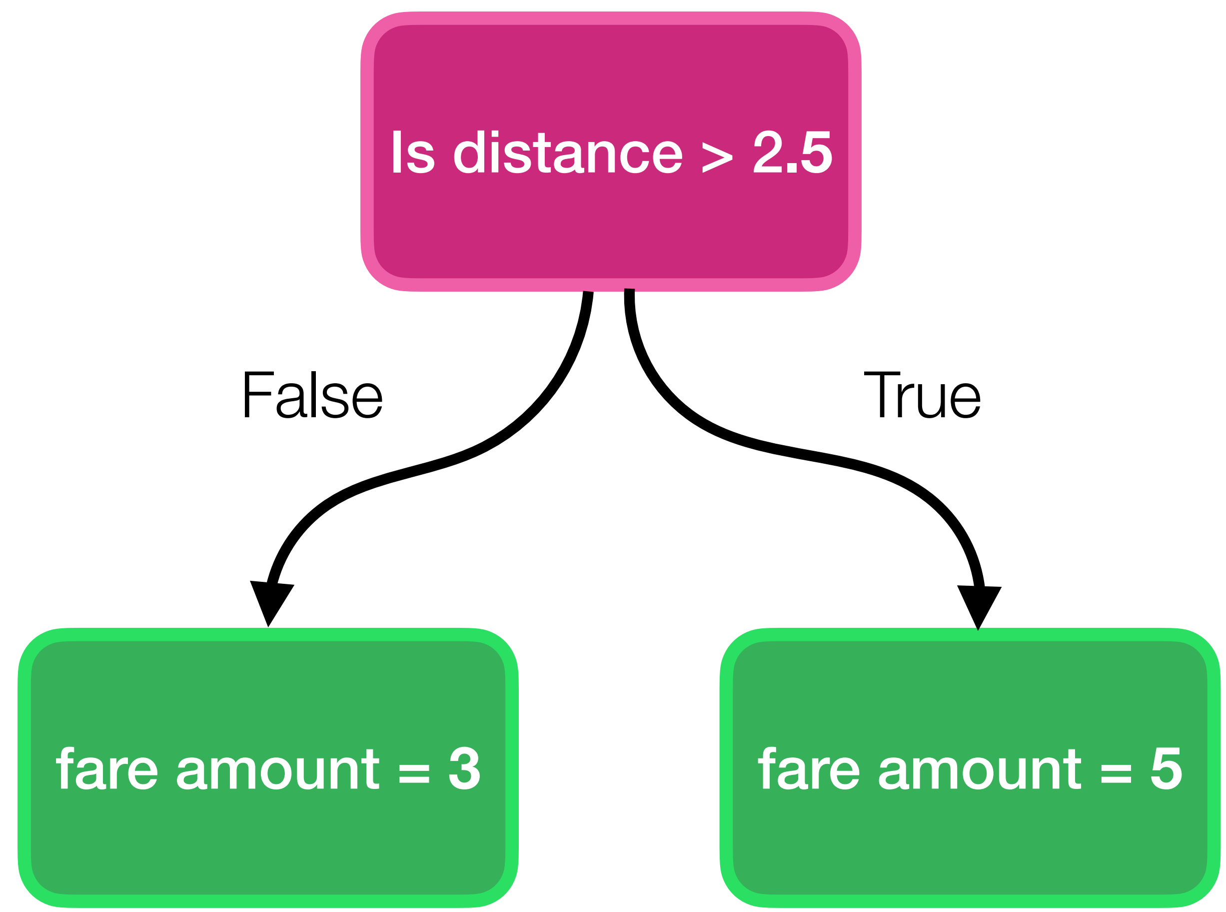
What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{3}{5} = 0.6$$

so, if **X** = **2.5**, **Y** = **3** and **Z** = **5**, MSE is **0.6**

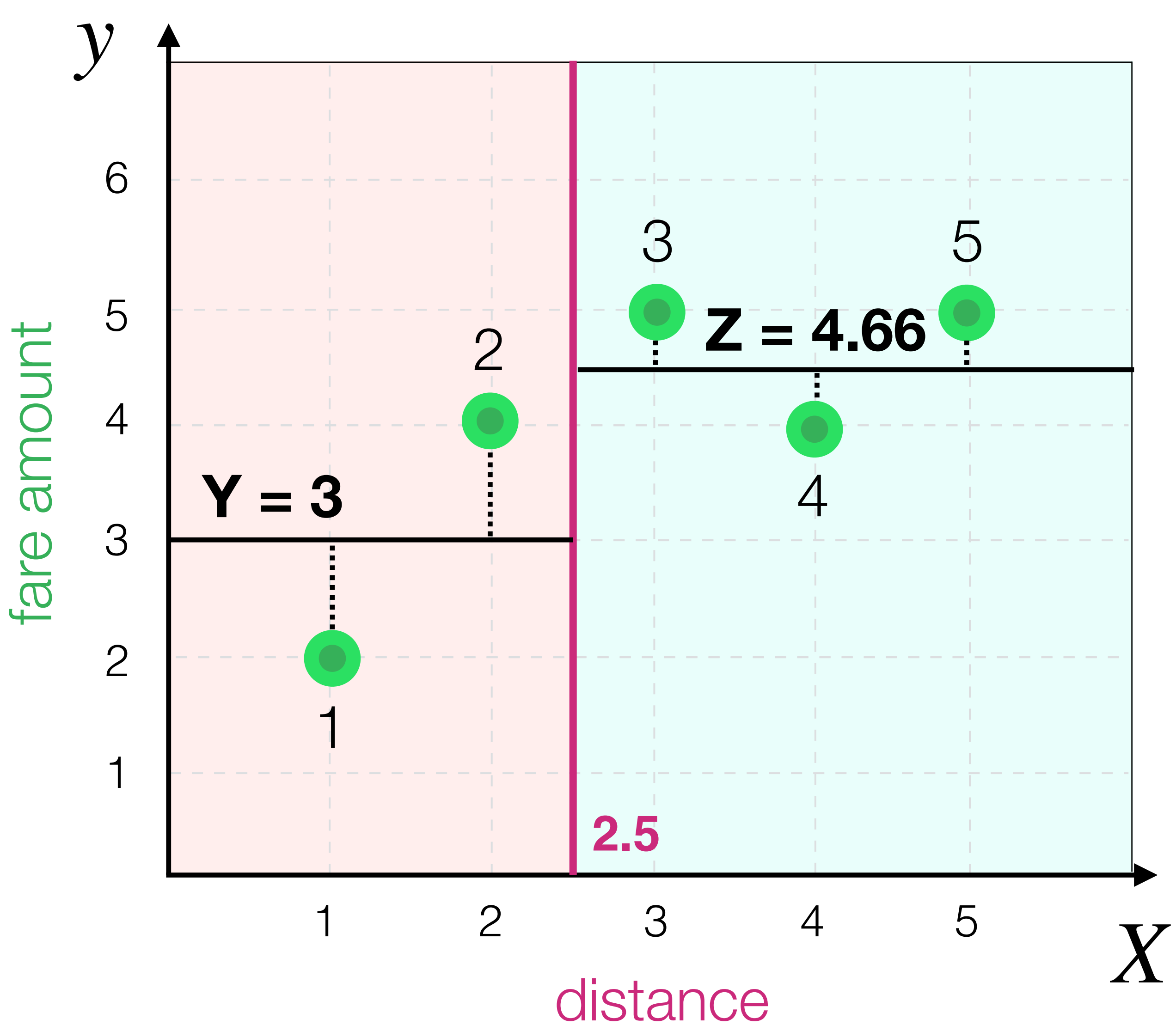


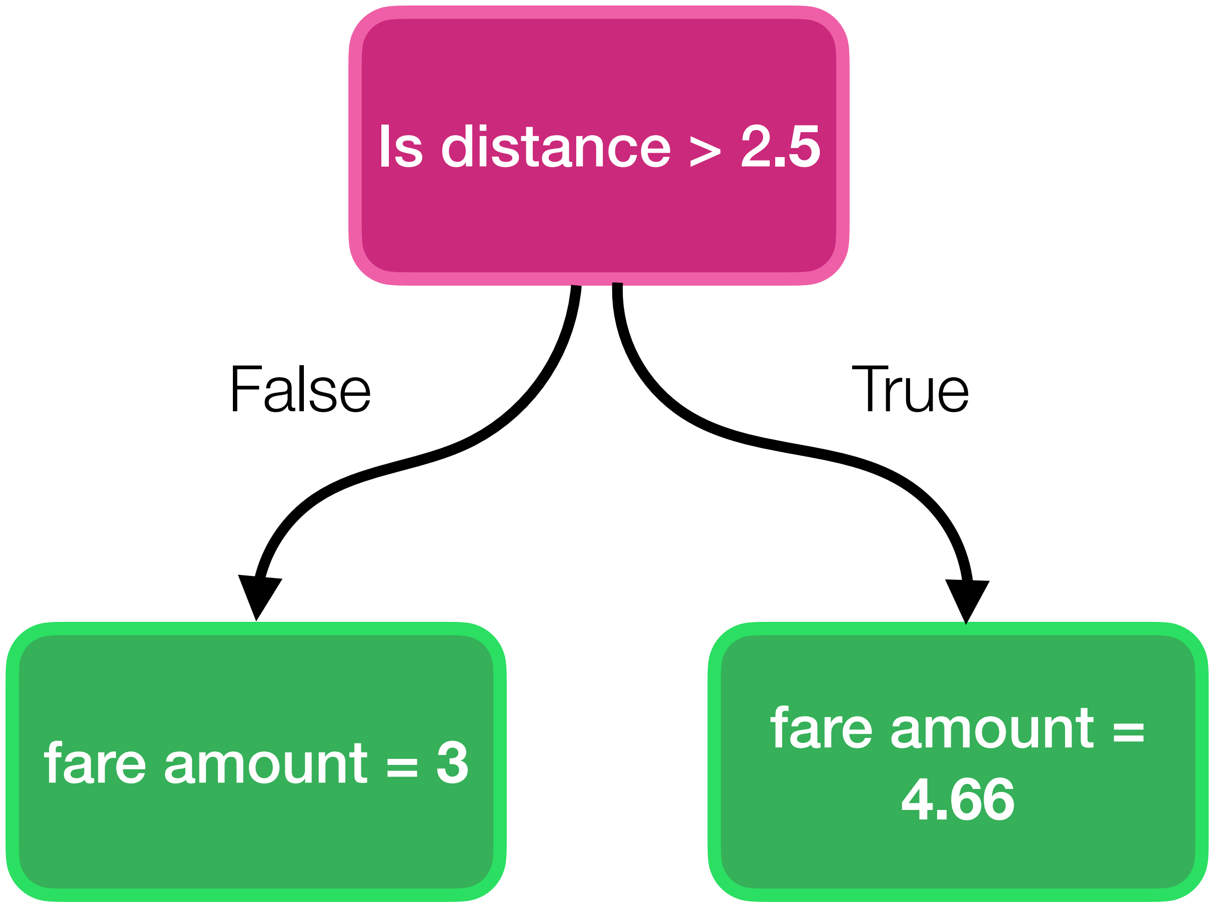


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{(2-3)^2+(4-3)^2+(5-4.66)^2+(4-4.66)^2+(5-4.66)^2}{5}$$

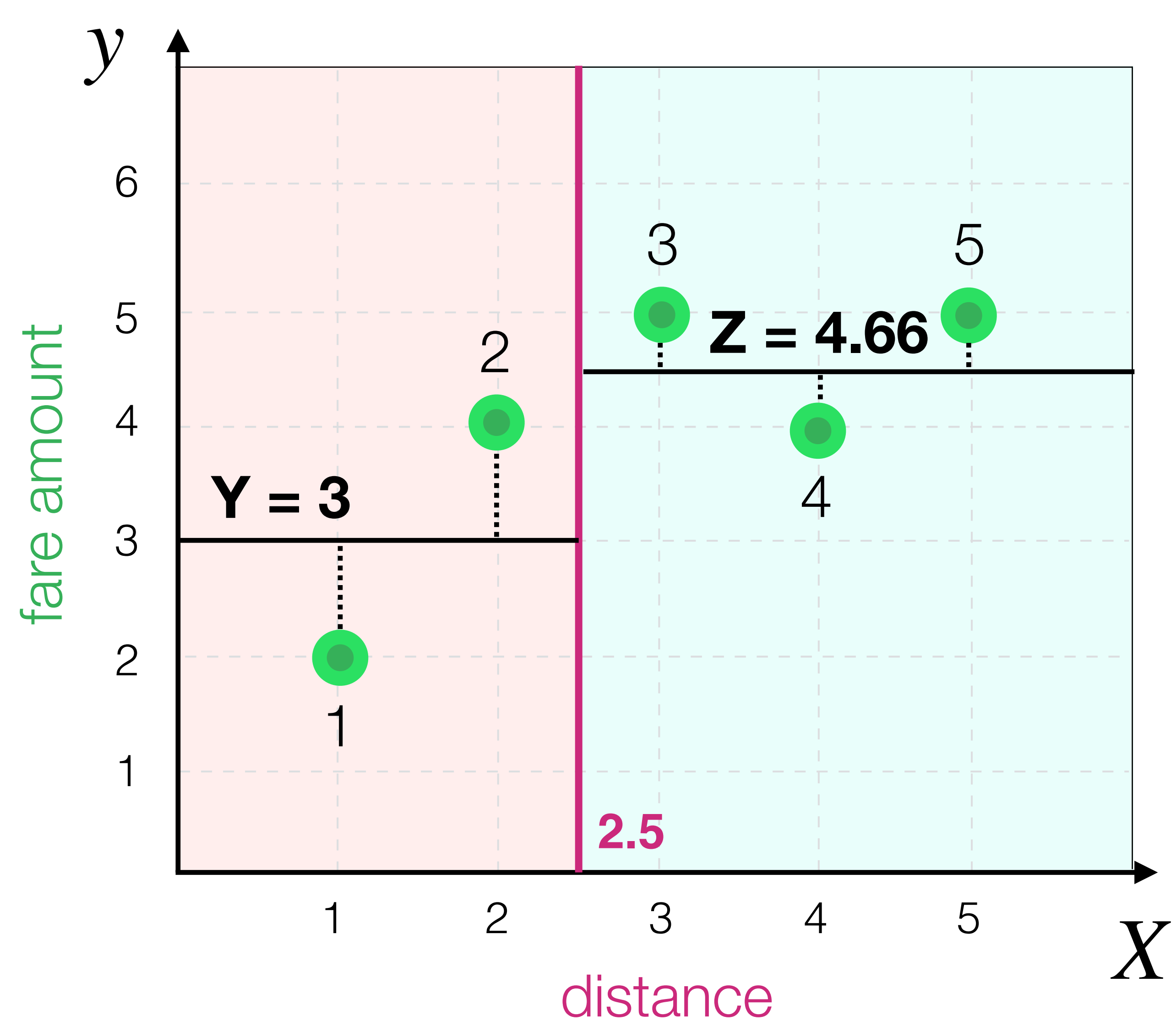


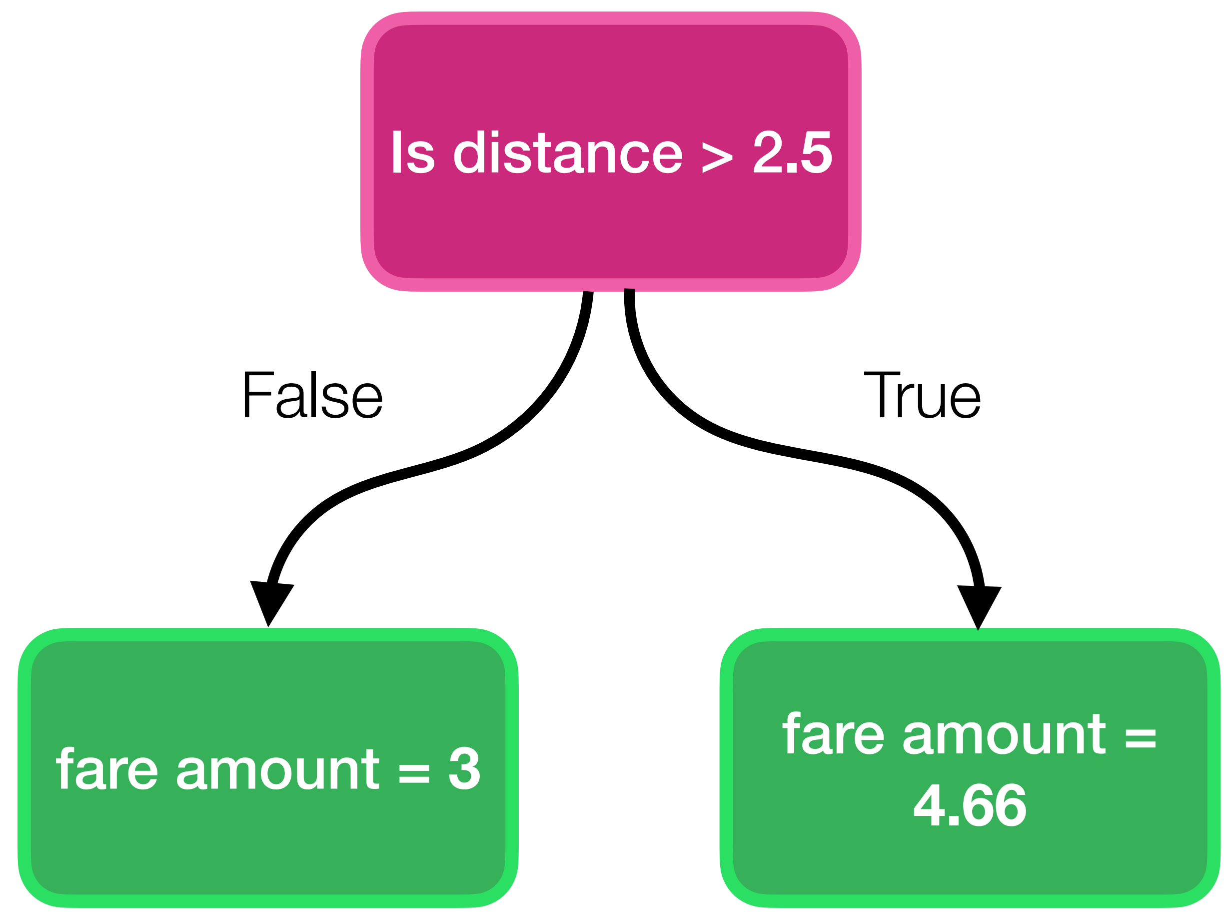


What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1 + 1 + 0.12 + 0.43 + 0.12}{5}$$



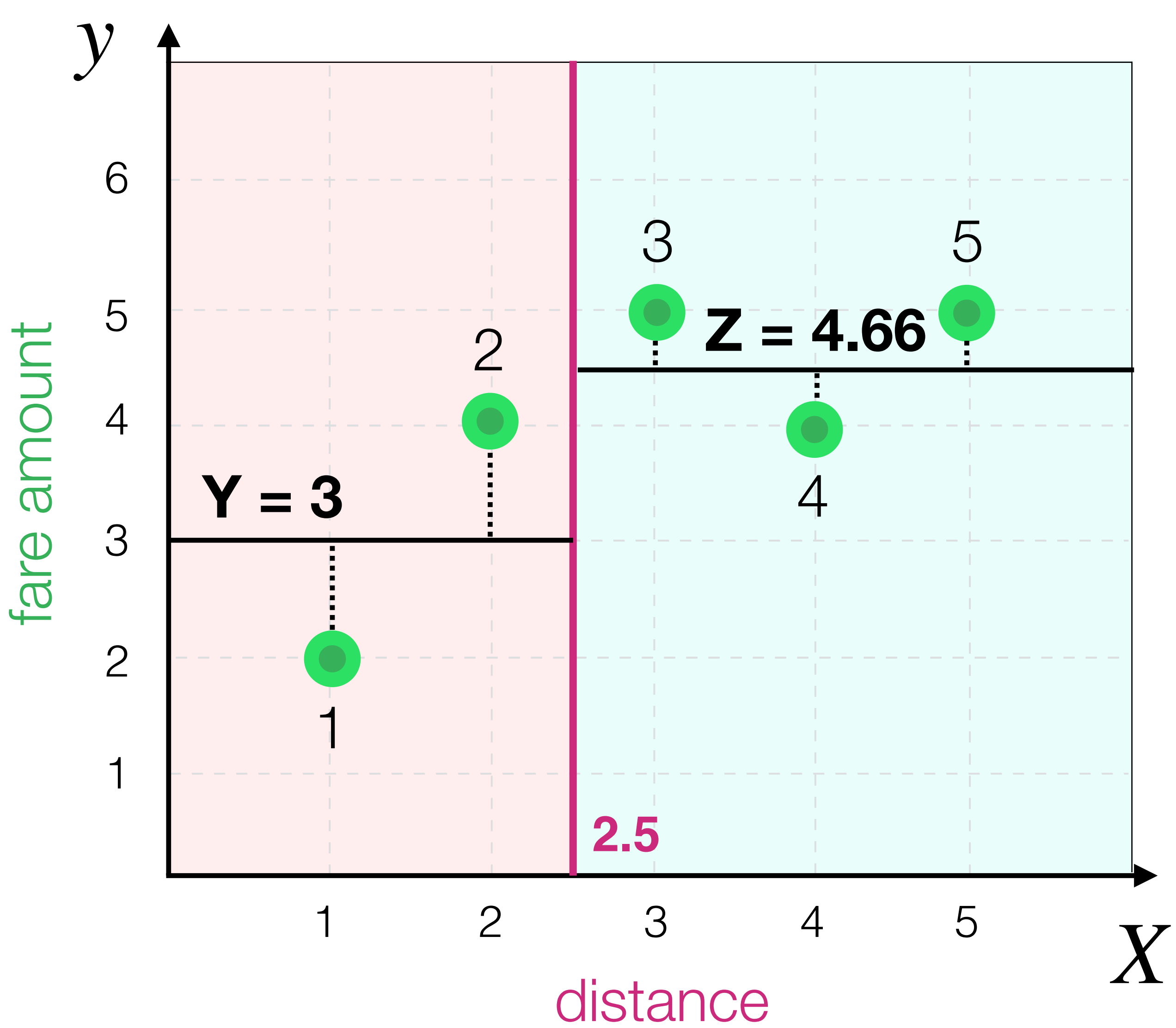


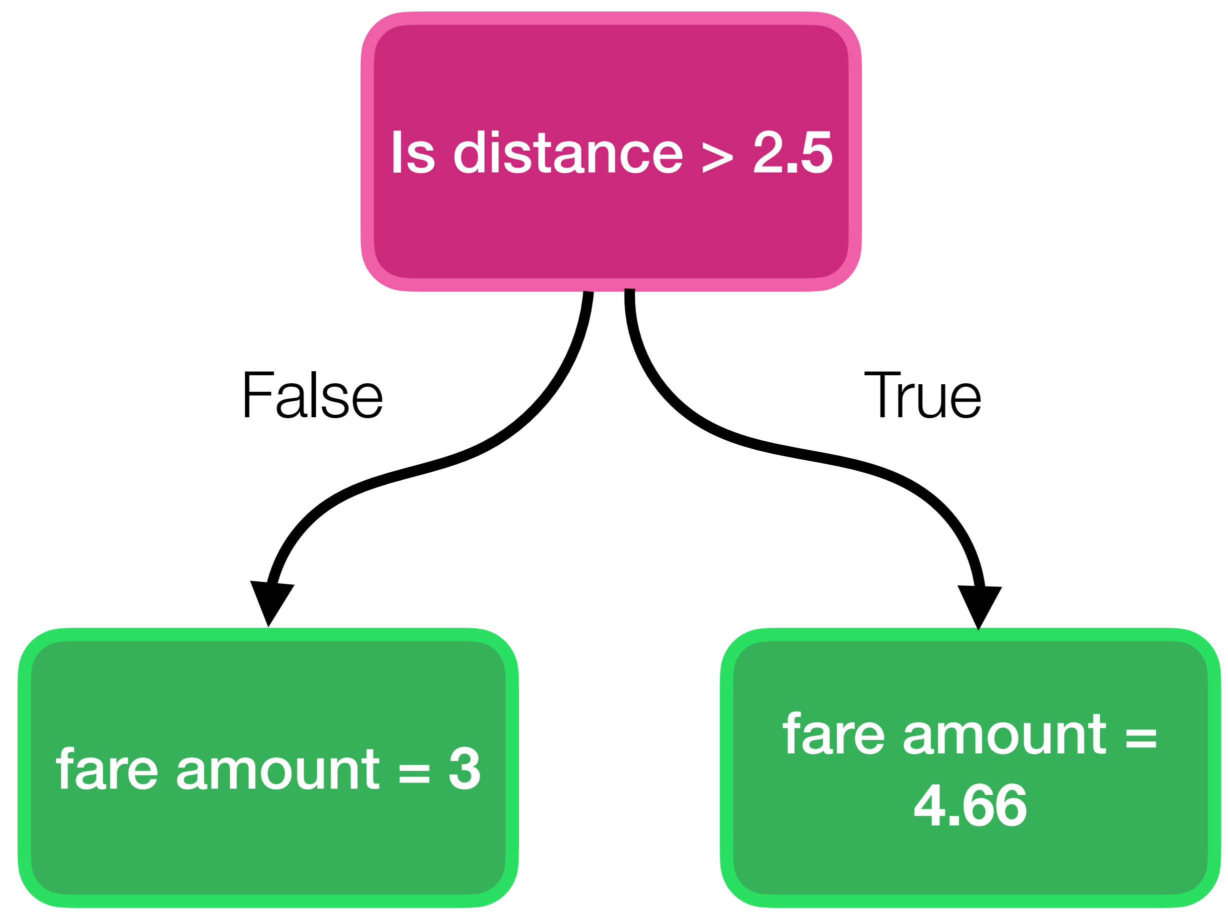
What are most reasonable values for **Y** and **Z (that minimise total MSE)**?

Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{2.67}{5} = 0.53$$

so, if **Y** = **3** and **Z** = **4.66**, MSE is **smallest**

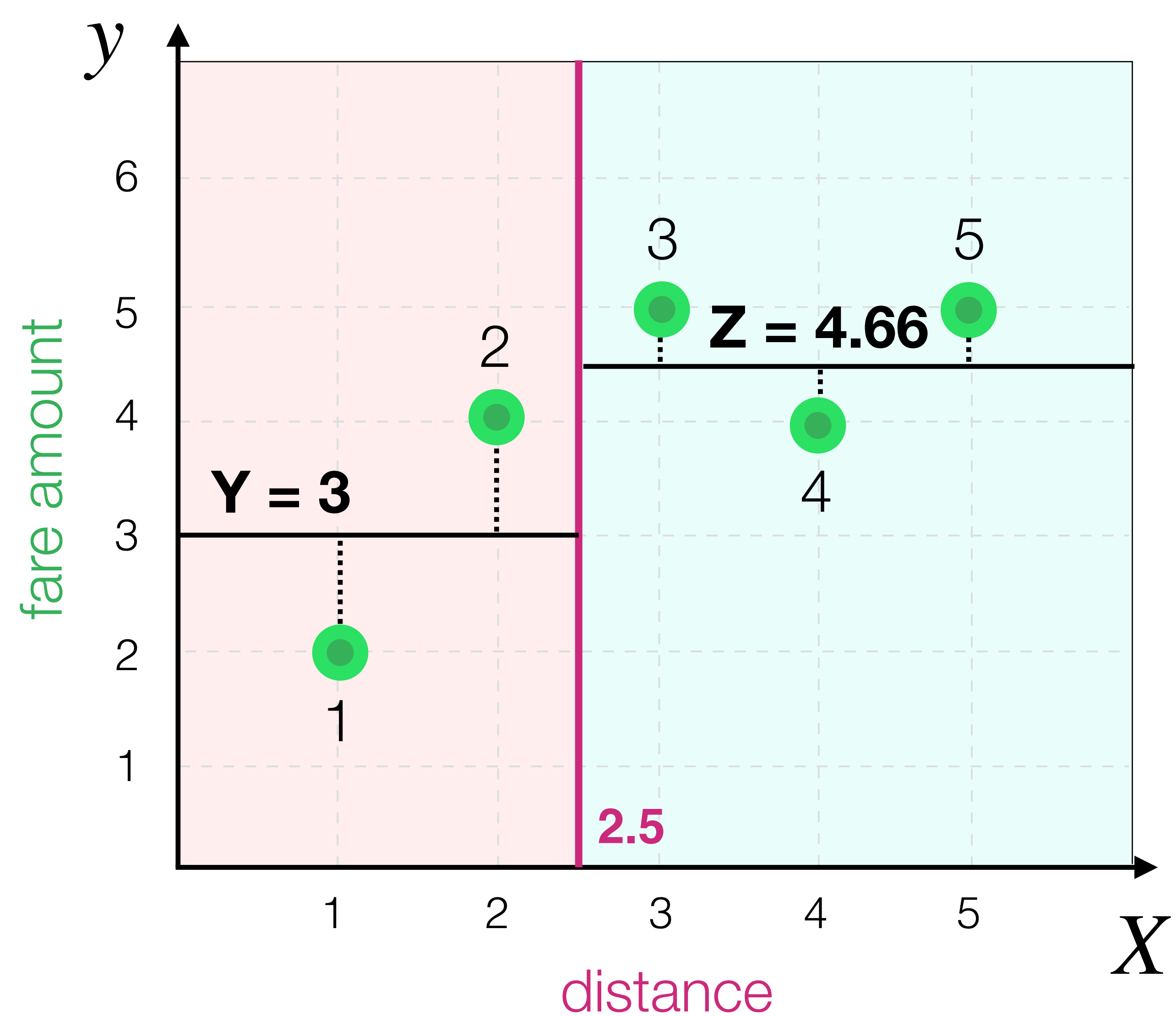


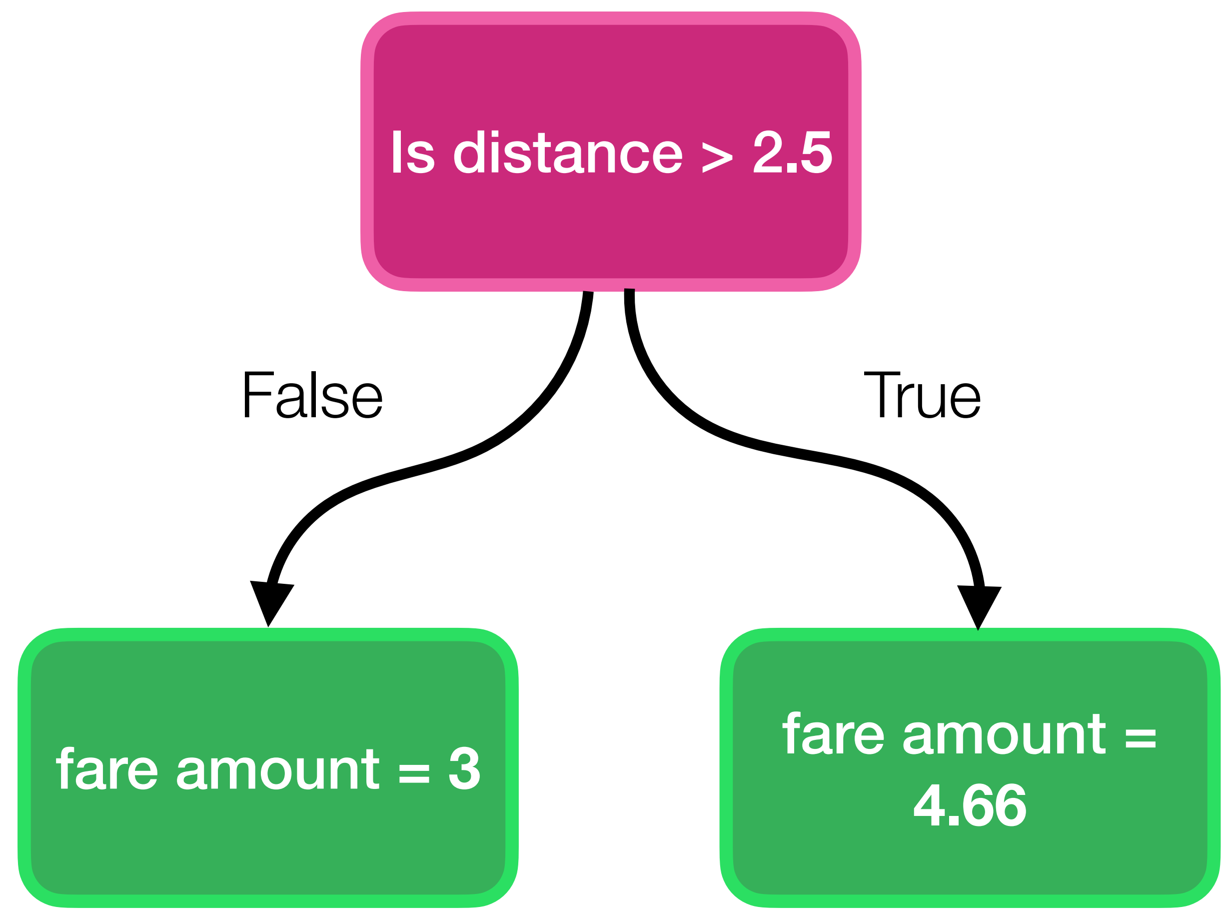


Decision Tree Algorithm

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{2.67}{5} = 0.53$$

so, if **Y** = **3** and **Z** = **4.66**,
MSE is **smallest**

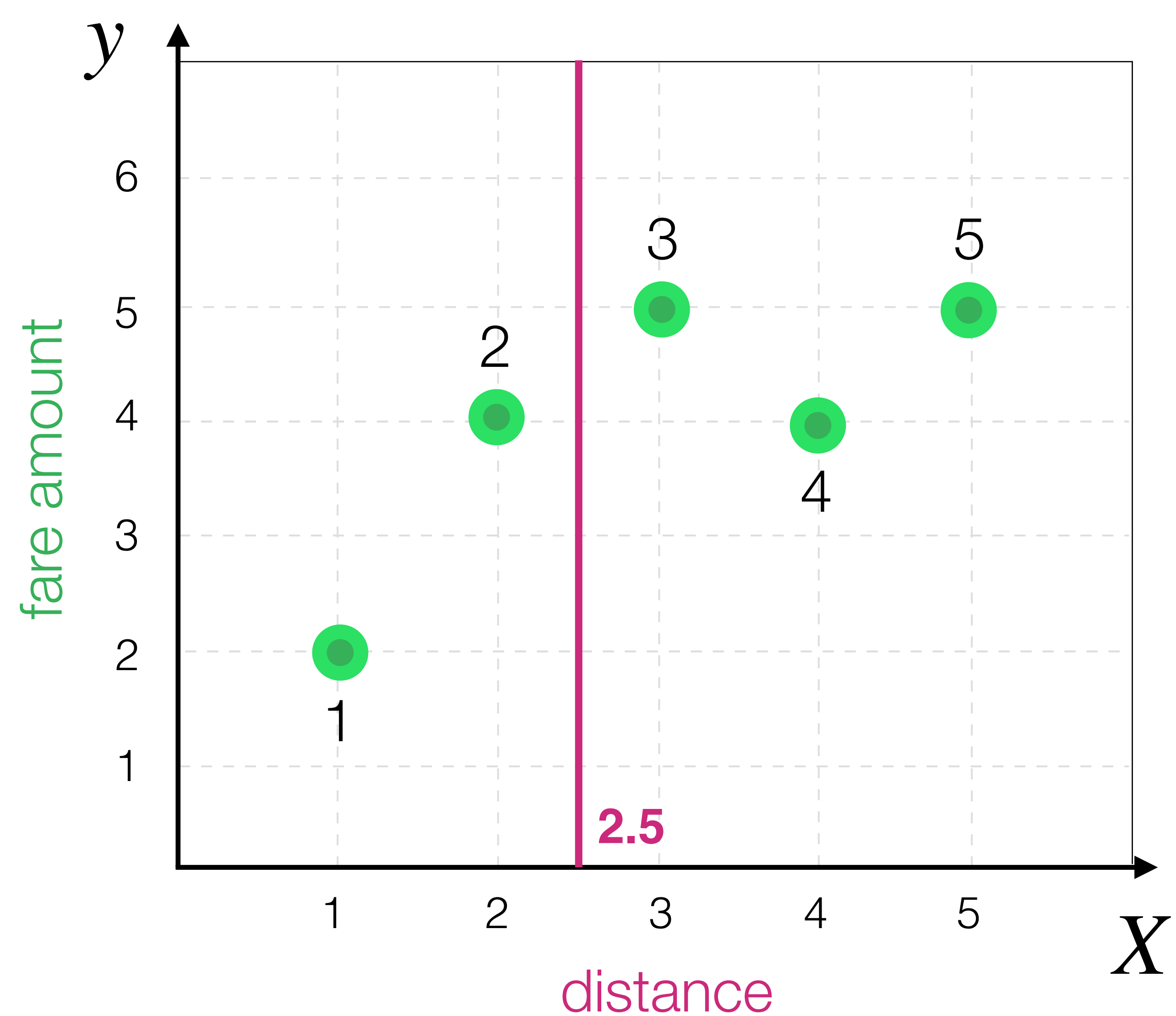


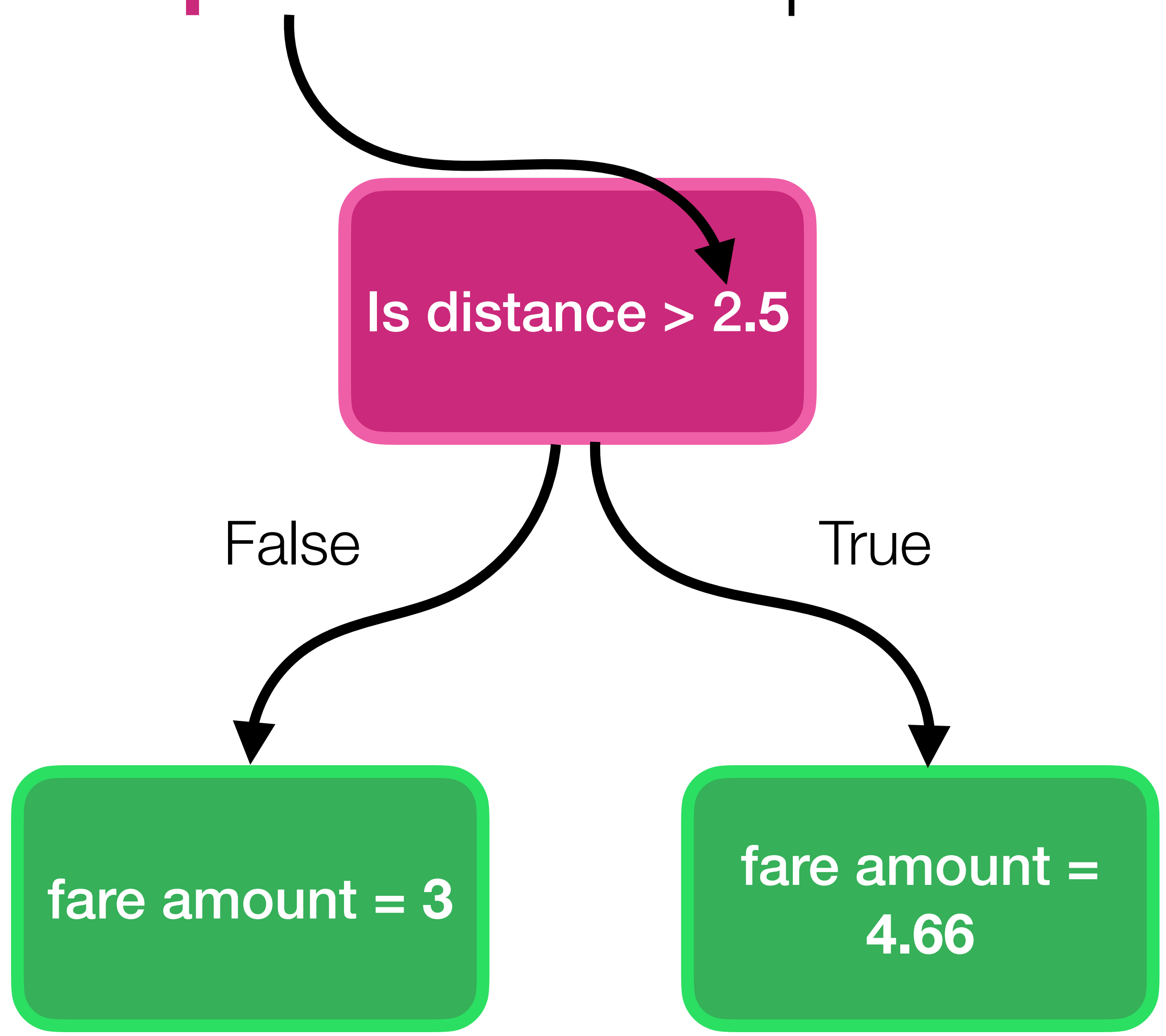


Are we happy?

Decision Tree Algorithm

Hold on, how did we choose this **split** on the first place?

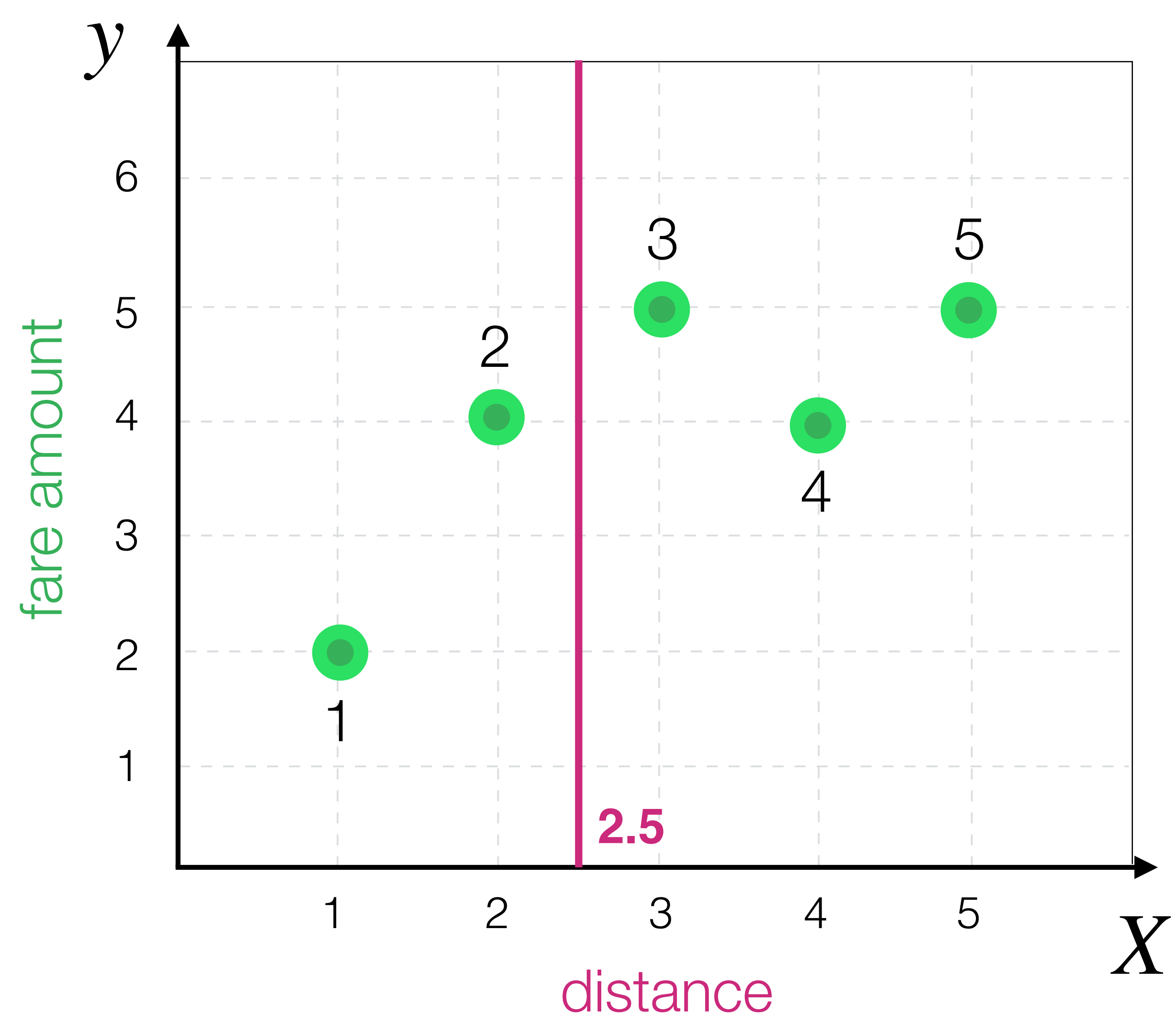


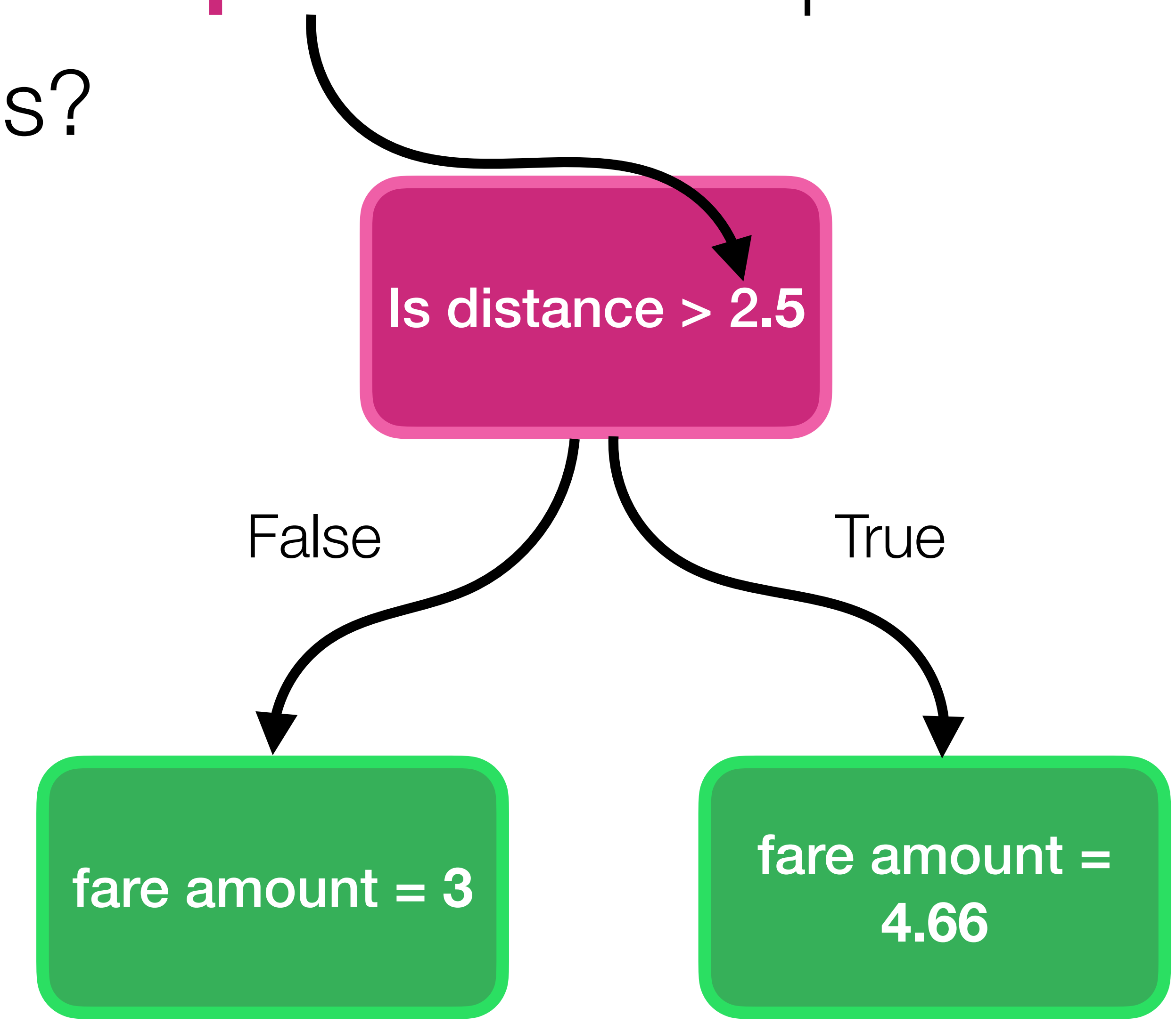


Decision Tree Algorithm

Hold on, how did we choose this **split** on the first place?

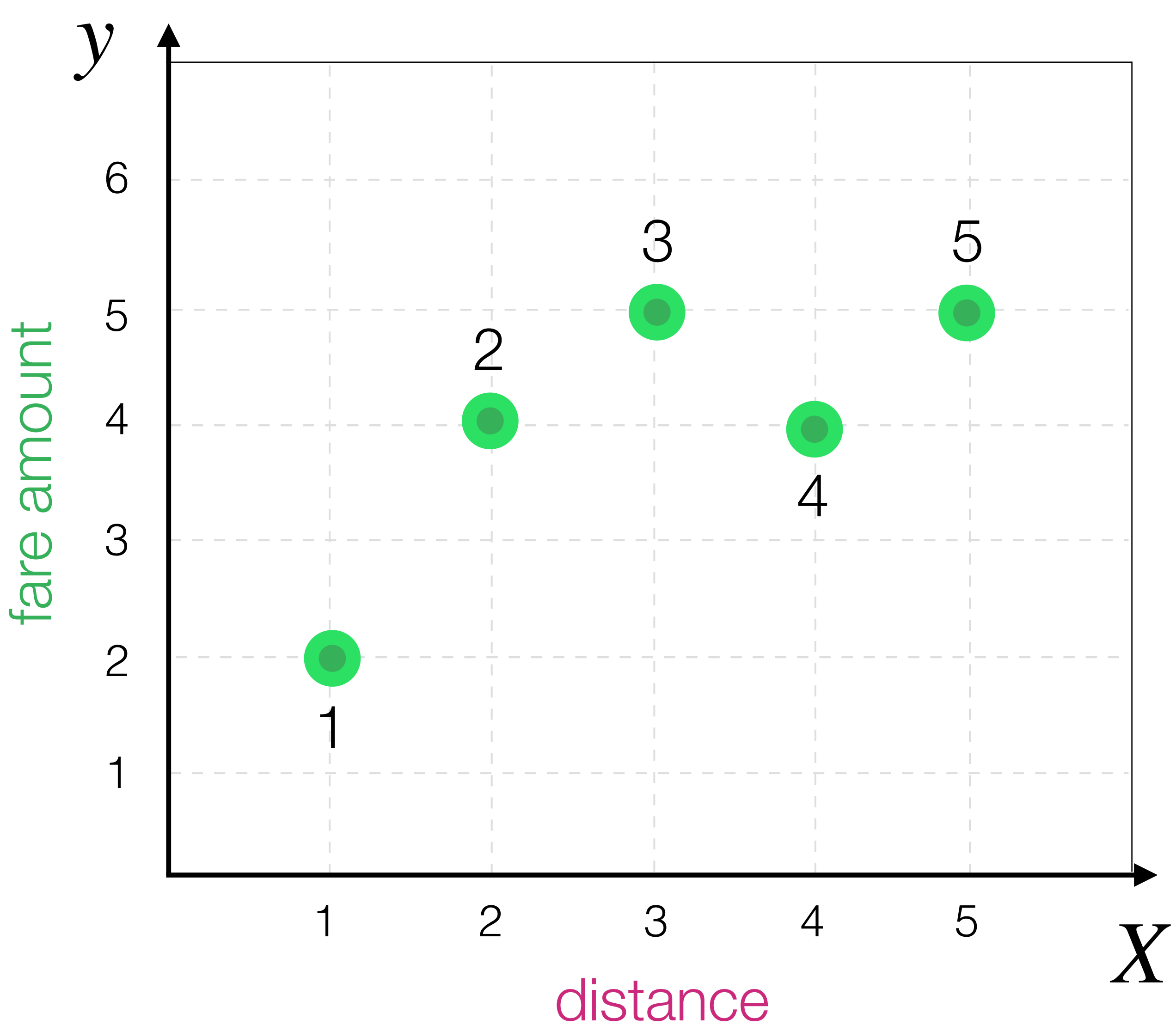
Maybe there are better options?

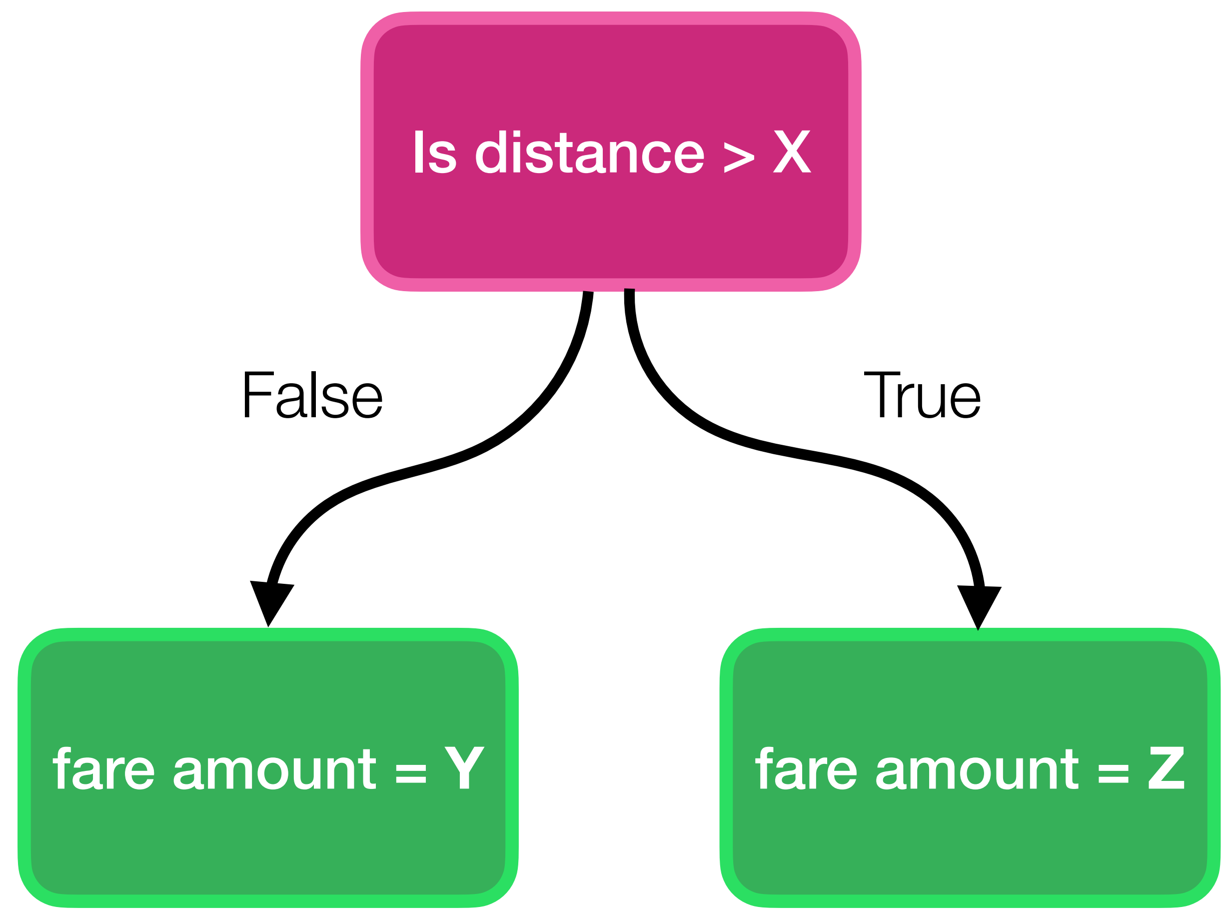




Decision Tree Algorithm

What are the possible **split** options in this case?





Decision Tree Algorithm

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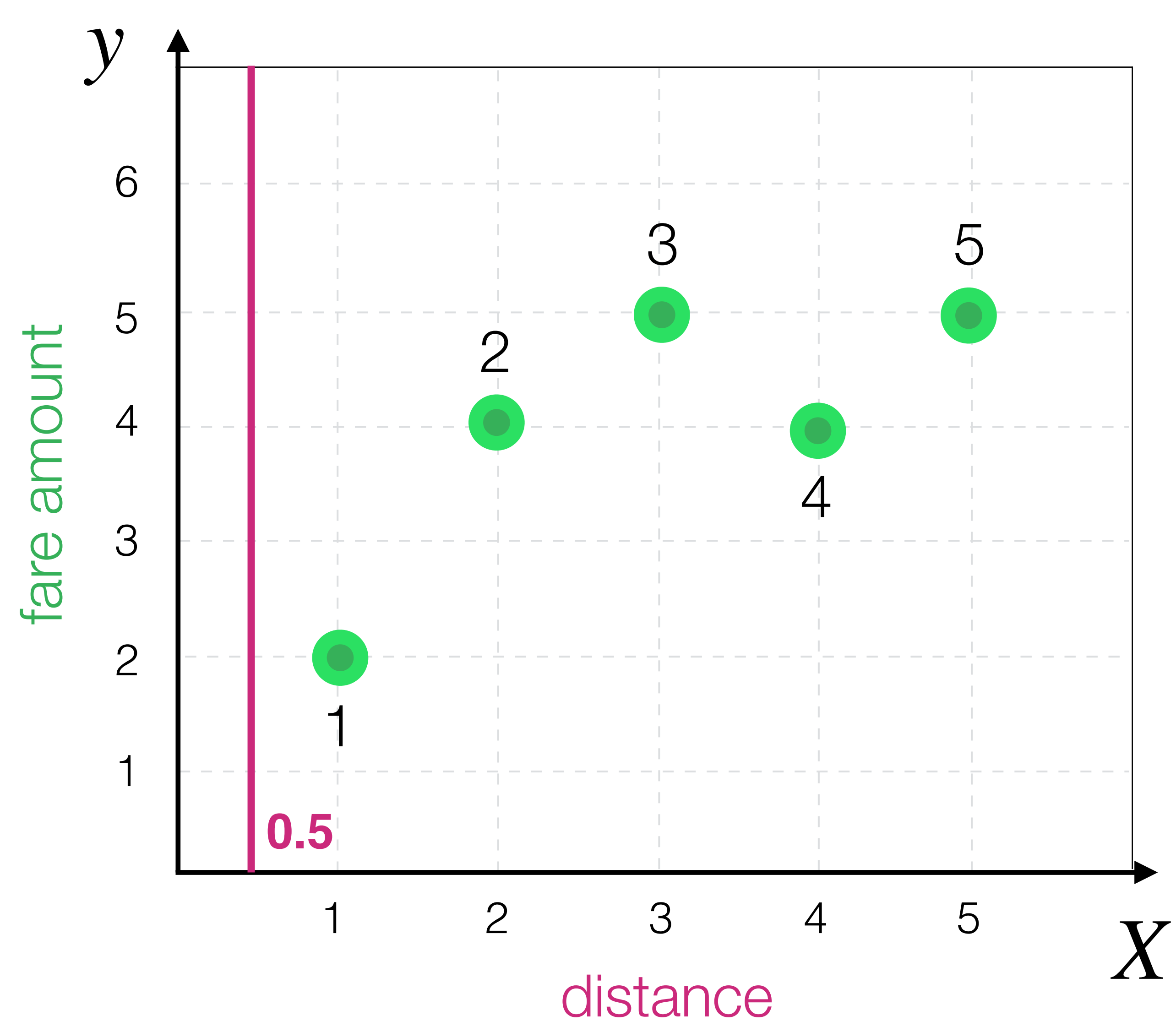

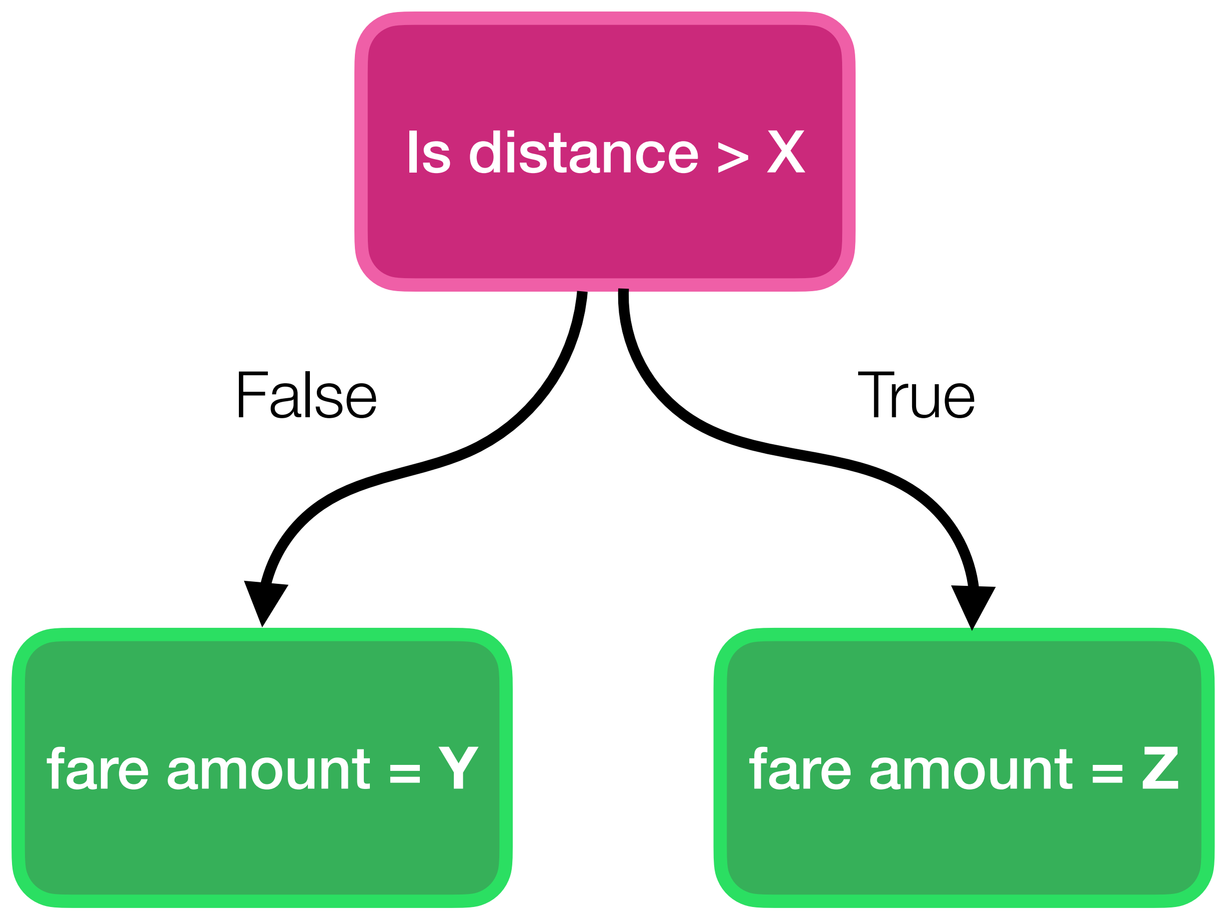

Decision Tree Algorithm

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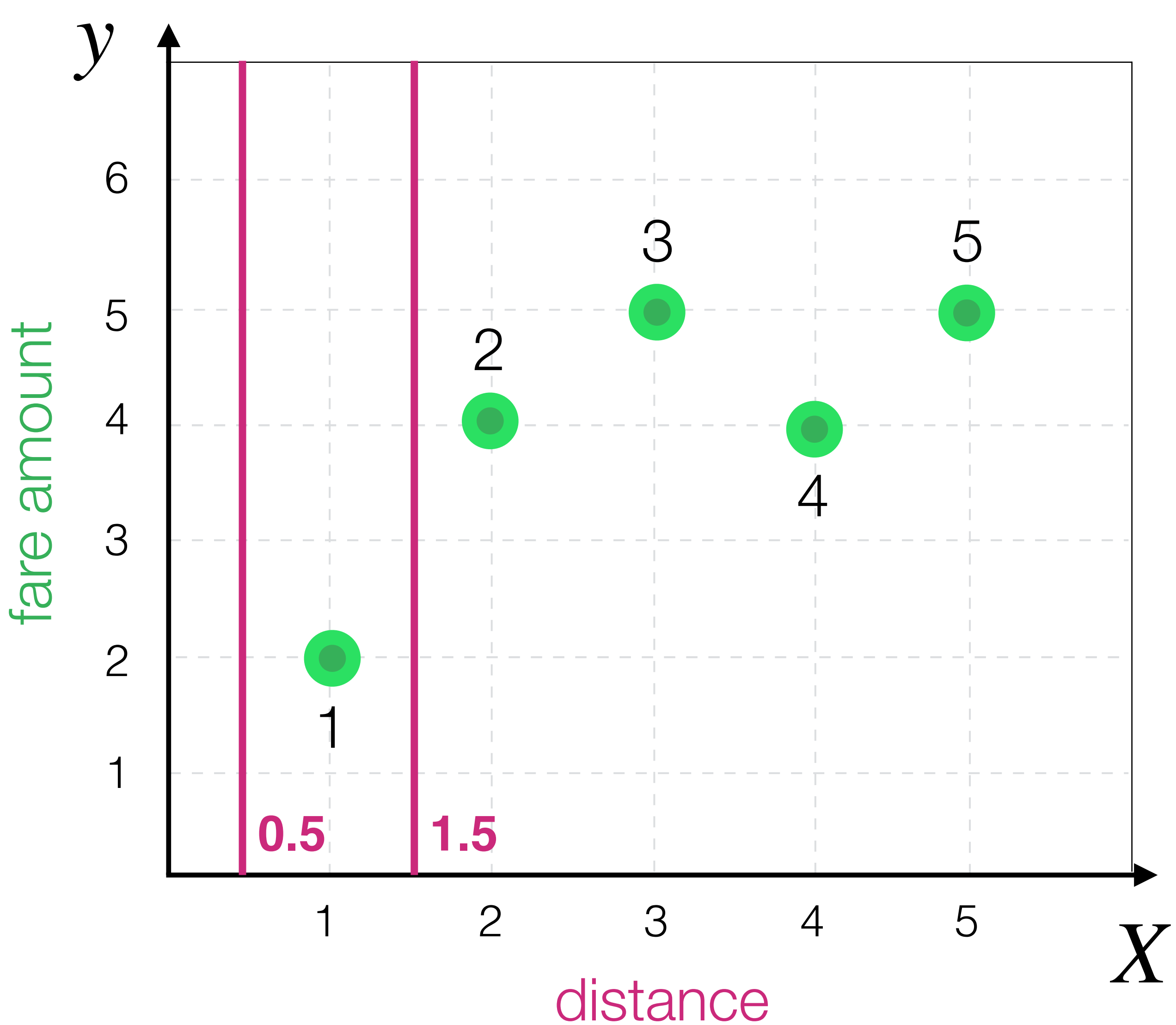

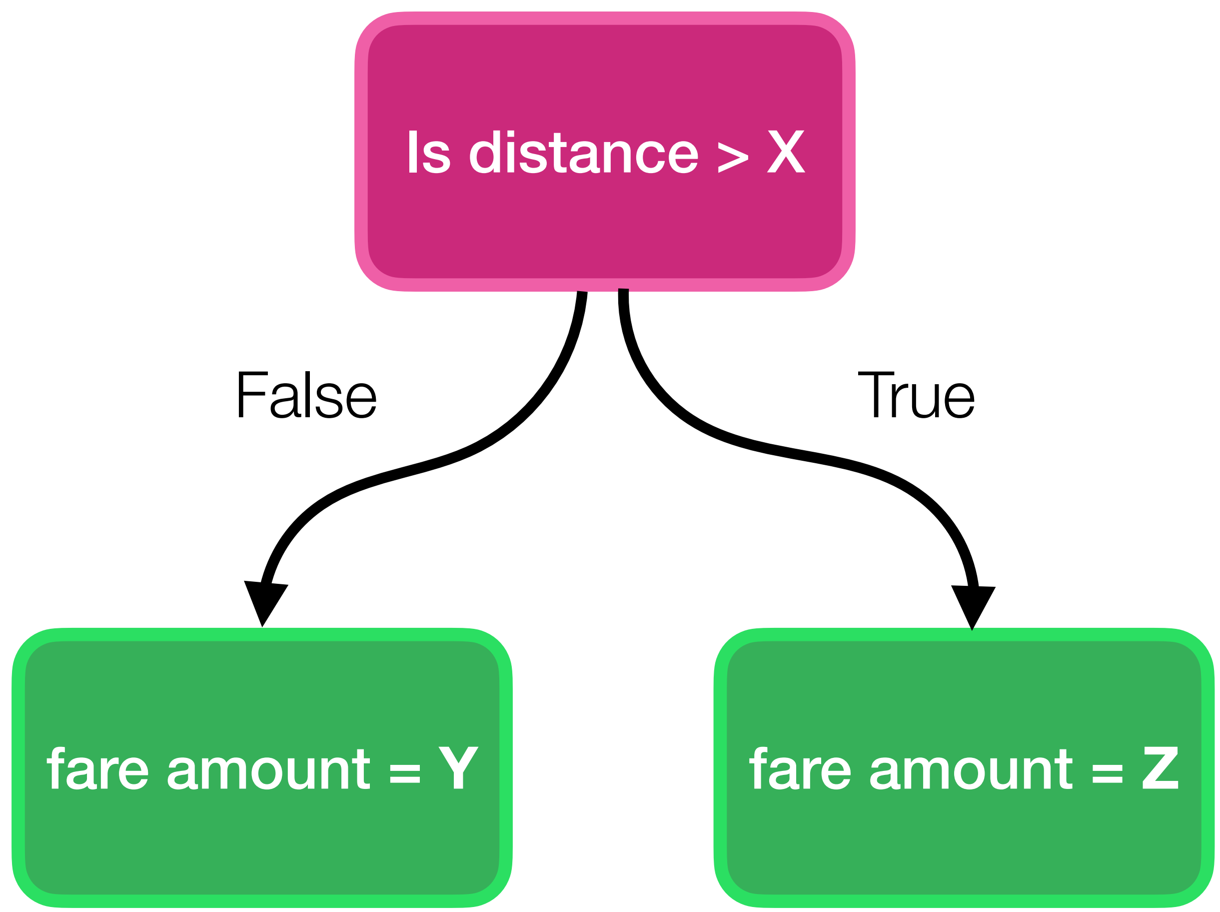

Decision Tree Algorithm

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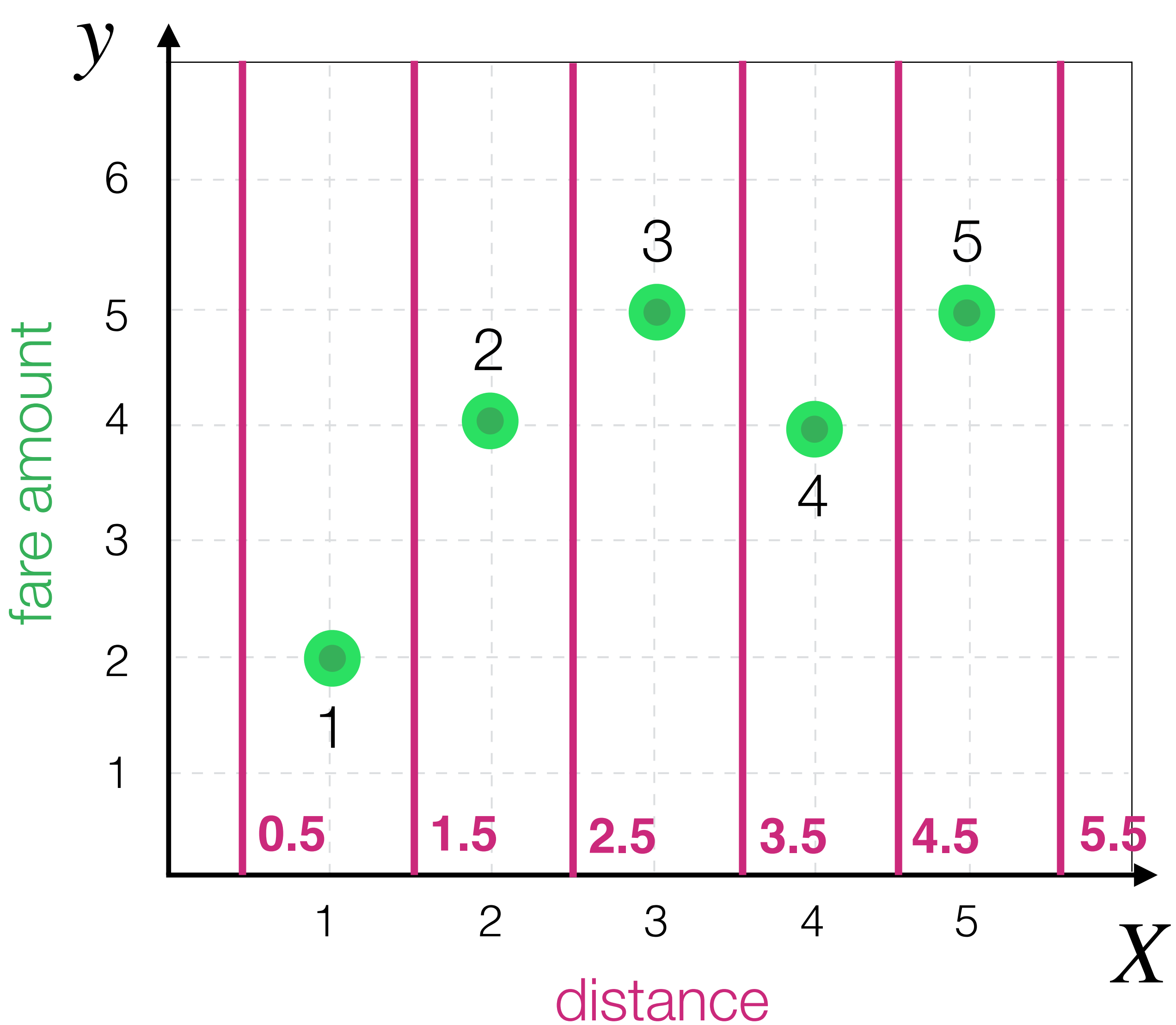
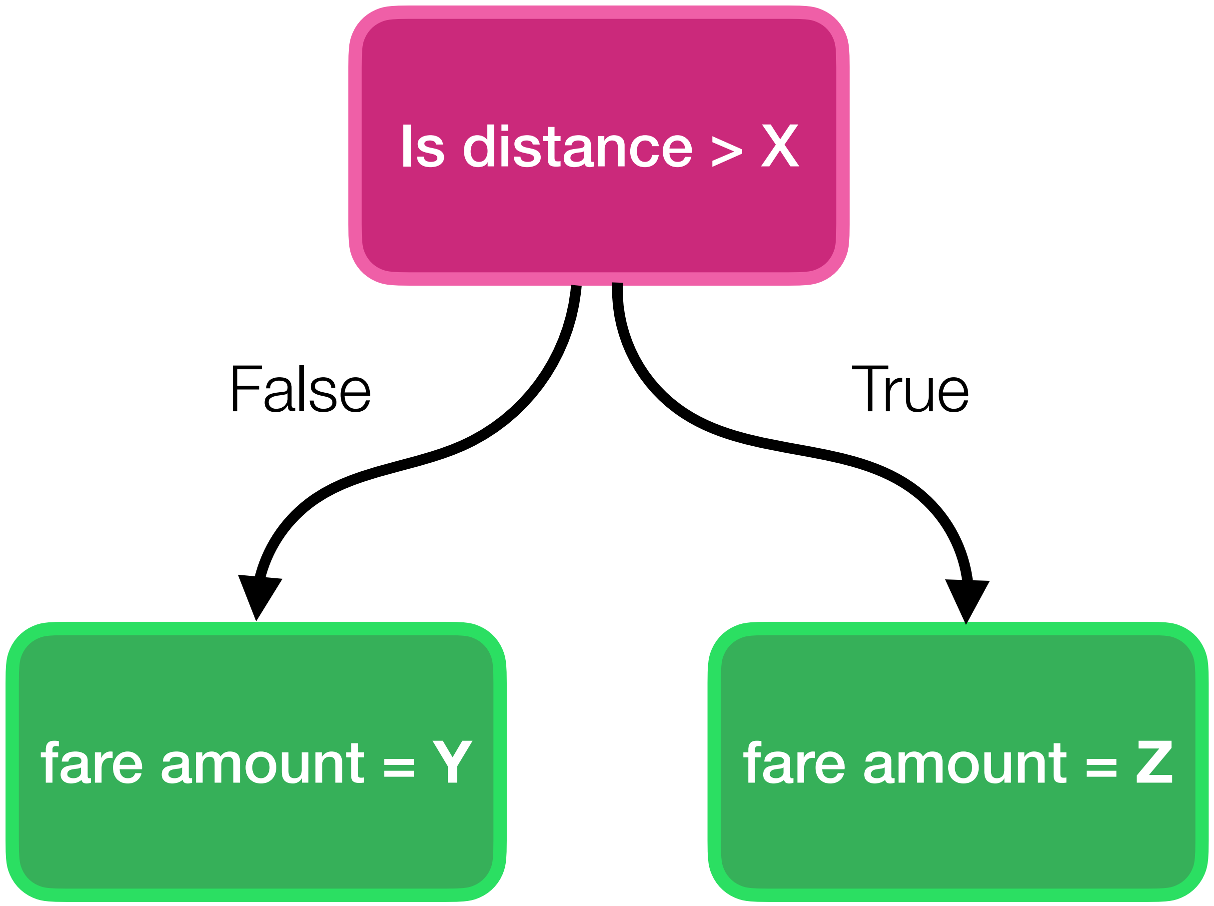
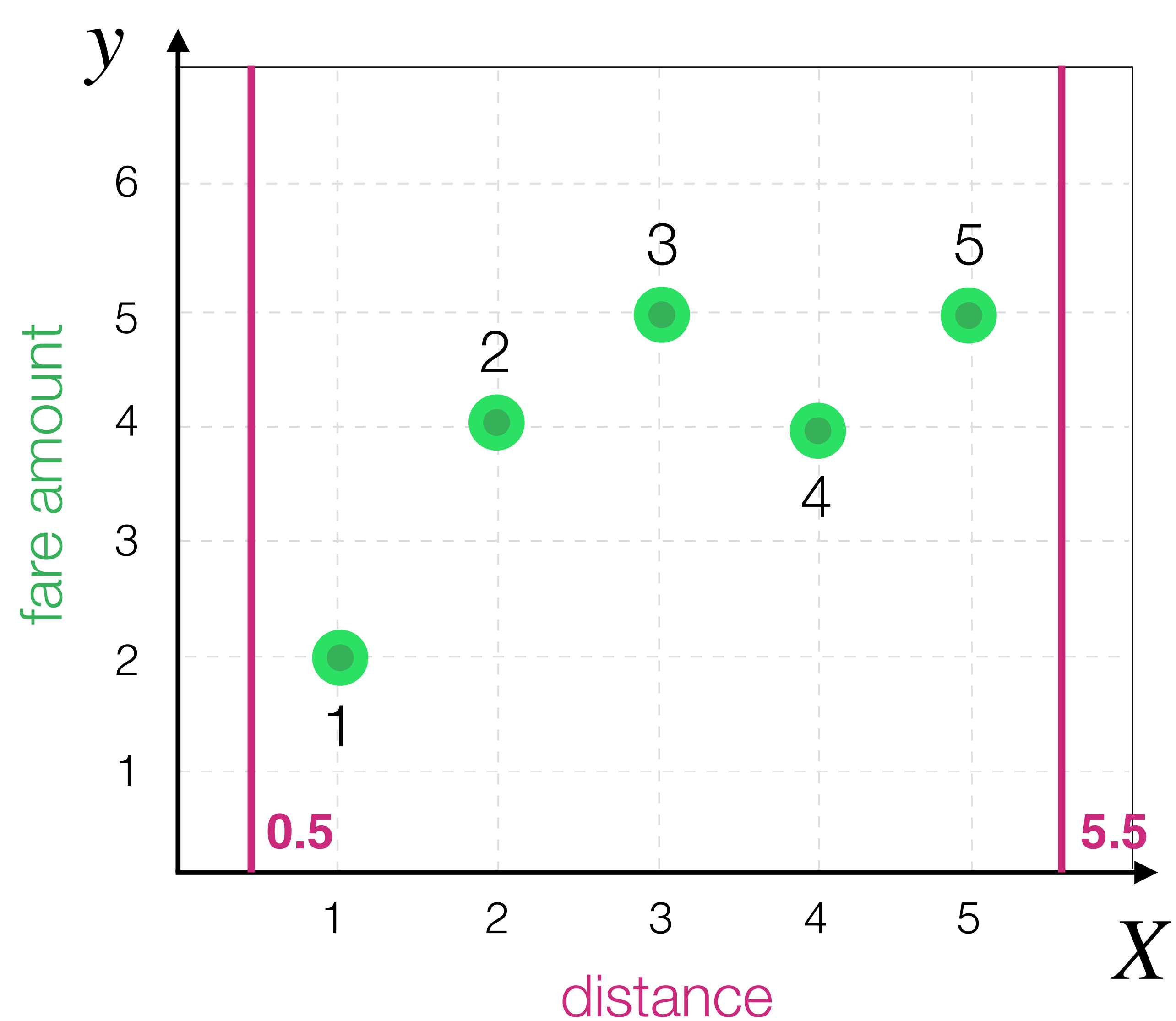
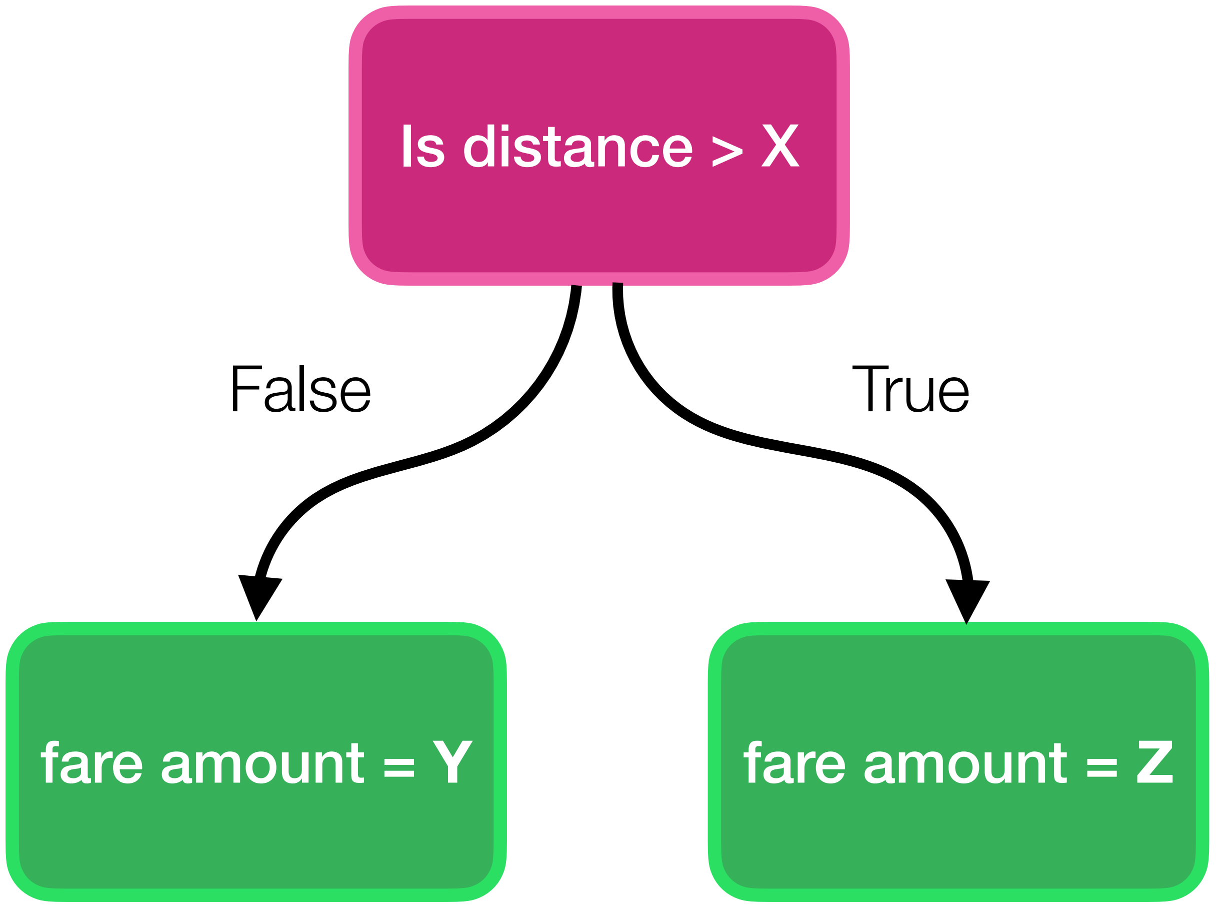




Decision Tree Algorithm

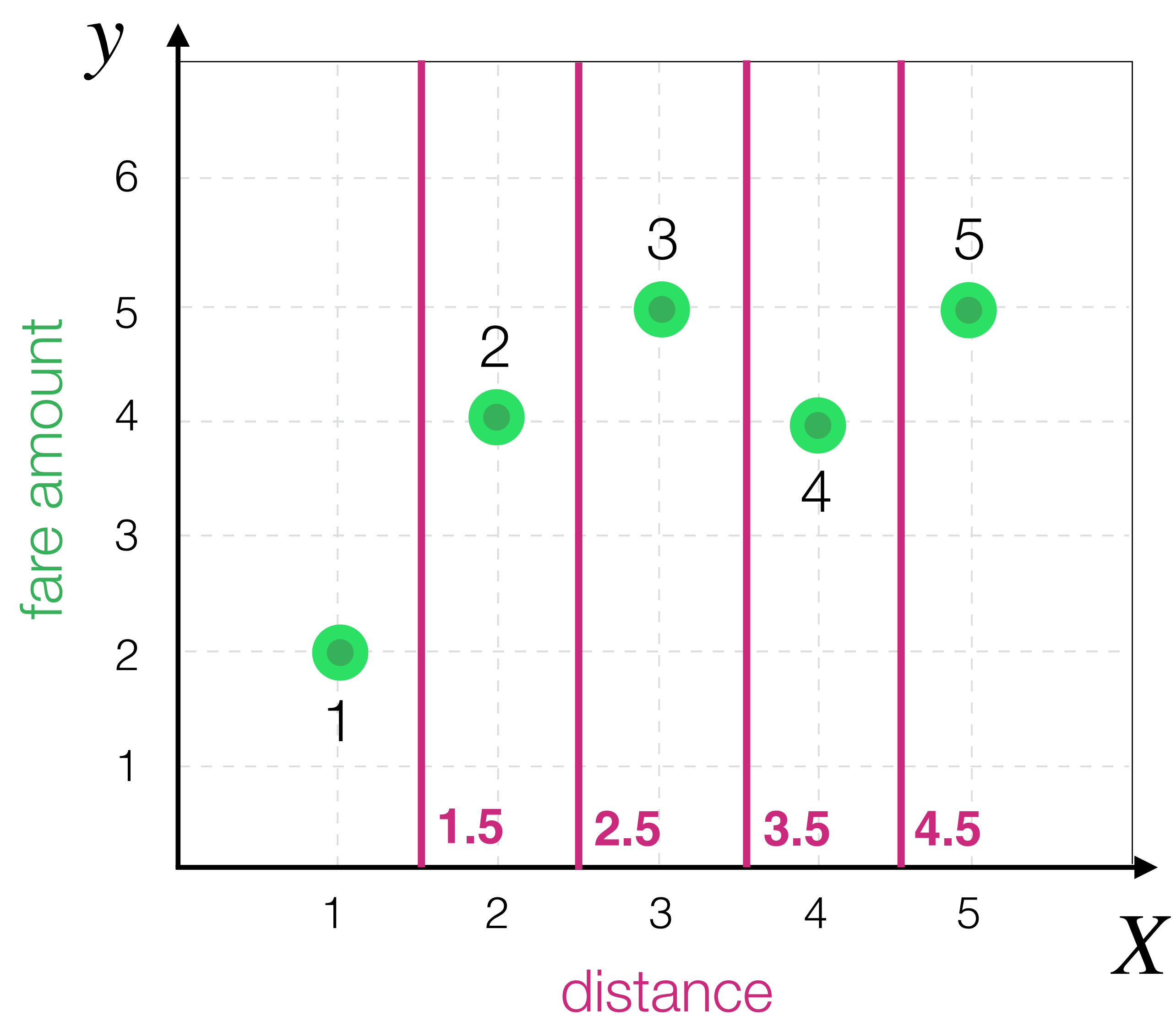
Are these meaningful?

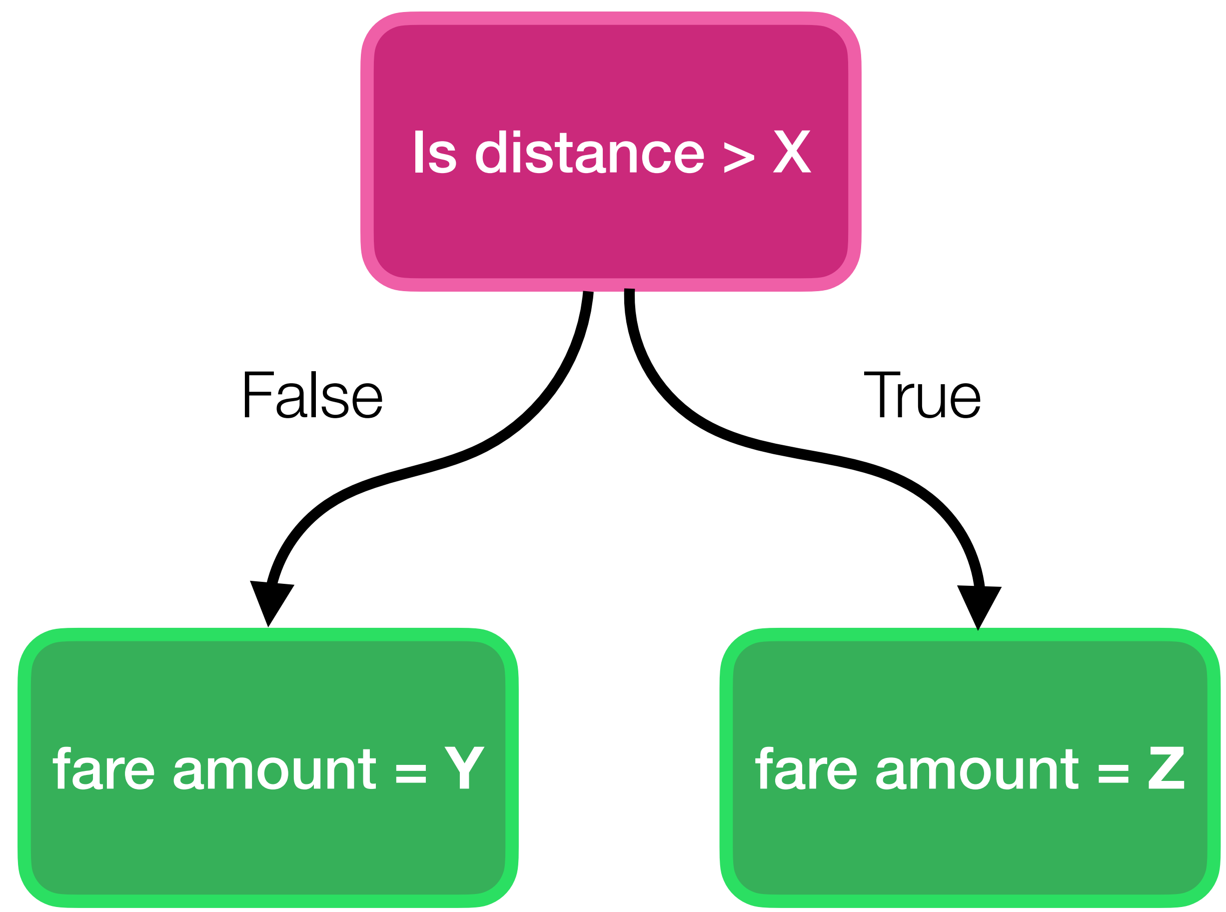




Decision Tree Algorithm

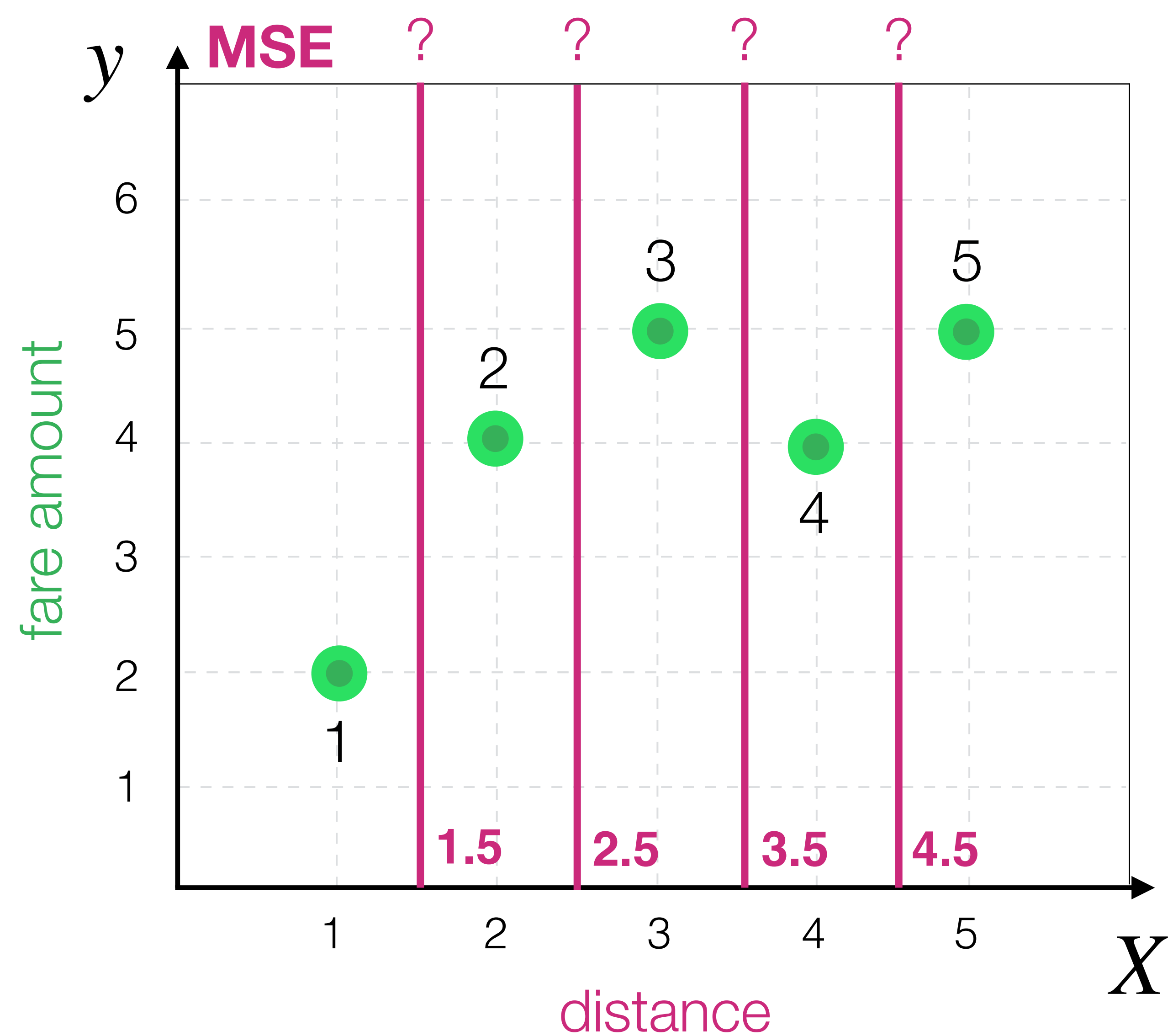
How to compare remaining?

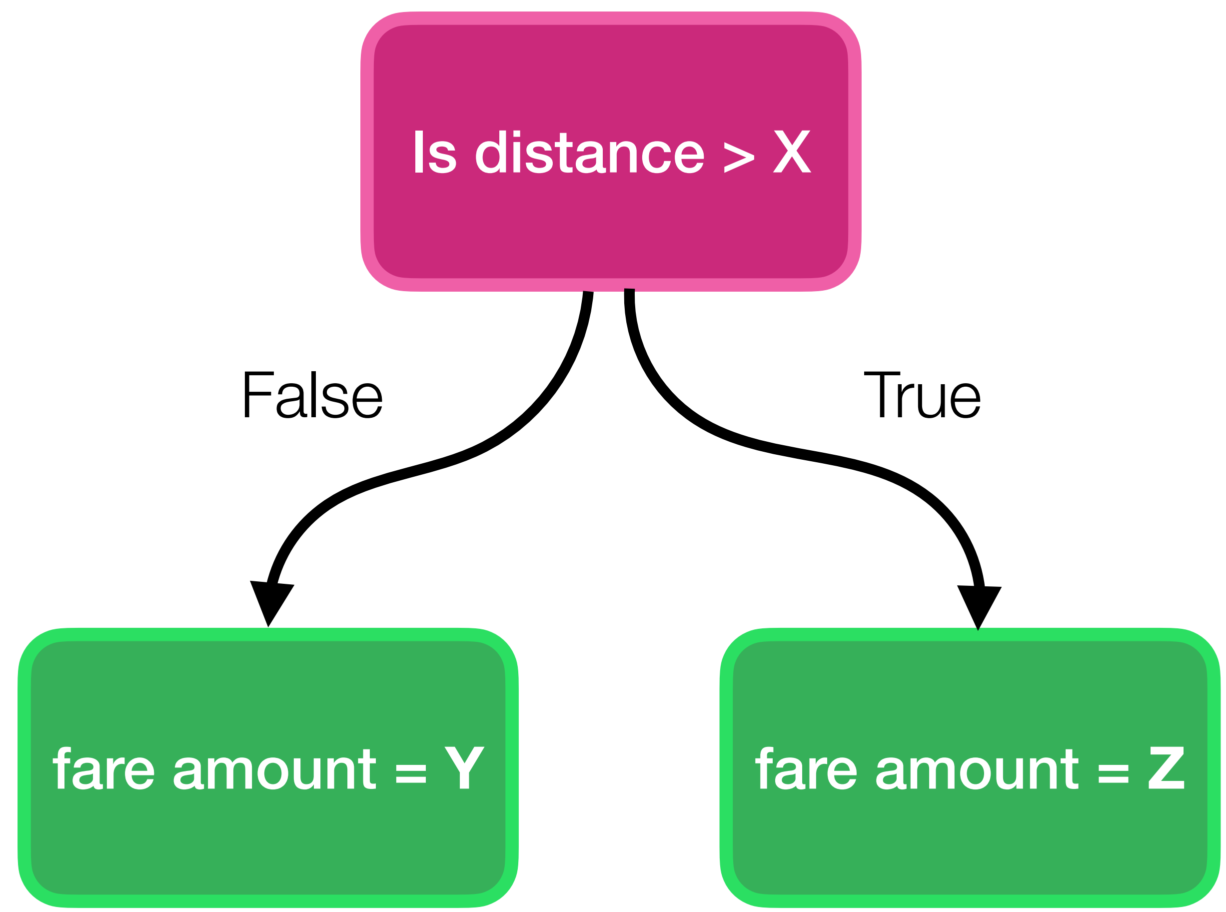




Decision Tree Algorithm

How to compare remaining?
For each one we can compute MSE





Decision Tree Algorithm

How to compare remaining?
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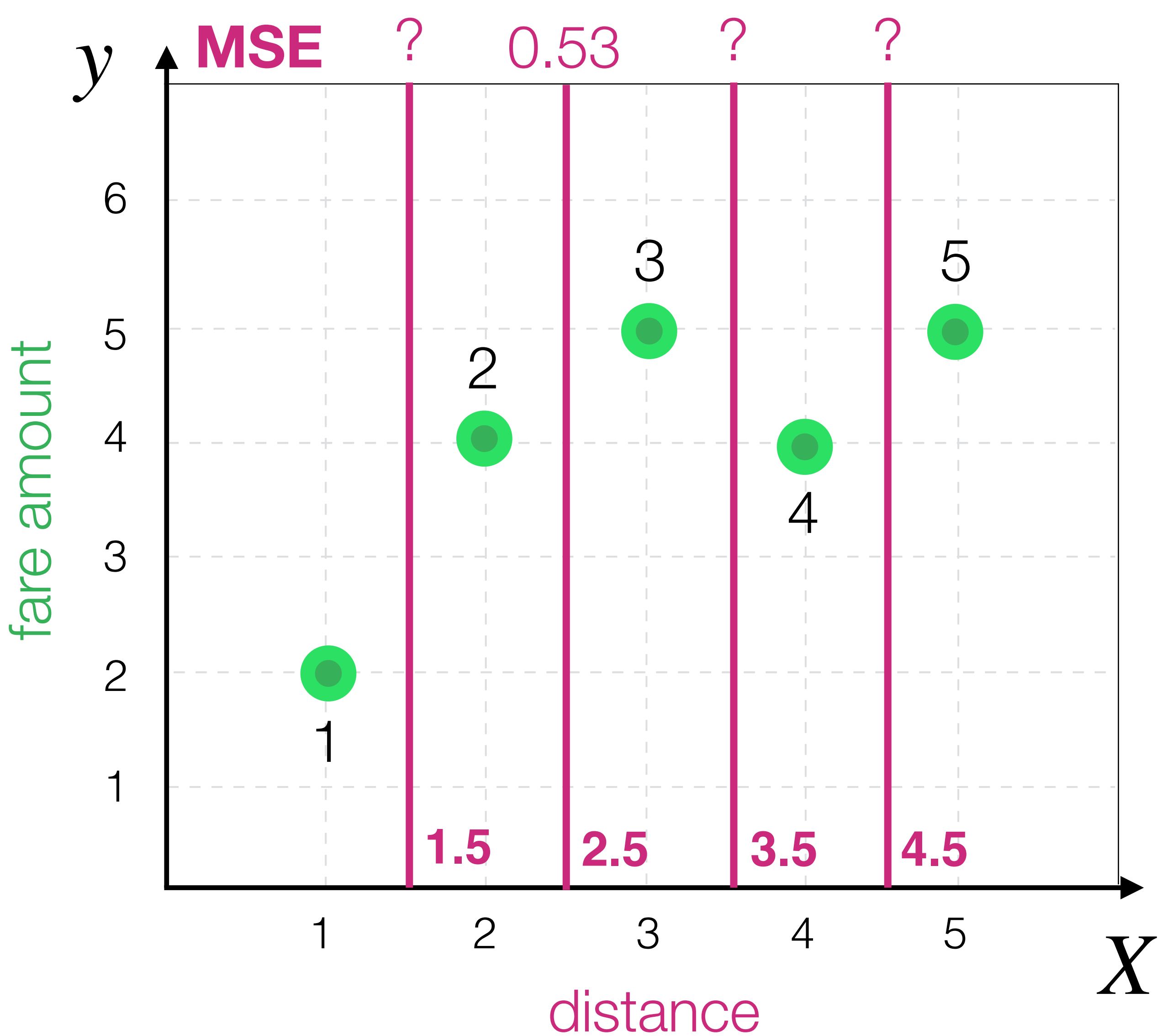
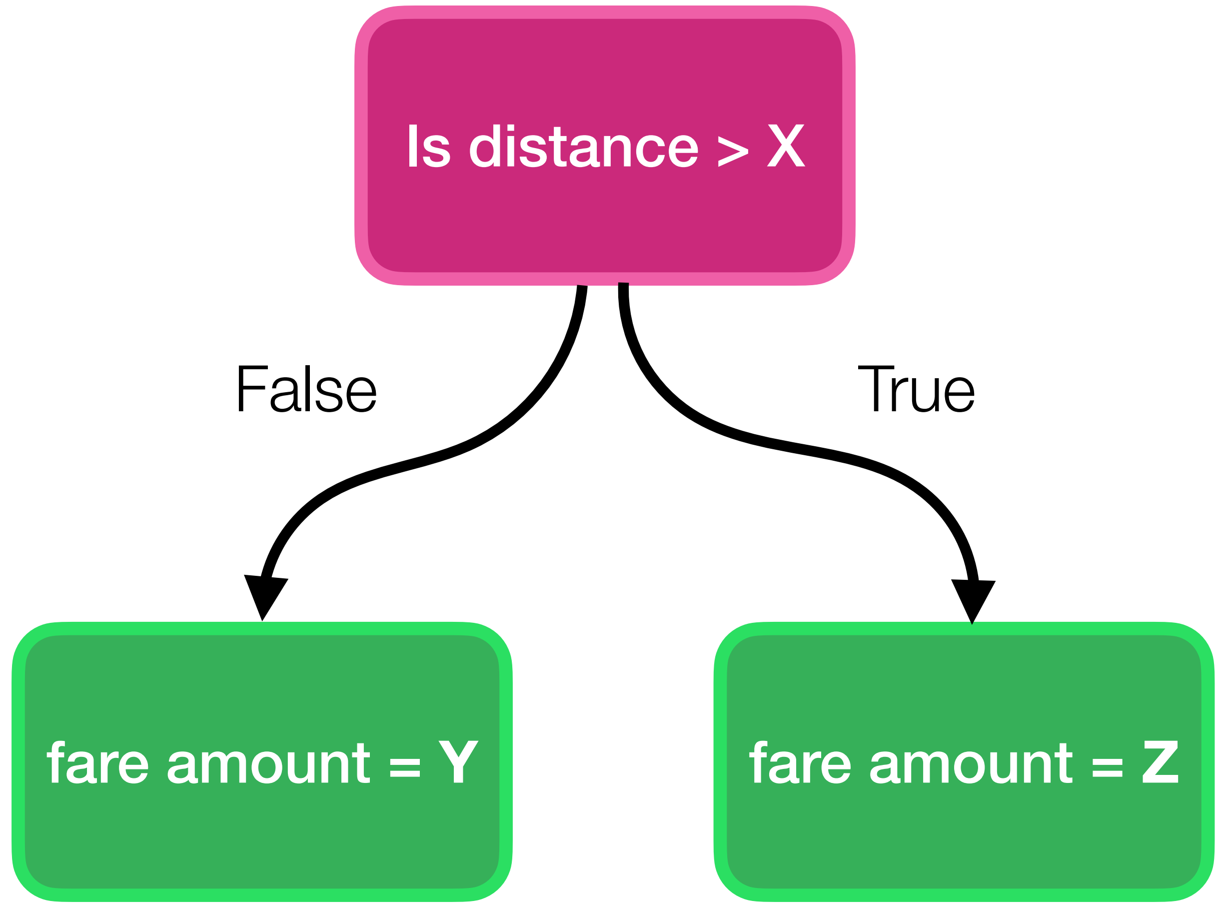
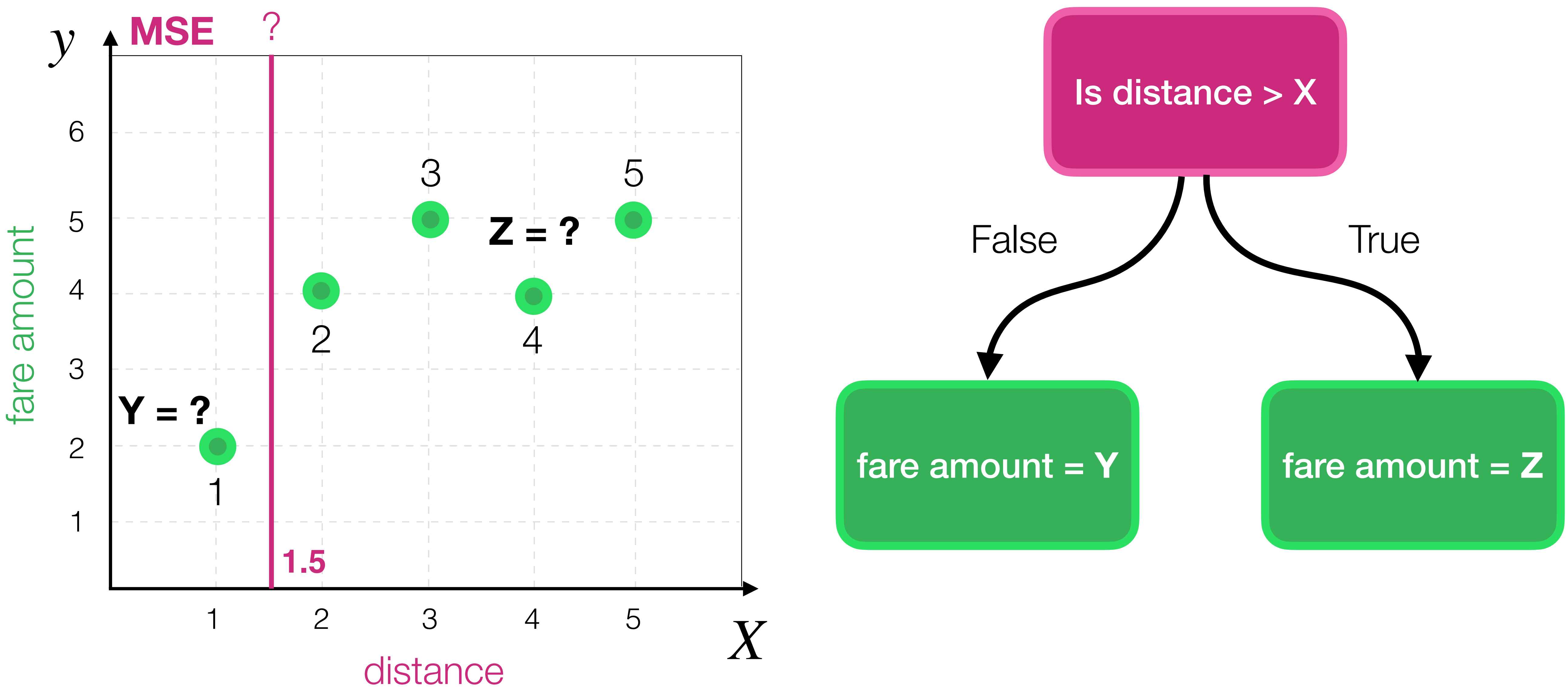
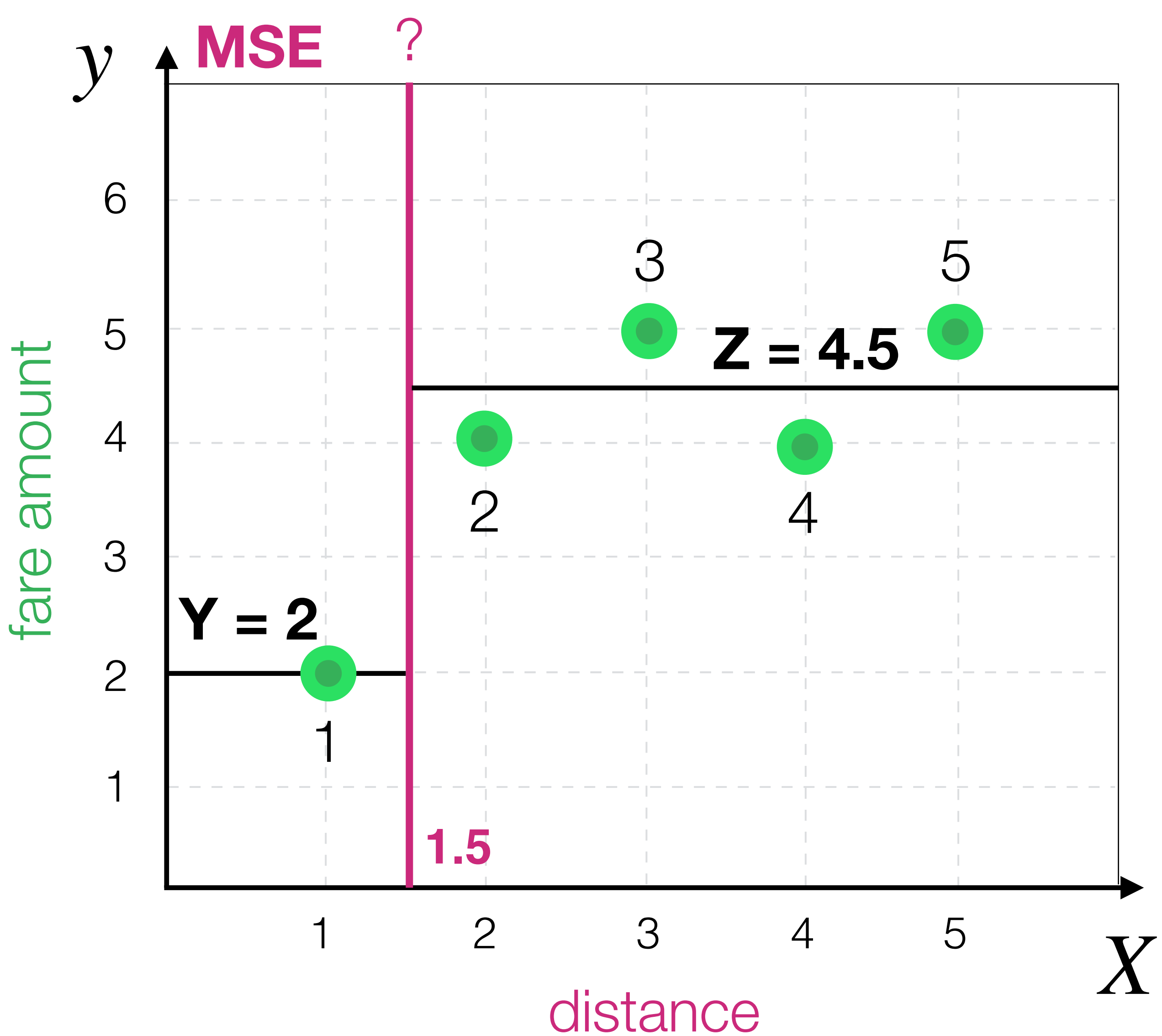


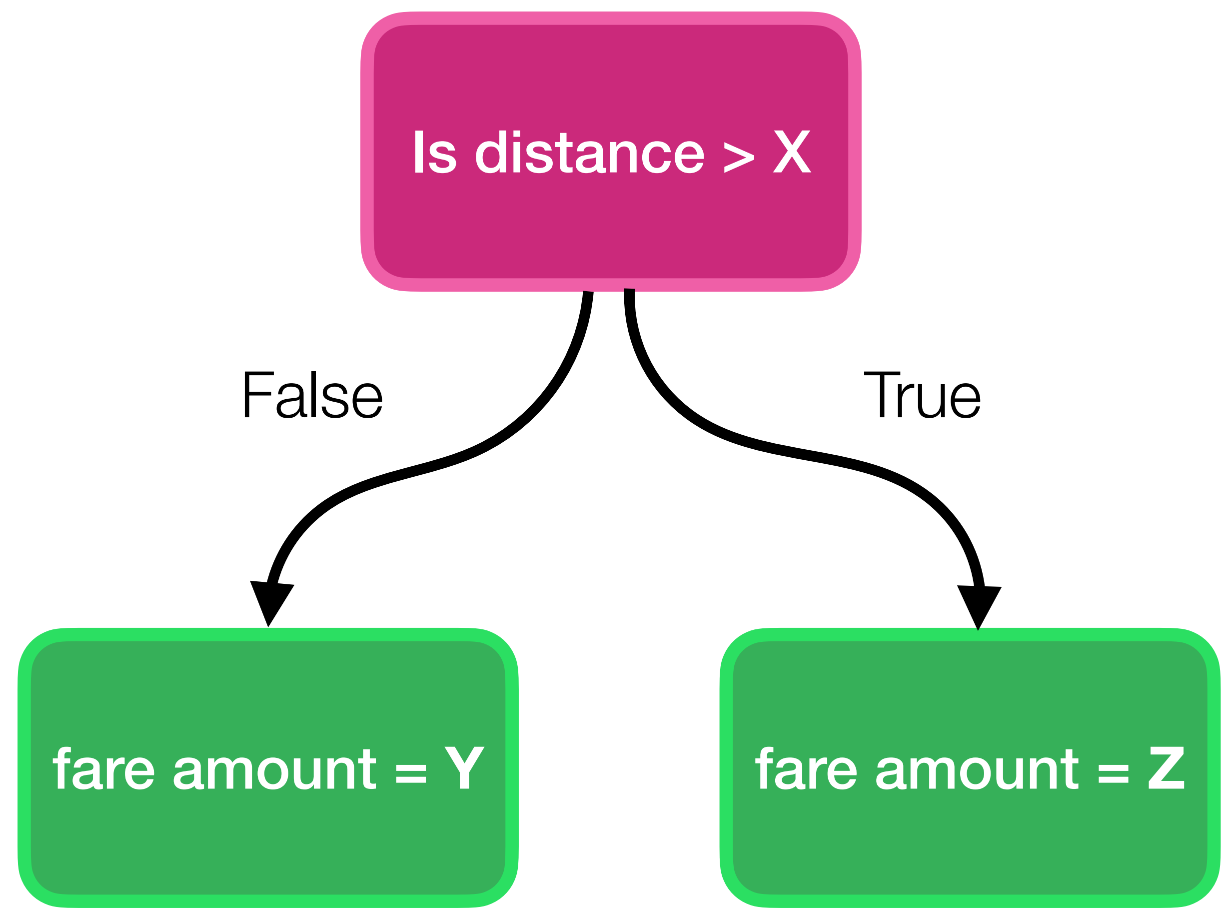


Decision Tree Algorithm

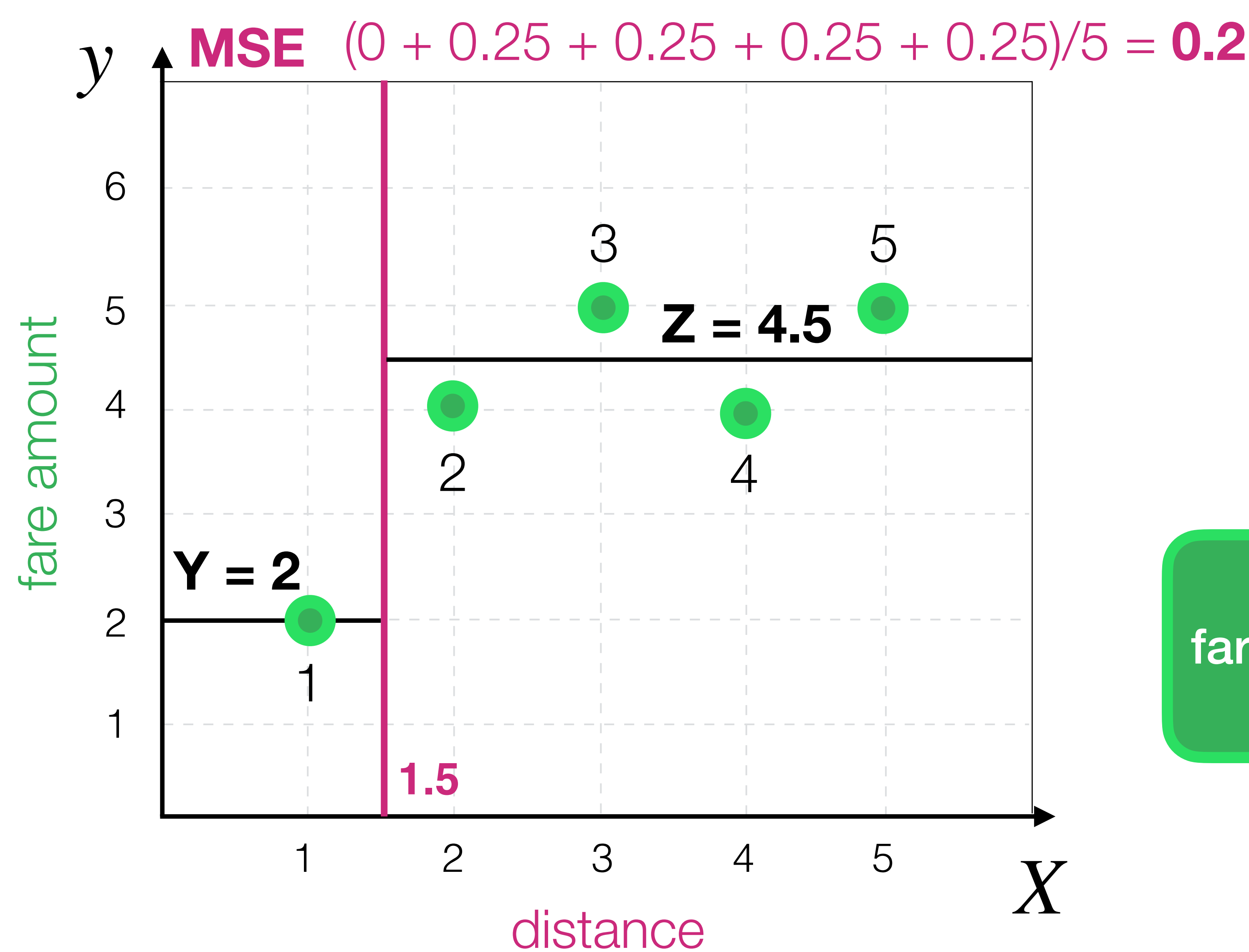


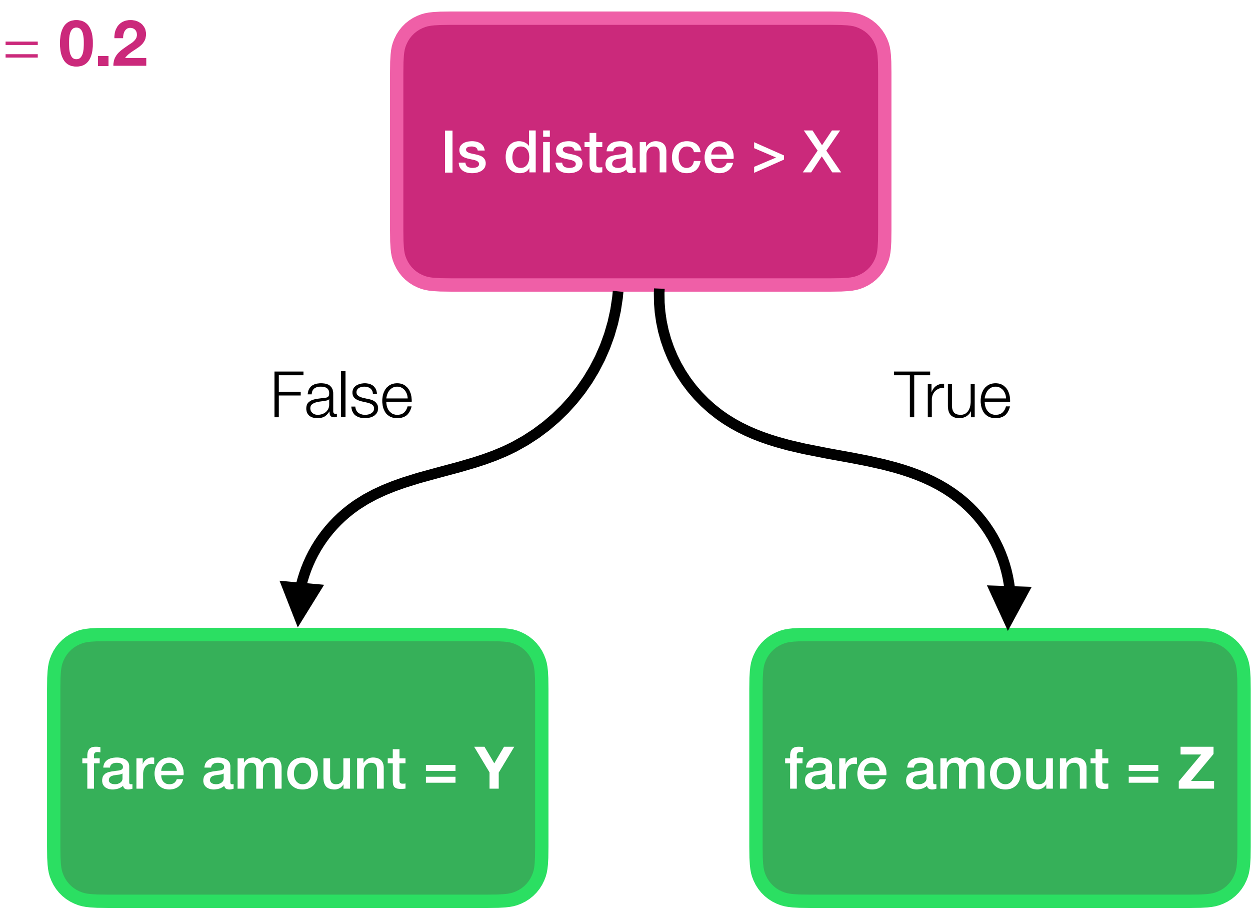
Decision Tree Algorithm





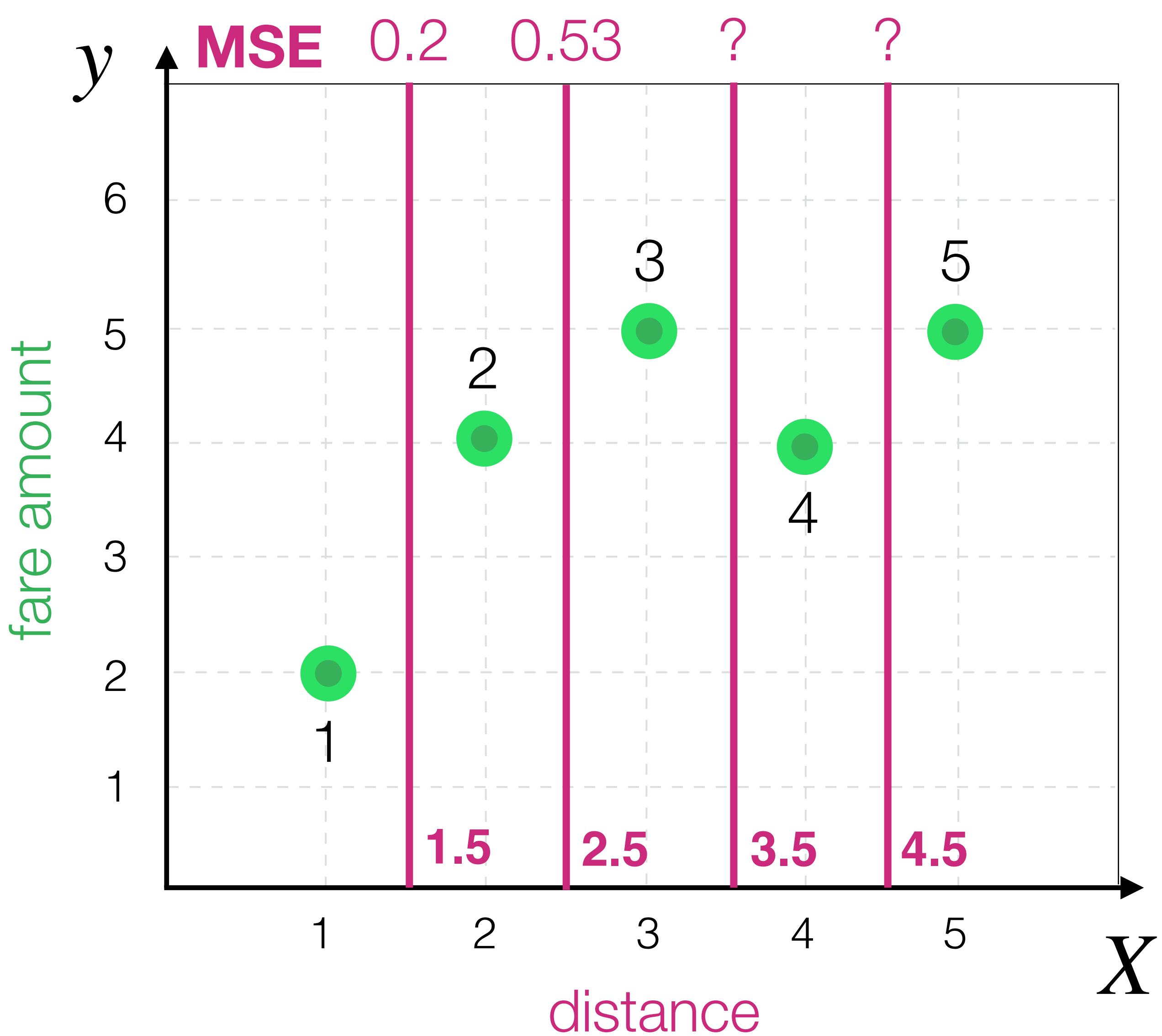
Decision Tree Algorithm

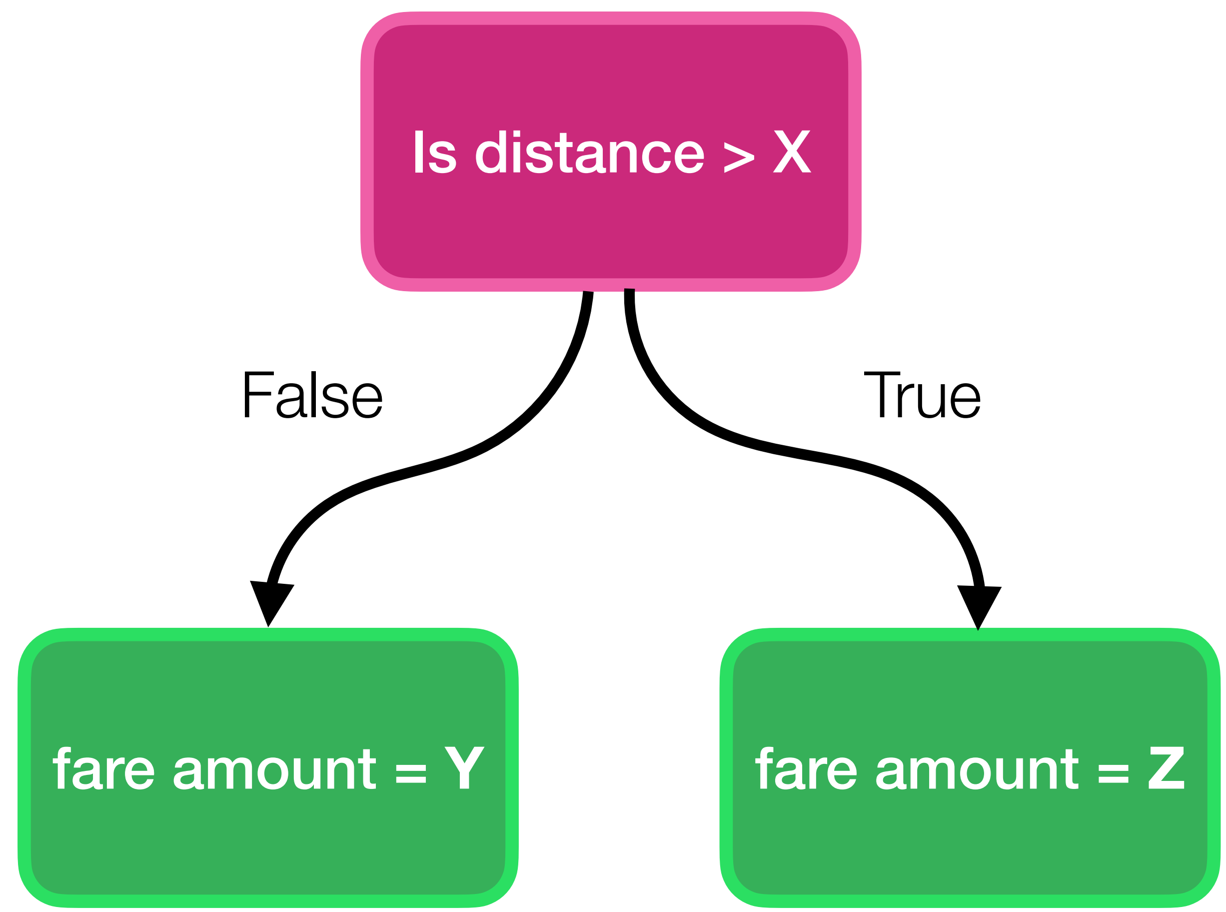




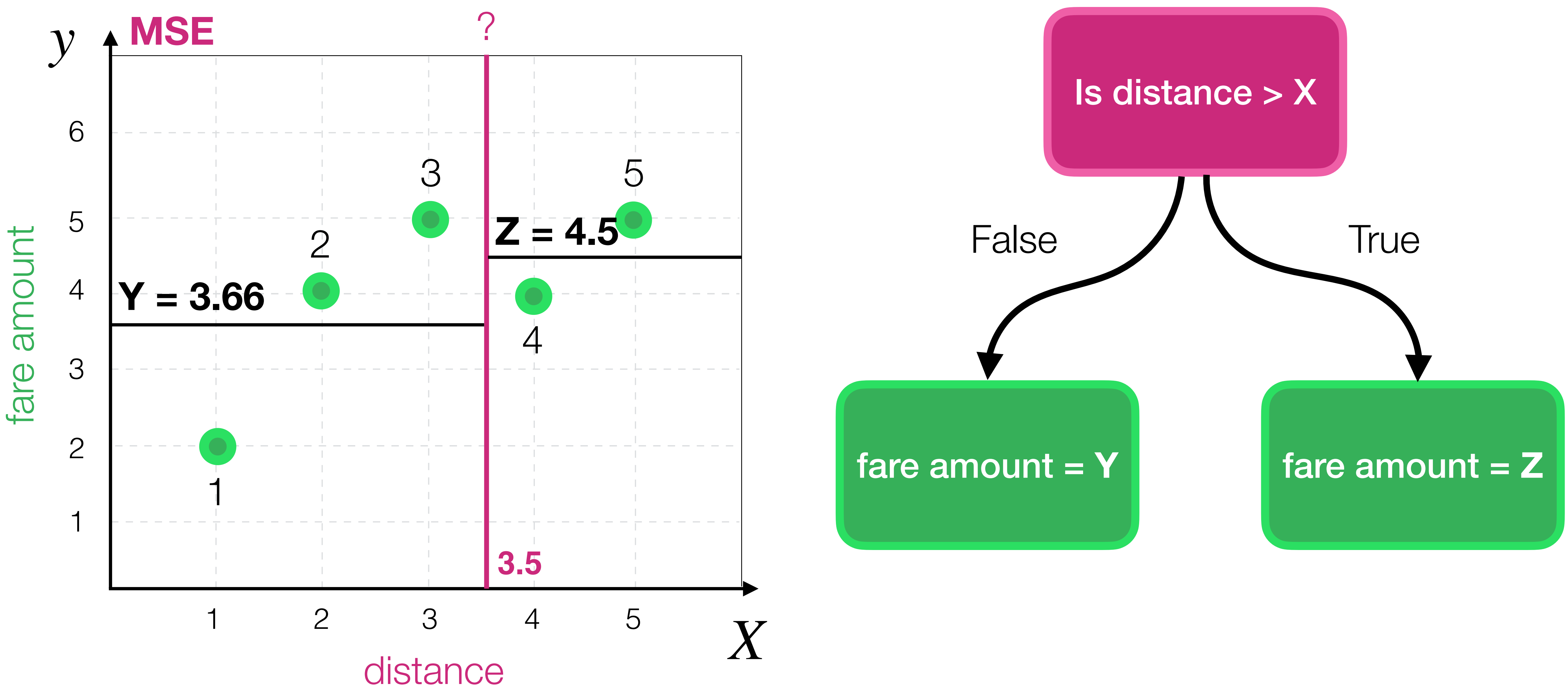
Decision Tree Algorithm

How to compare remaining?
For each one we can compute MSE

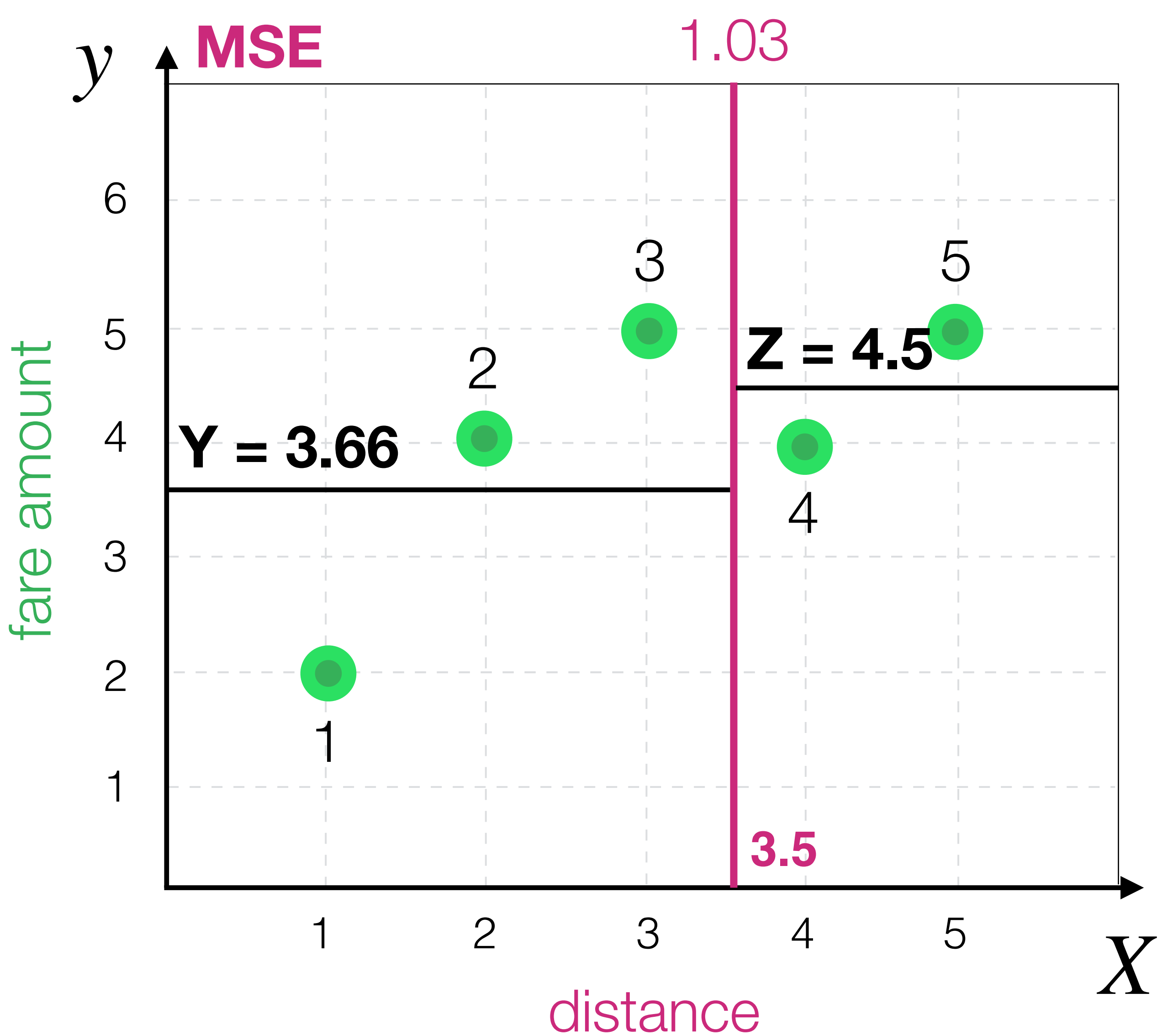


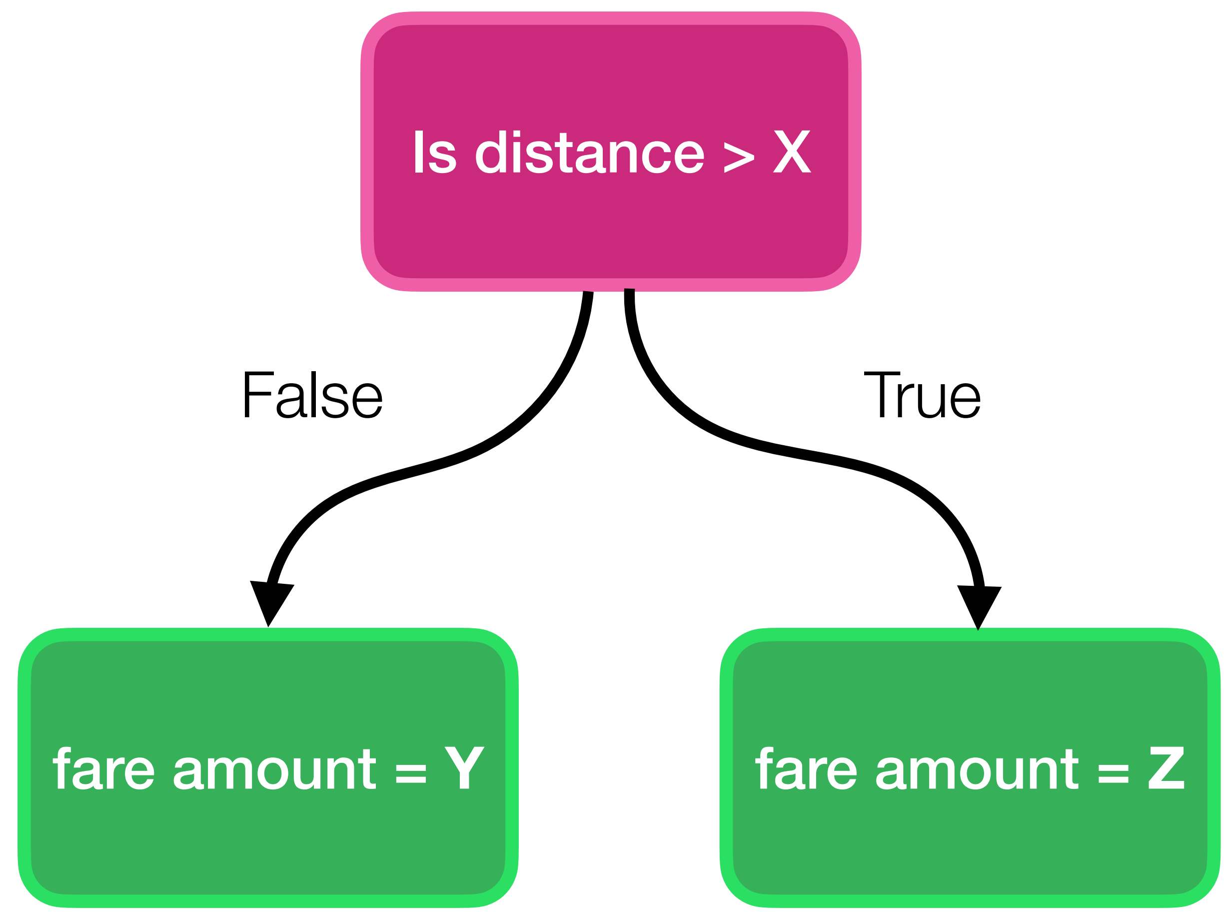


Decision Tree Algorithm



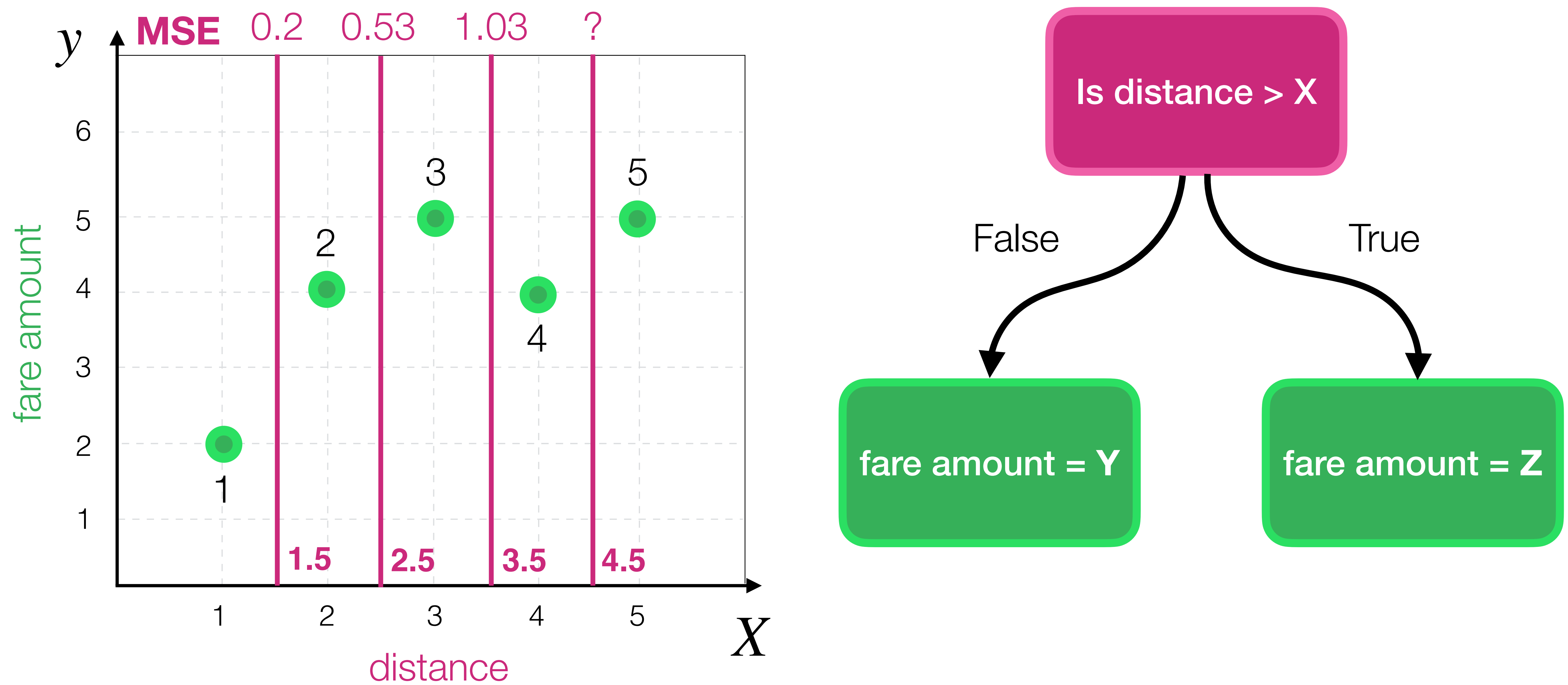
Decision Tree Algorithm



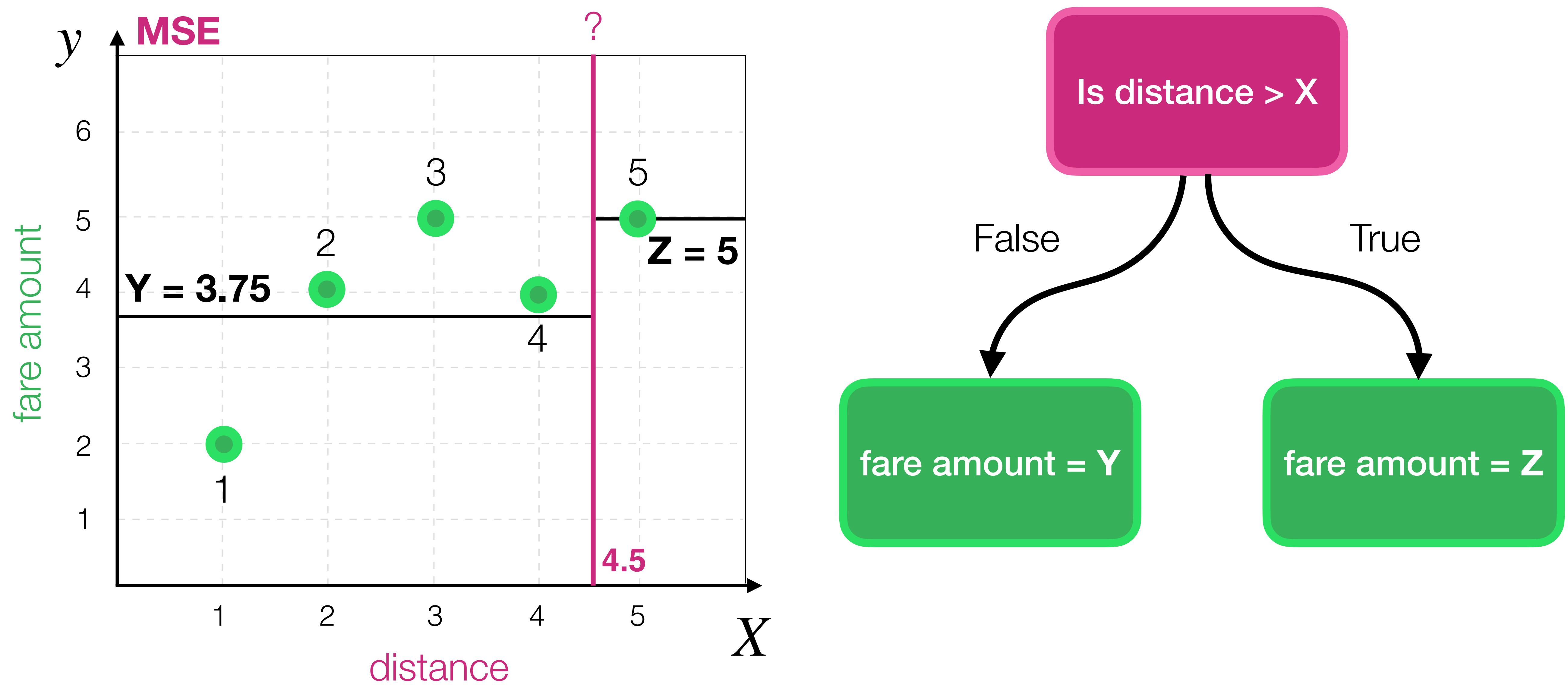


Decision Tree Algorithm

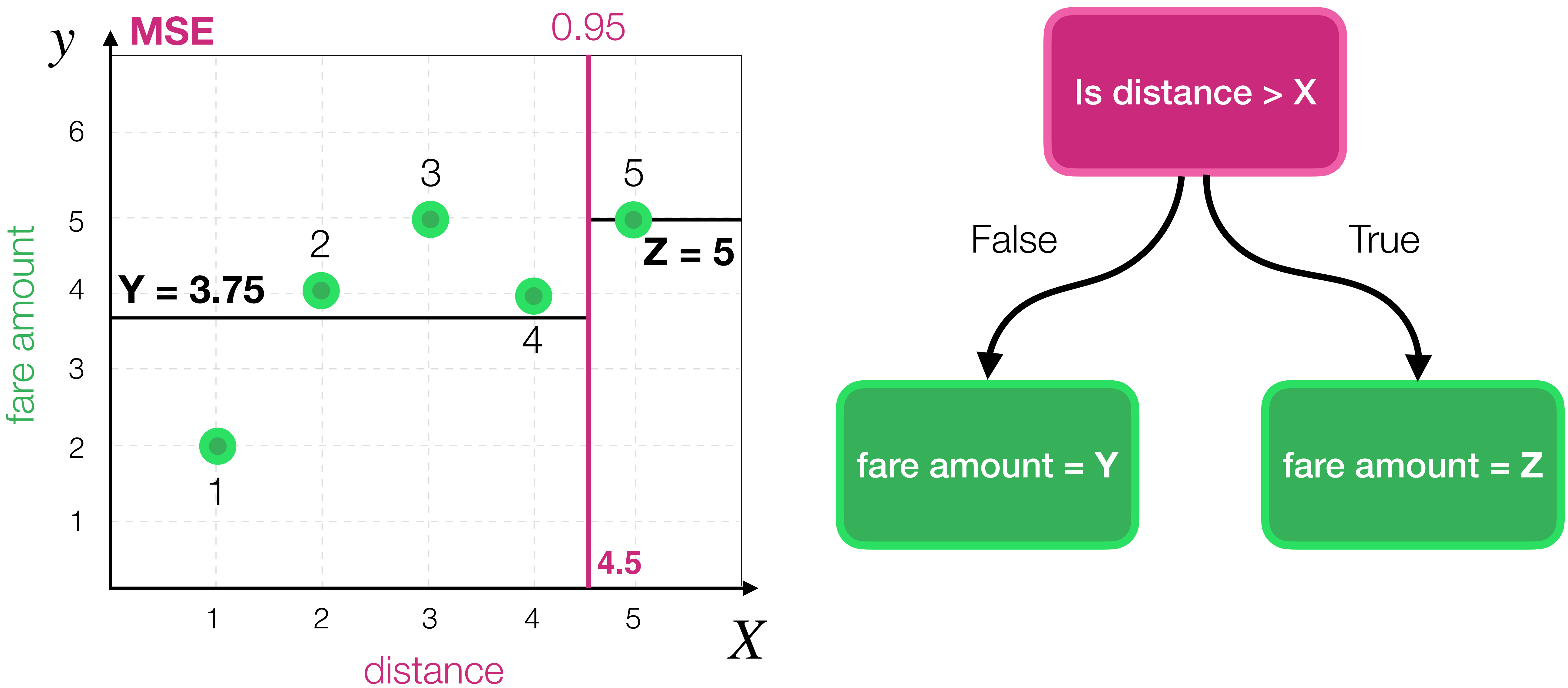
How to compare remaining?
For each one we can compute MSE



Decision Tree Algorithm

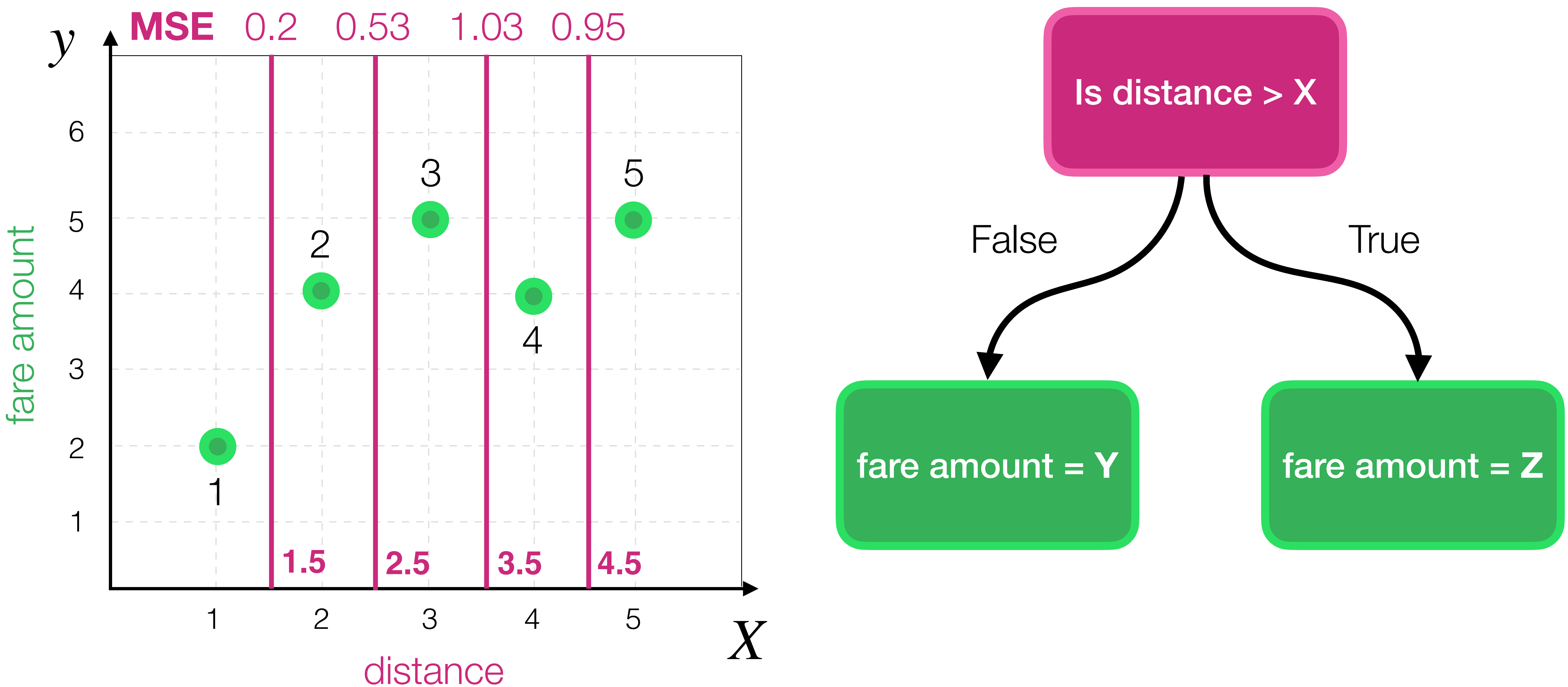


Decision Tree Algorithm



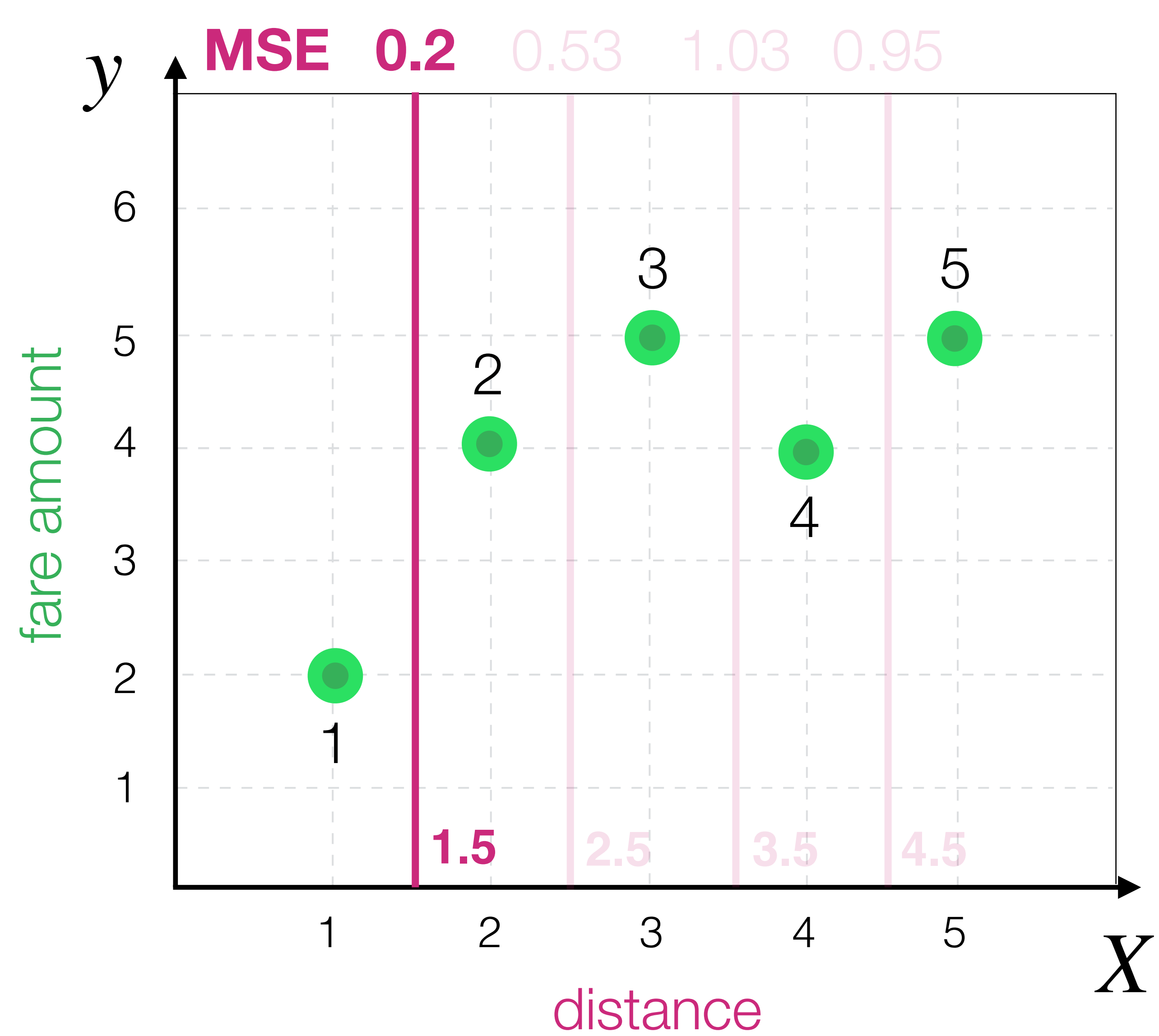
Decision Tree Algorithm

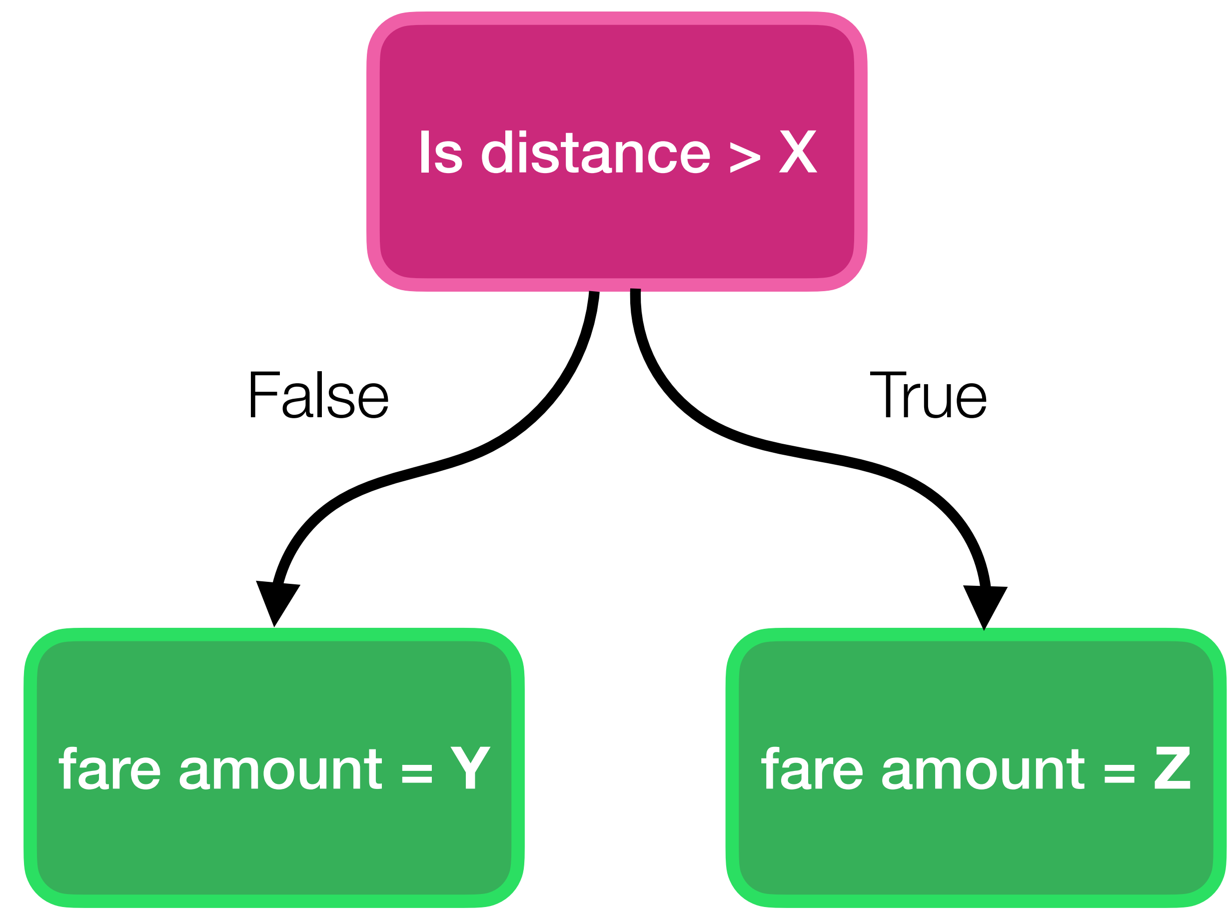
How to compare remaining?
For each one we can compute MSE



Decision Tree Algorithm

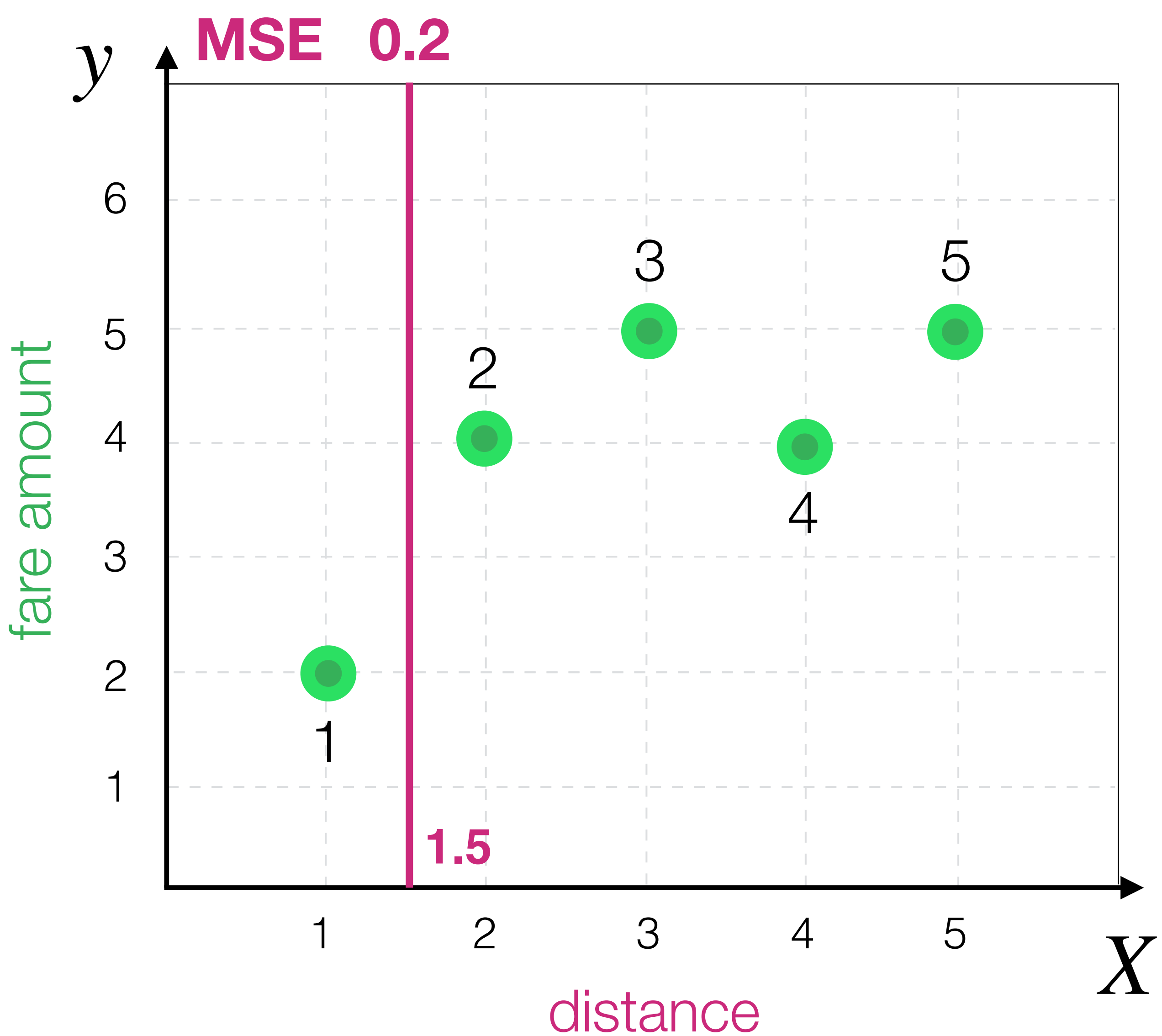
We choose the **split** that minimises total MSE

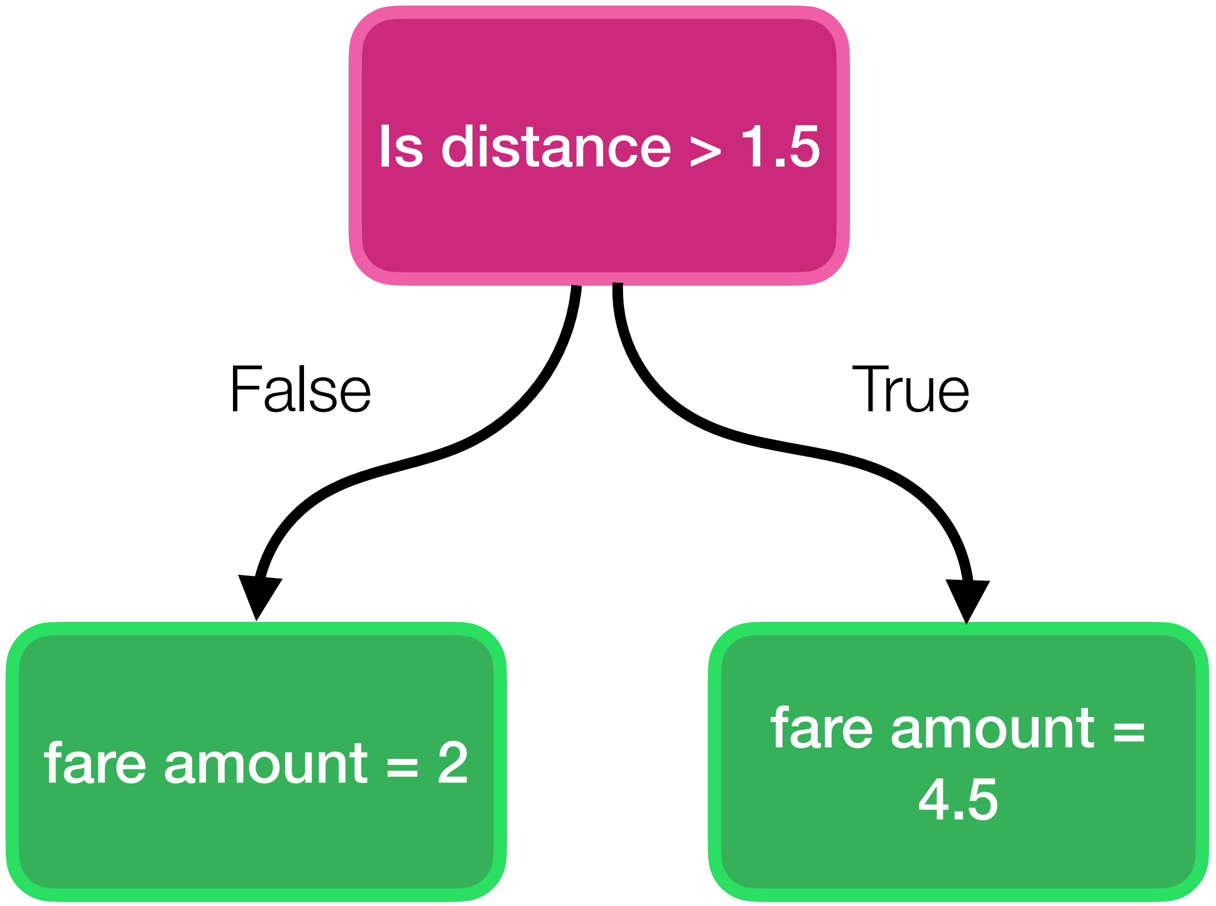




Decision Tree Algorithm

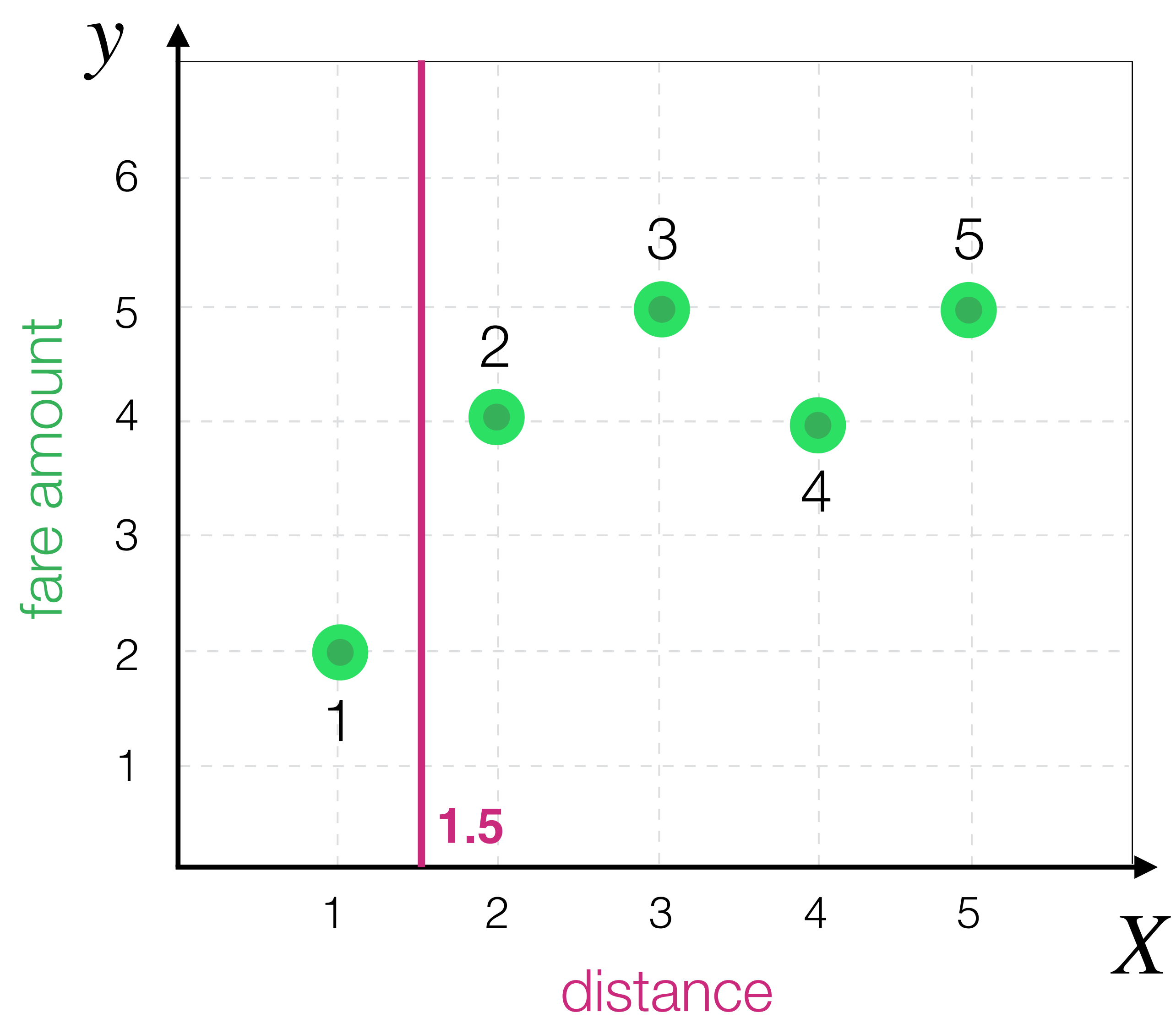
Thus, the resulting tree:

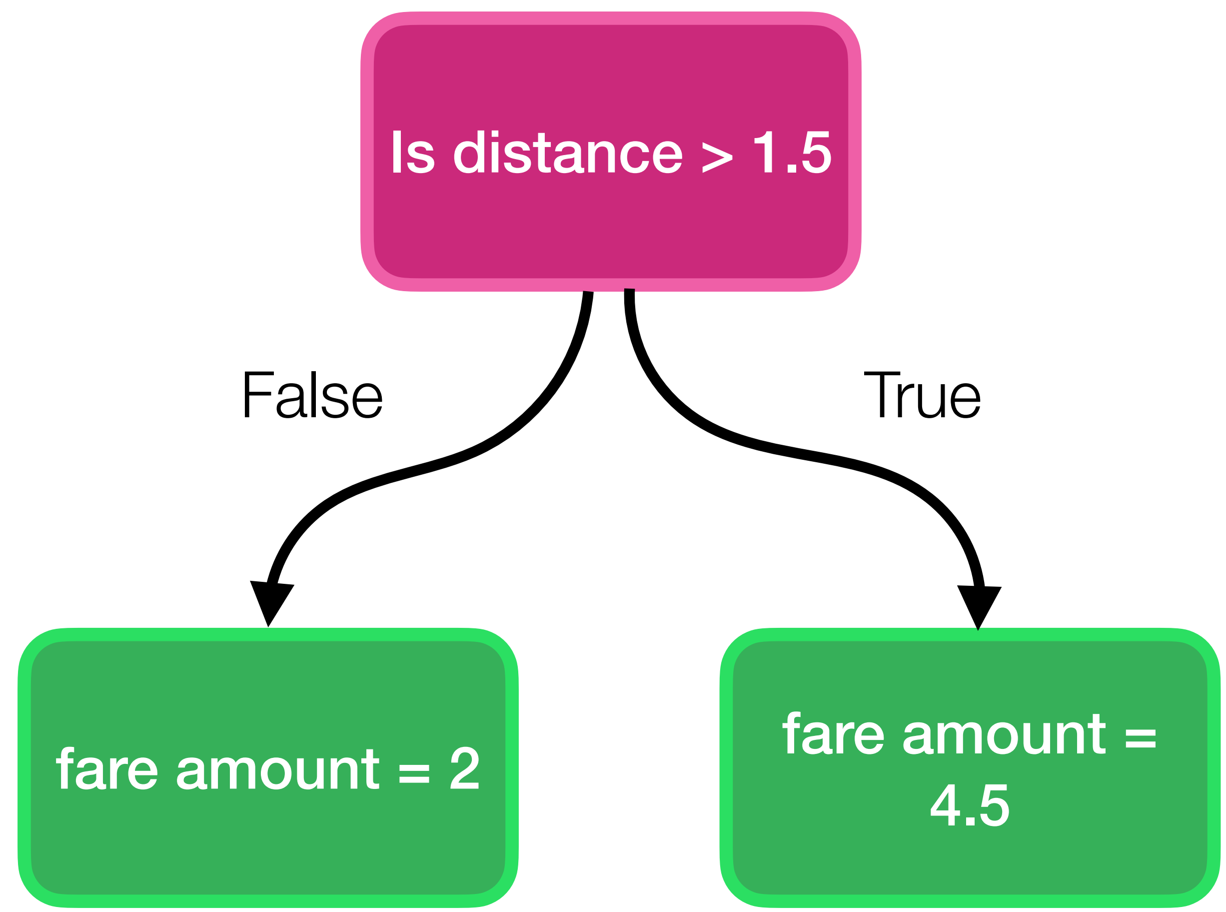




Decision Tree Algorithm

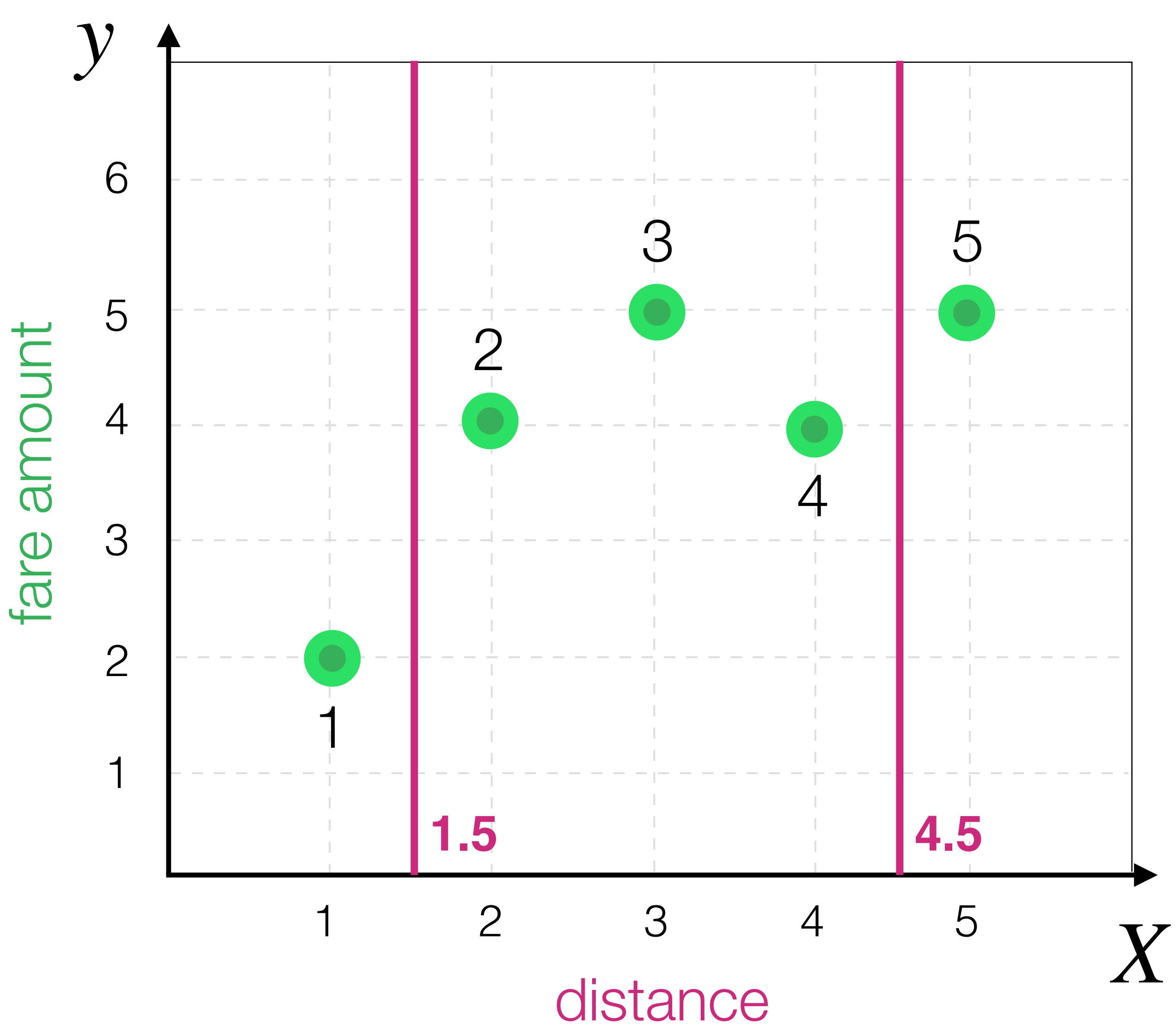
Can we make our decision tree more accurate?

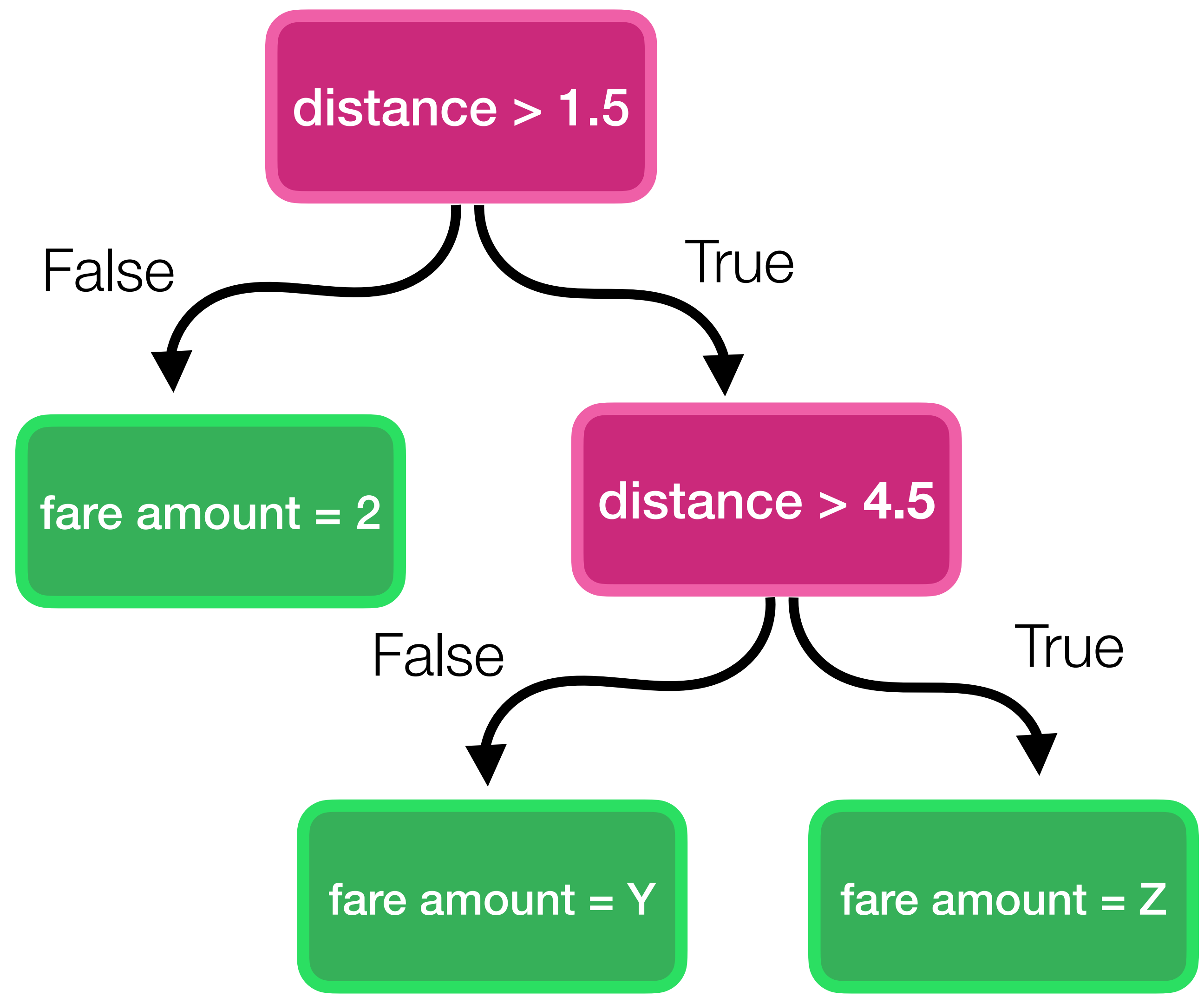




Decision Tree Algorithm

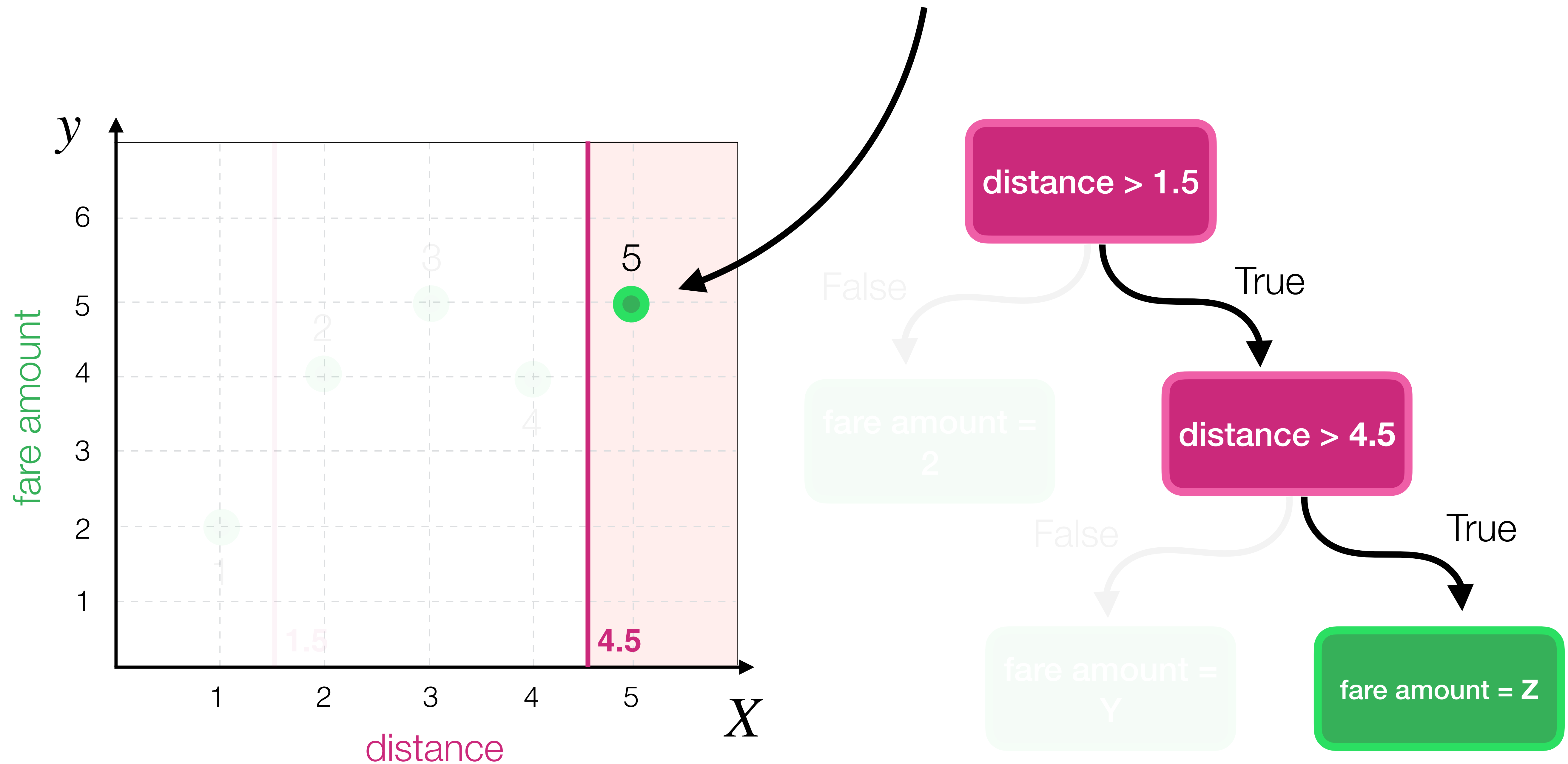
Can we make our decision tree more accurate?
Yes, by going **deeper**!





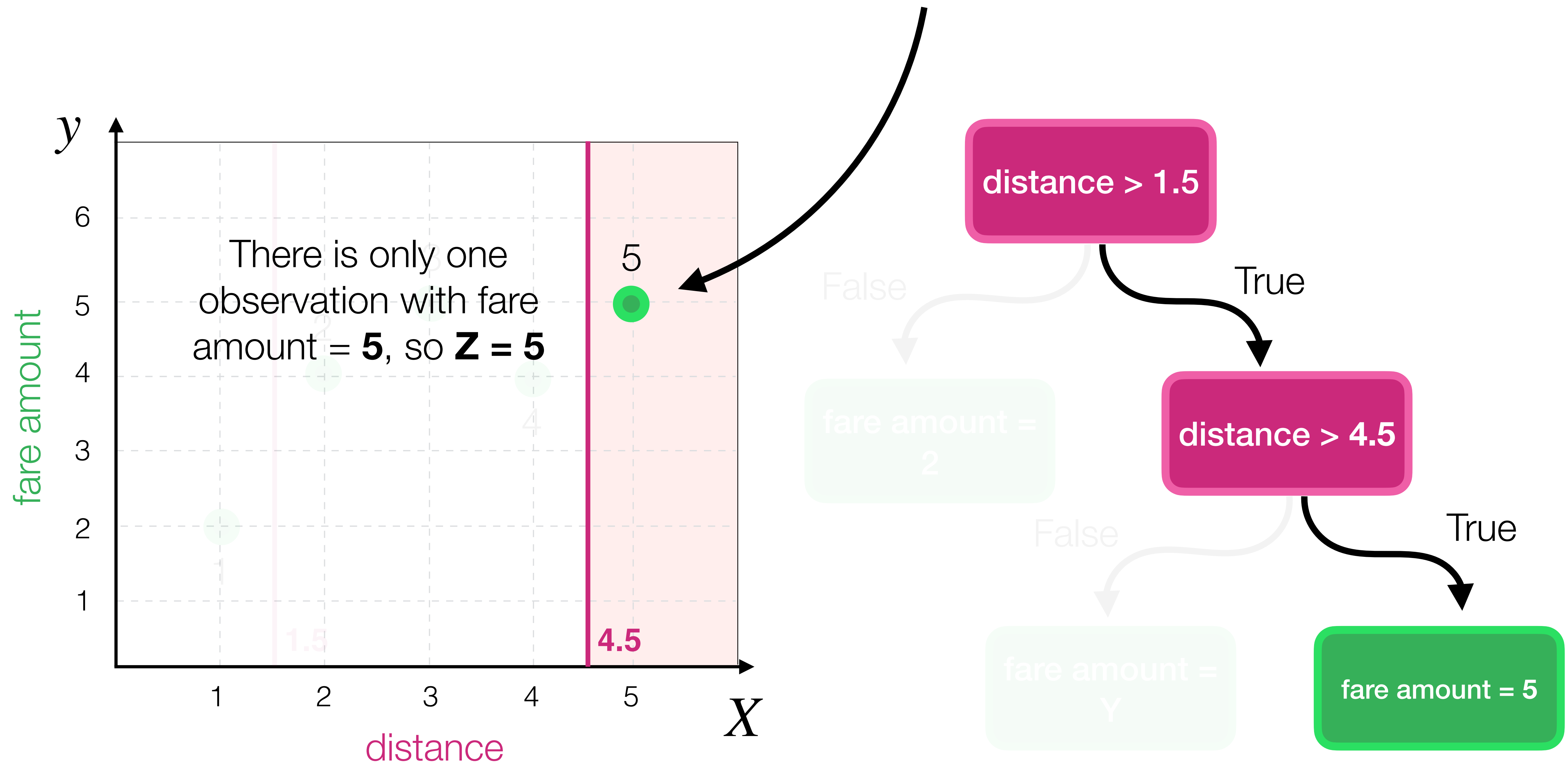
Decision Tree Algorithm

What is the best value for **Z** in this case?

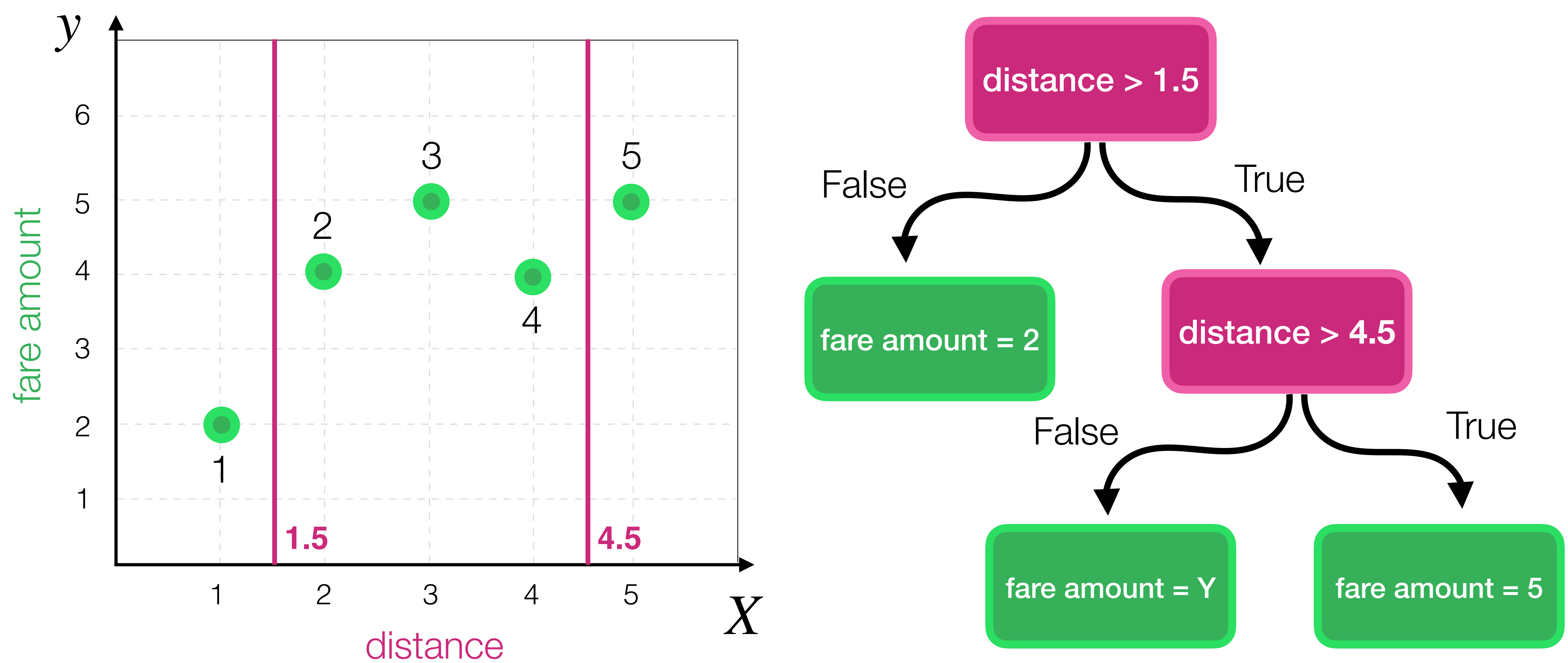


Decision Tree Algorithm

What is the best value for **Z** in this case?

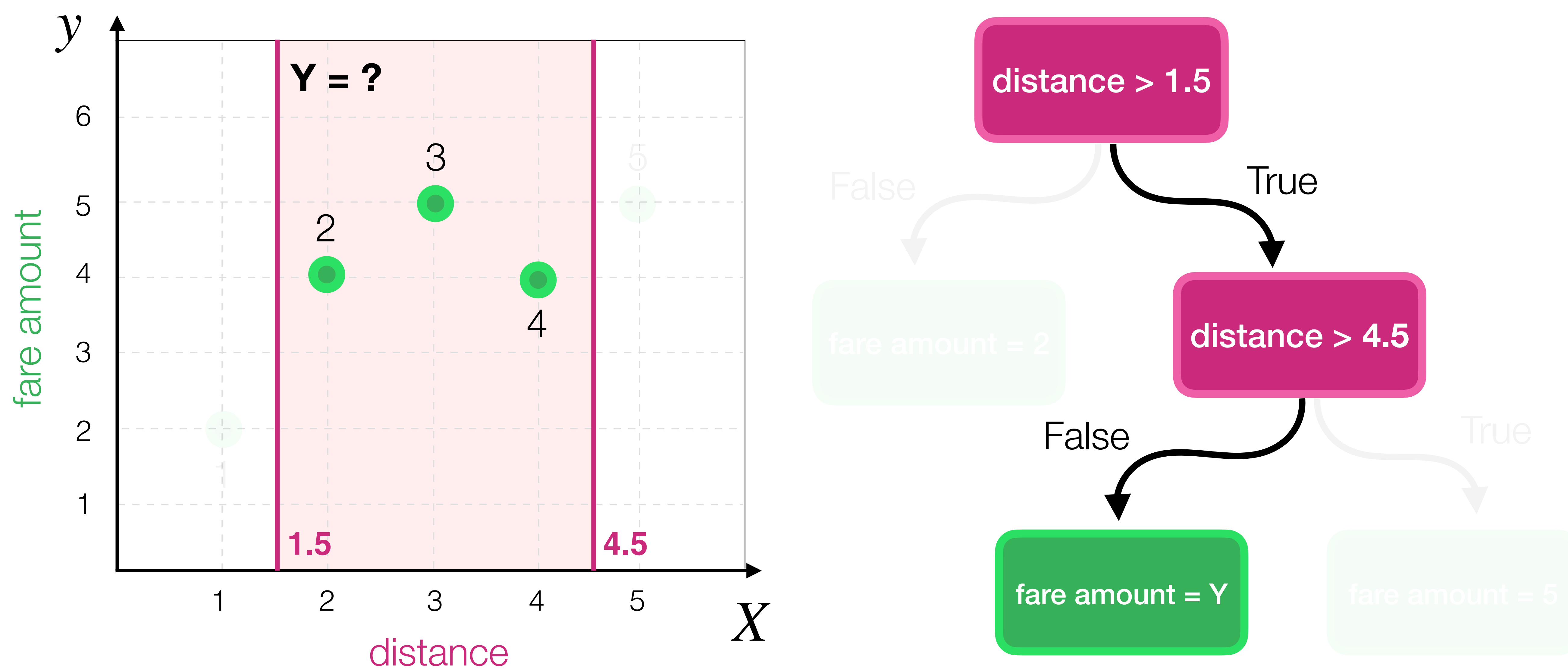


Decision Tree Algorithm



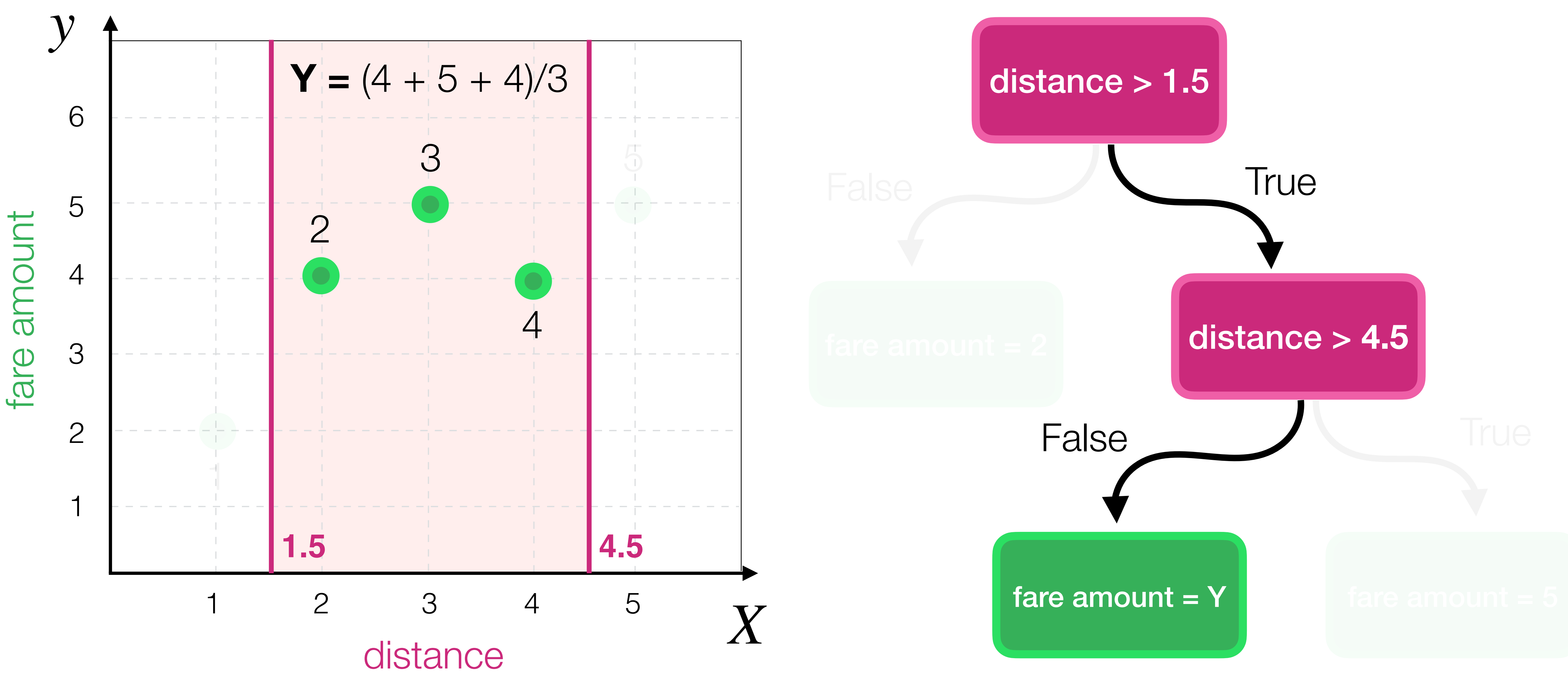
Decision Tree Algorithm

What is the best value for **Y** in this case?



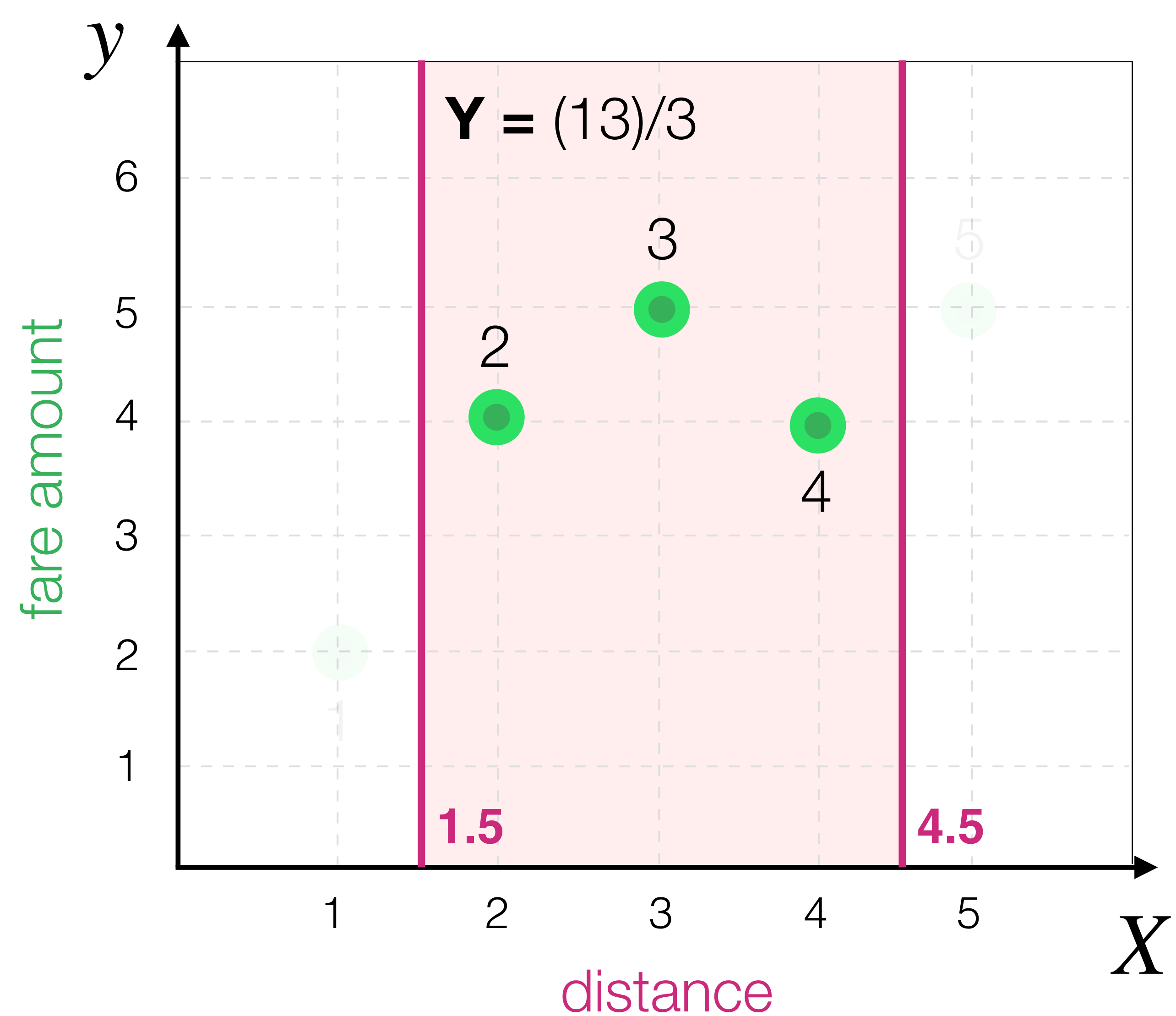
Decision Tree Algorithm

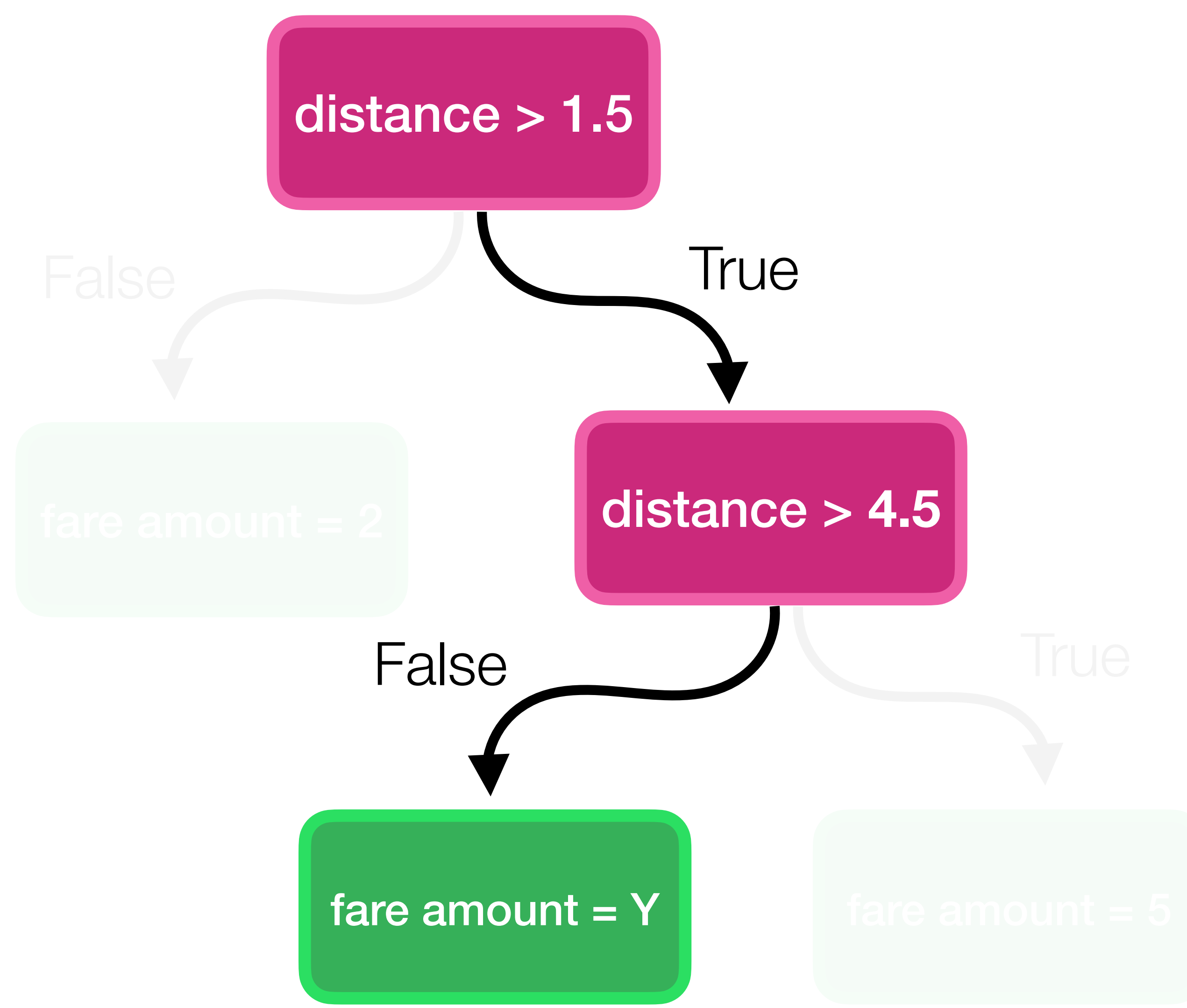
What is the best value for **Y** in this case?



Decision Tree Algorithm

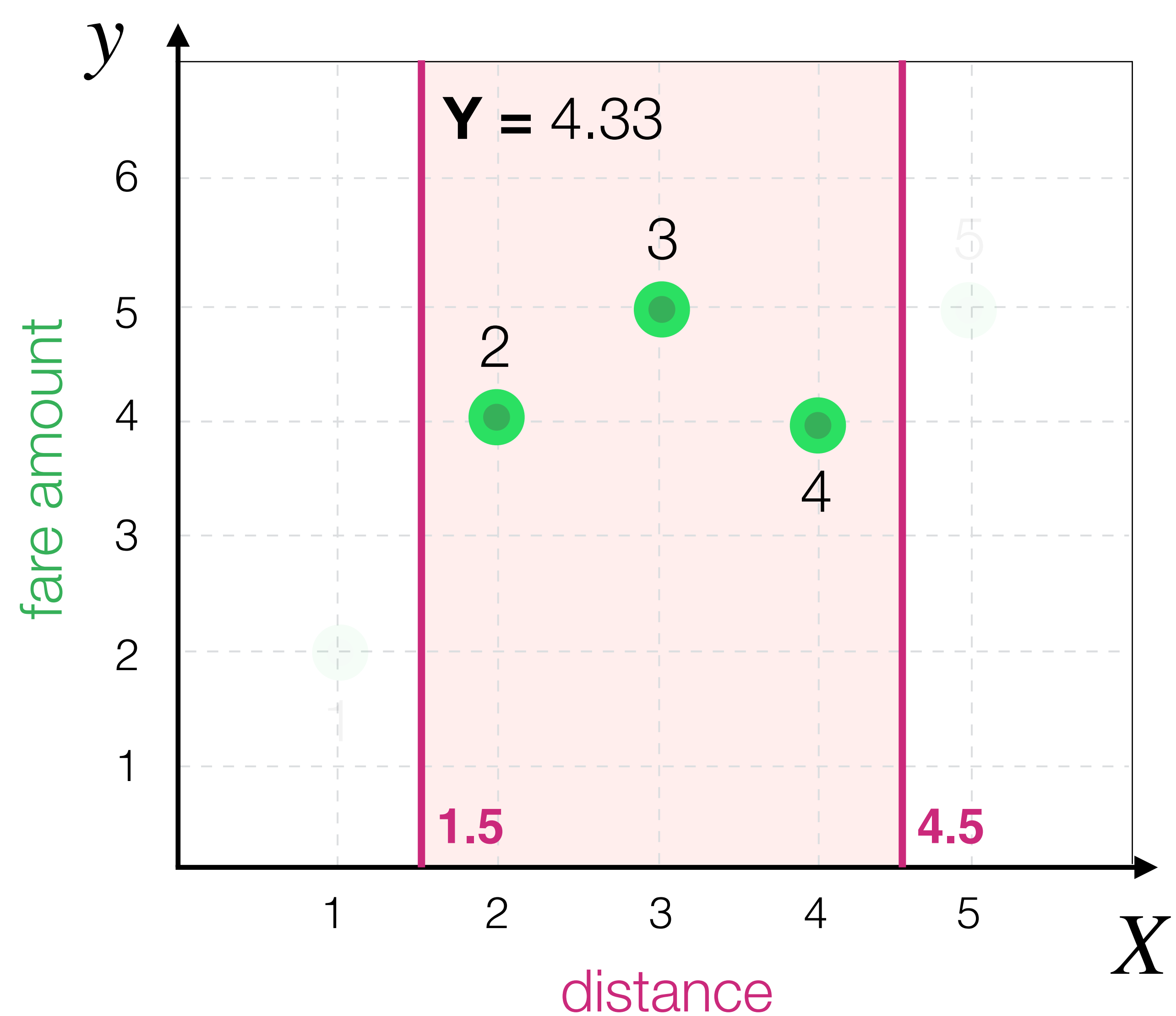
What is the best value for **Y** in this case?

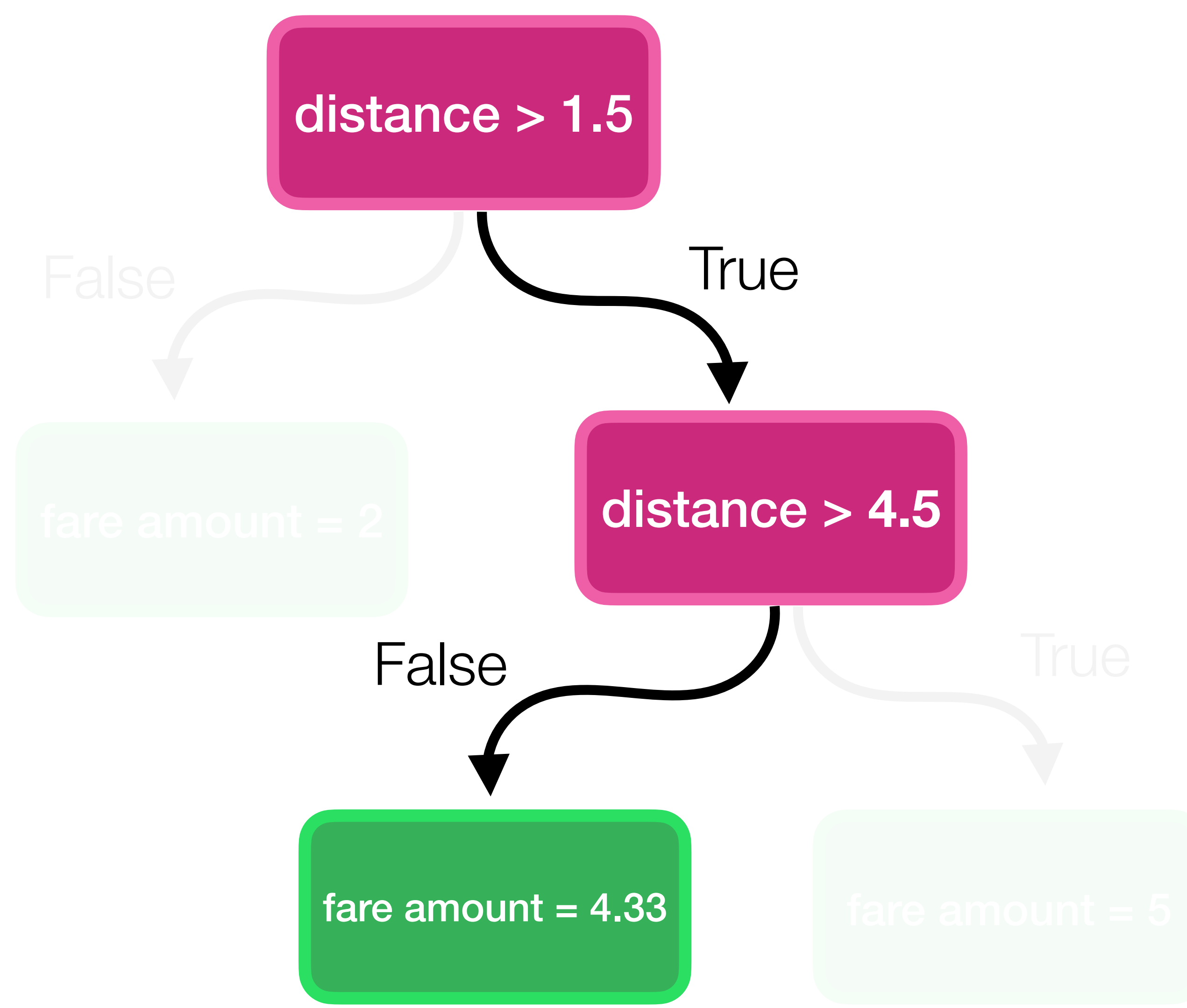




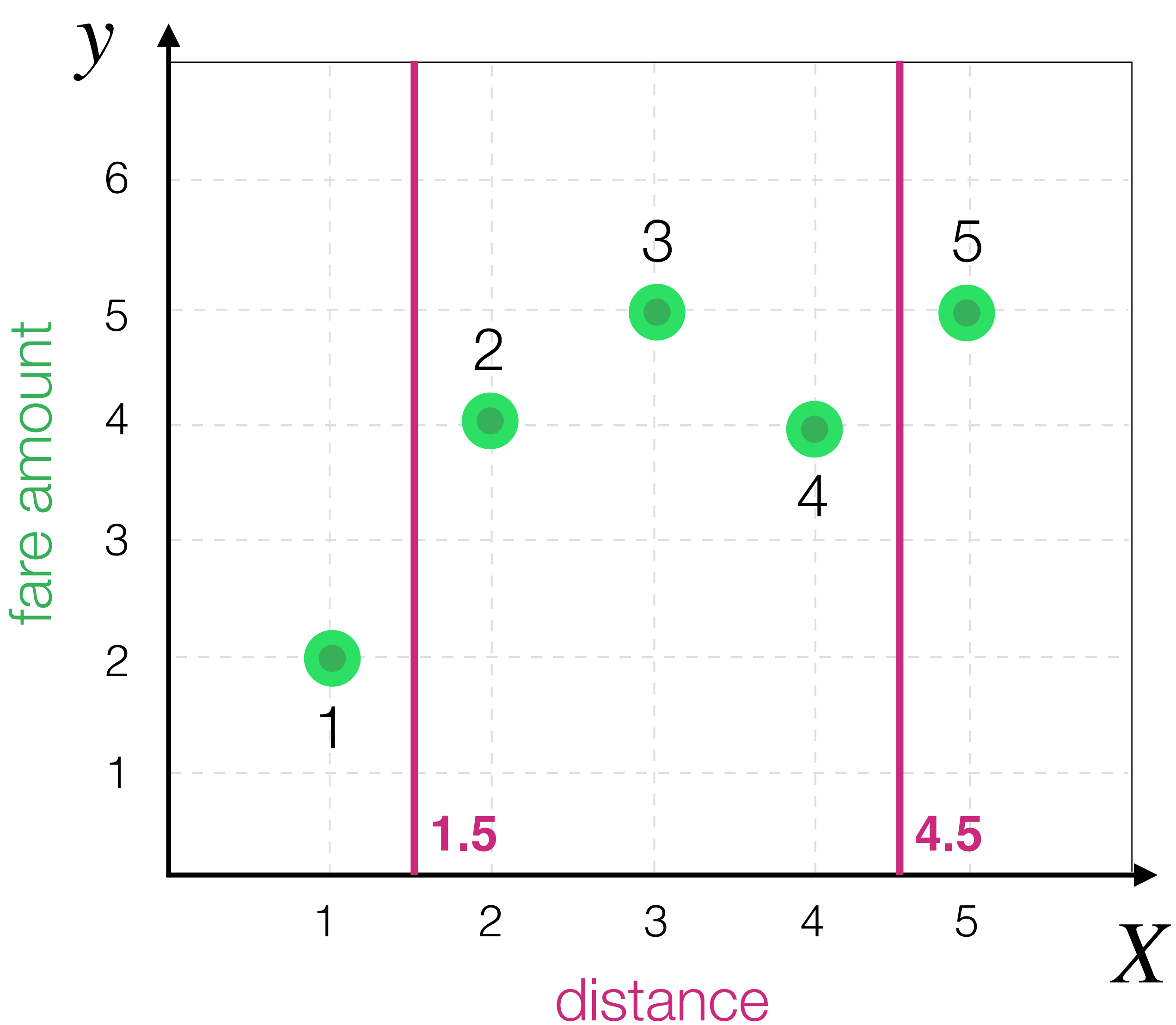
Decision Tree Algorithm

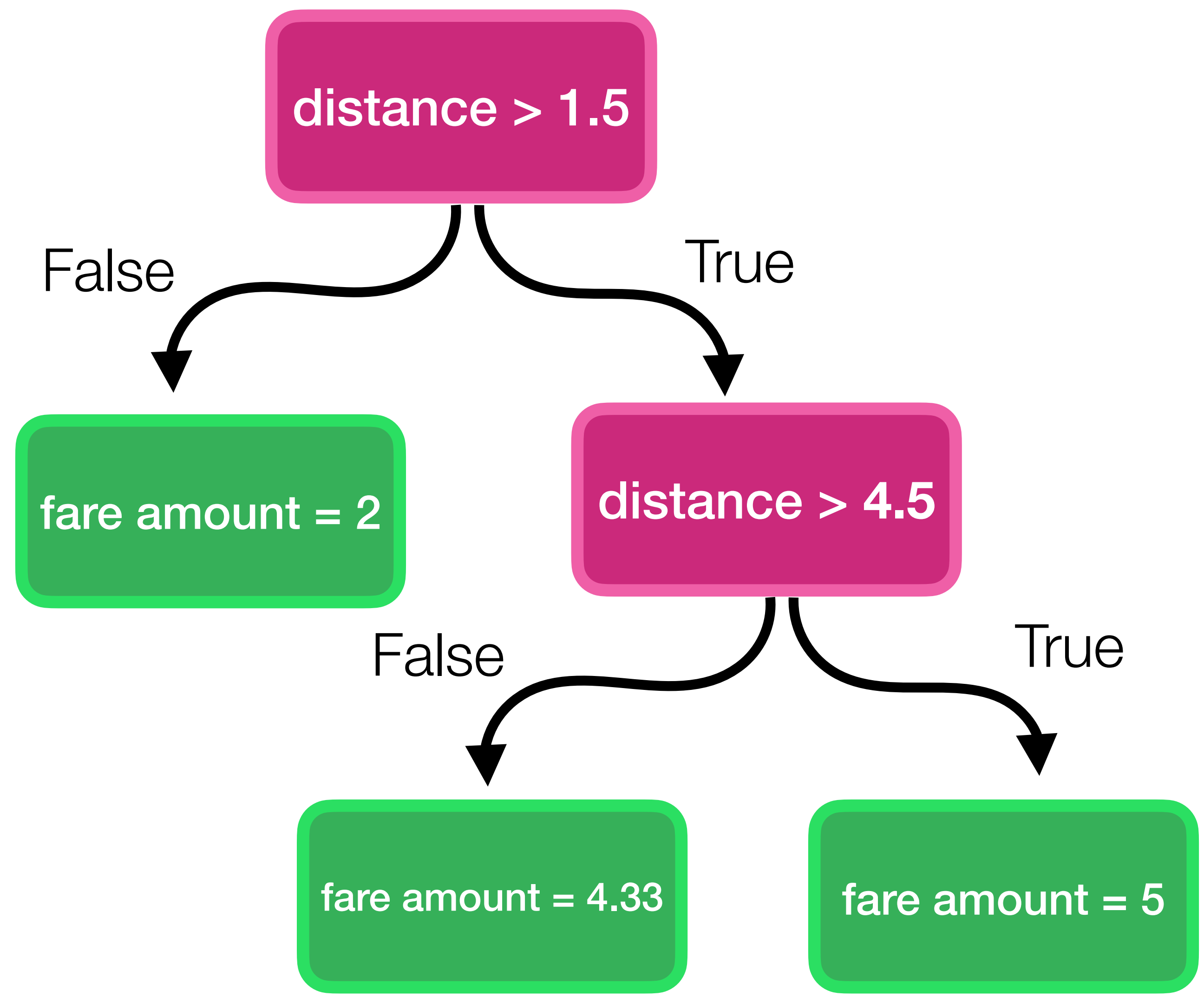
What is the best value for **Y** in this case?





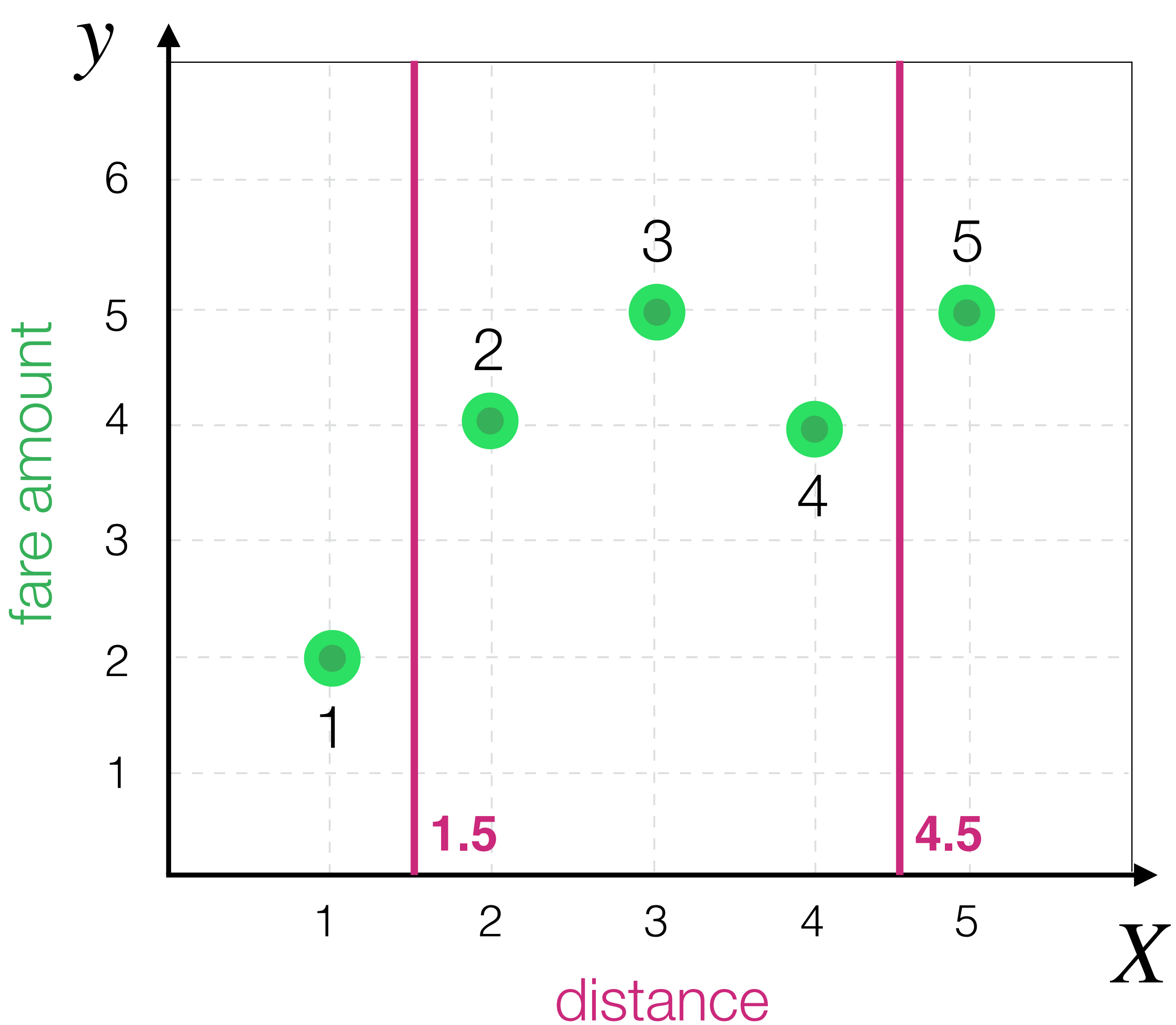
Decision Tree Algorithm

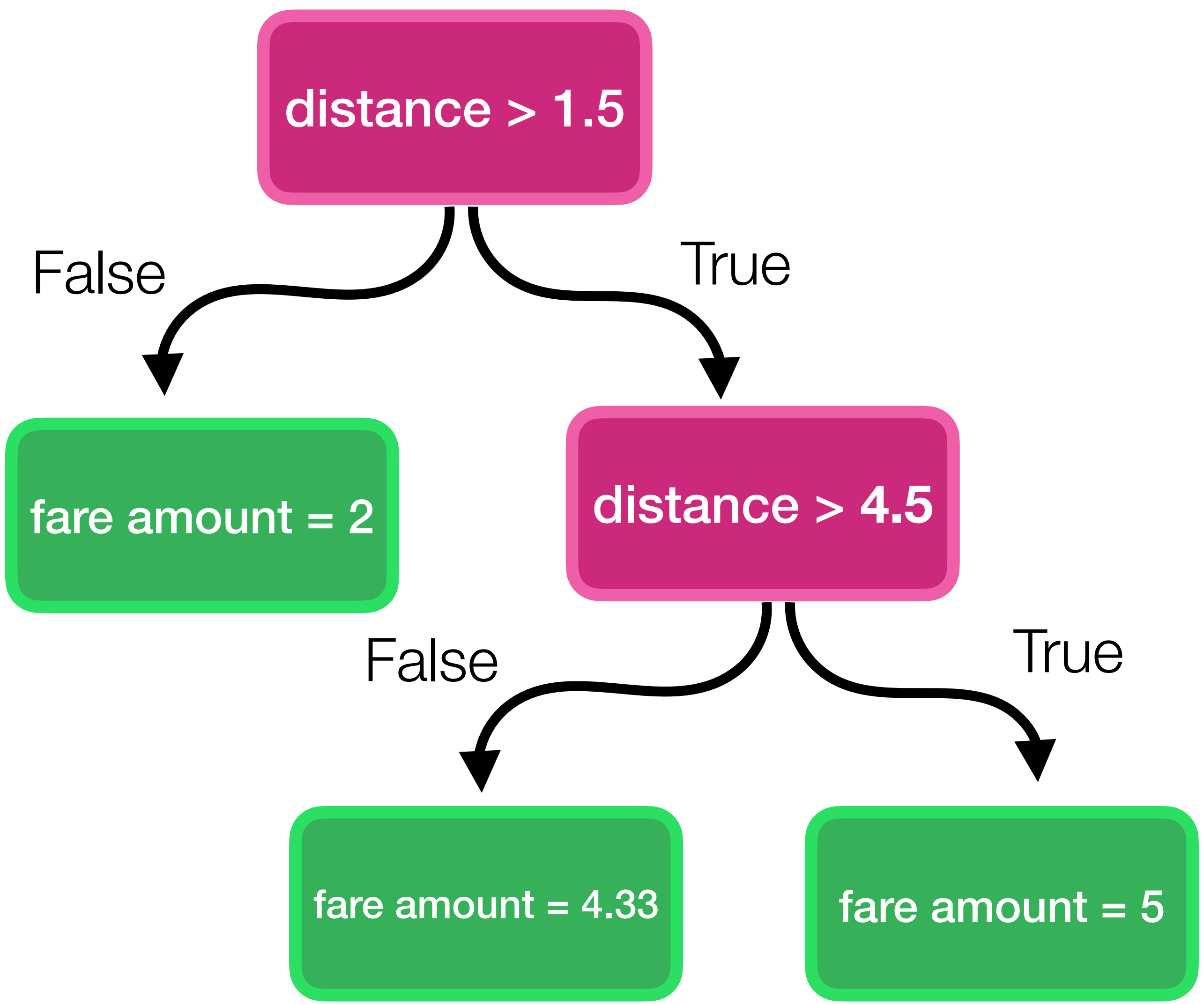




Decision Tree Algorithm

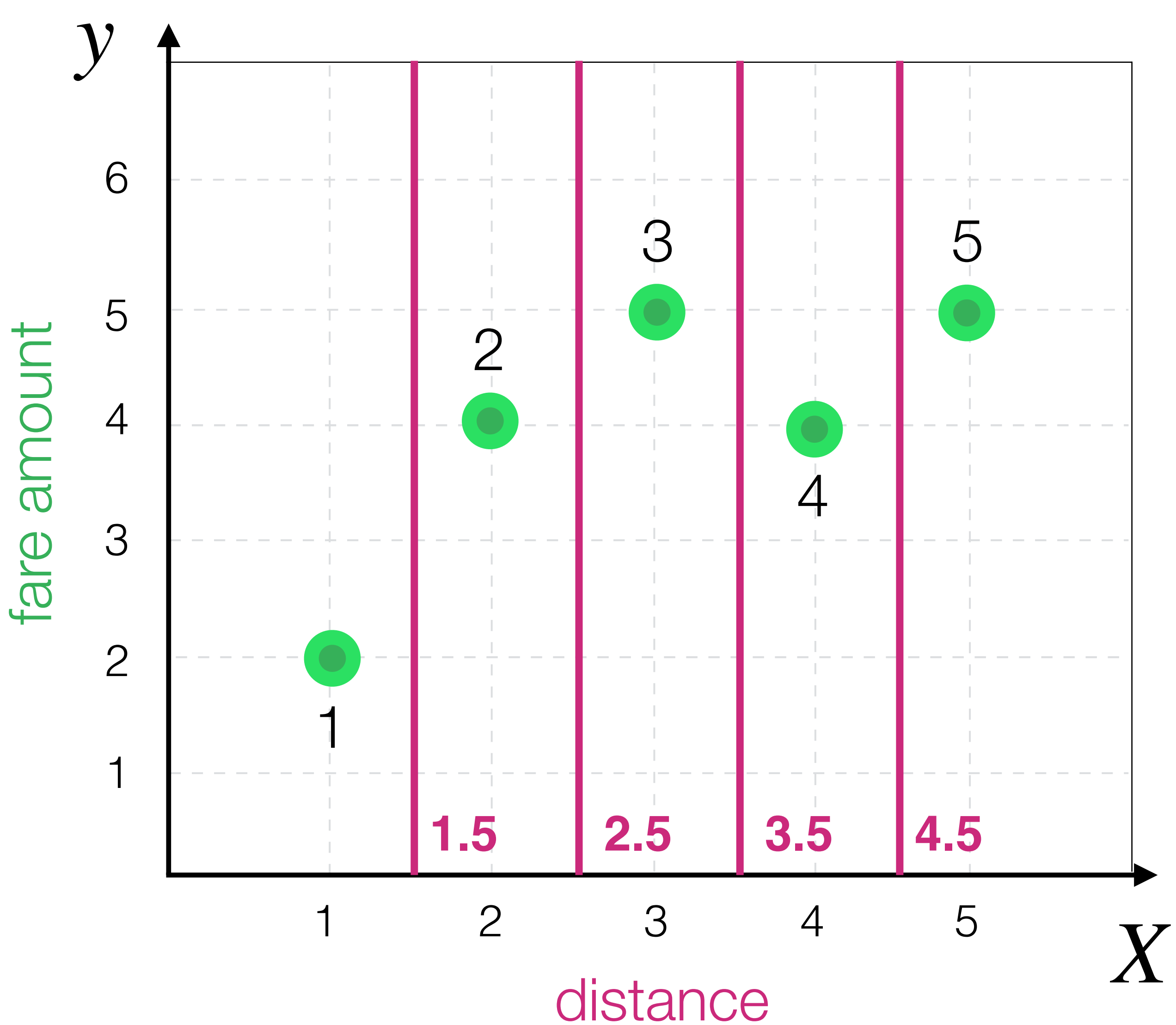
In principle nobody can stop us from…

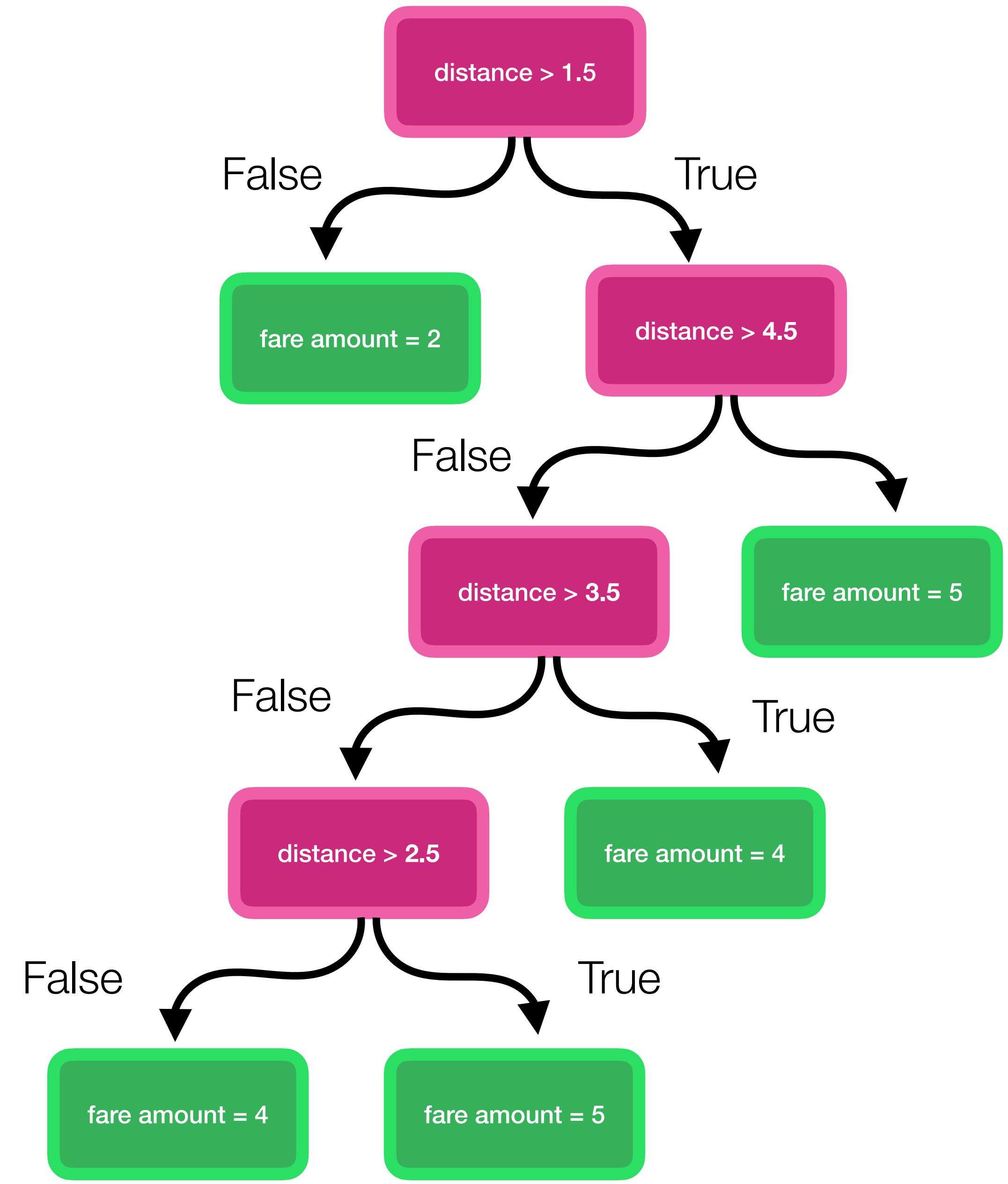




Decision Tree Algorithm

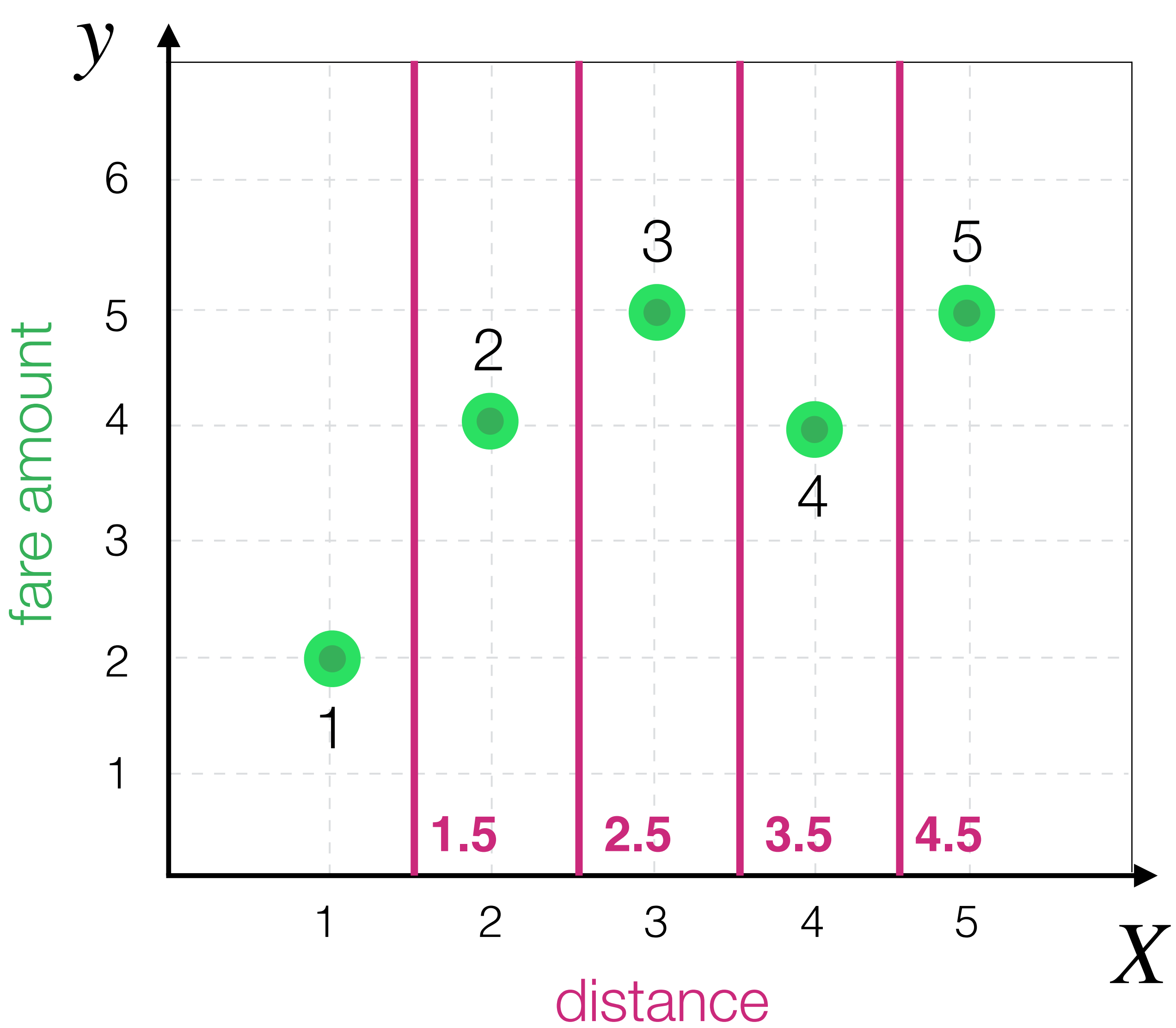
In principle nobody can stop us from…





Decision Tree Algorithm

In principle nobody can stop us from…

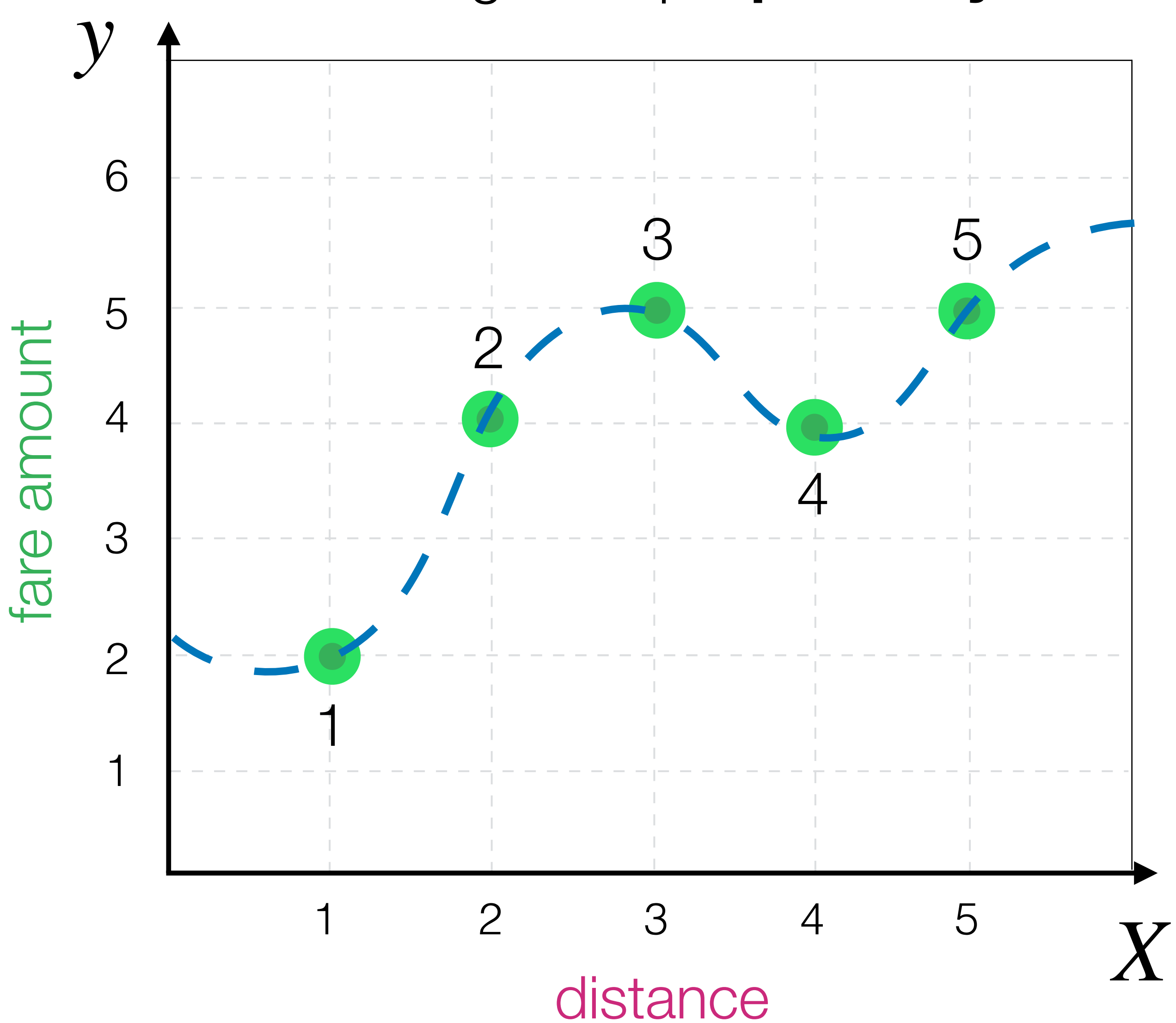


Every observation is in its own leaf of the tree

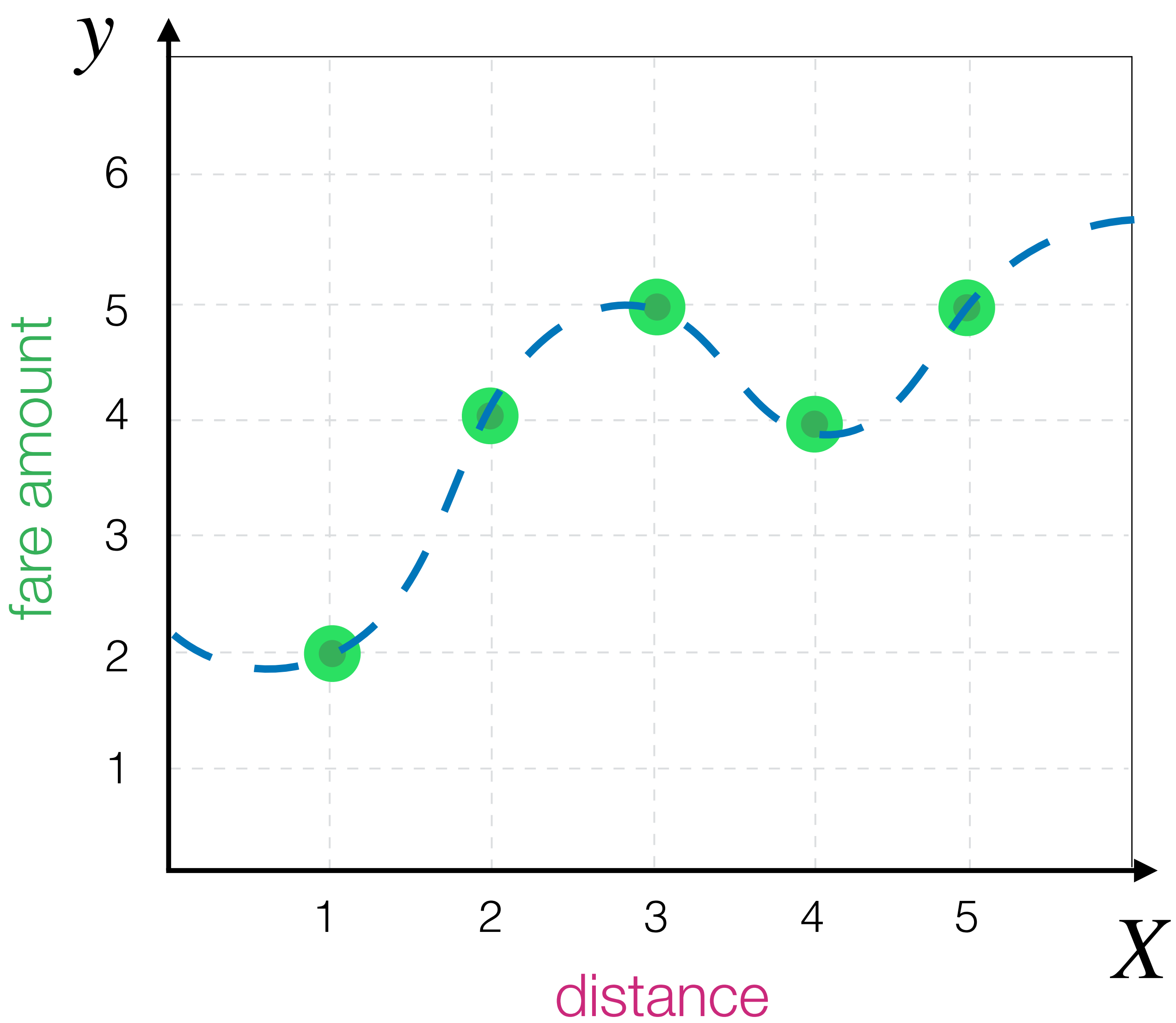
MSE = 0.0

Decision Tree Algorithm

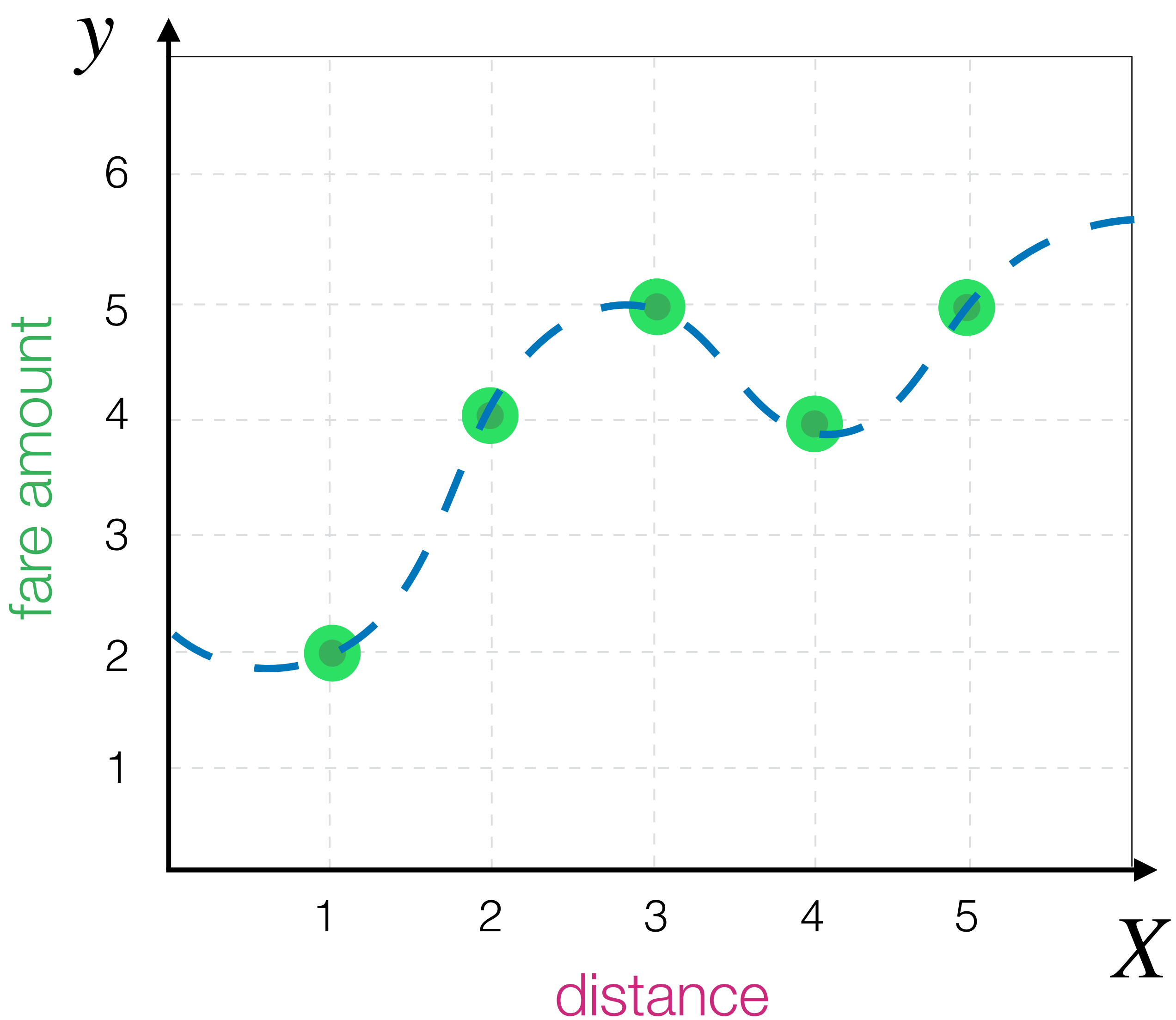
You can think about this as super **powerful** model that predicts all training example **perfectly**



Every observation is in its own leaf of the tree



y
6
5
4
3
2
1
fare amount
1
2
3
4
5
X
distance



y
6
5
4
3
2
1
fare amount
1
2
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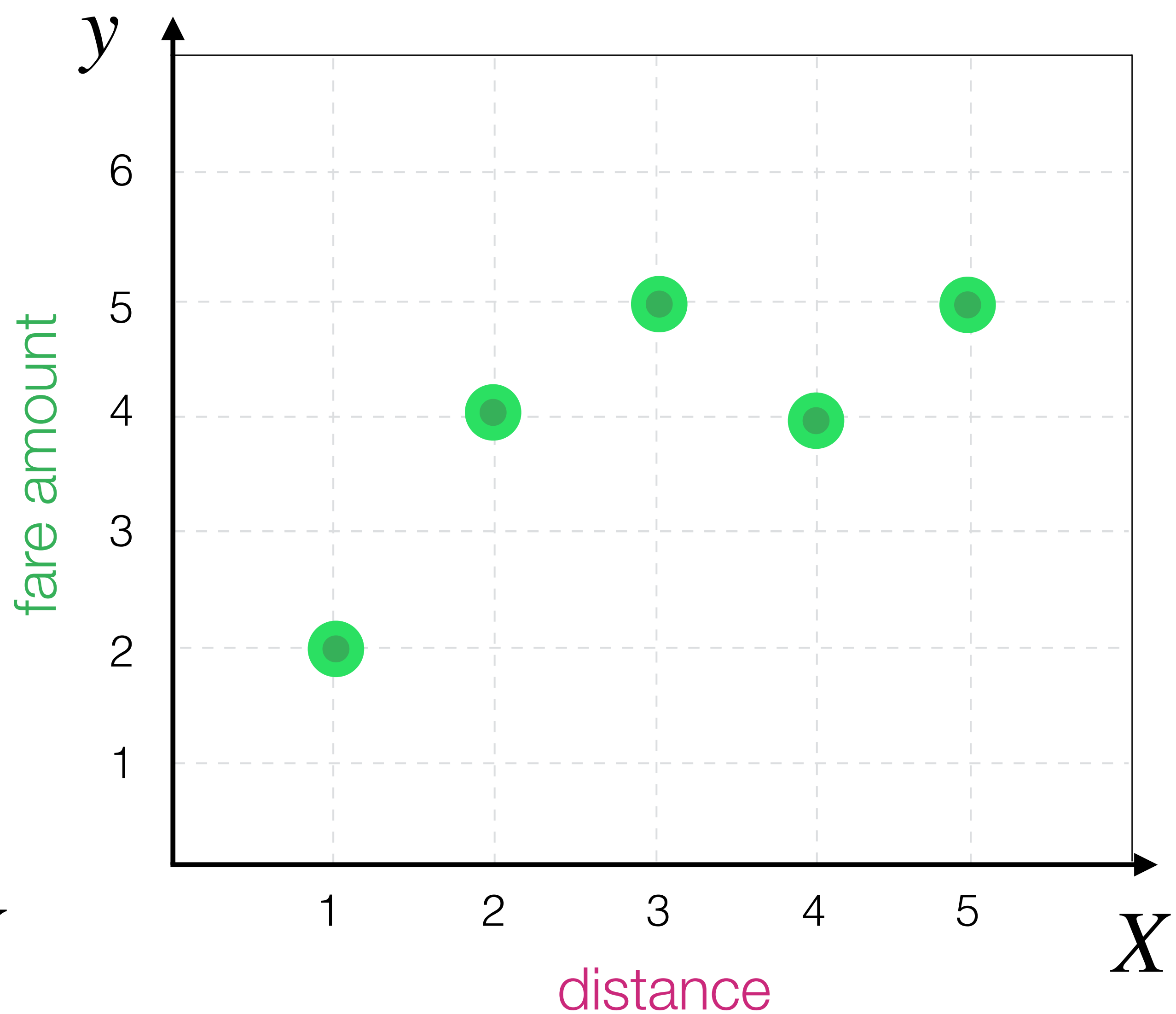
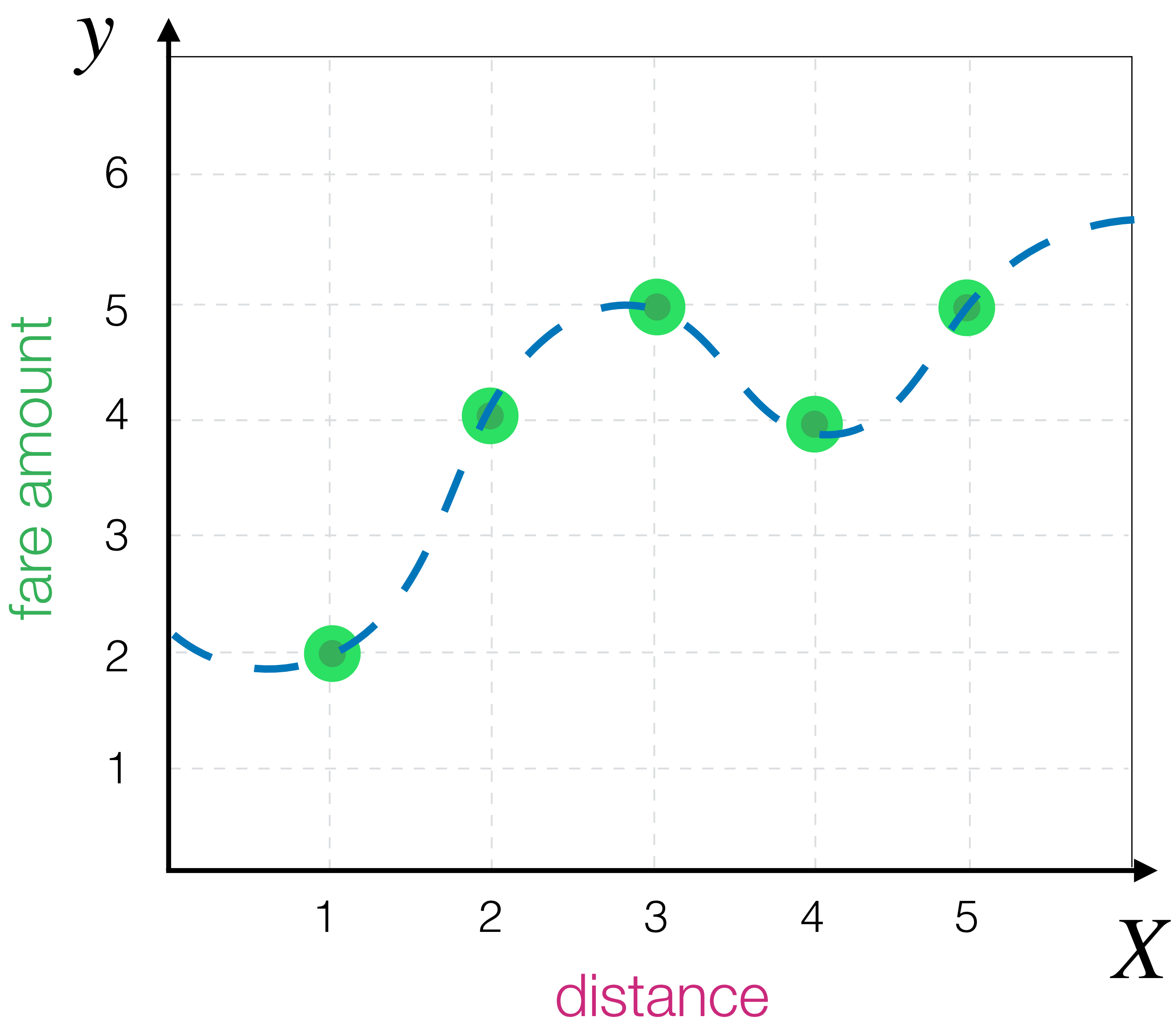
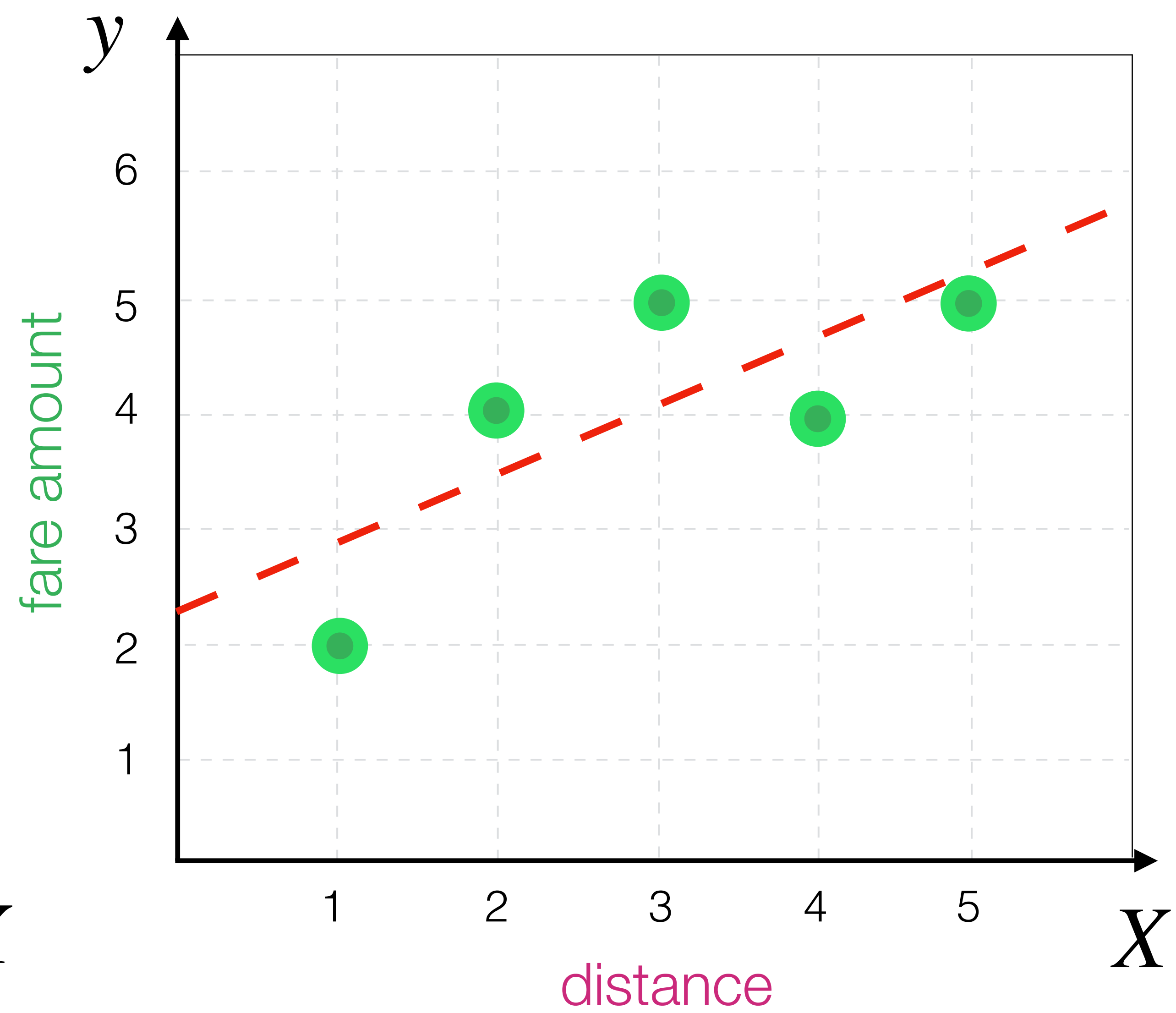
4
5
X
distance



y
6
5
4
3
2
1
fare amount
1
2
3
4
5
X
distance

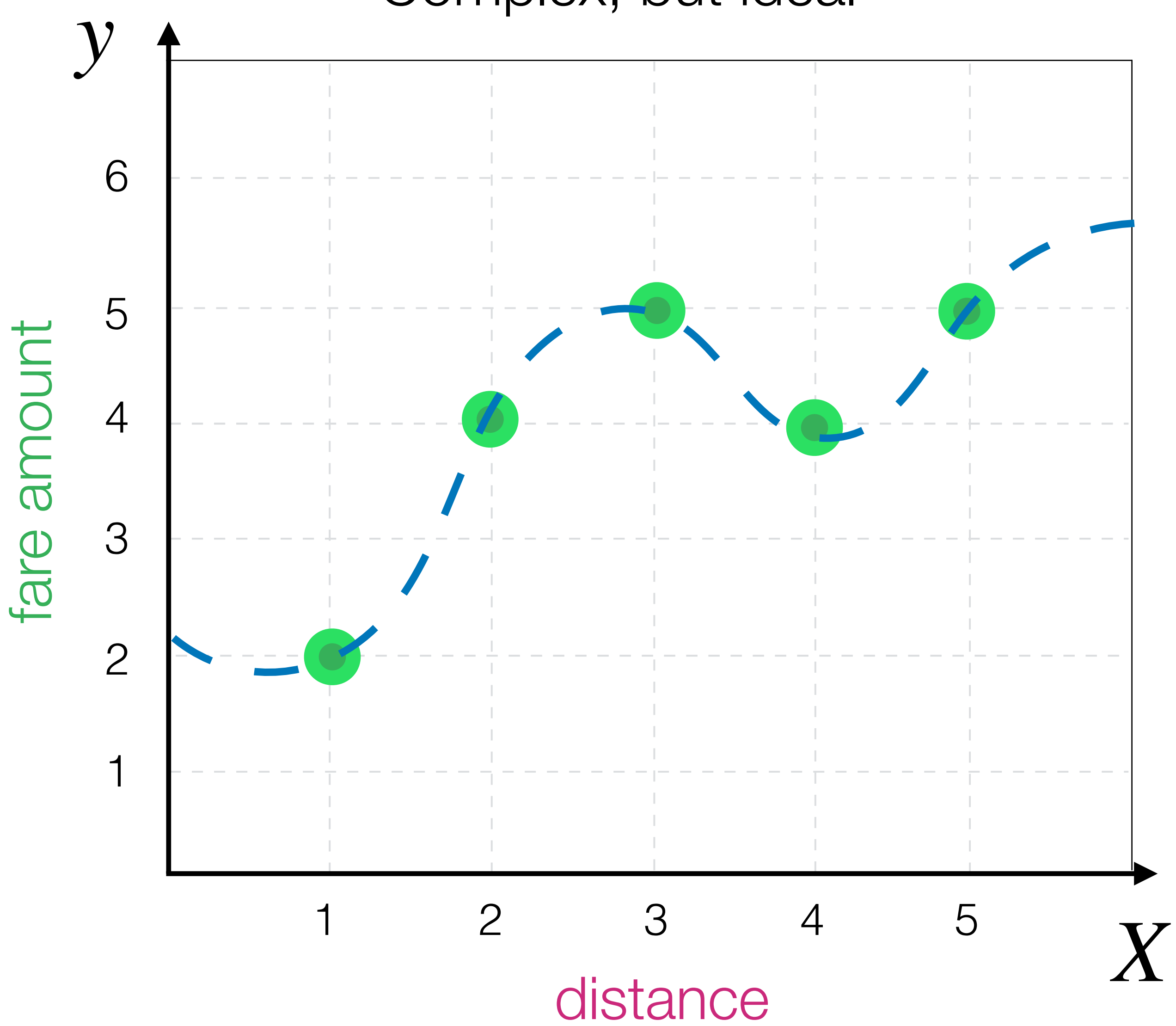


y
6
5
4
3
2
1
fare amount
1
2
3
4
5
X
distance

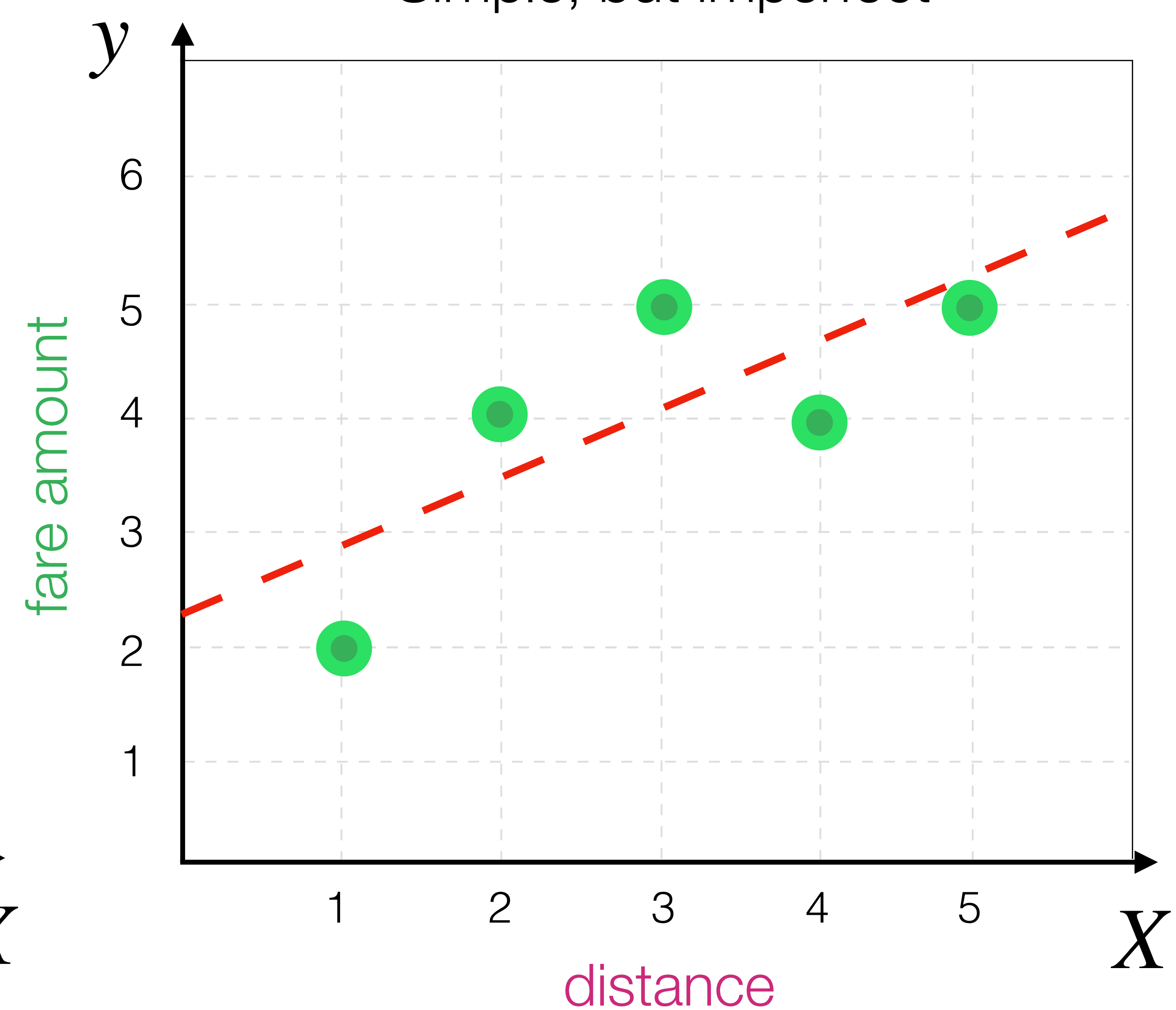


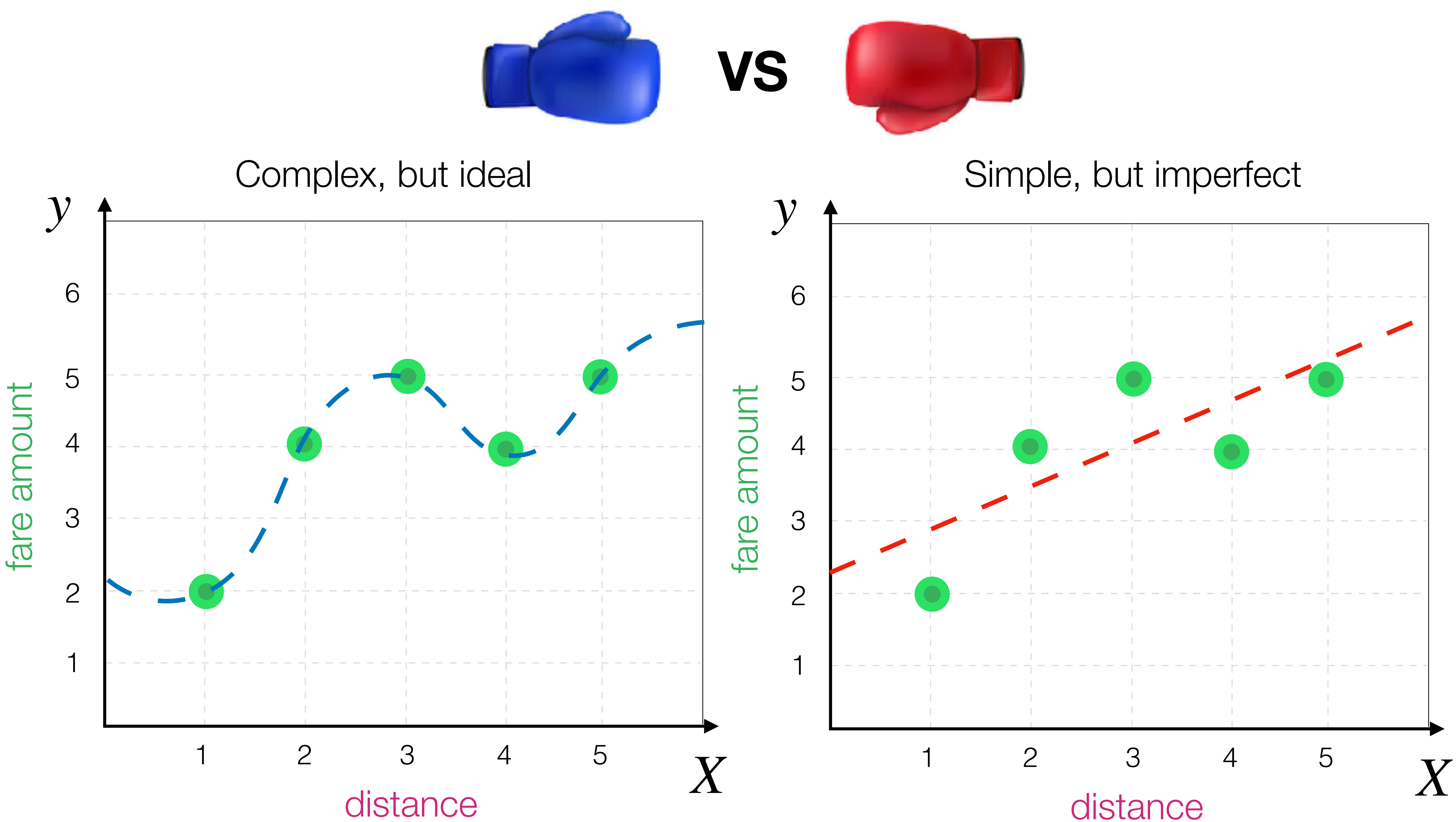
y
6
5
4
3
2
1
fare amount
1
2
3
4
5
X
distance

Complex, but ideal



Simple, but imperfect

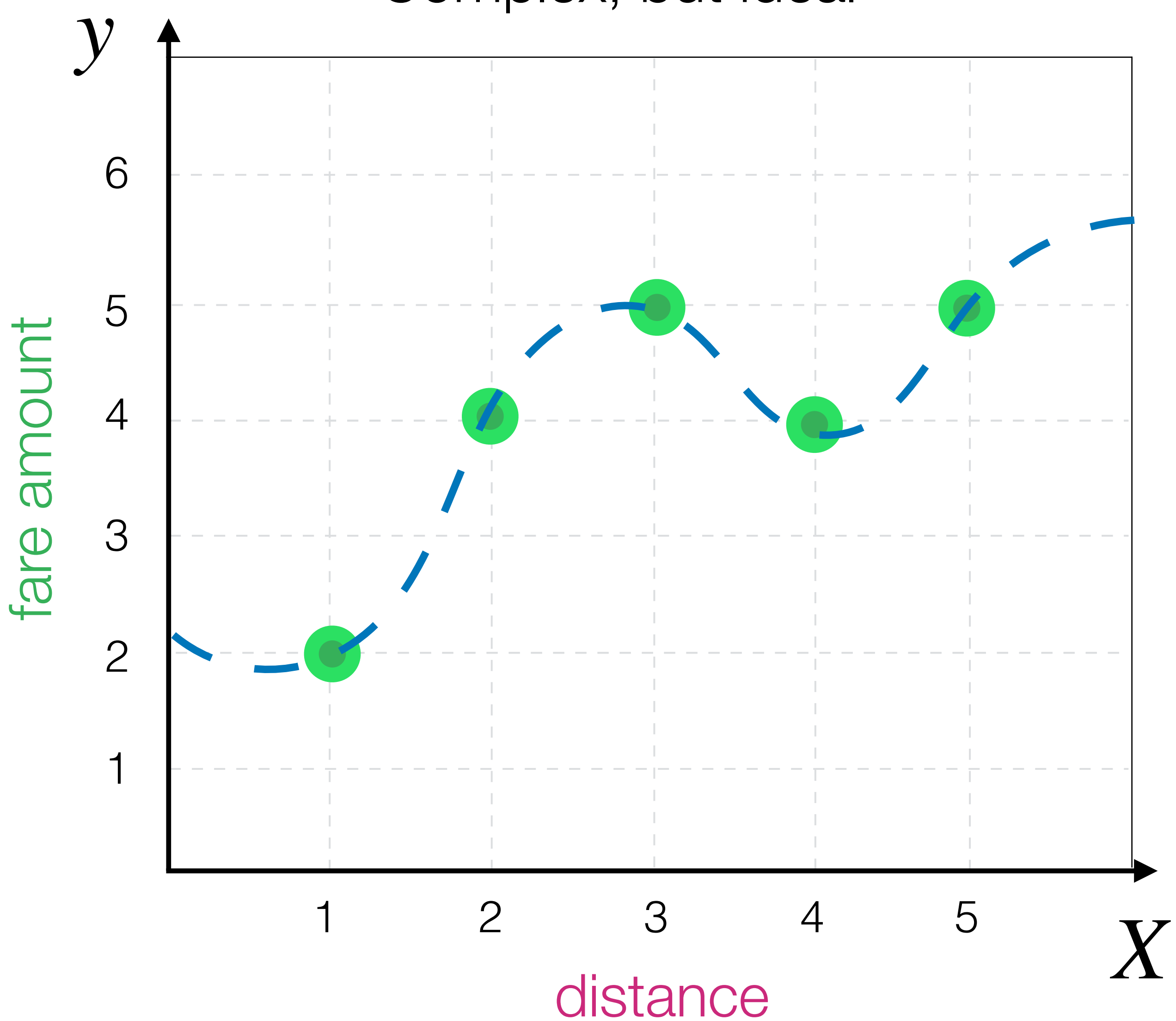



VS
Complex, but ideal
Simple, but imperfect
y
fare amount
6
5
4
3
2
1
1
2
3
4
5
X
distance
y
fare amount
6
5
4
3
2
1
1
2
3
4
5
X
distance

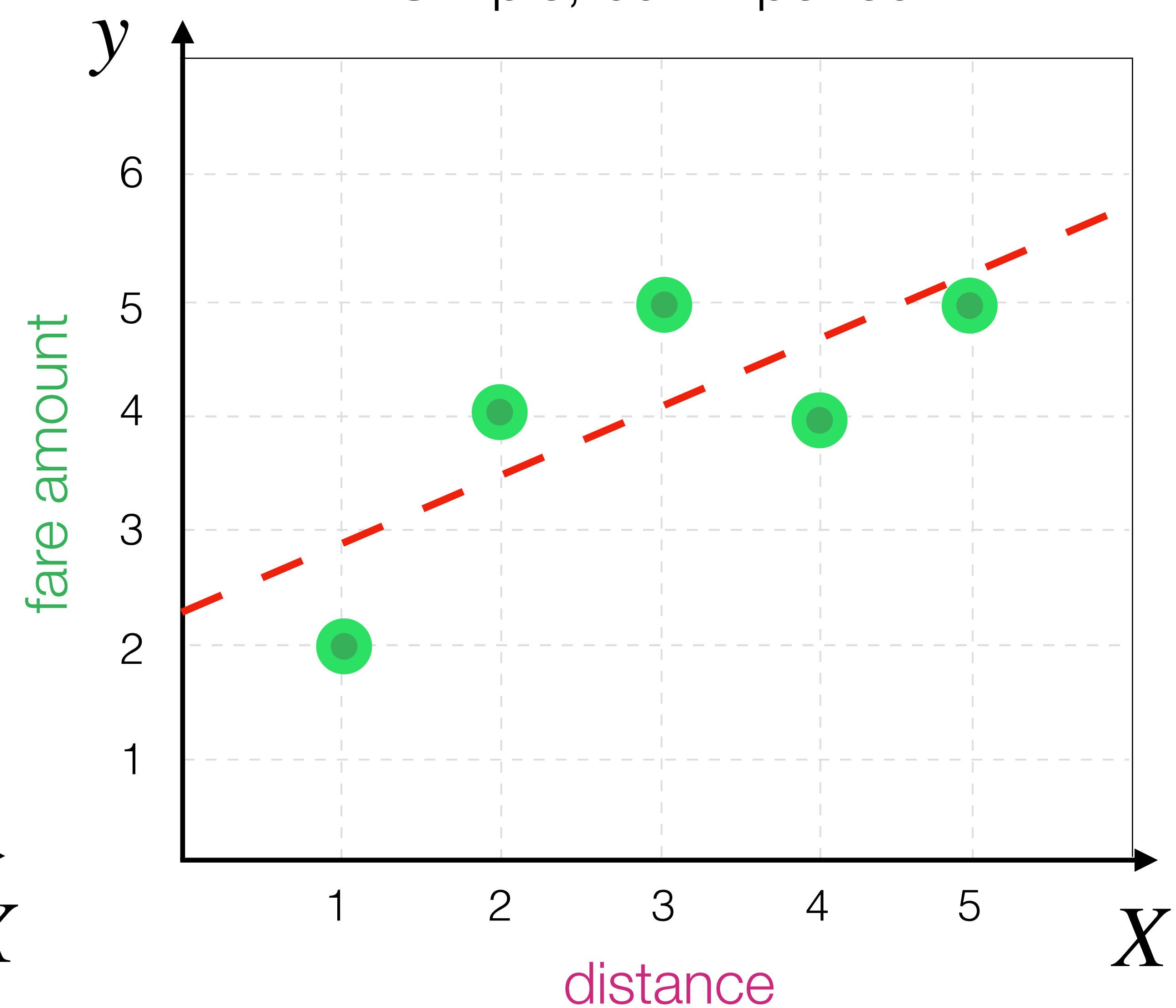
How do we know which one to choose?

VS

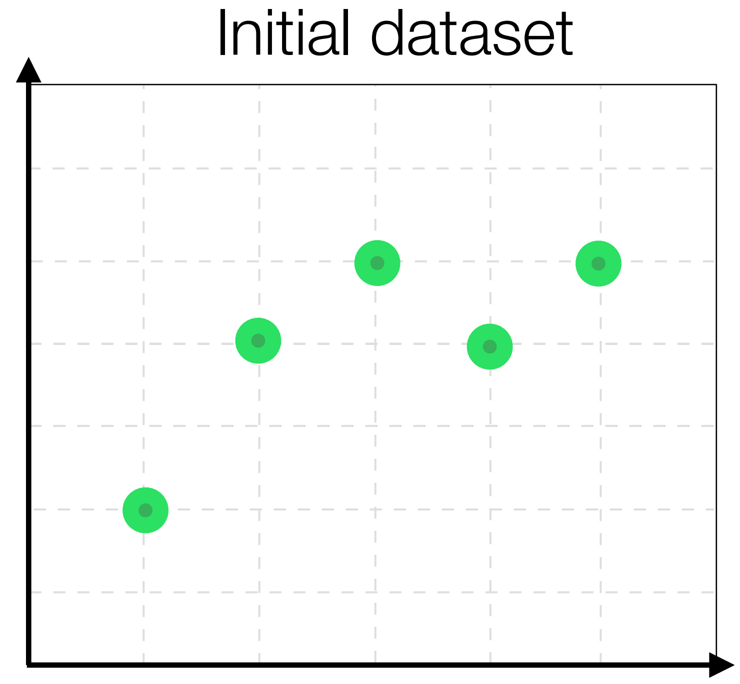
Complex, but ideal



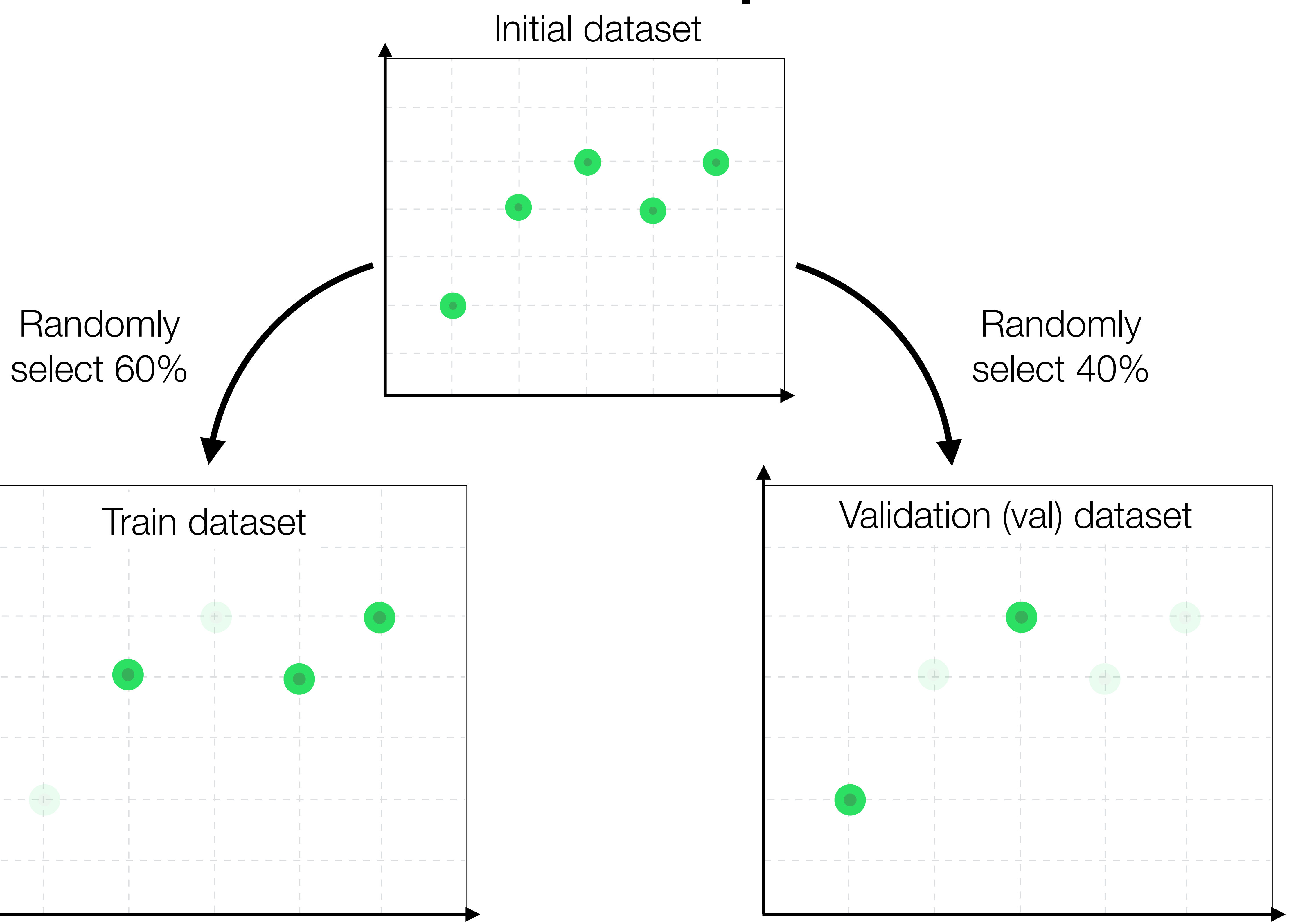
Simple, but imperfect



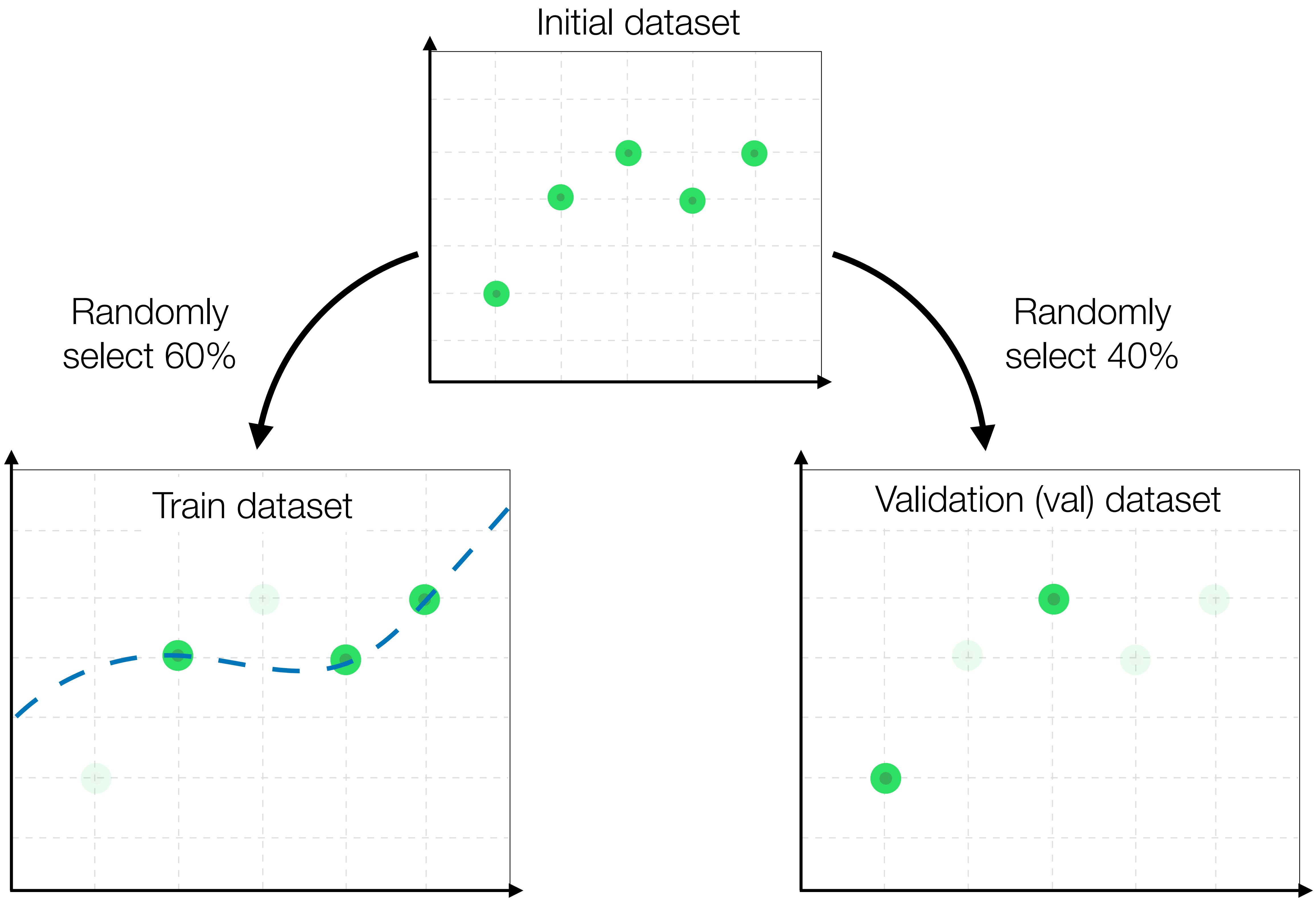
Train/val split



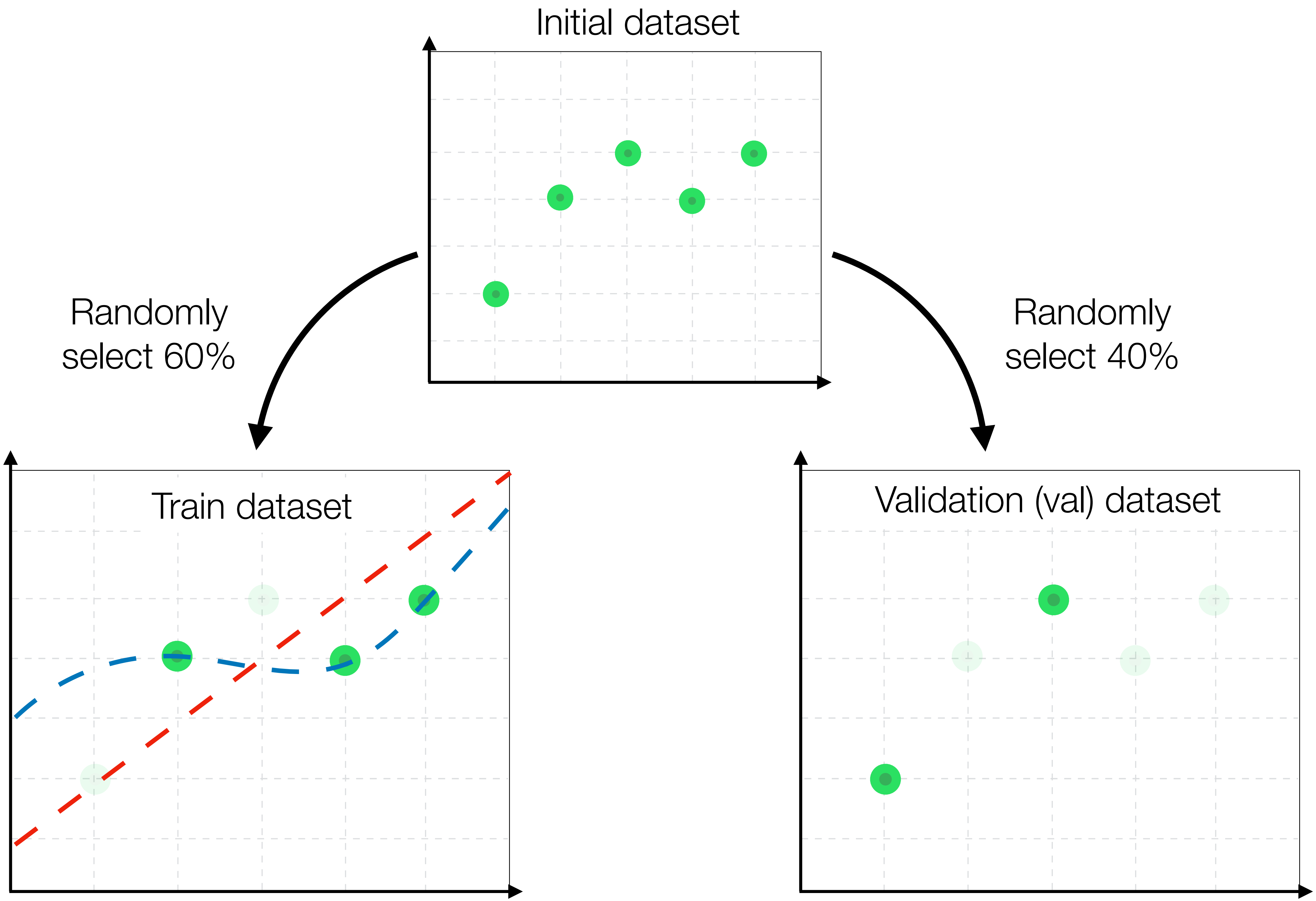
Train/val split



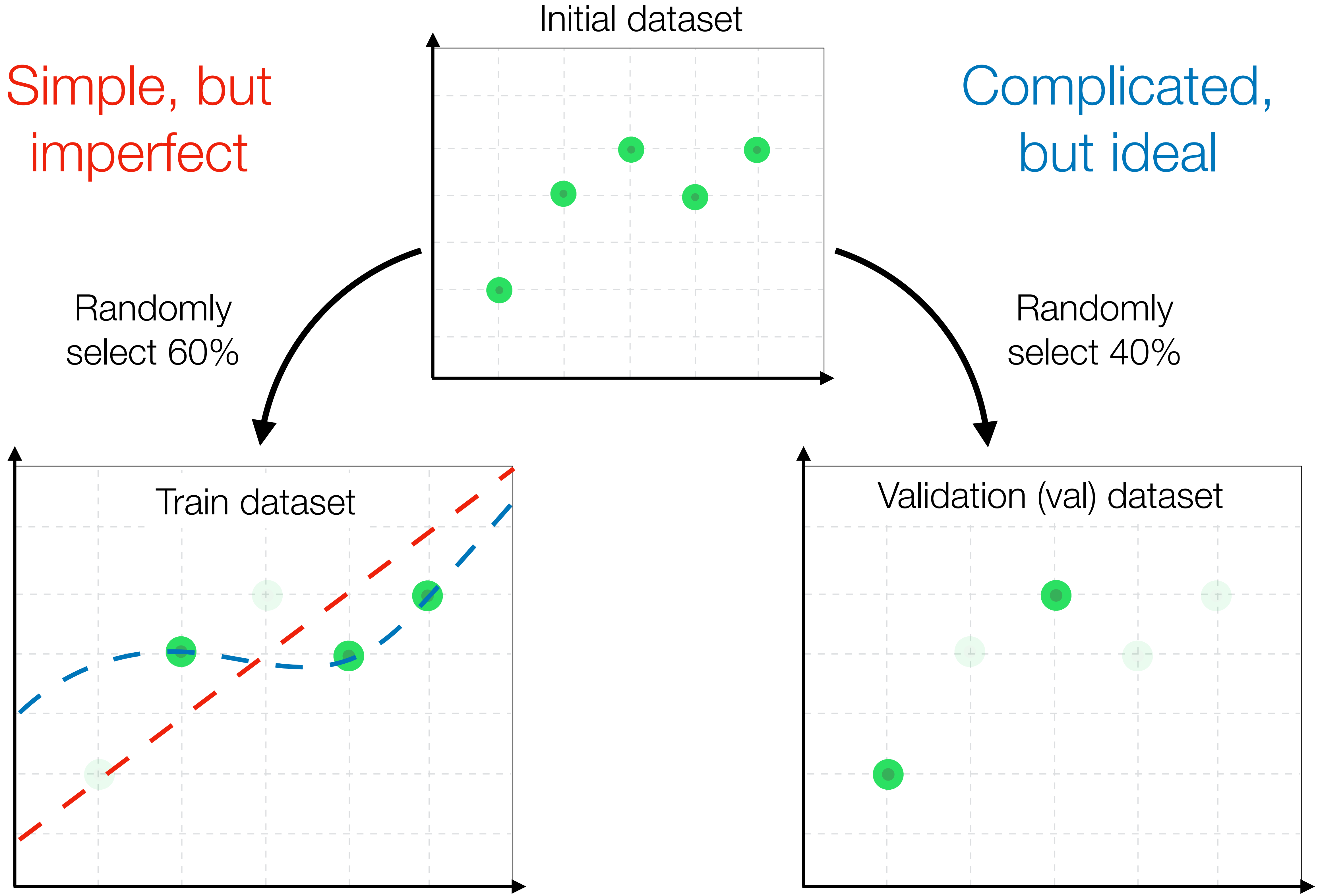
Train/val split



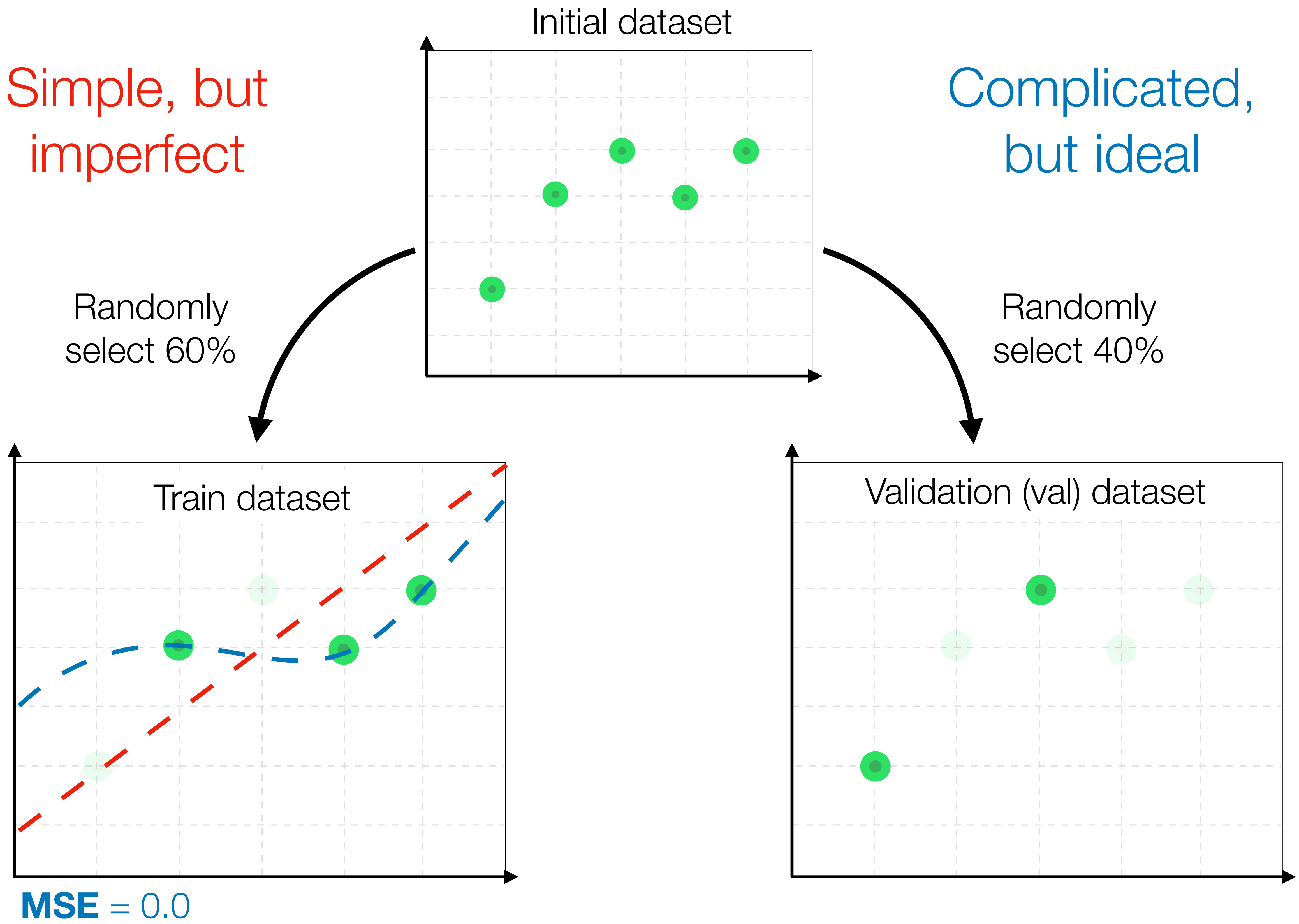
Train/val split



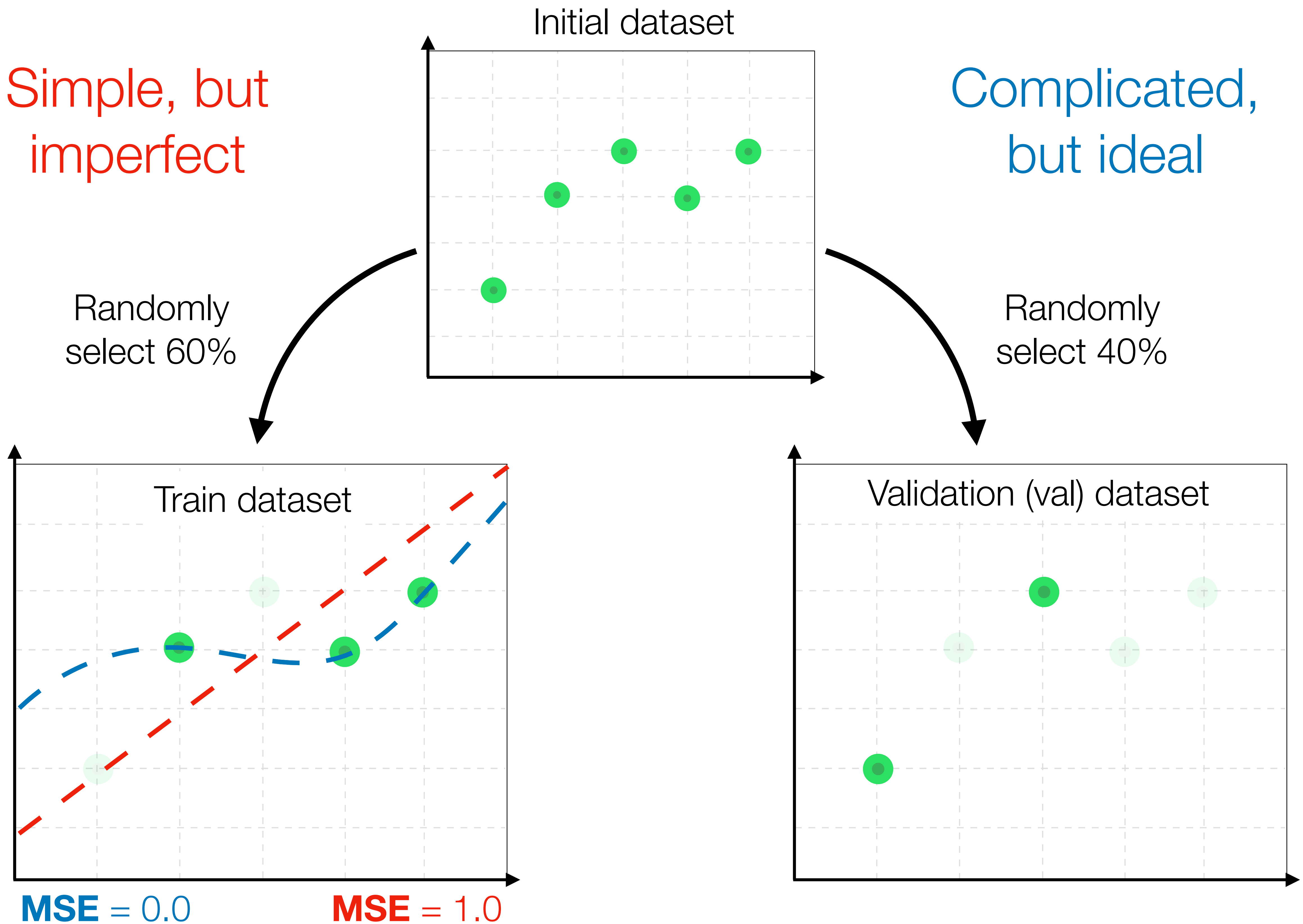
Train/val split



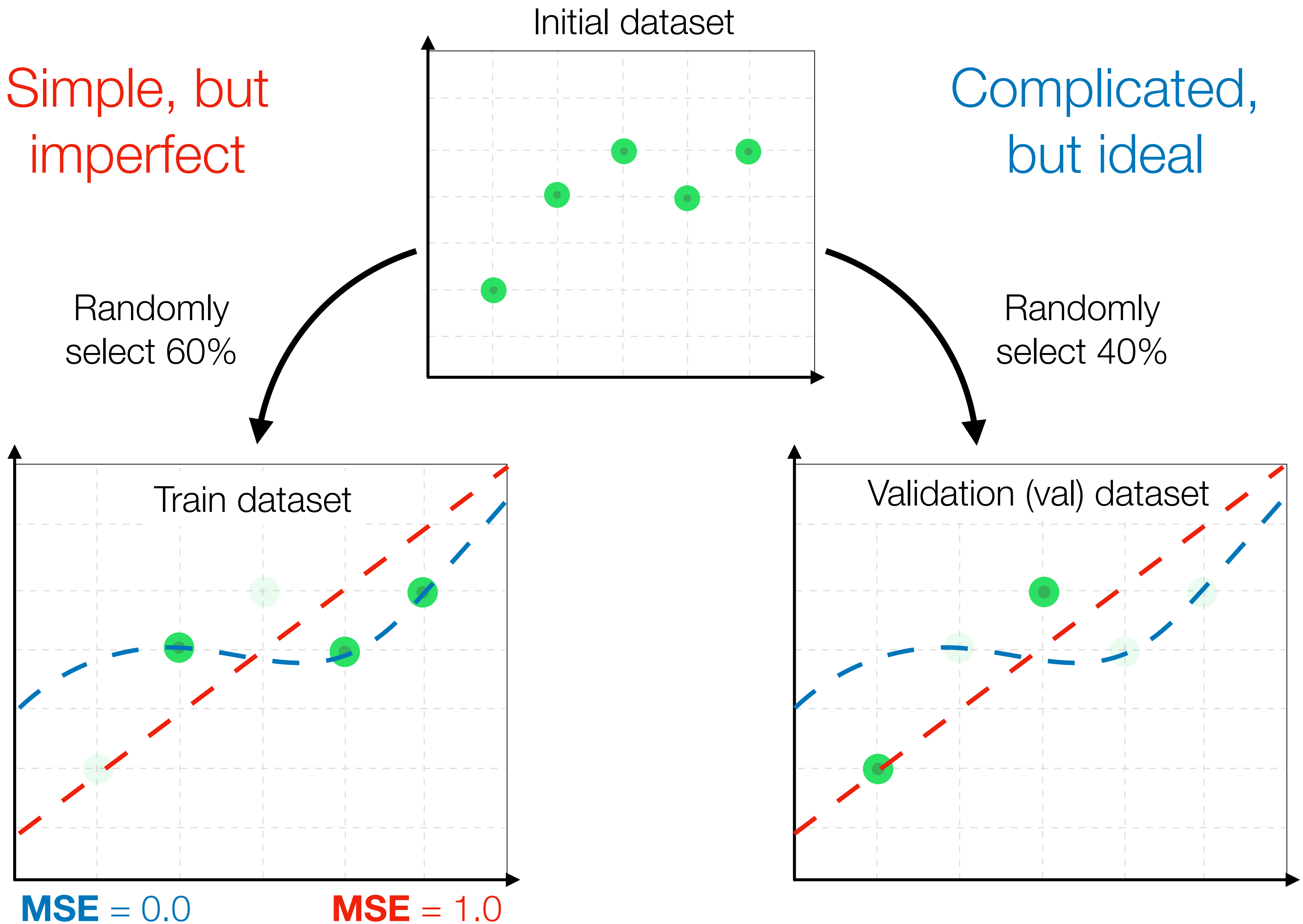
Train/val split



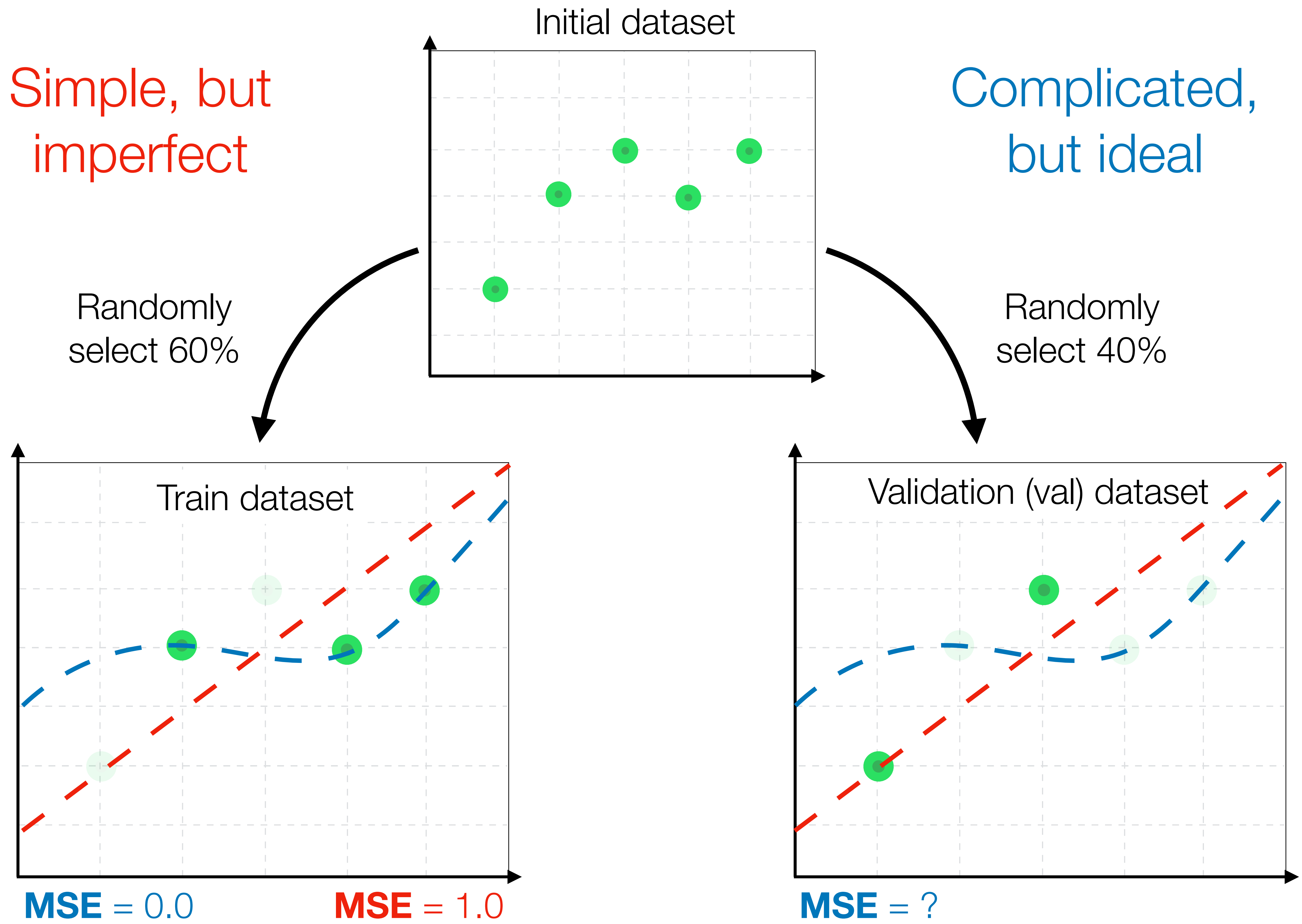
Train/val split



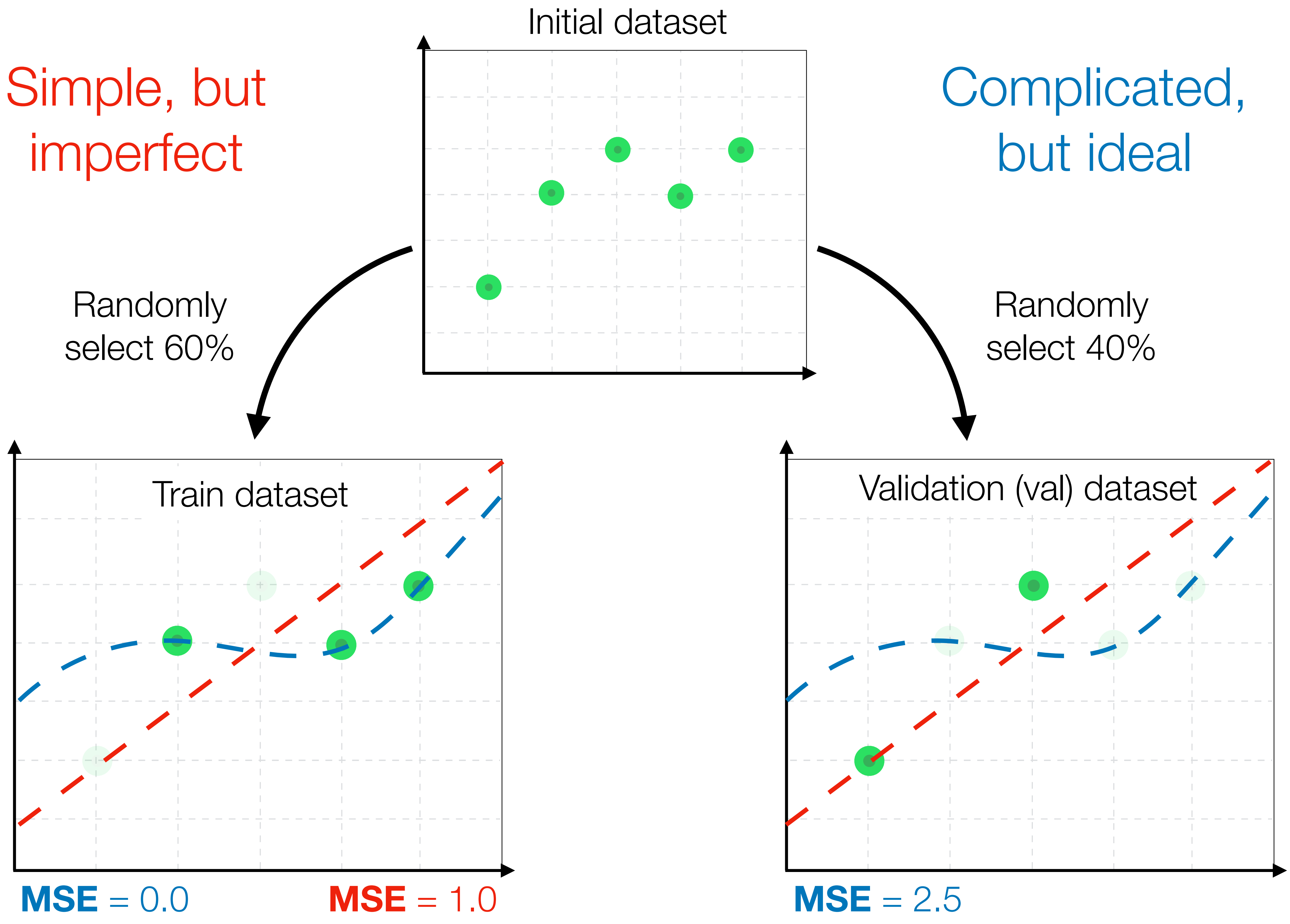
Train/val split



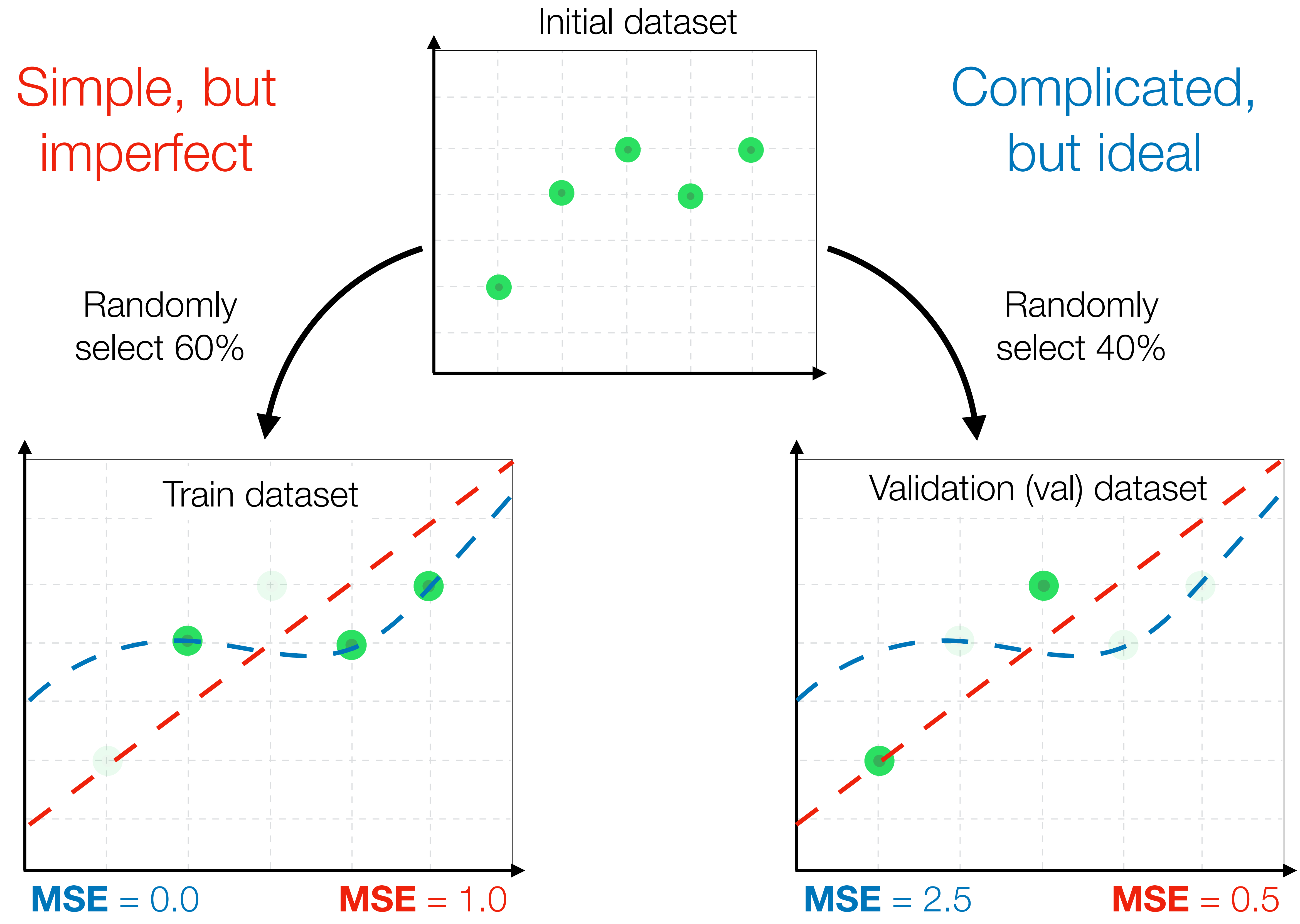
Train/val split



Train/val split



Train/val split



Train/val split

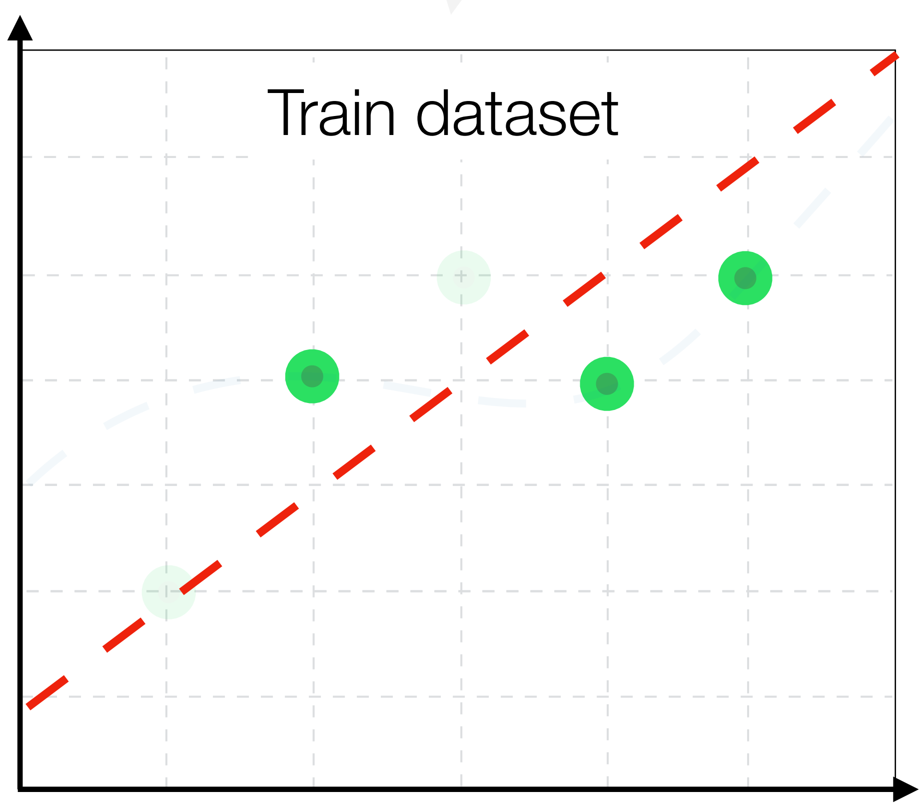
Initial dataset

Simple, but imperfect

Complicated, but ideal

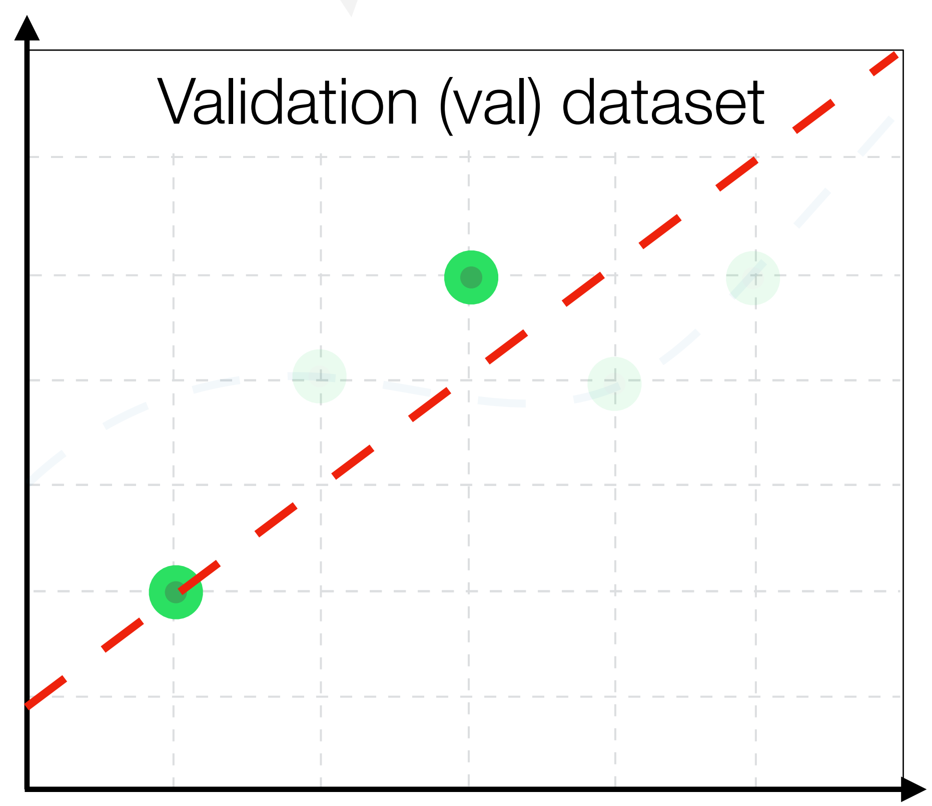
Randomly select 60%

Randomly select 40%



MSE = 0.0

MSE = 1.0



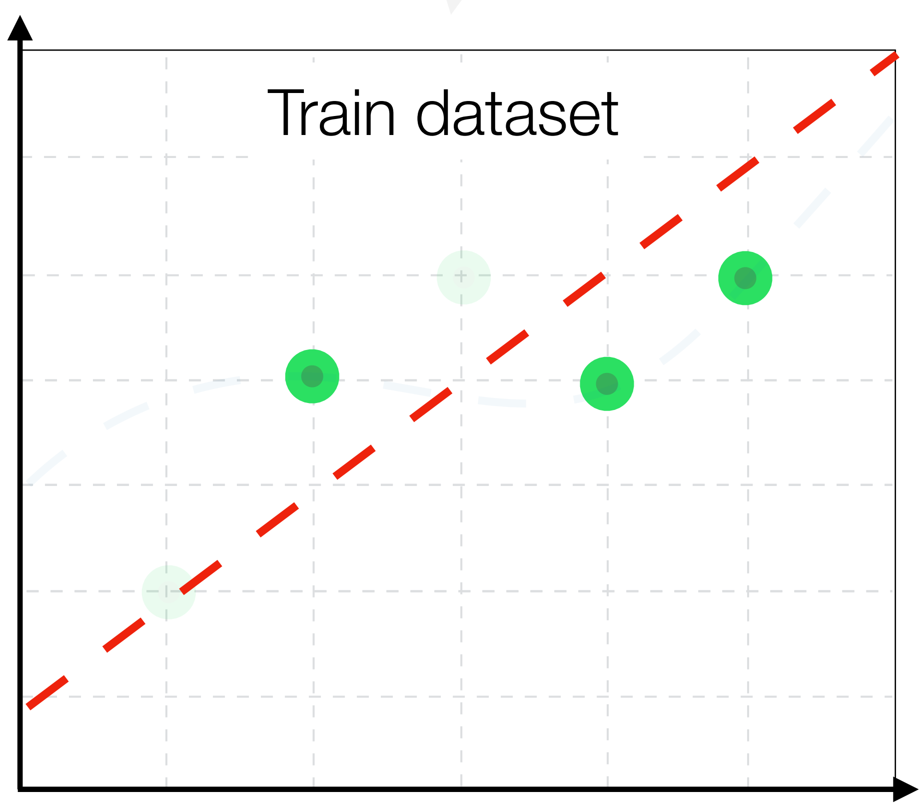
MSE = 2.5

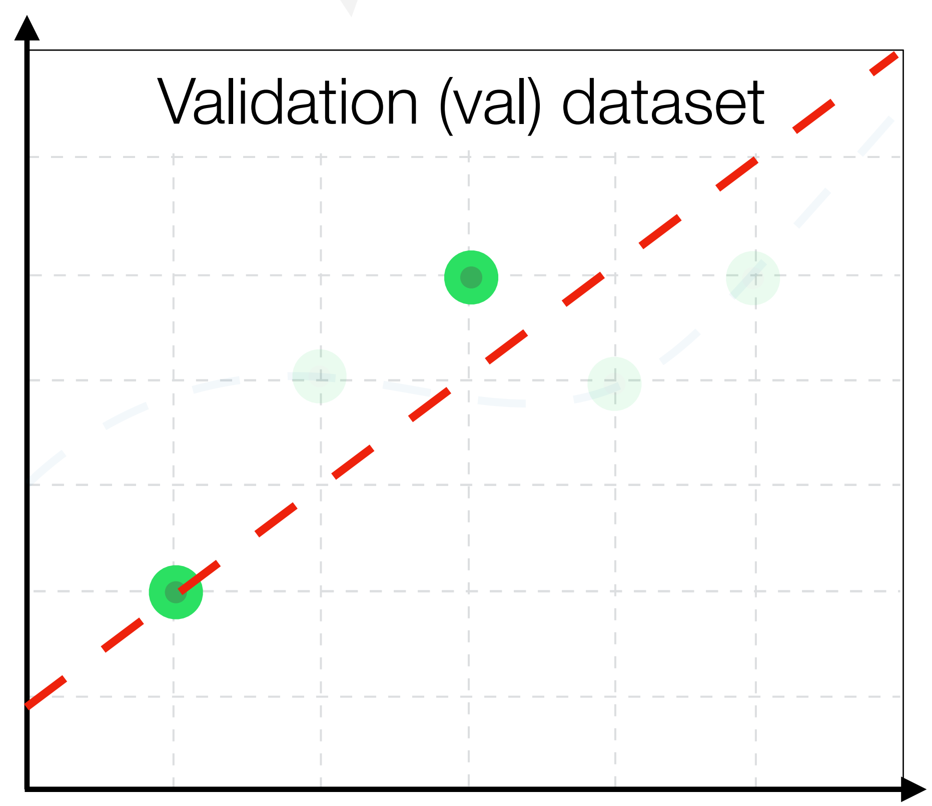
MSE = 0.5

Simple, but imperfect

The **gap** between **training** and **validation** performance is not large

MSE = 1.0 **MSE** = 0.5





Train/val split

Initial dataset

Simple, but imperfect

Complicated, but ideal

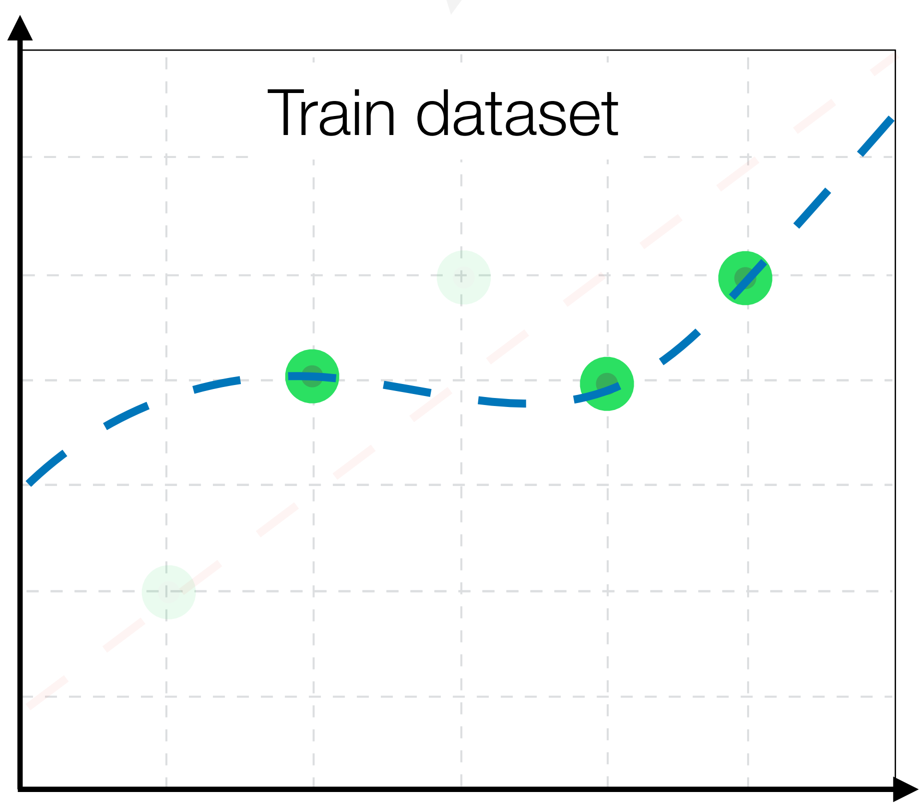
Large gap between training and validation

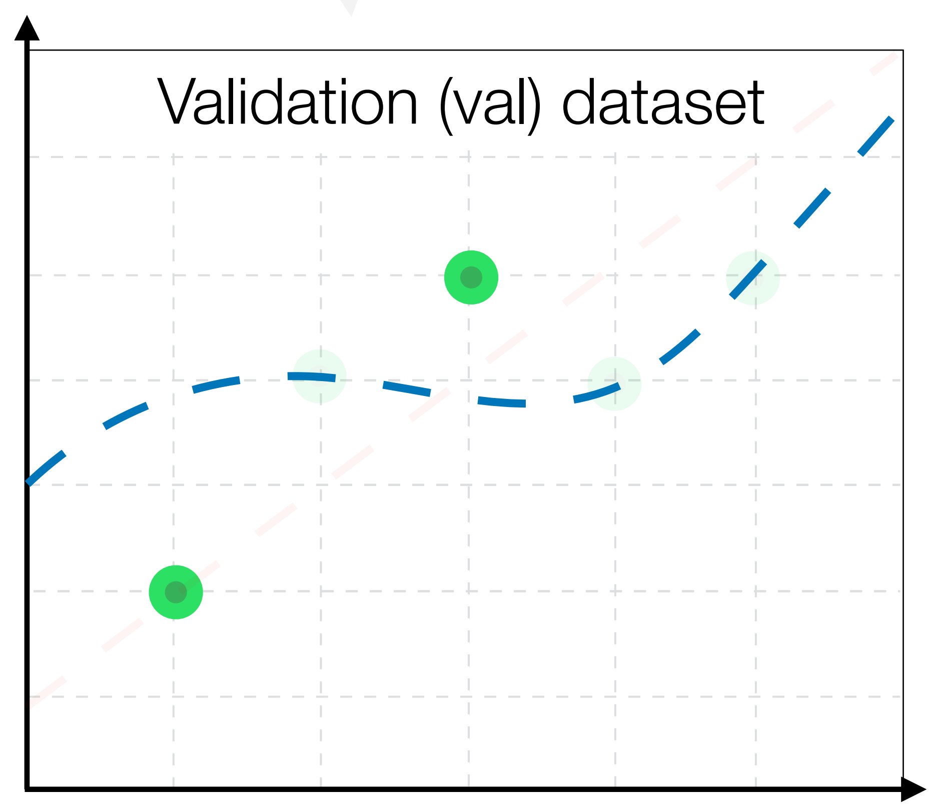
Randomly select 60%

Randomly select 40%

MSE = 0.0

MSE = 2.5



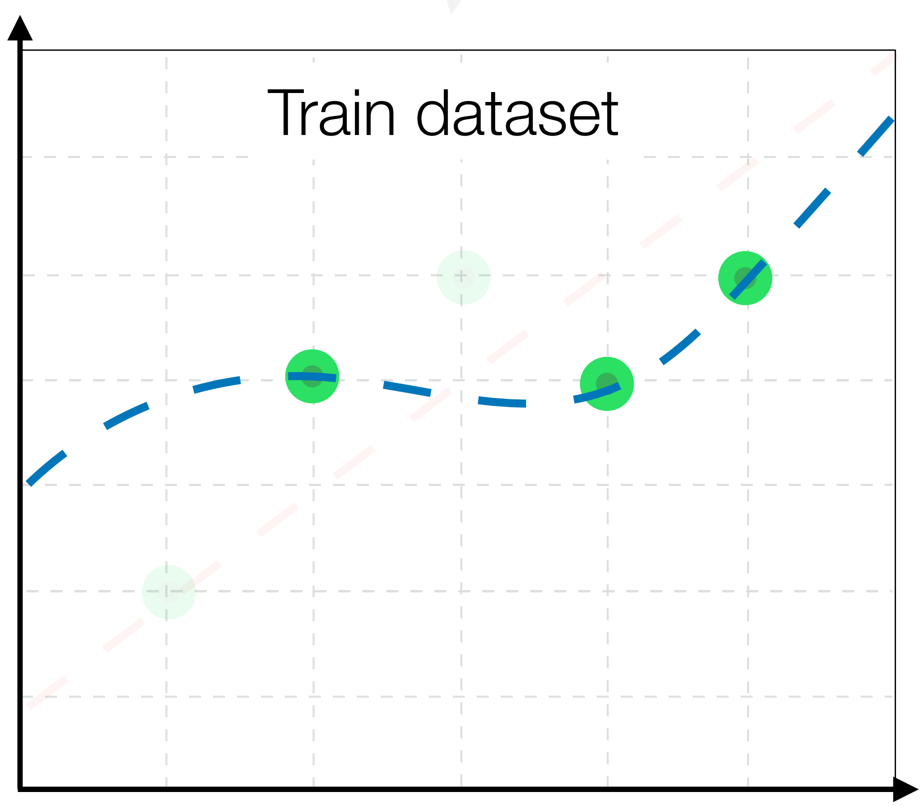


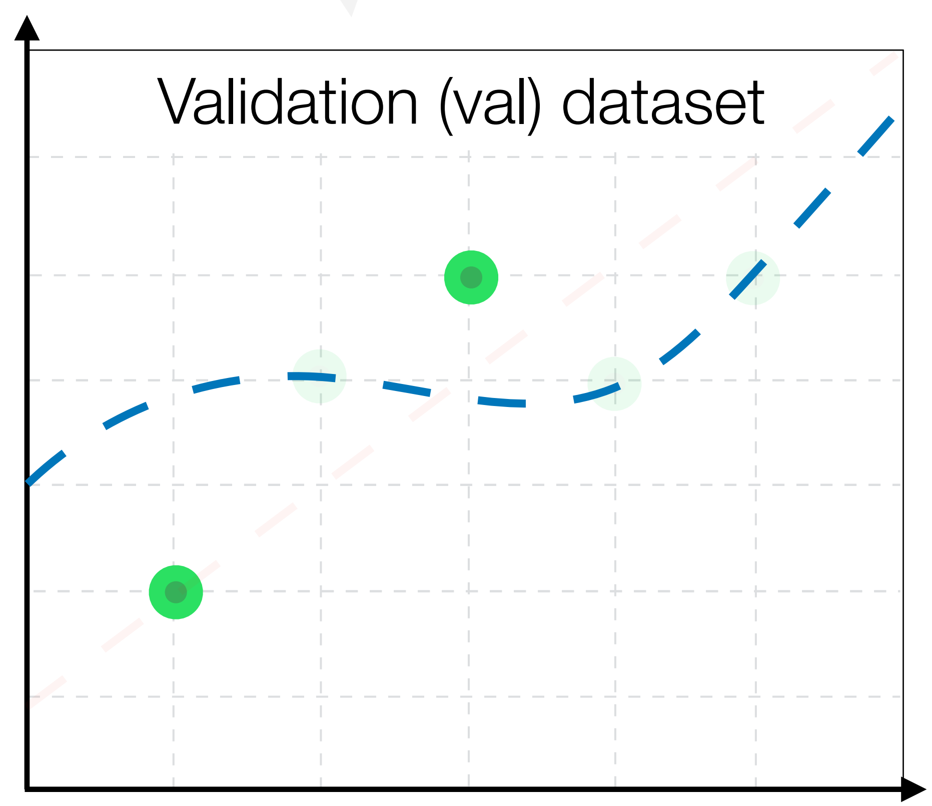
Overfitting

Simple, but imperfect

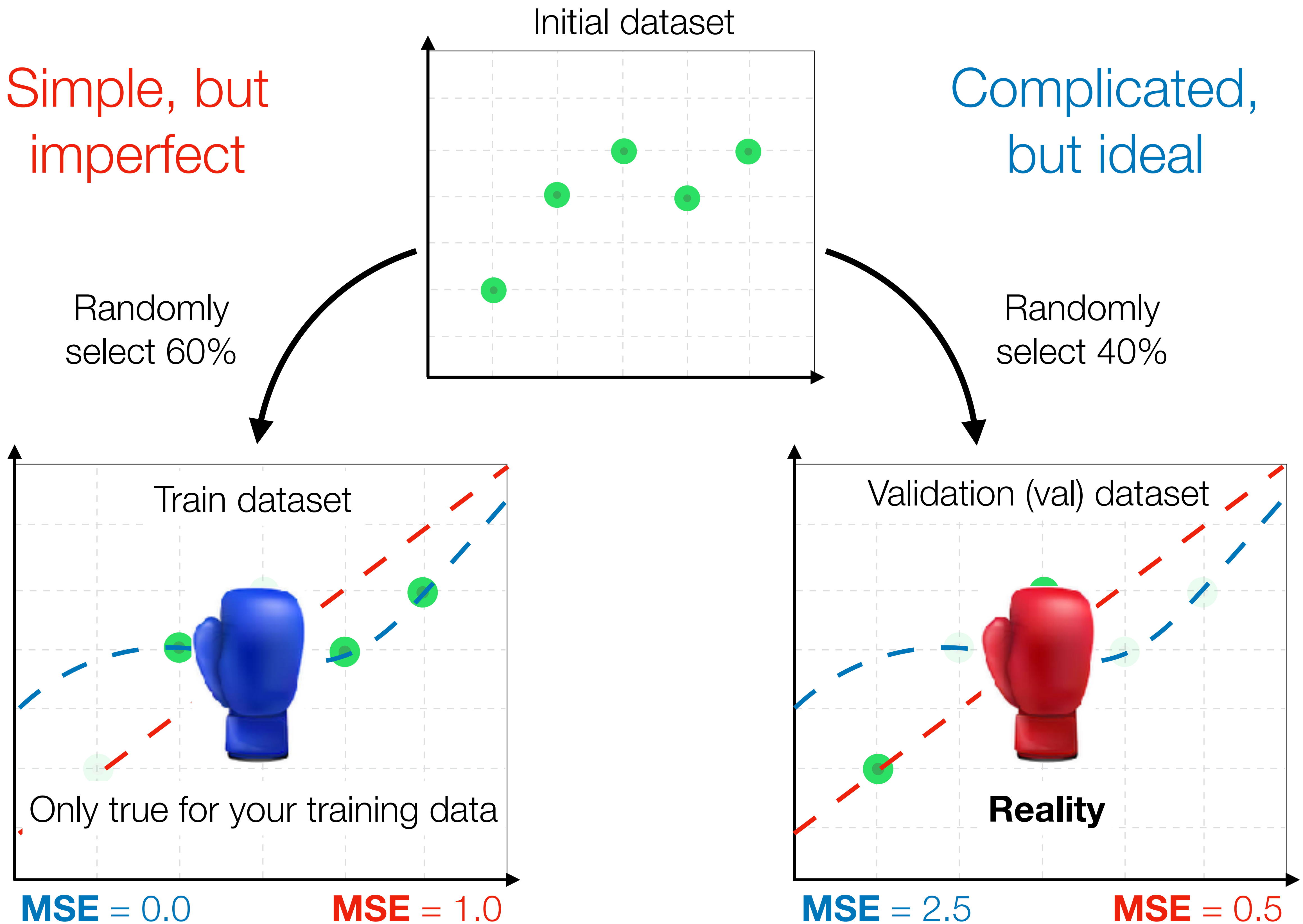
Large gap between training and validation

MSE = 0.0 ⟷ **MSE** = 2.5



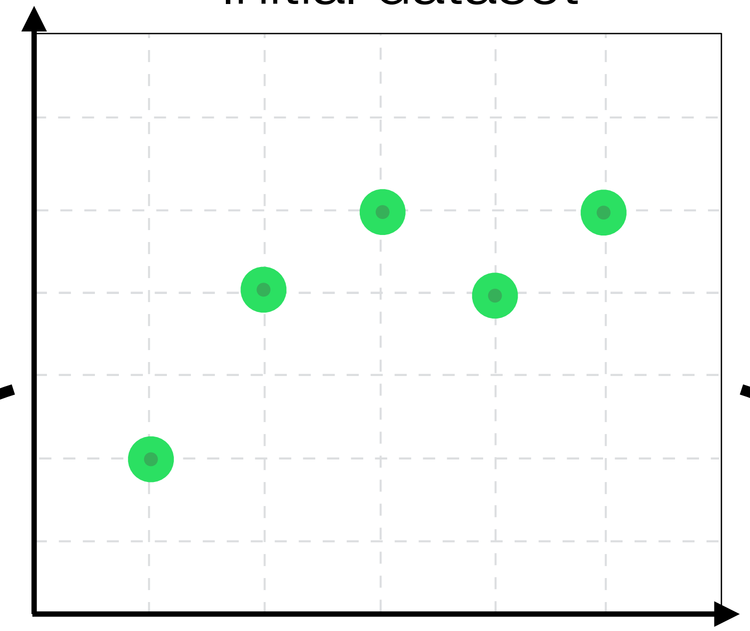


Train/val split



Train/val split

Initial dataset



Simple, but imperfect

Randomly select 60%

Complicated, but ideal

Randomly select 40%

What if the split was **lucky** and validation set turned out to be too simple for one of the models?

Cross Validation (CV) Algorithm

Cross Validation (CV) Algorithm

The whole dataset **100%**

Cross Validation (CV) Algorithm

Training data **80%**	Test **20%**

Cross Validation (CV) Algorithm

Training data **80%**

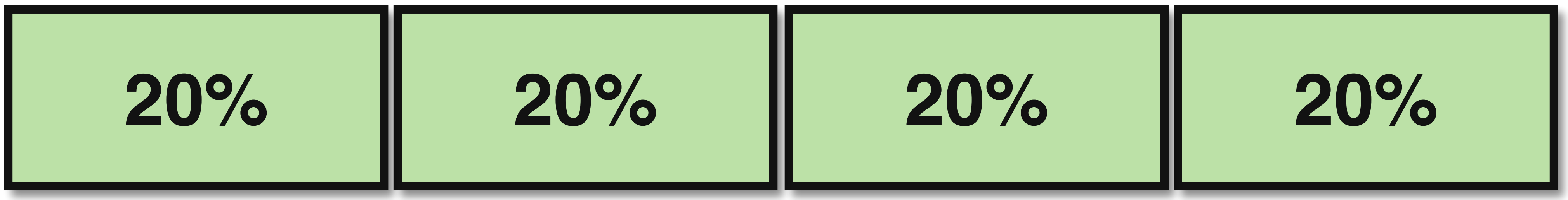
Test **20%**

Test set is used later for the **final evaluation**

Cross Validation (CV) Algorithm

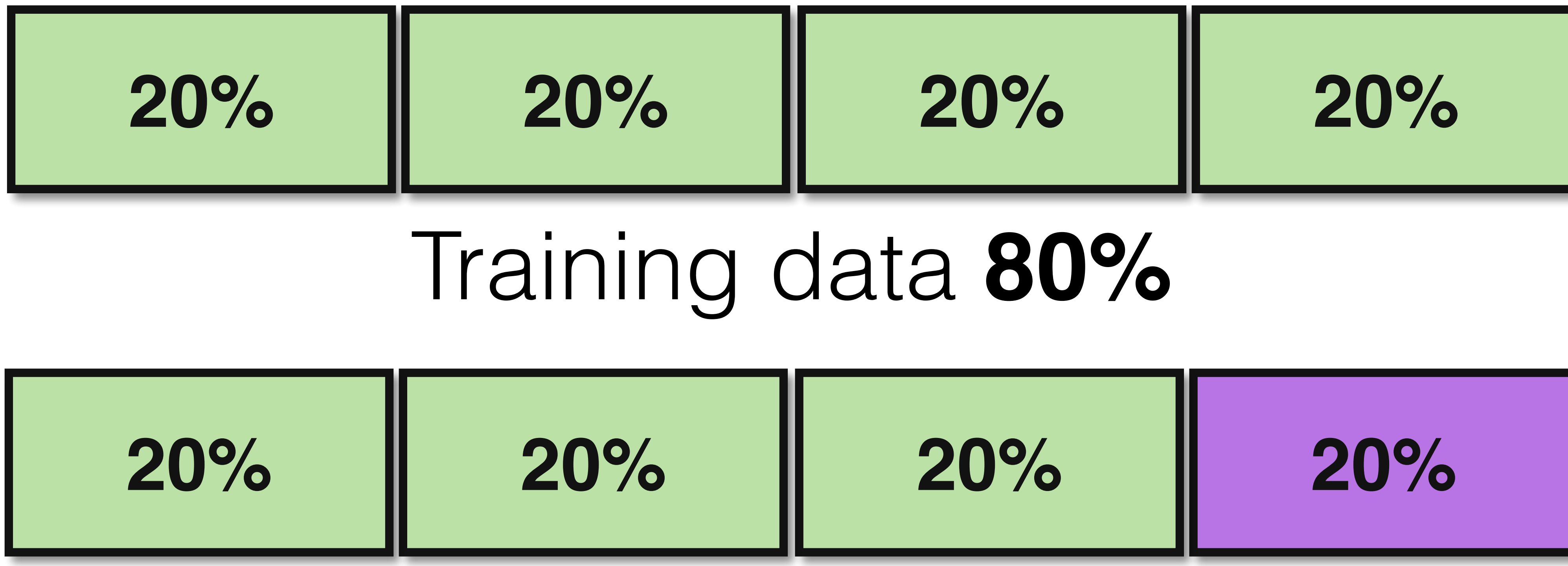
Training data **80%**

Cross Validation (CV) Algorithm



Training data **80%**

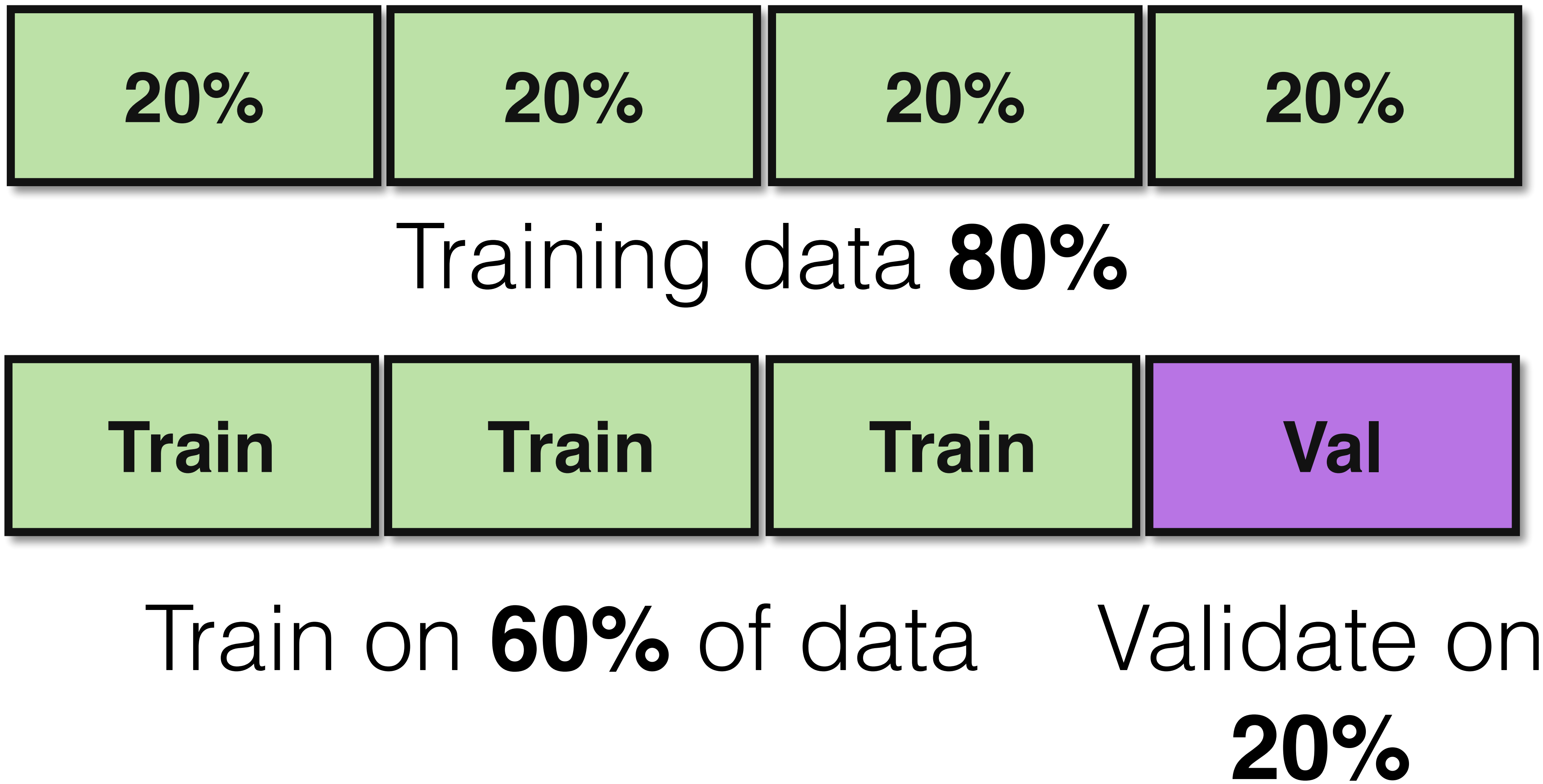
Cross Validation (CV) Algorithm



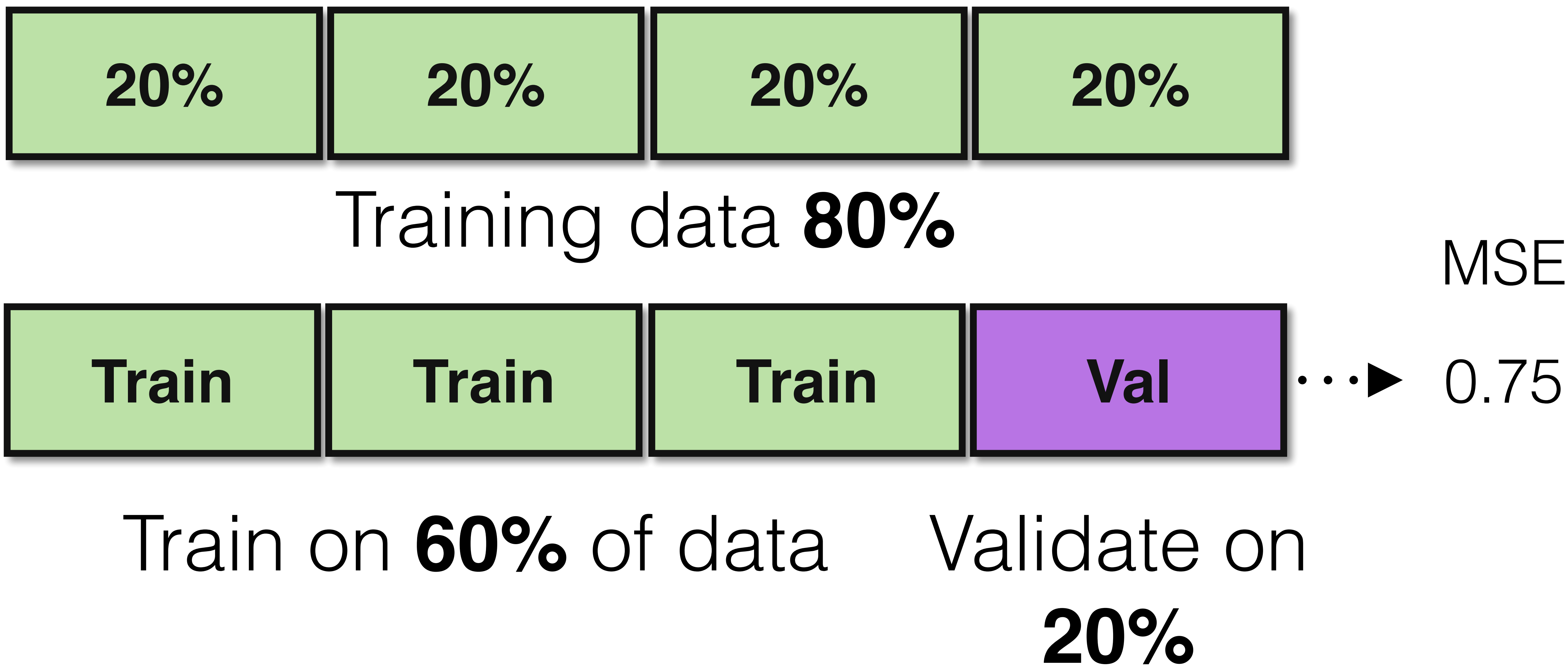
Train on **60%** of data

Validate on **20%**

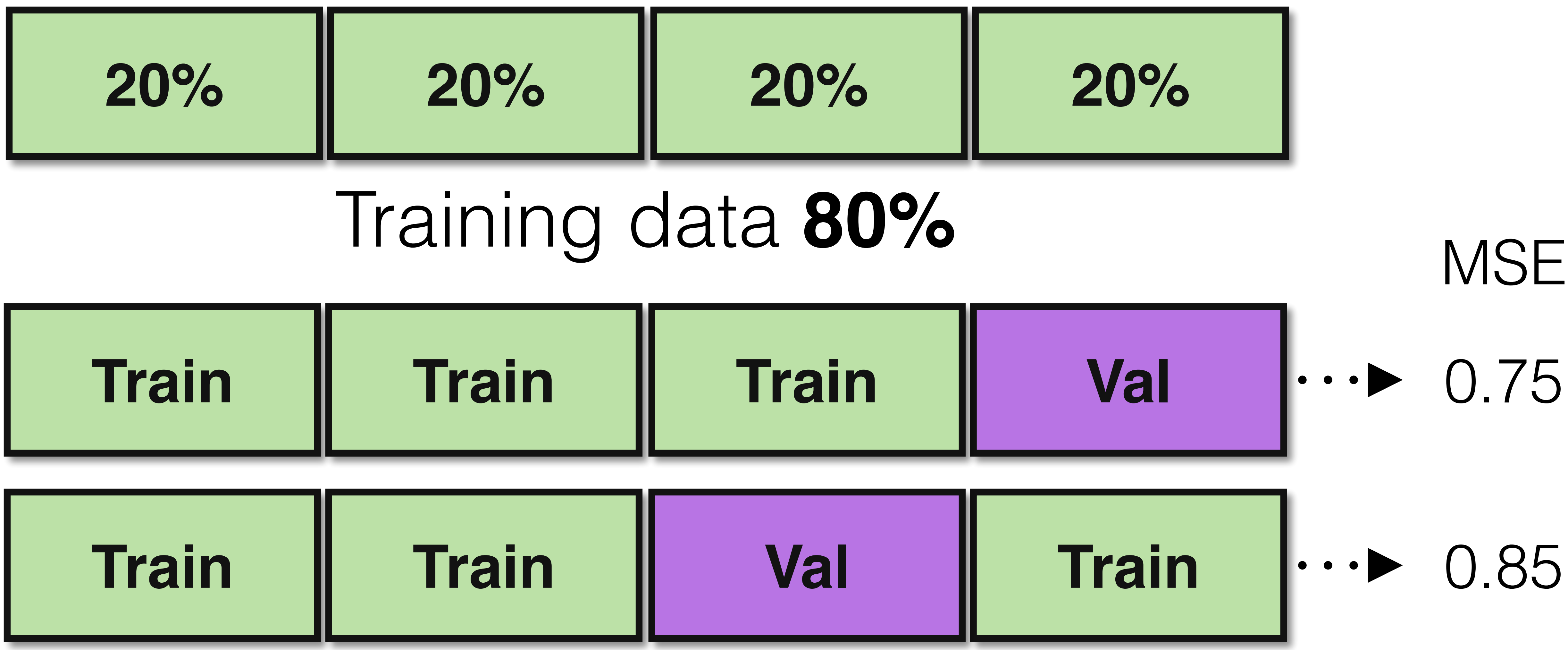
Cross Validation (CV) Algorithm



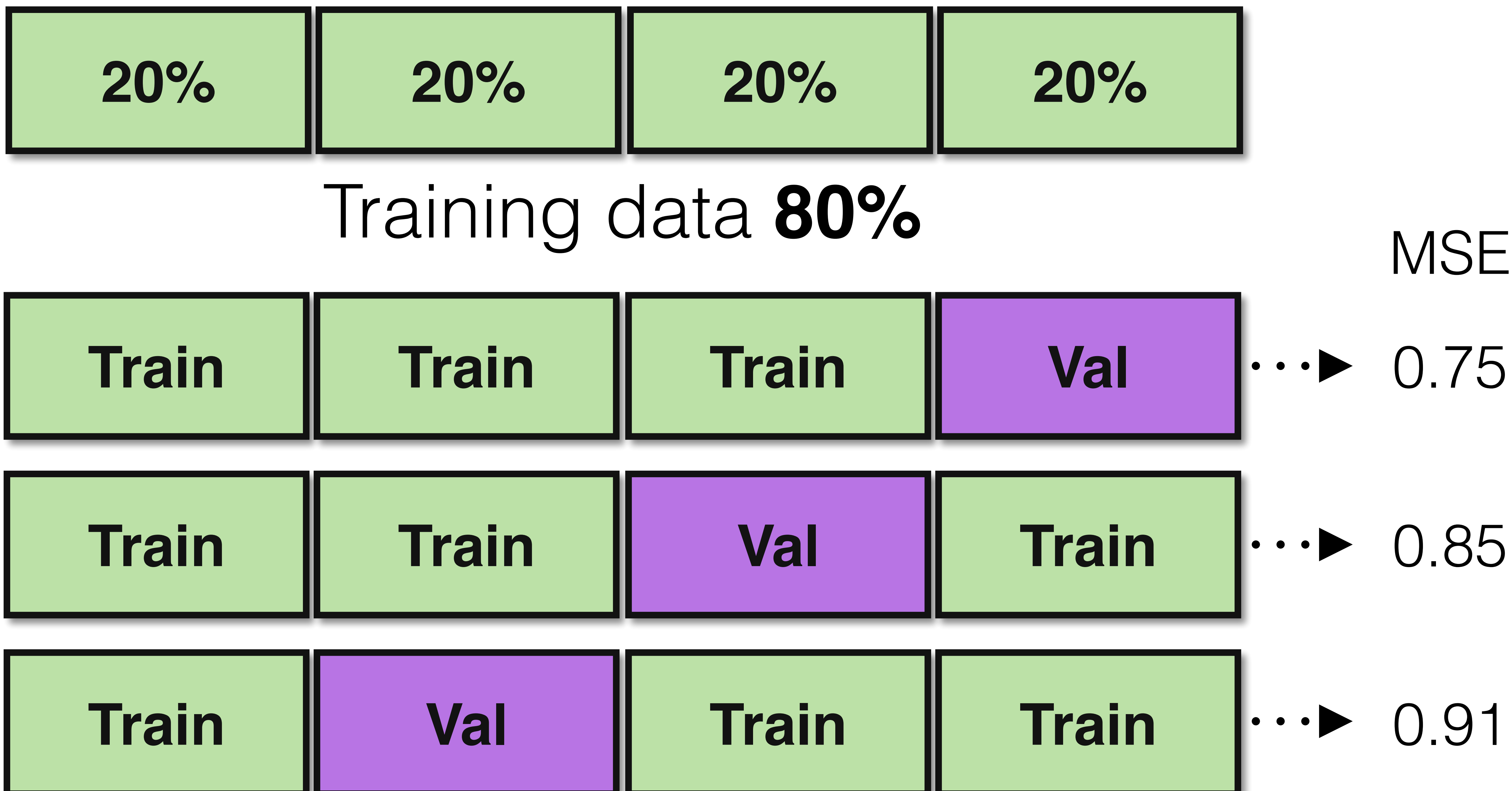
Cross Validation (CV) Algorithm



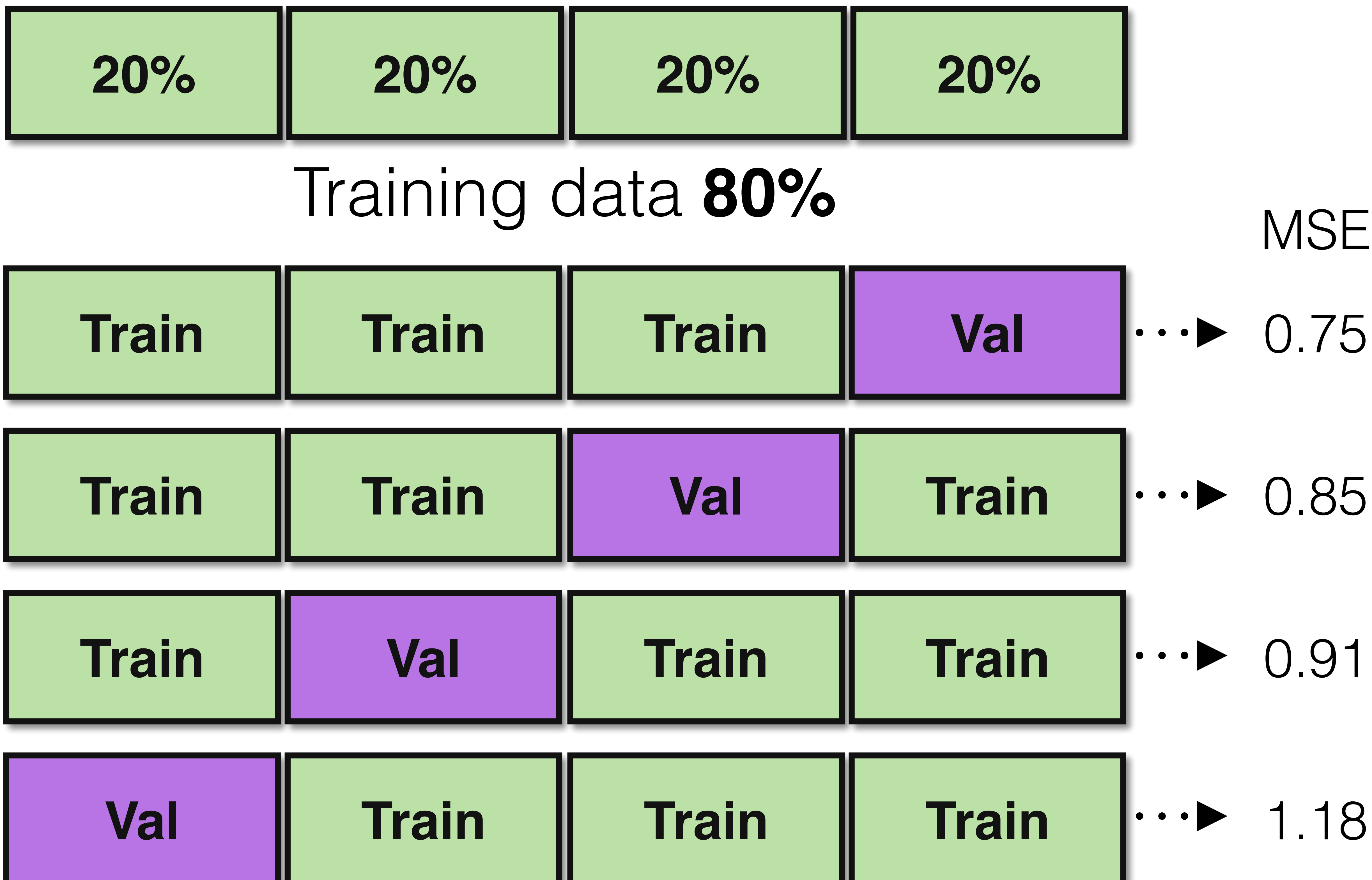
Cross Validation (CV) Algorithm



Cross Validation (CV) Algorithm

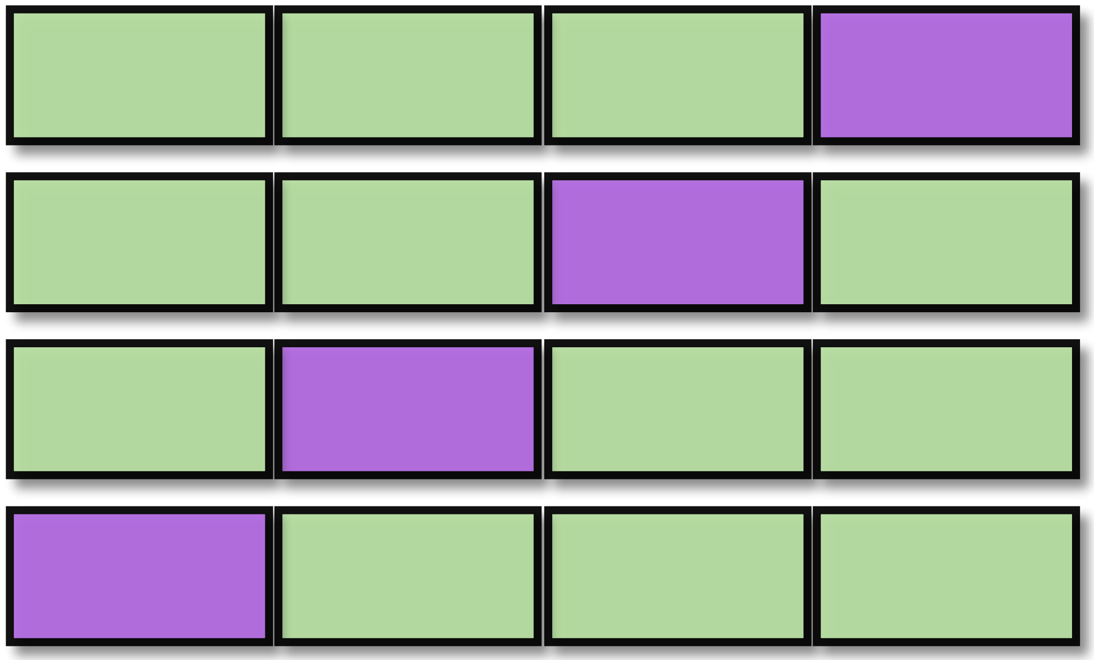


Cross Validation (CV) Algorithm



Cross Validation (CV) Algorithm

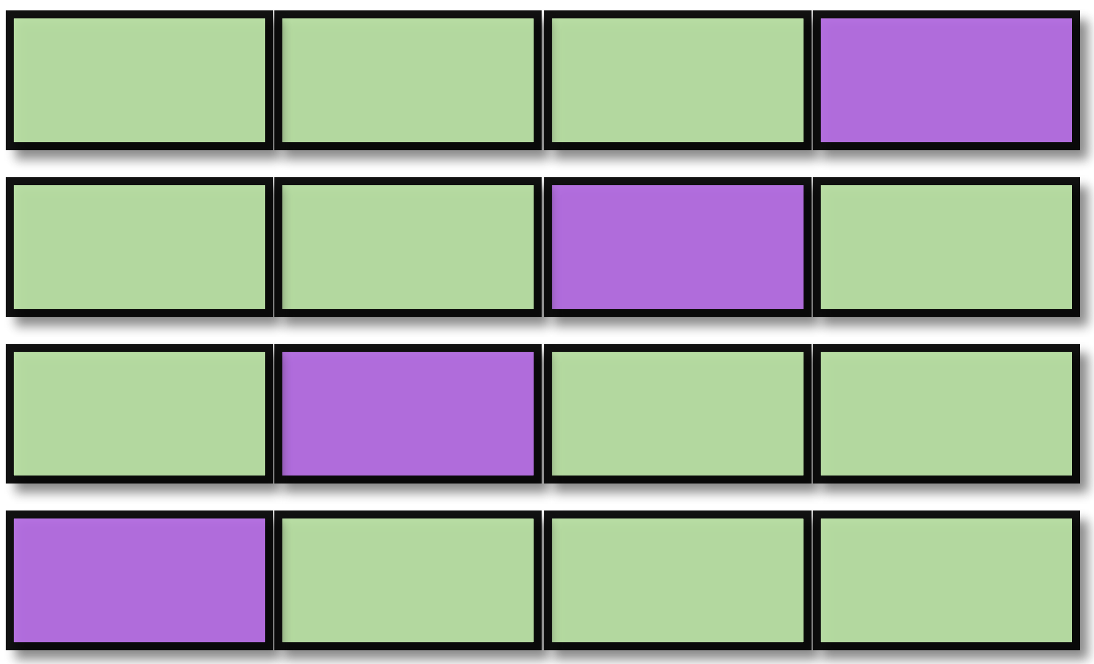
Simple, but imperfect



Average MSE = **0.65**

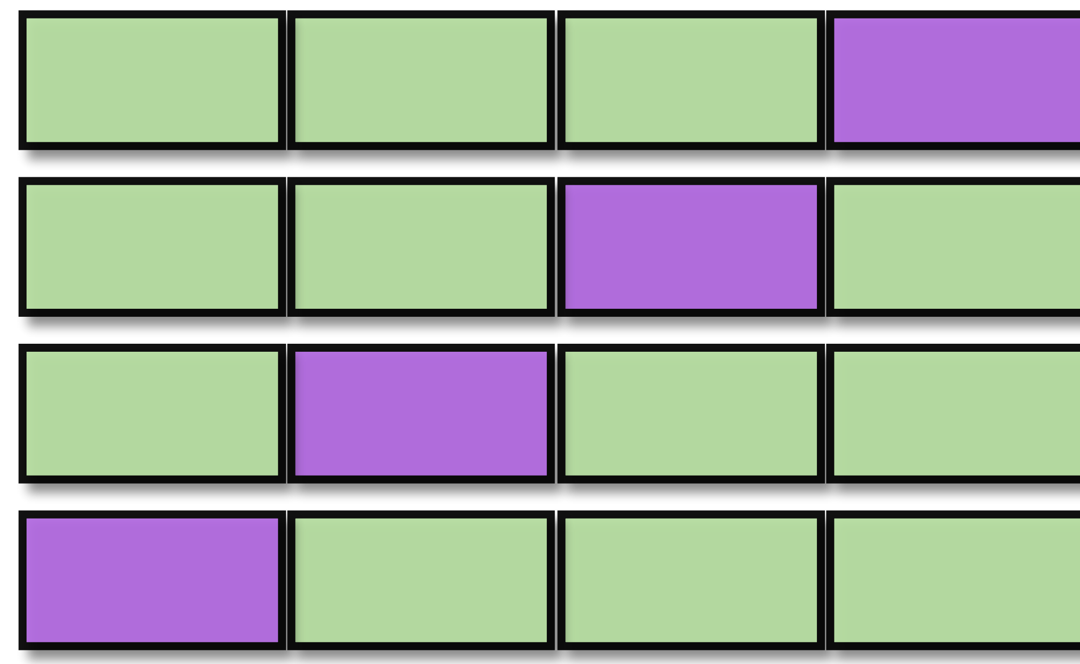
Cross Validation (CV) Algorithm

Simple, but imperfect



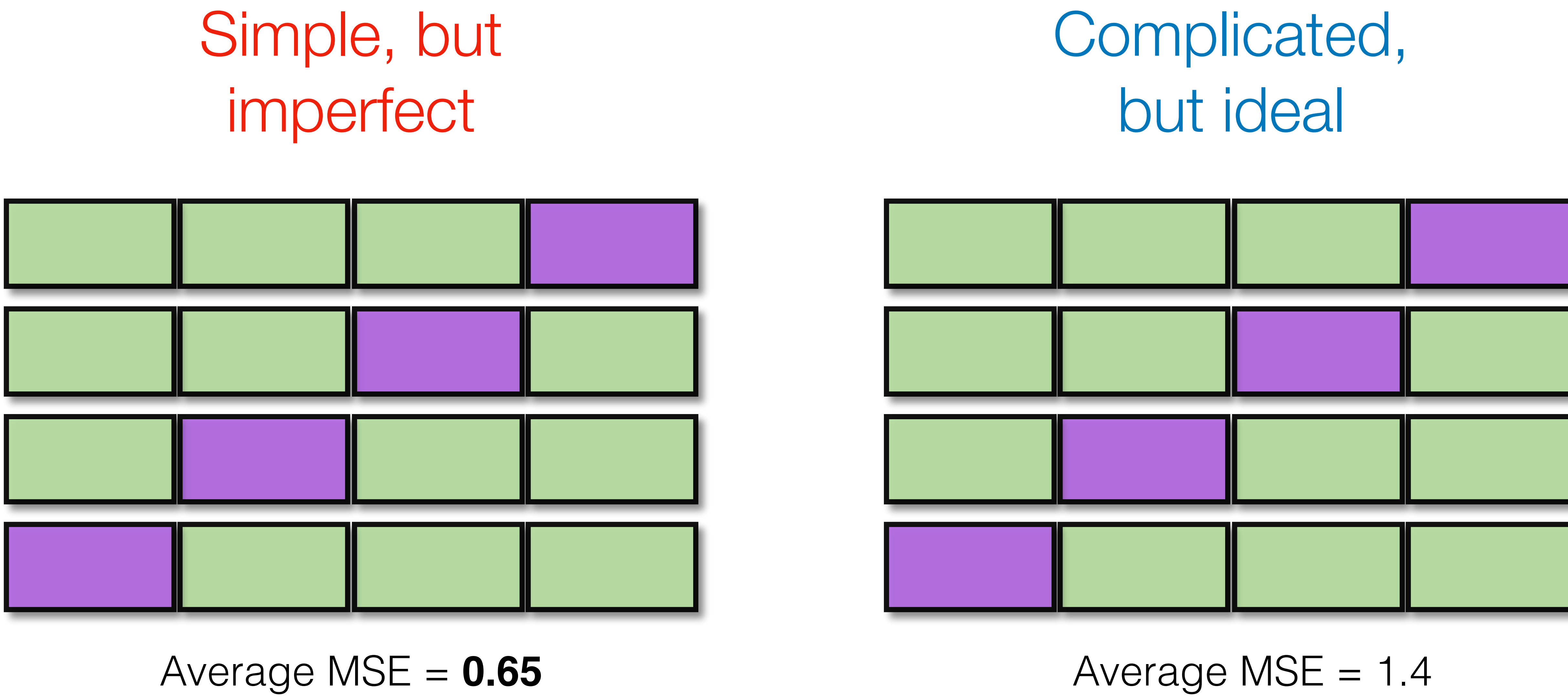
Average MSE = **0.65**

Complicated, but ideal



Average MSE = 1.4

Cross Validation (CV) Algorithm

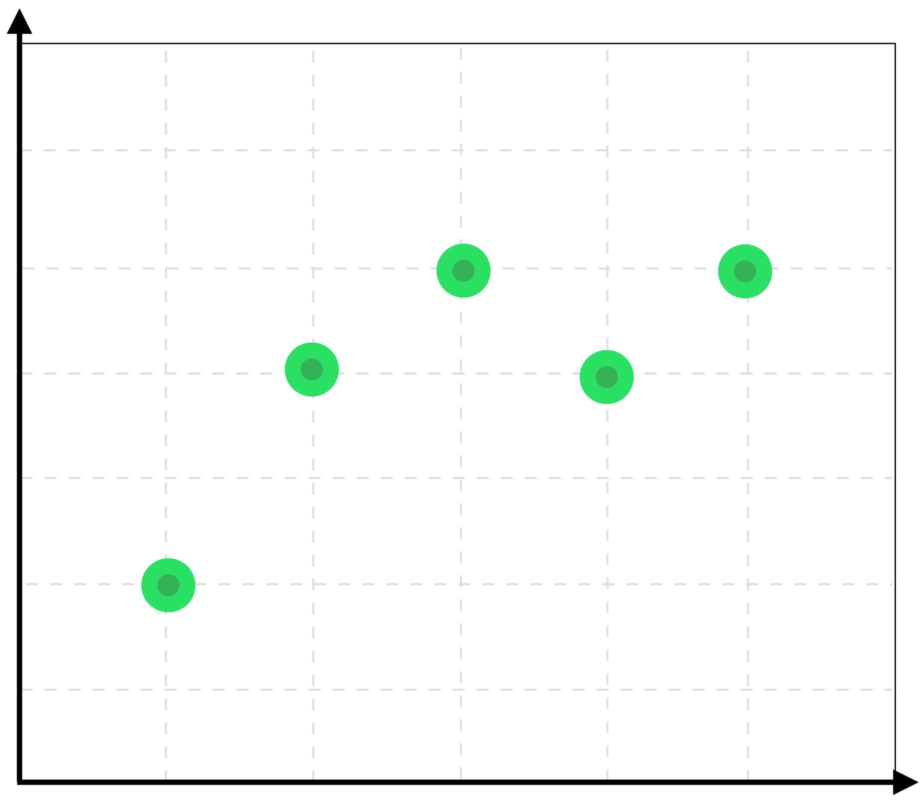


Choose the **best** model

Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

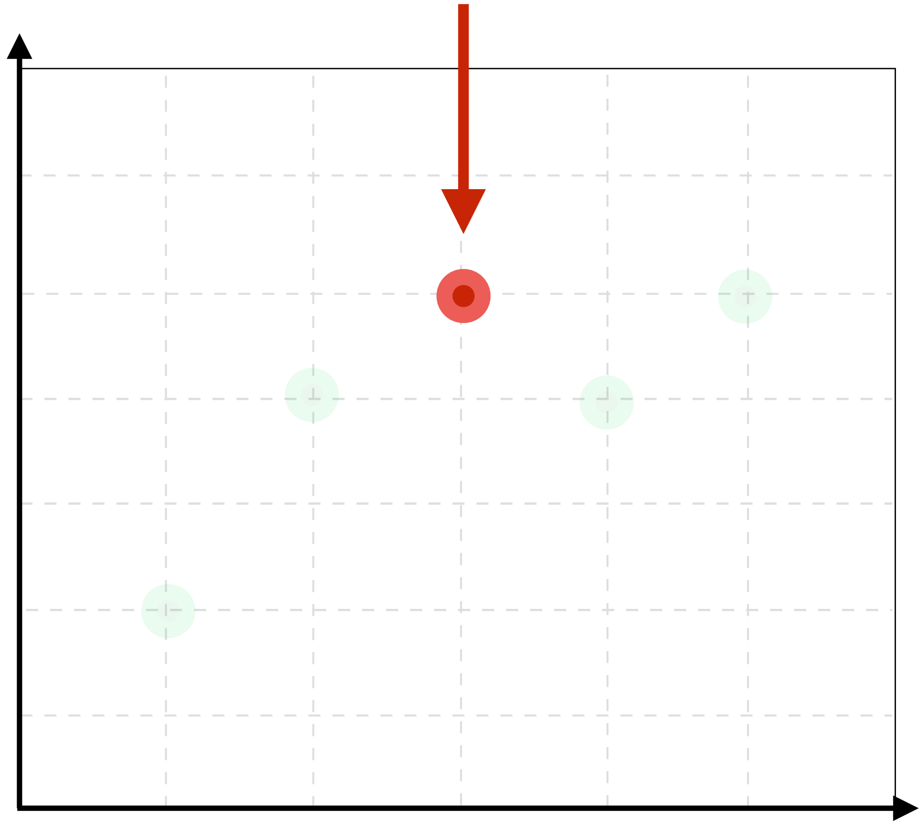


Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

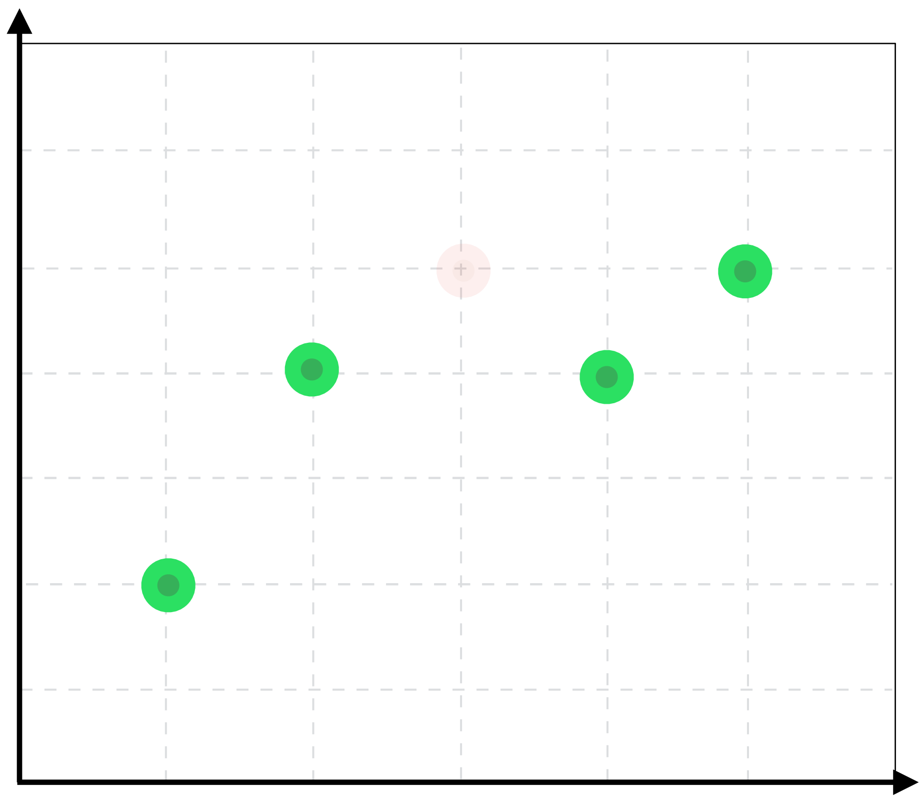
Choose a **test set**
(20% at random)



Cross Validation (CV) Algorithm

Simple, but imperfect

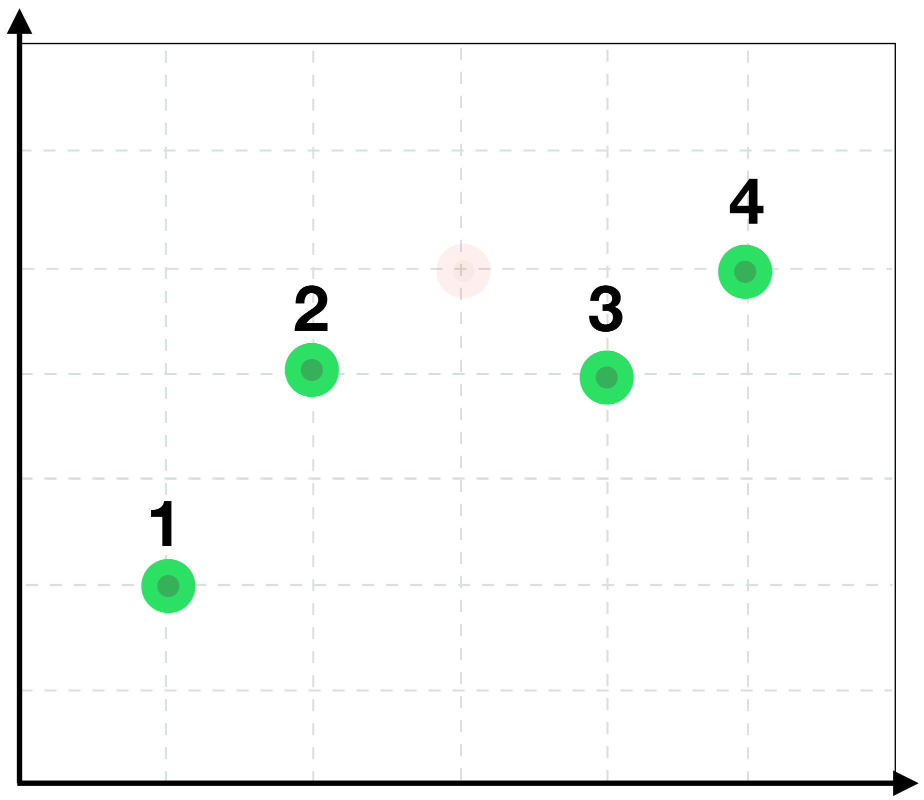
Complicated, but ideal



Cross Validation (CV) Algorithm

Simple, but imperfect

We have **4 folds** by **20%** of data each

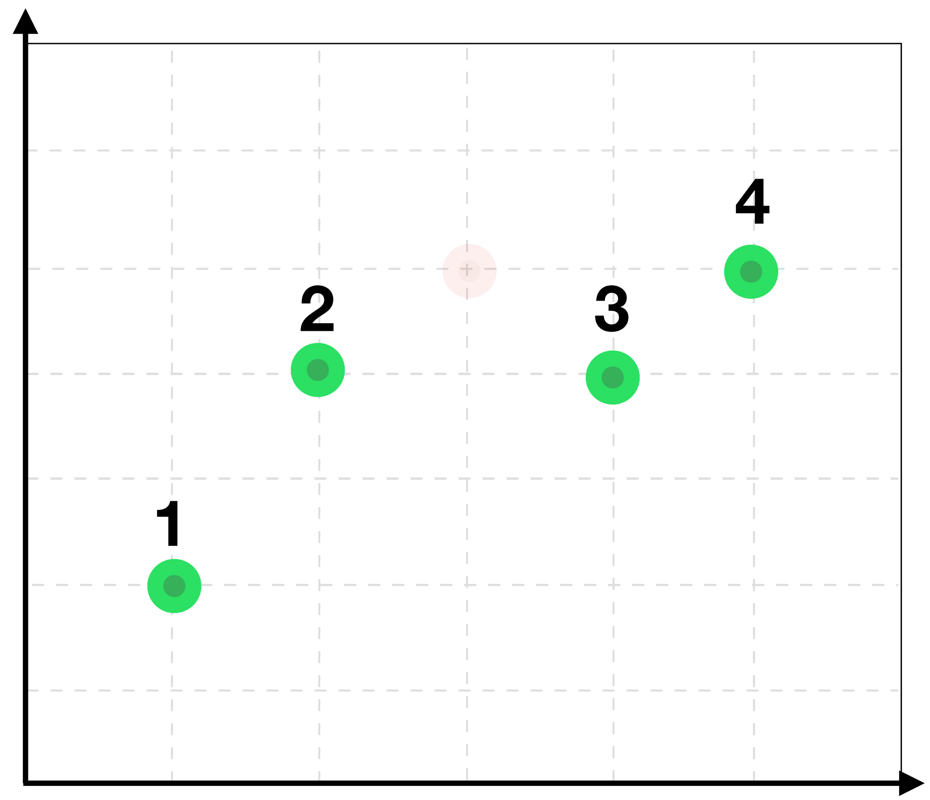


Complicated, but ideal

Cross Validation (CV) Algorithm

Simple, but imperfect

We have **4 folds** by **20%** of data each



Every iteration one of this points will be a **validation set**

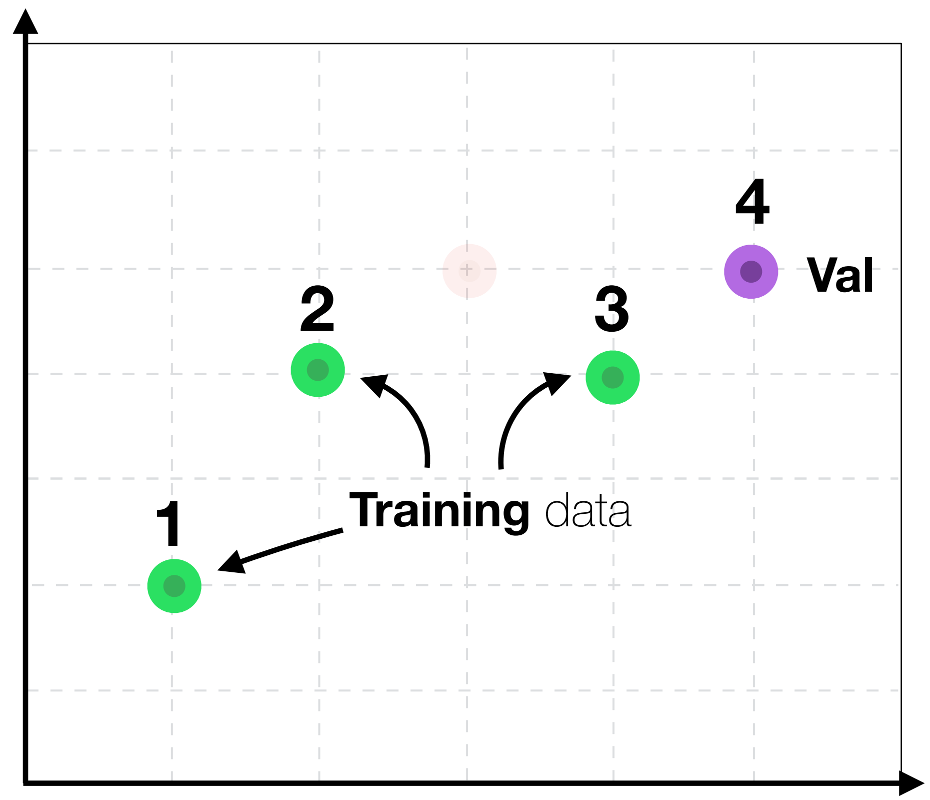
Complicated, but ideal

Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

Iteration I

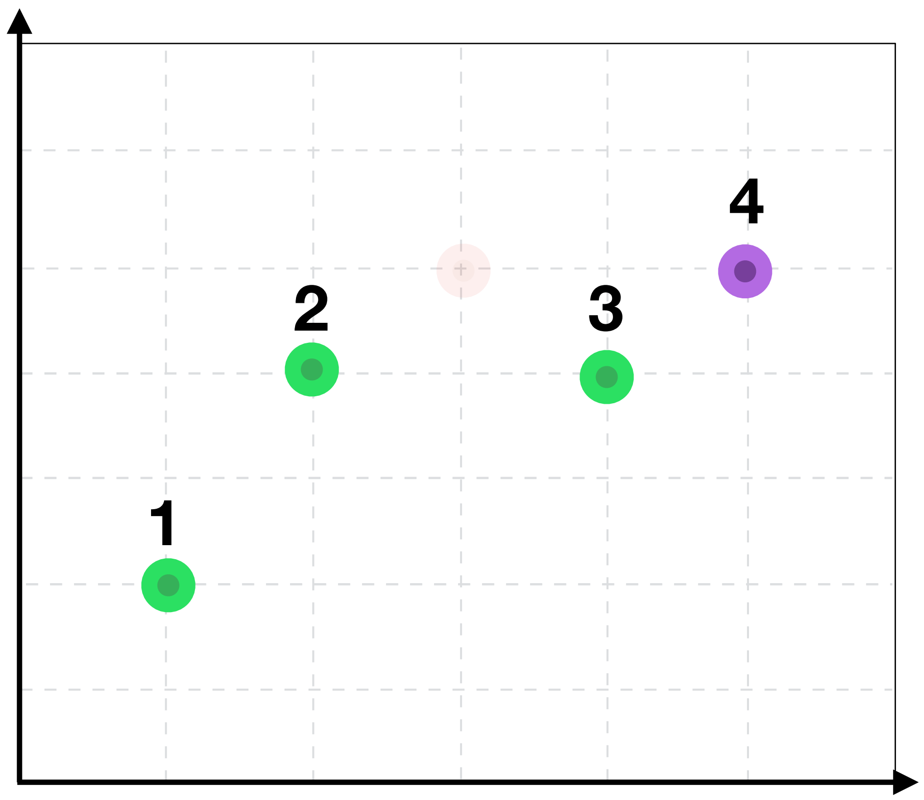


Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

Iteration I



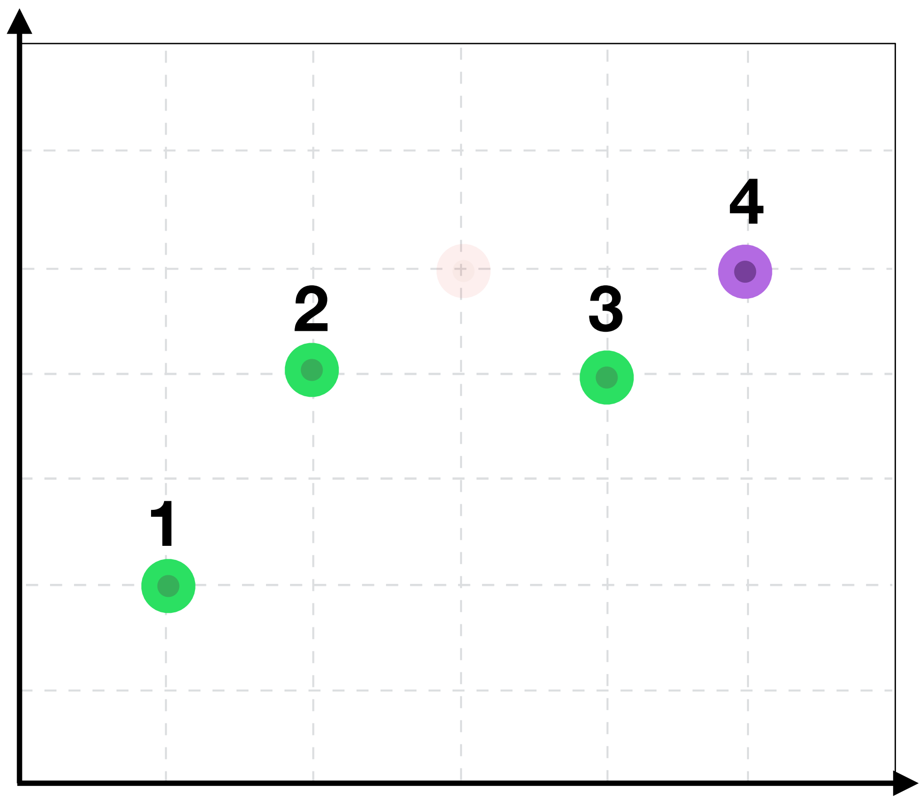
Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

Iteration **I**

	Training	Val
I		
II		
III		
IV		
Average		



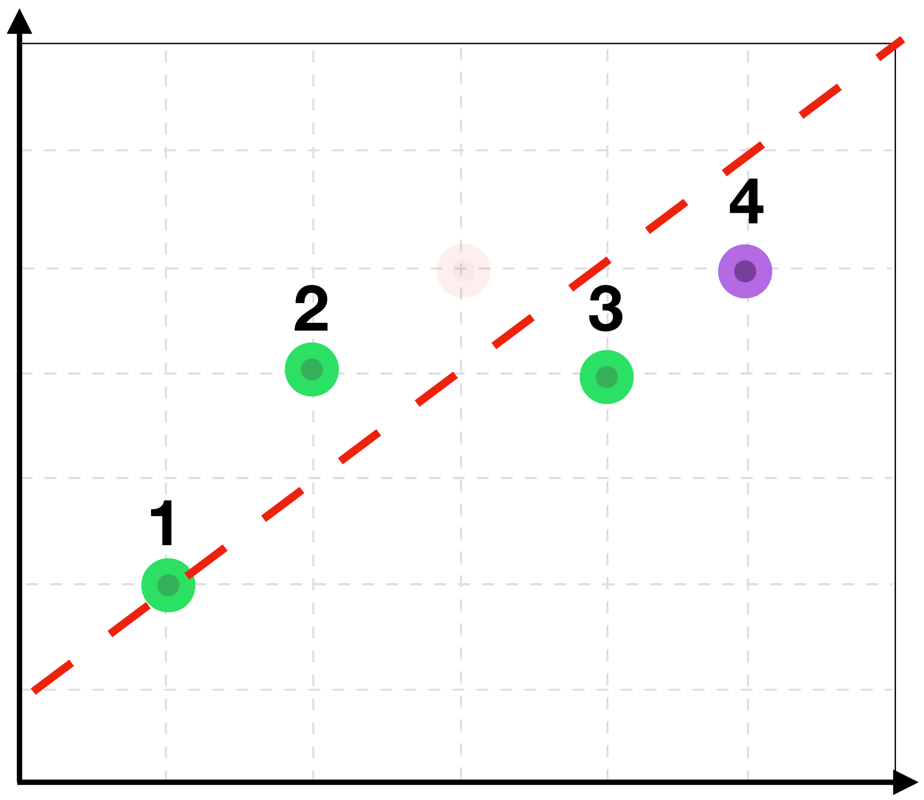
Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

Iteration I

	Training	Val
I		
II		
III		
IV		
Average		



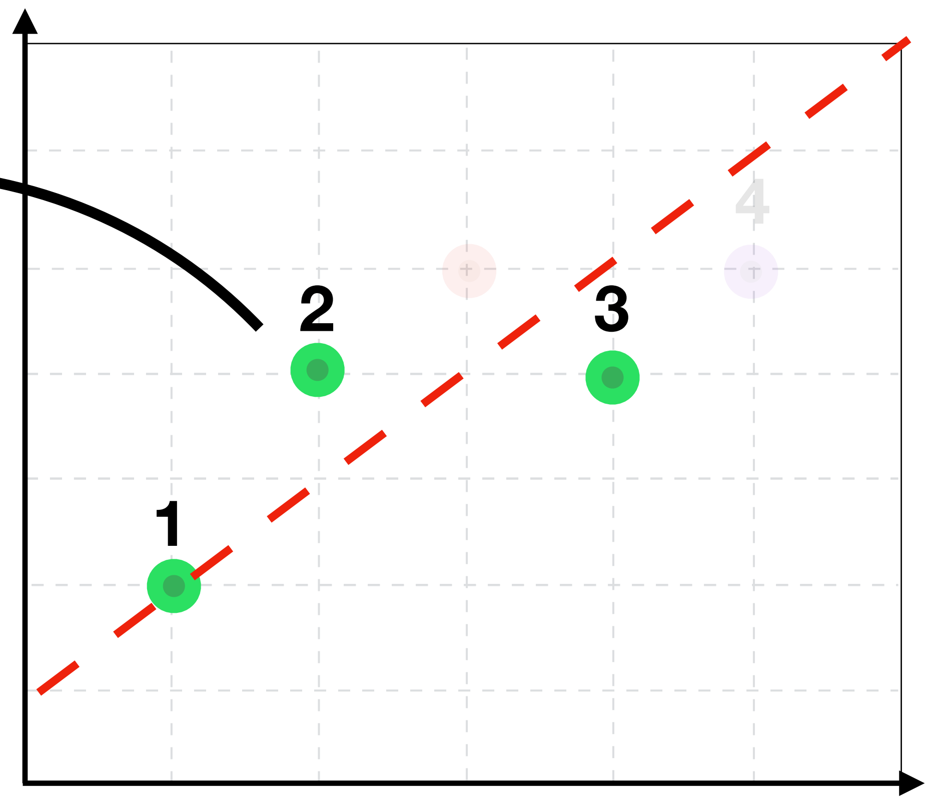
Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

Iteration **I**

	Training	Val
I		
II		
III		
IV		
Average		



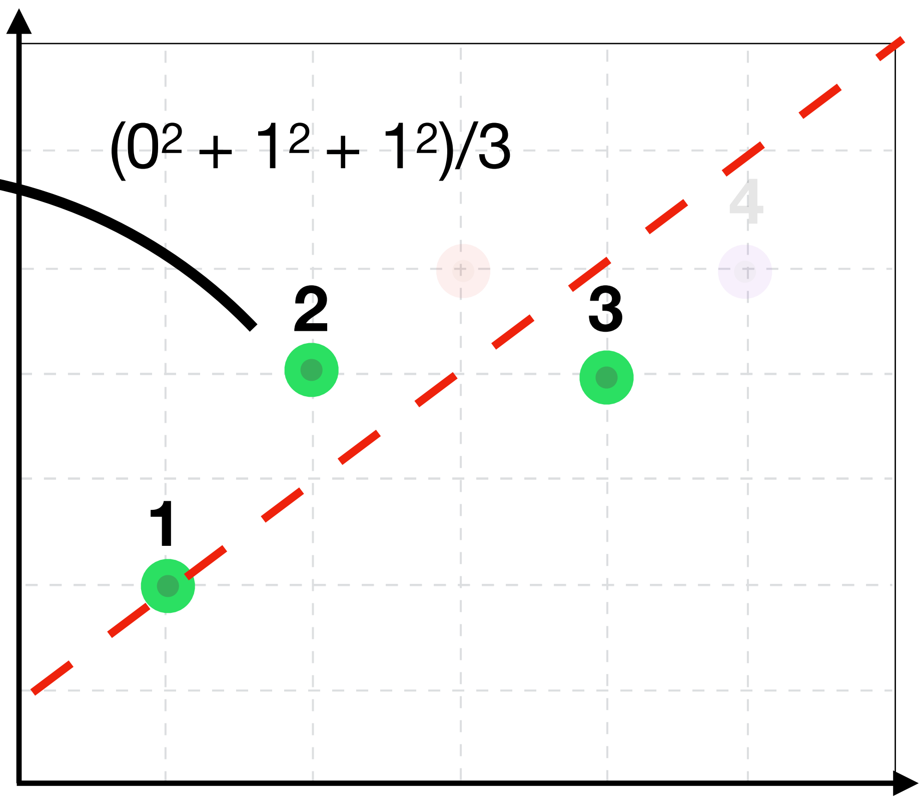
Cross Validation (CV) Algorithm

Simple, but imperfect

Iteration **I**

Complicated, but ideal

	Training	Val
I		
II		
III		
IV		
Average		



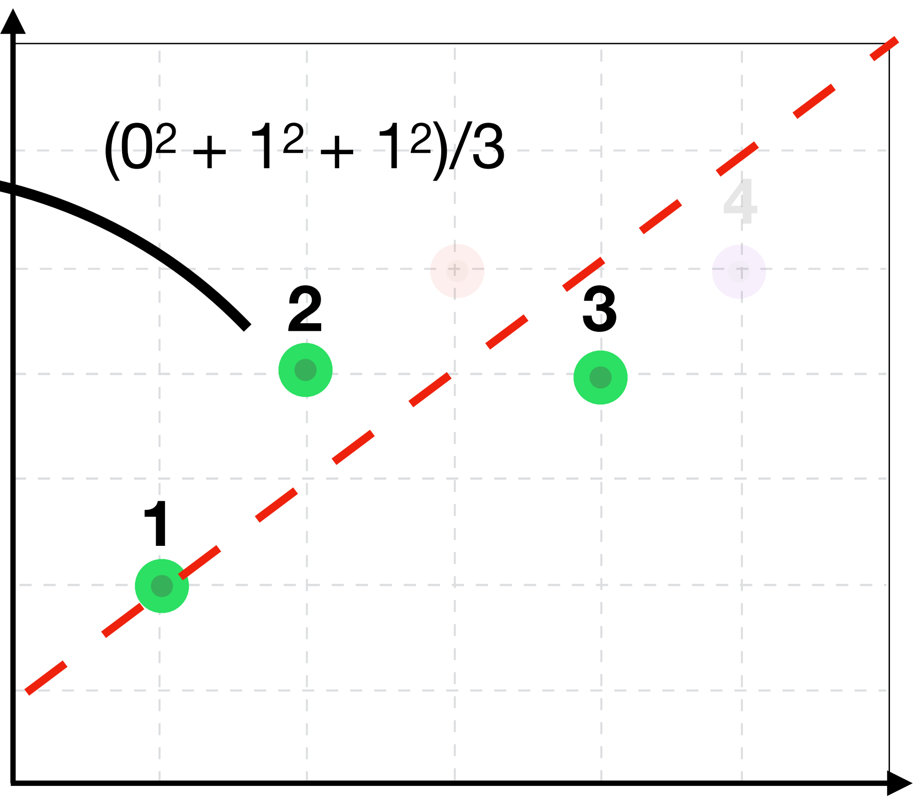
Cross Validation (CV) Algorithm

Simple, but imperfect

Iteration **I**

Complicated, but ideal

	Training	Val
I	0.66	
II		
III		
IV		
Average		



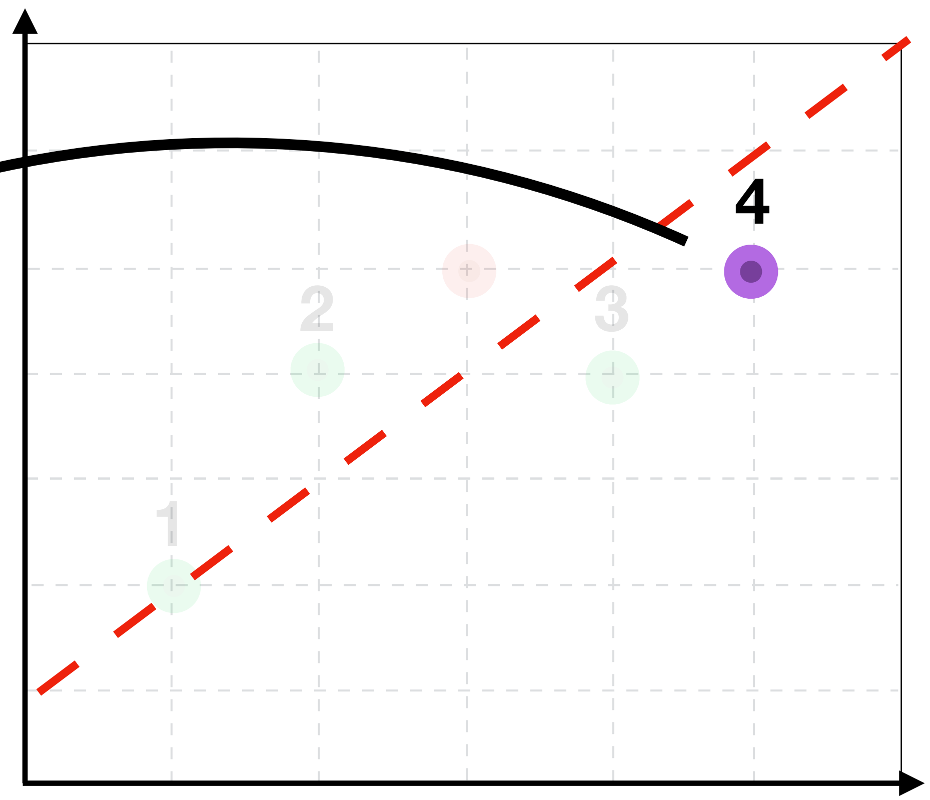
Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

Iteration **I**

	Training	**Val**
I	0.66	
II		
III		
IV		
Average		



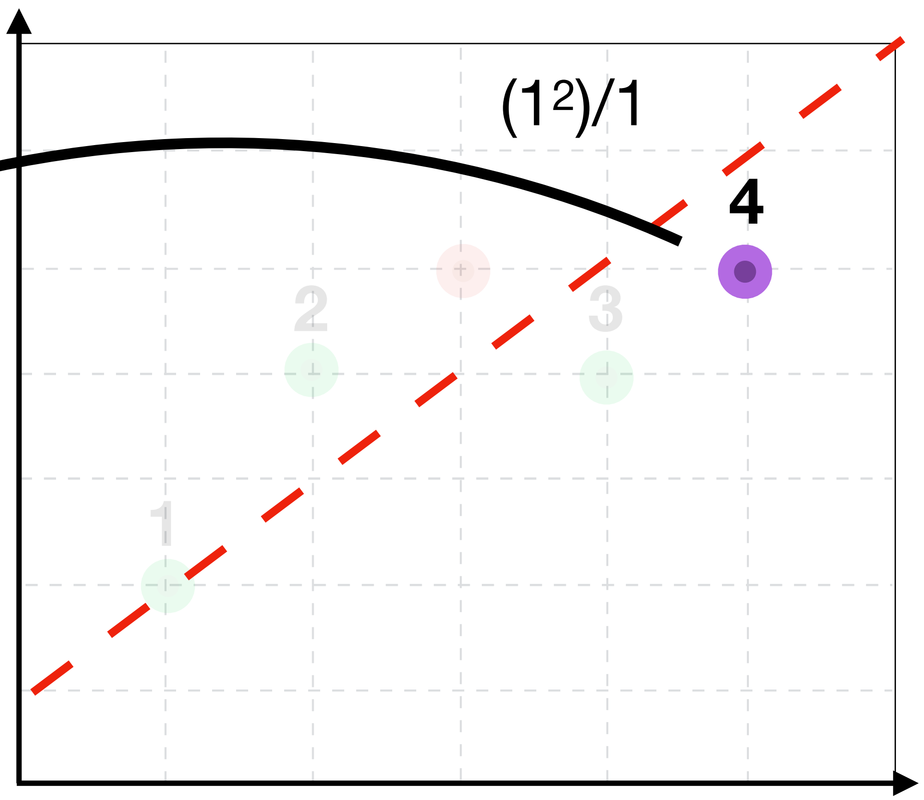
Cross Validation (CV) Algorithm

Simple, but imperfect

Complicated, but ideal

Iteration I

	Training	Val
I	0.66	
II		
III		
IV		
Average		



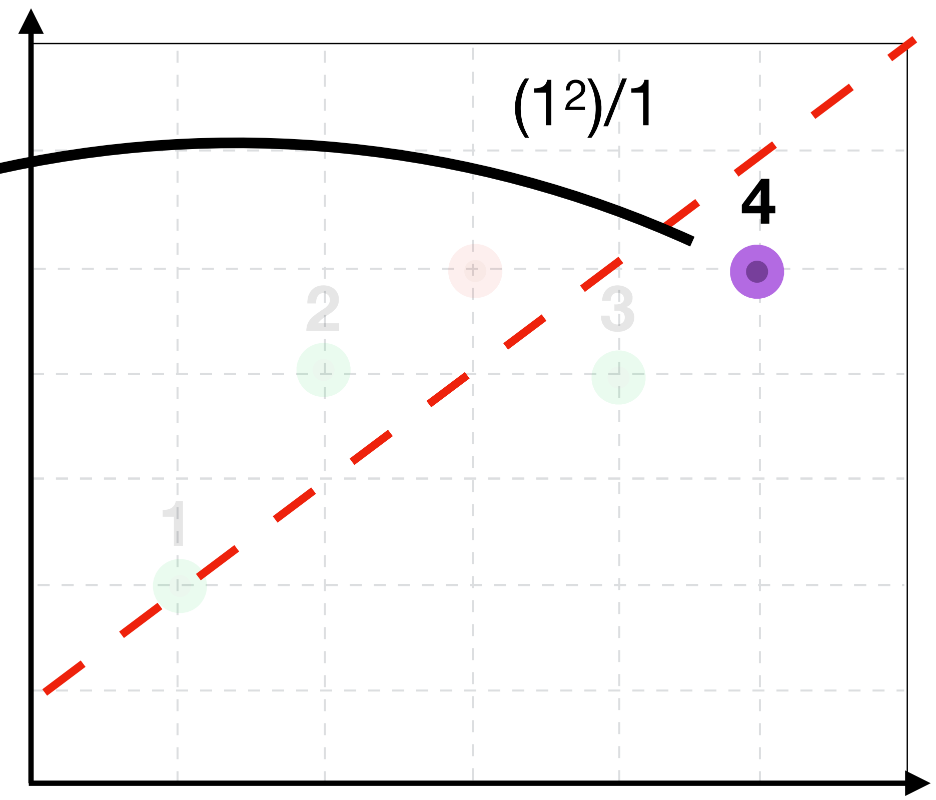
Cross Validation (CV) Algorithm

Simple, but imperfect

Iteration I

Complicated, but ideal

	Training	Val
I	0.66	1
II		
III		
IV		
Average		



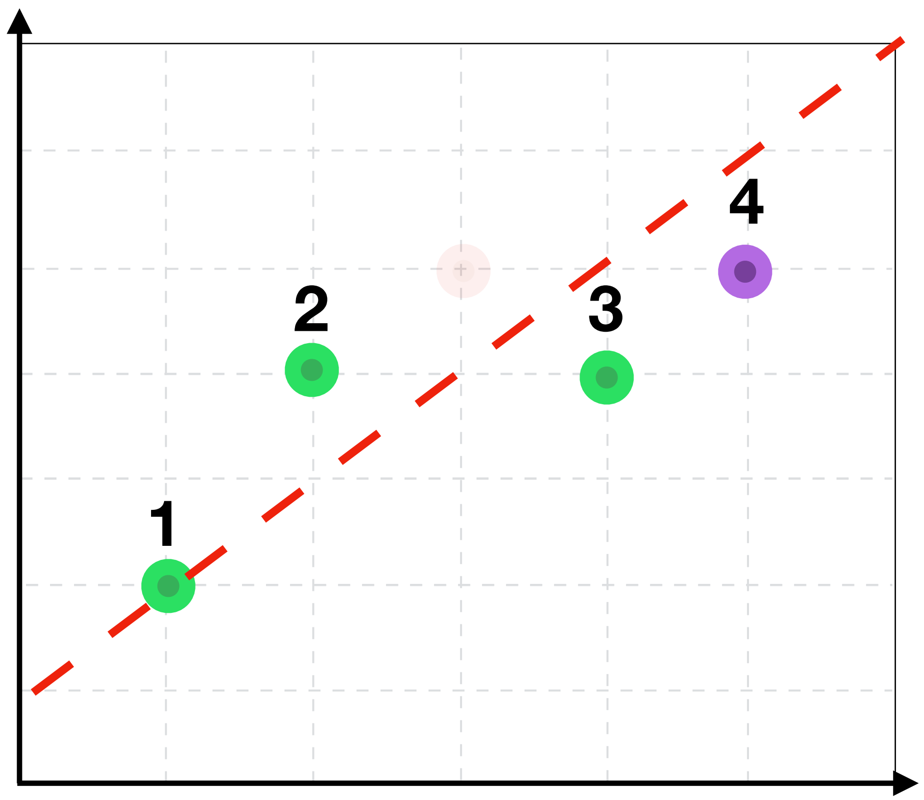
Cross Validation (CV) Algorithm

Simple, but imperfect

Iteration **I**

Complicated, but ideal

	Training	**Val**
I	0.66	1
II		
III		
IV		
Average		

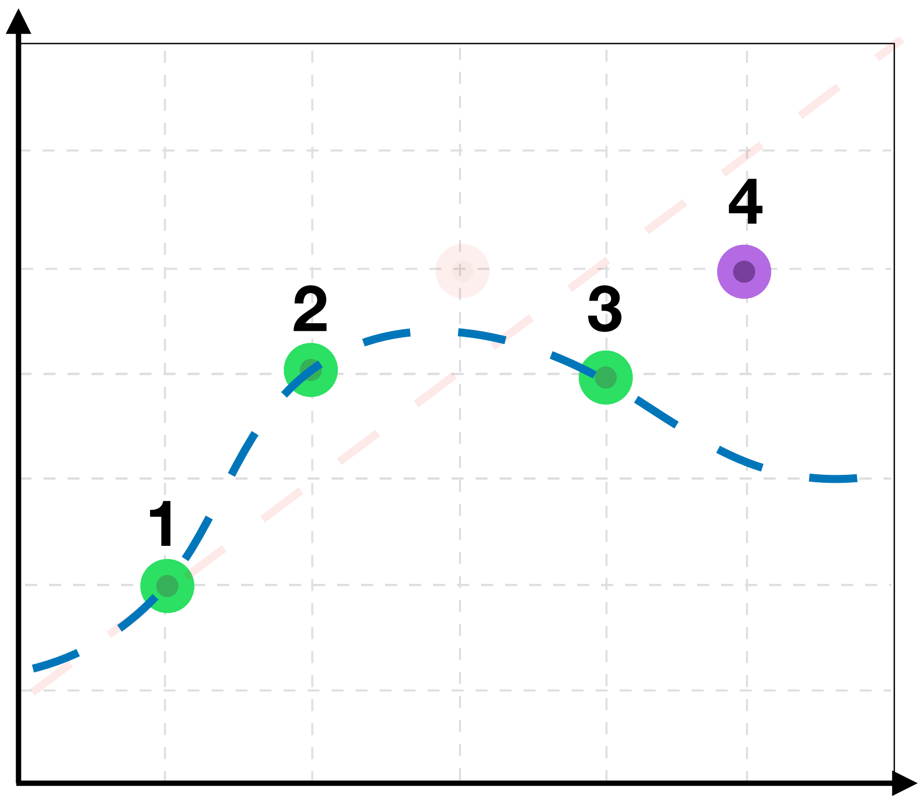


Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

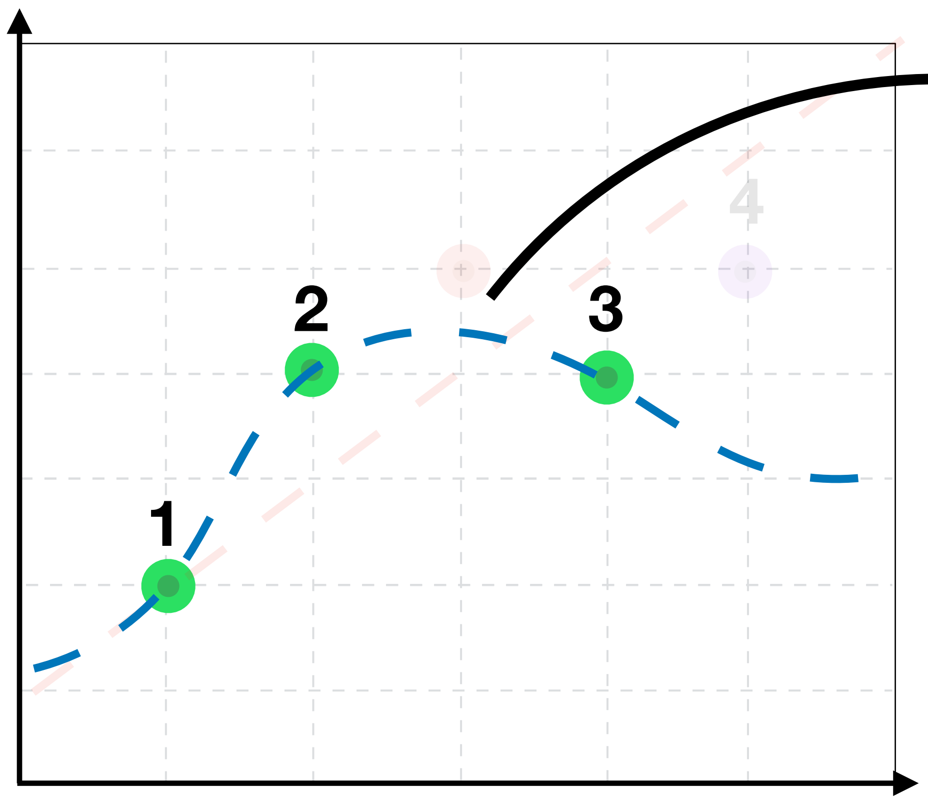
	Training	Val
I		
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

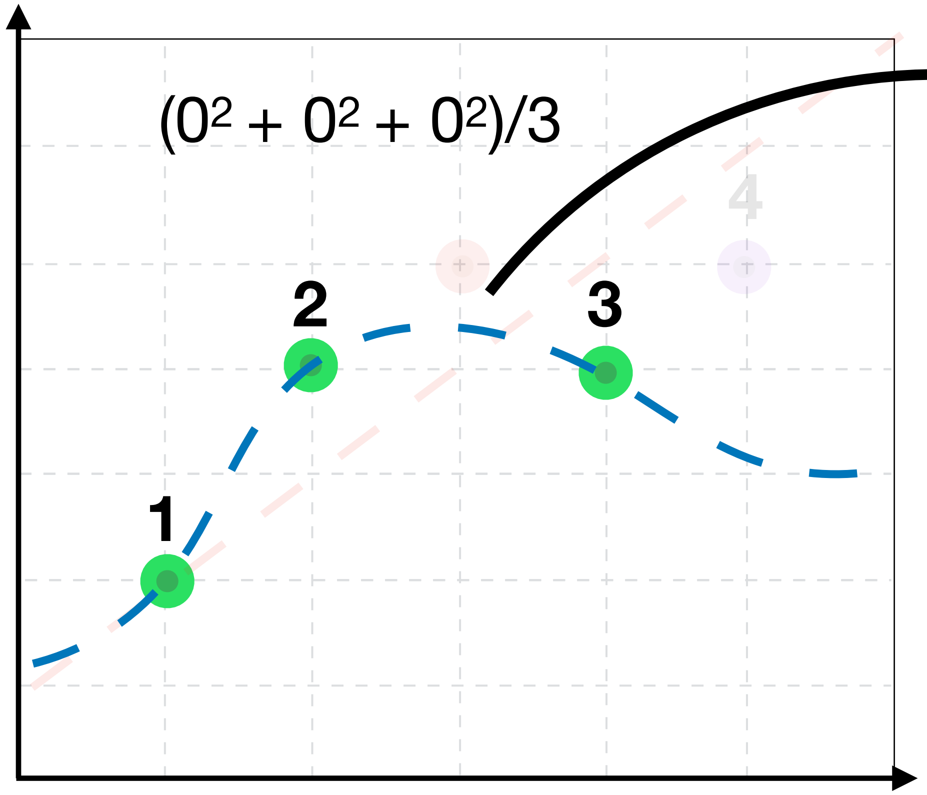
	Training	Val
I		
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

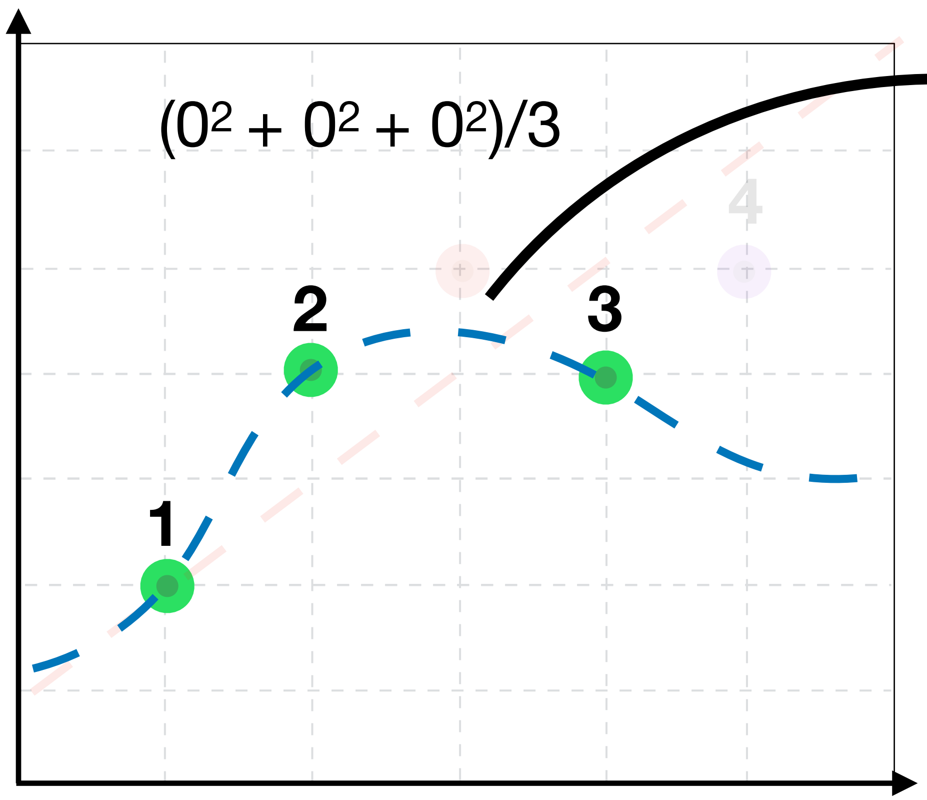
	Training	Val
I		
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

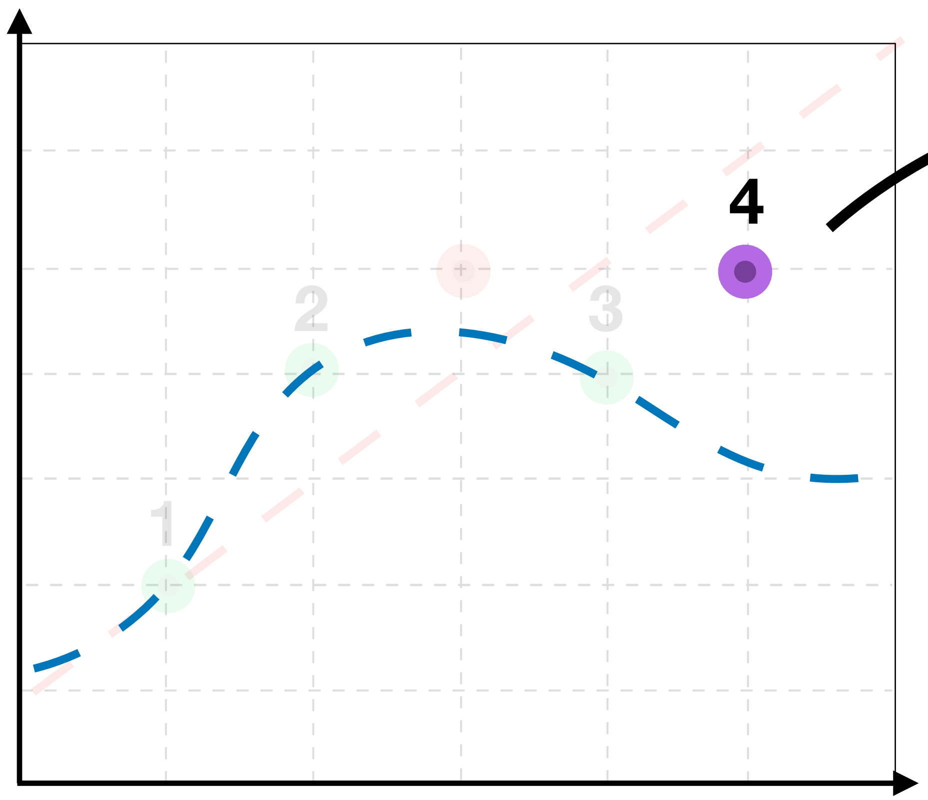
	Training	Val
I	0	
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

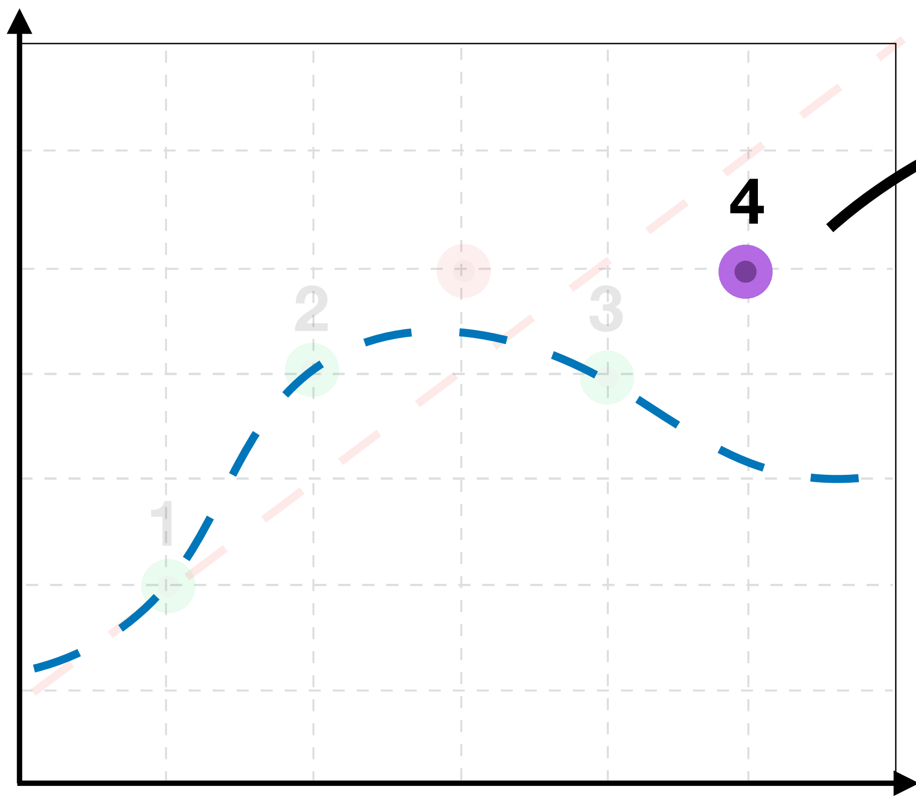
	Training	Val
I	0	
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

$(2^2)/1$

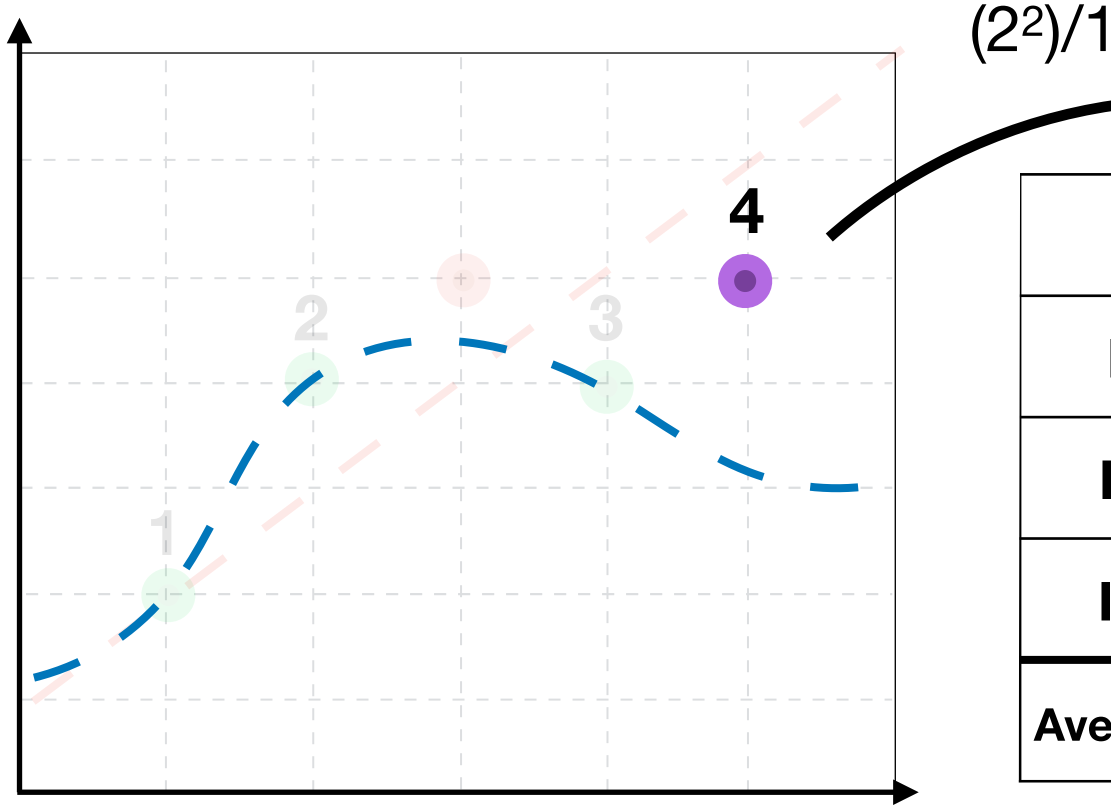
	Training	Val
I	0	
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

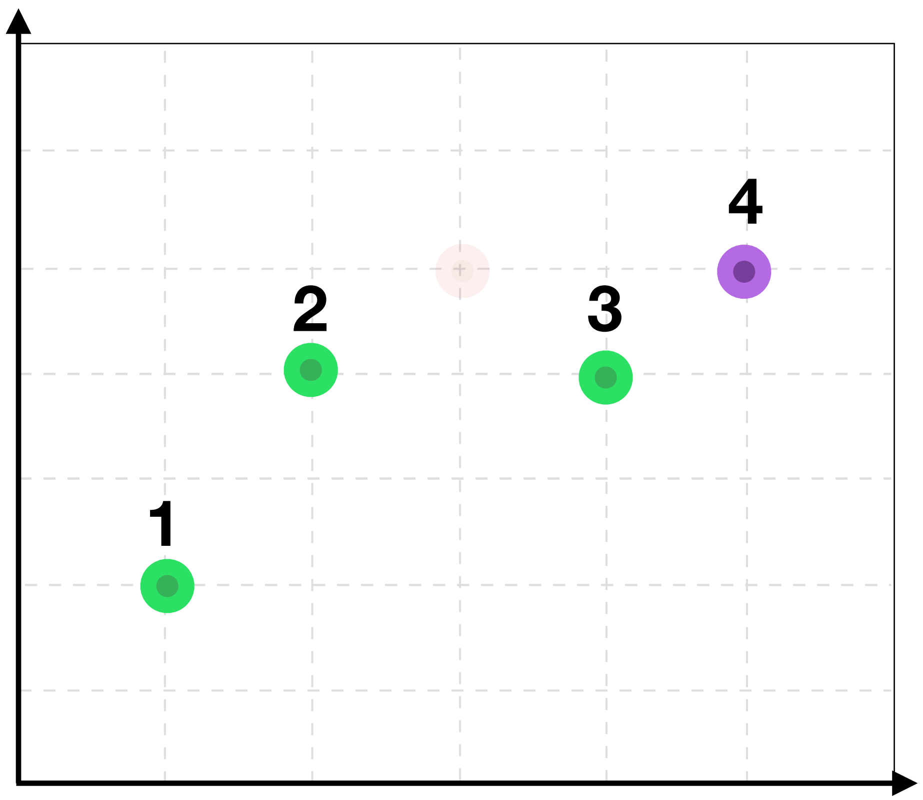
	Training	Val
I	0	4
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **I**



Complicated, but ideal

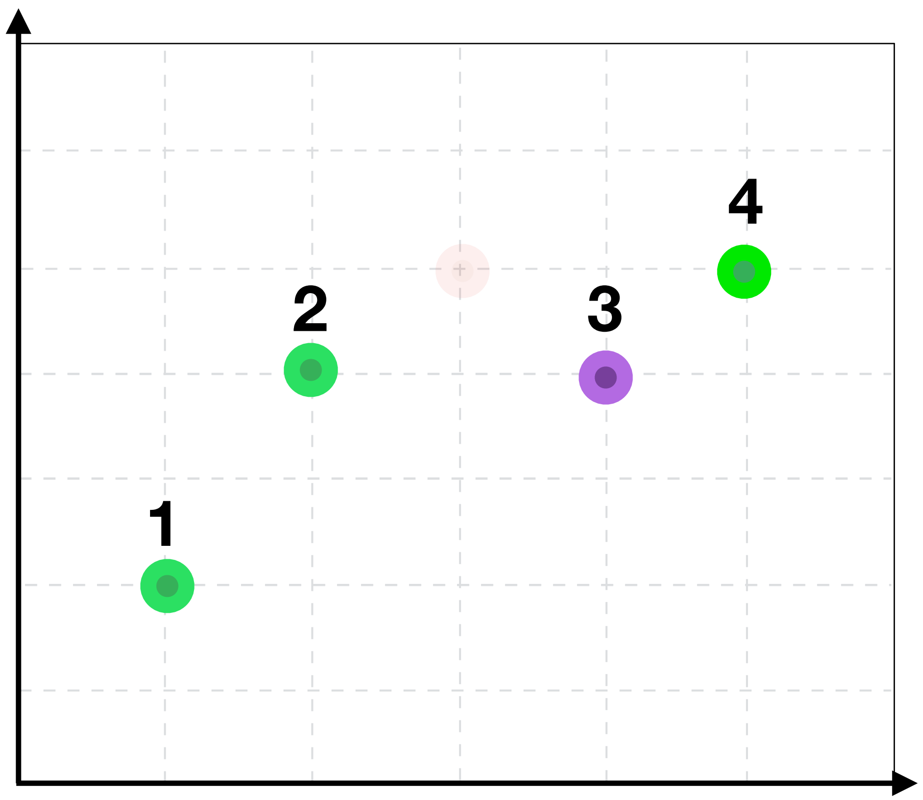
	Training	Val
I	0	4
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **II**



Complicated, but ideal

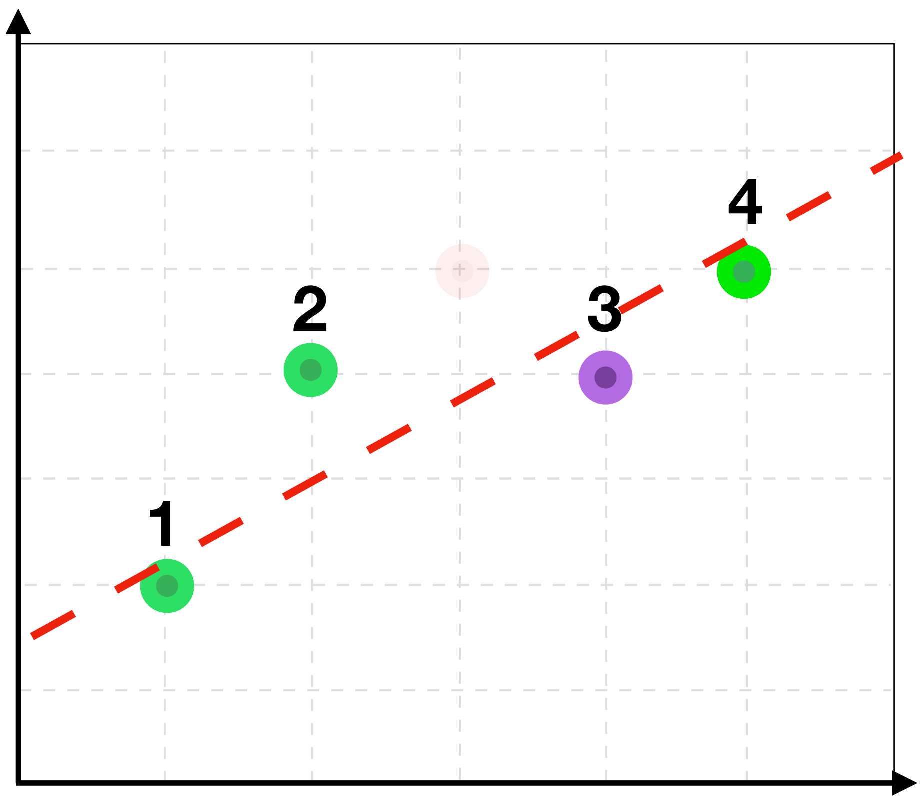
	Training	Val
I	0	4
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II		
III		
IV		
Average		

Iteration **II**



Complicated, but ideal

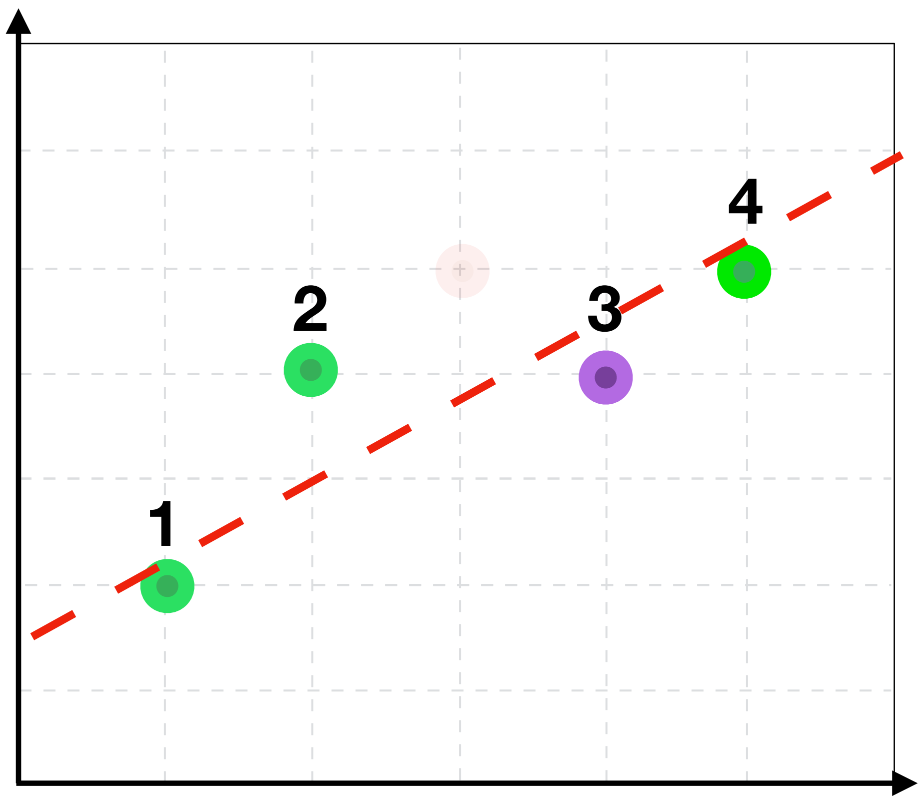
	Training	Val
I	0	4
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III		
IV		
Average		

Iteration **II**



Complicated, but ideal

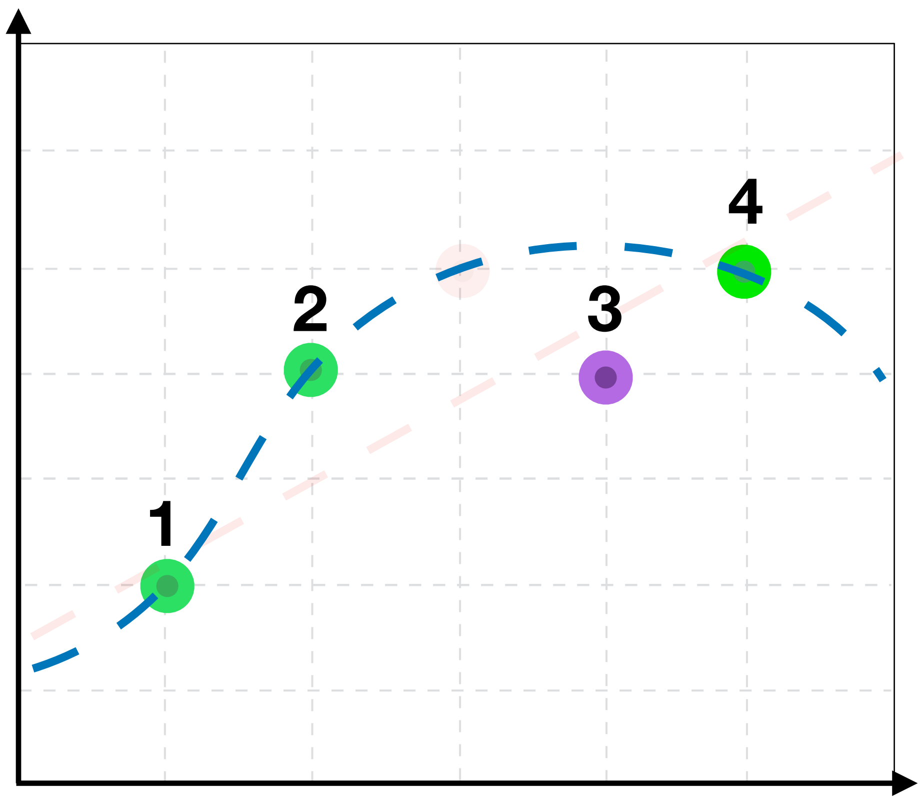
	Training	Val
I	0	4
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III		
IV		
Average		

Iteration **II**



Complicated, but ideal

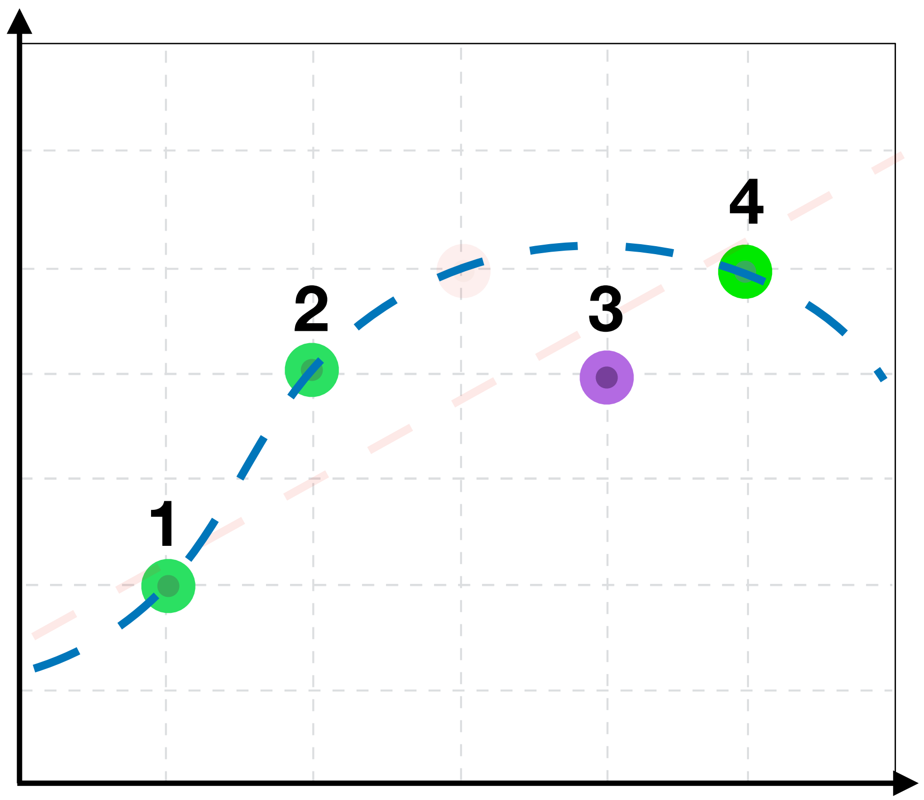
	Training	Val
I	0	4
II		
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III		
IV		
Average		

Iteration **II**



Complicated, but ideal

	Training	Val
I	0	4
II	0	1.25
III		
IV		
Average		

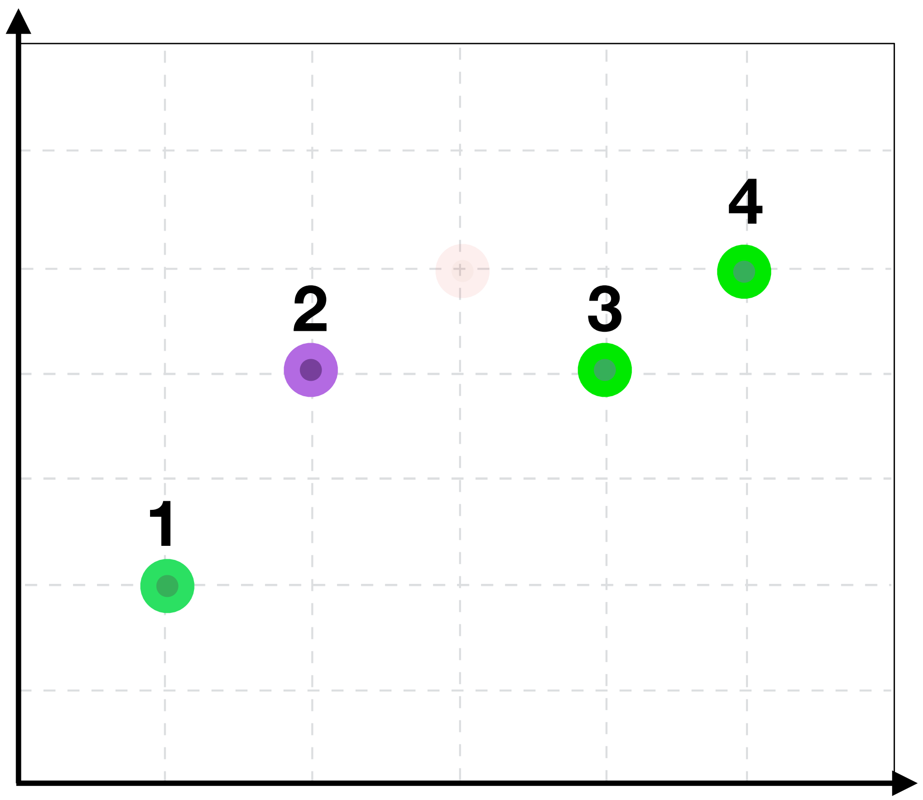
Cross Validation (CV) Algorithm

Simple, but imperfect

Iteration **III**

Complicated, but ideal

	Training	Val
I	0.66	1
II	0.37	0.25
III		
IV		
Average		



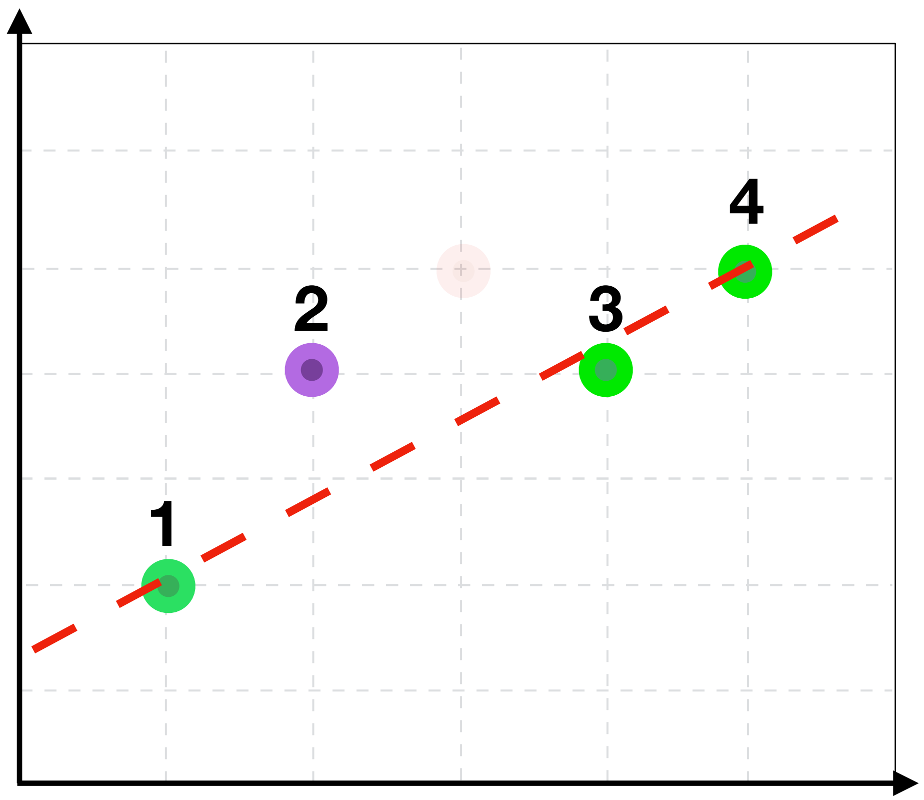
	Training	Val
I	0	4
II	0	1.25
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III		
IV		
Average		

Iteration **III**



Complicated, but ideal

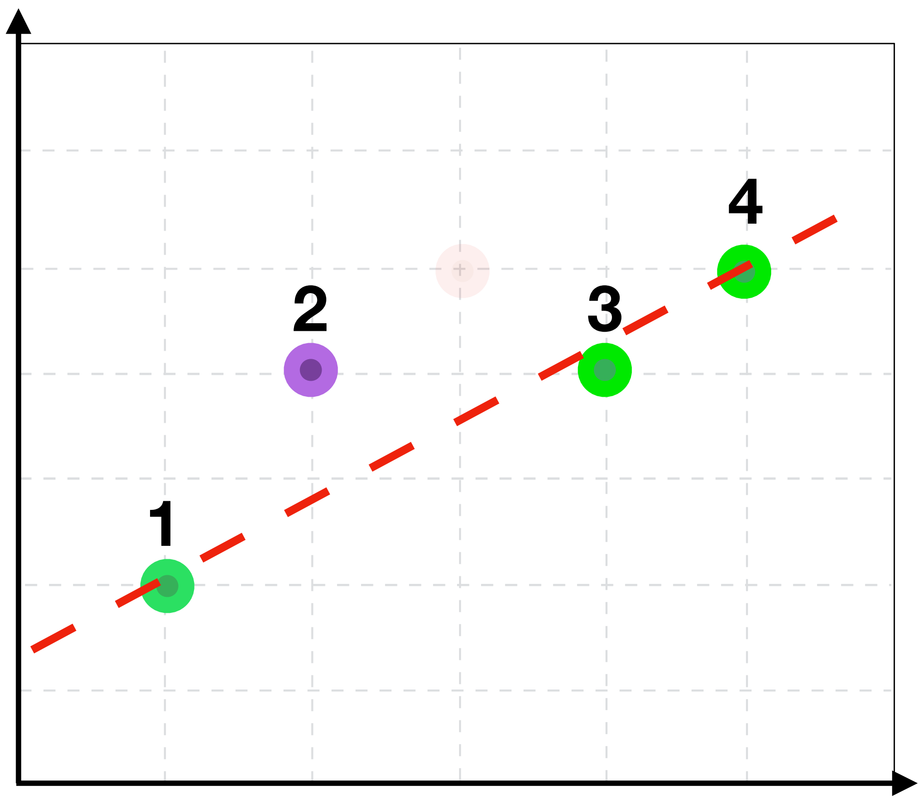
	Training	Val
I	0	4
II	0	1.25
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV		
Average		

Iteration **III**



Complicated, but ideal

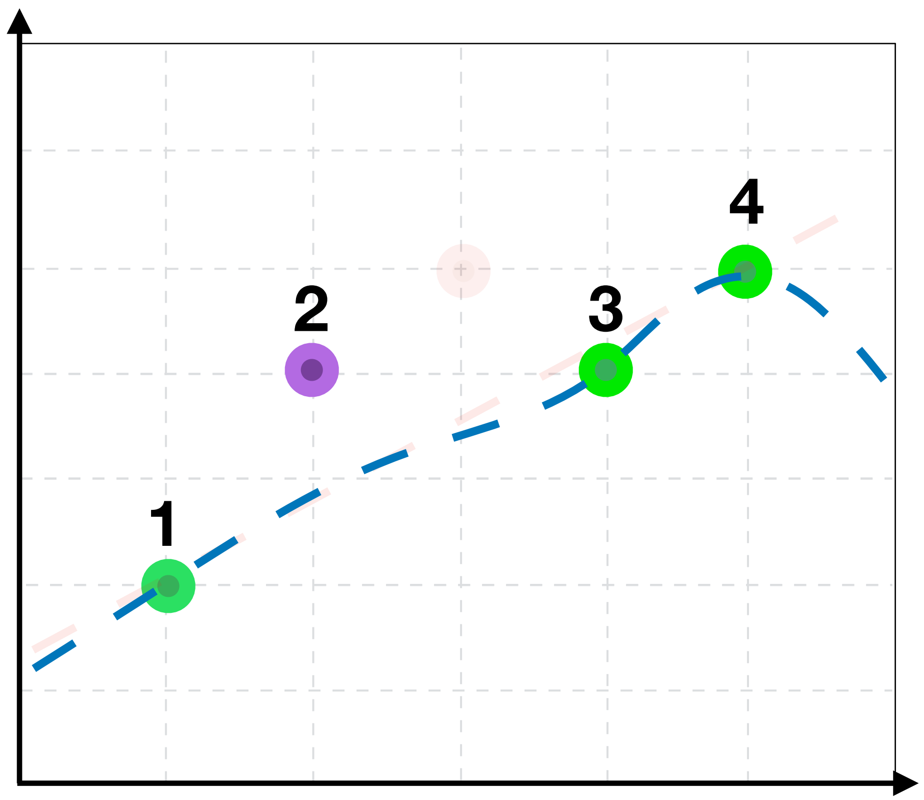
	Training	Val
I	0	4
II	0	1.25
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV		
Average		

Iteration **III**



Complicated, but ideal

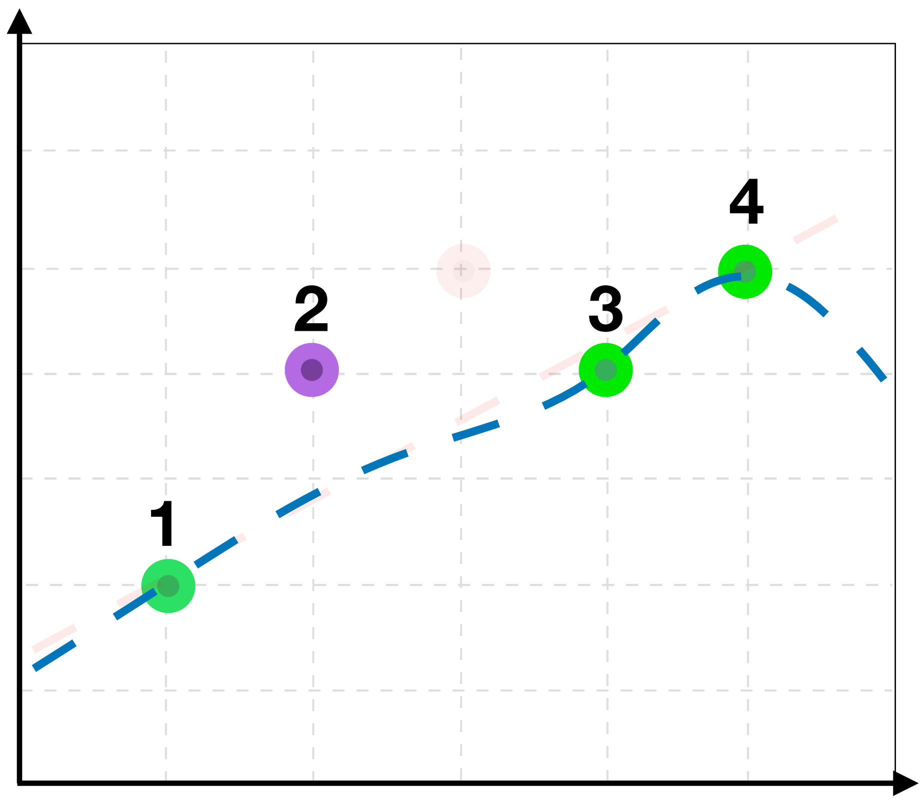
	Training	Val
I	0	4
II	0	1.25
III		
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV		
Average		

Iteration **III**



Complicated, but ideal

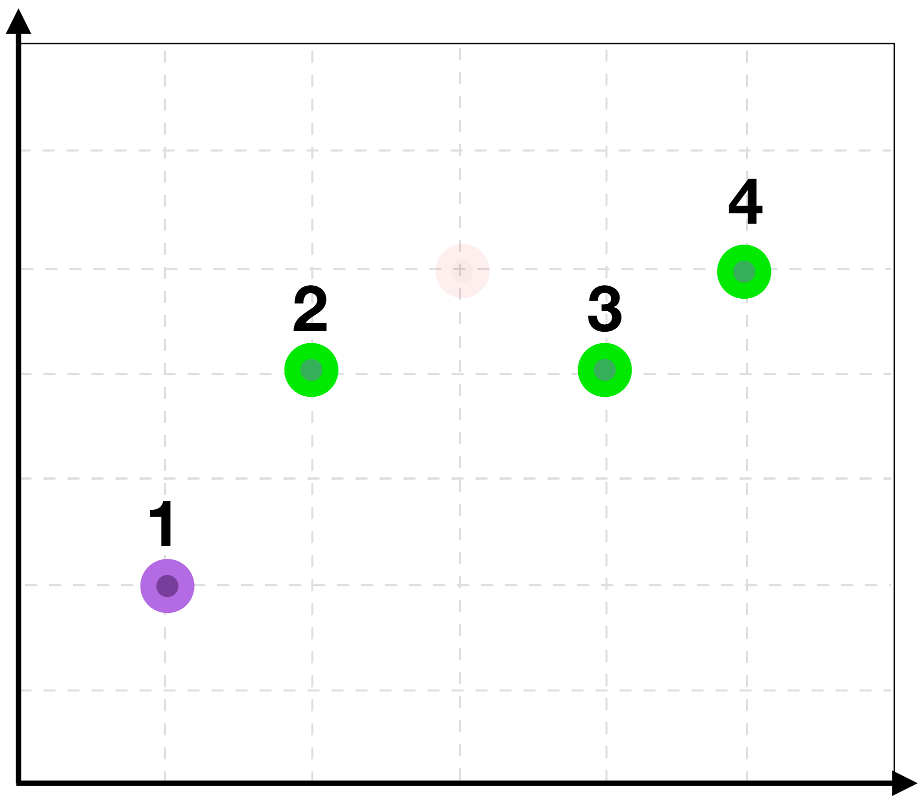
	Training	Val
I	0	4
II	0	1.25
III	0	1.25
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV		
Average		

Iteration **IV**



Complicated, but ideal

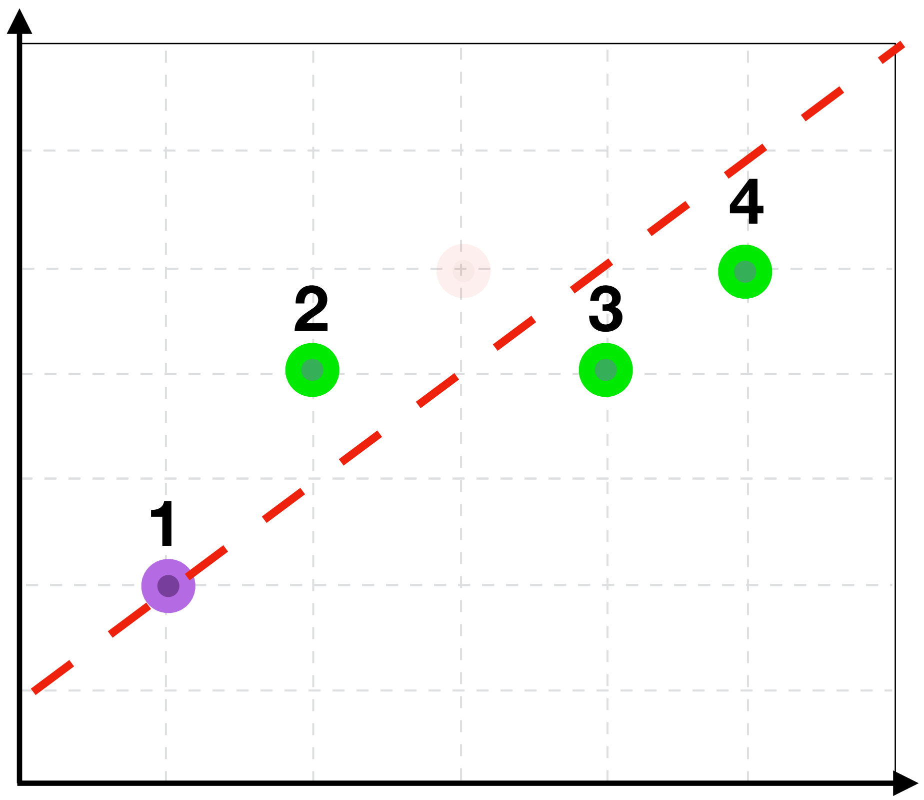
	Training	Val
I	0	4
II	0	1.25
III	0	1.25
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average		

Iteration **IV**



Complicated, but ideal

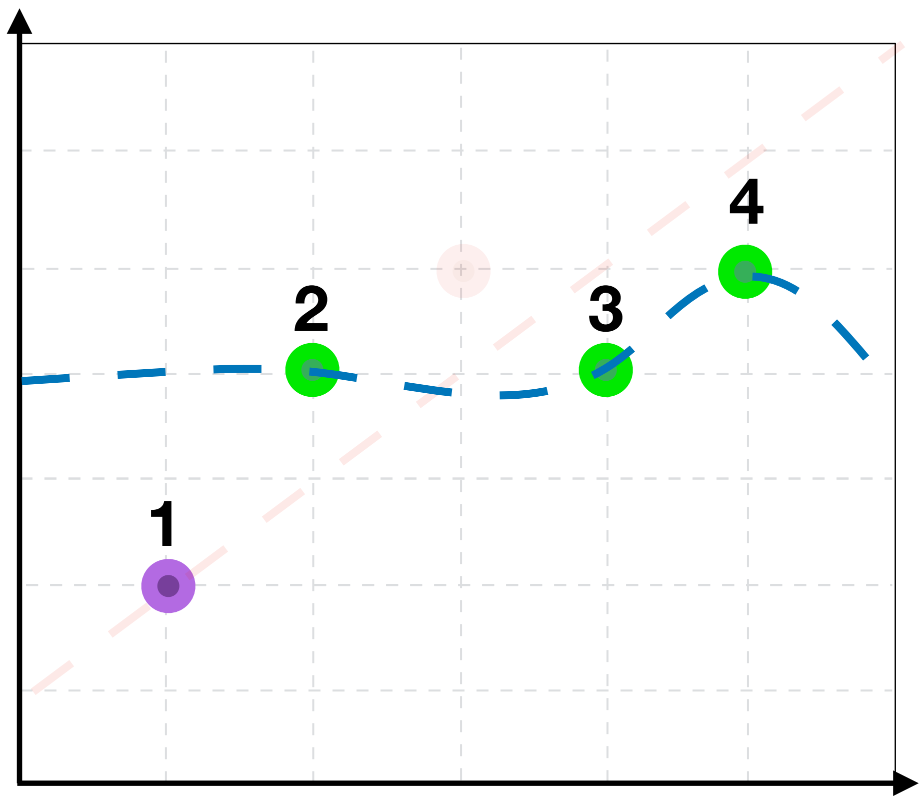
	Training	Val
I	0	4
II	0	1.25
III	0	1.25
IV		
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average		

Iteration **IV**



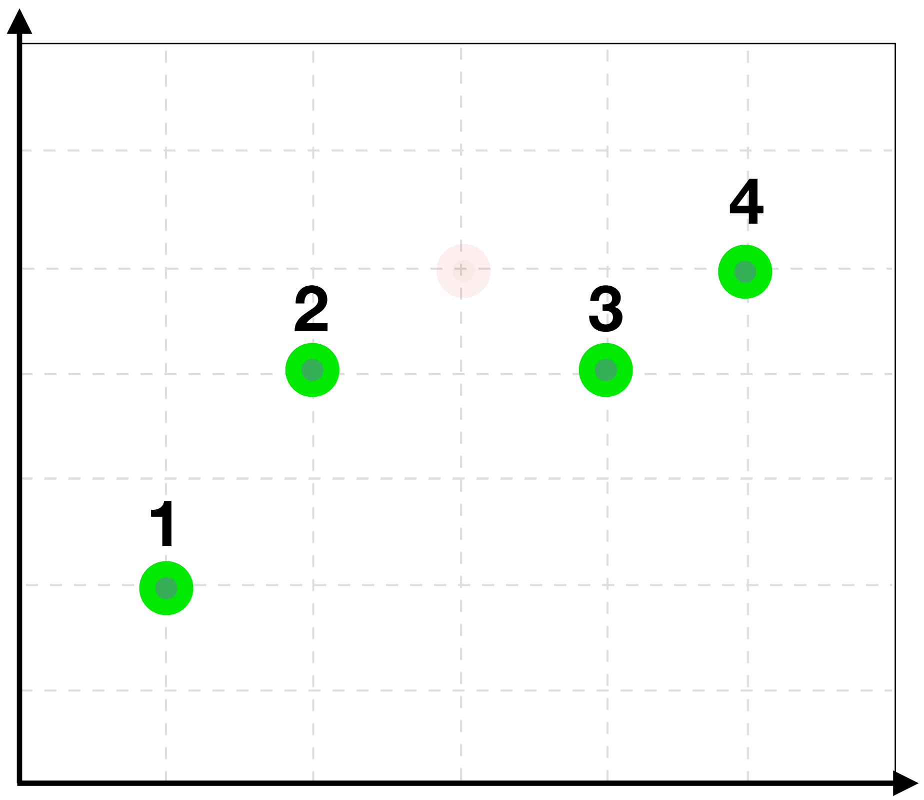
Complicated, but ideal

	Training	Val
I	0	4
II	0	1.25
III	0	1.25
IV	0	4
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average		



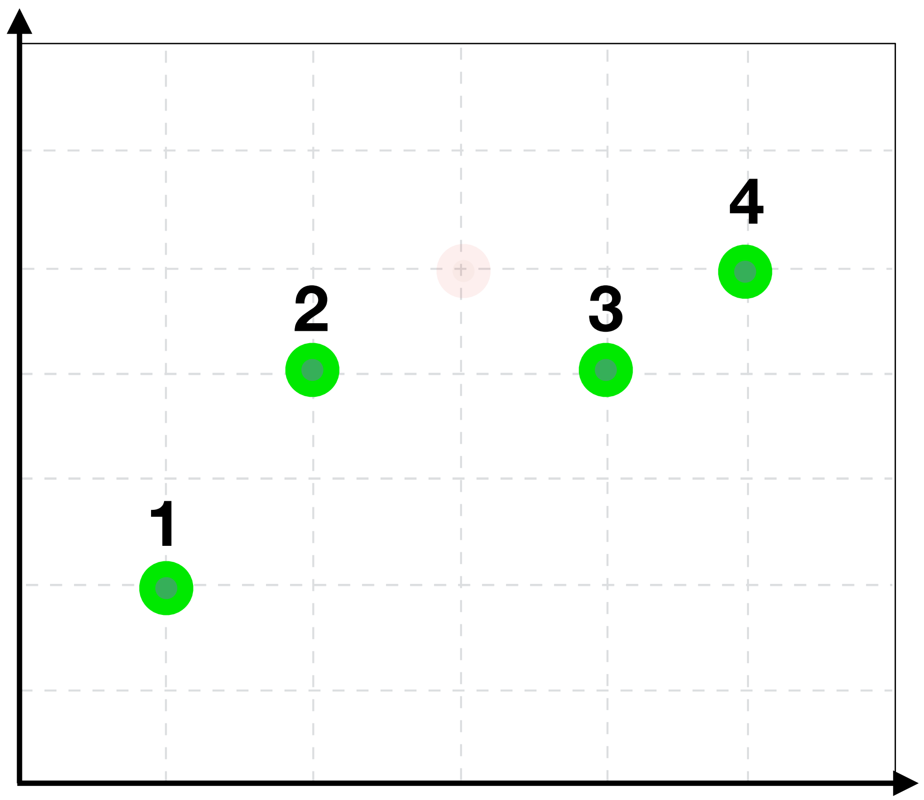
Complicated, but ideal

	Training	Val
I	0	4
II	0	1.25
III	0	1.25
IV	0	4
Average		

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average	0.511	0.625



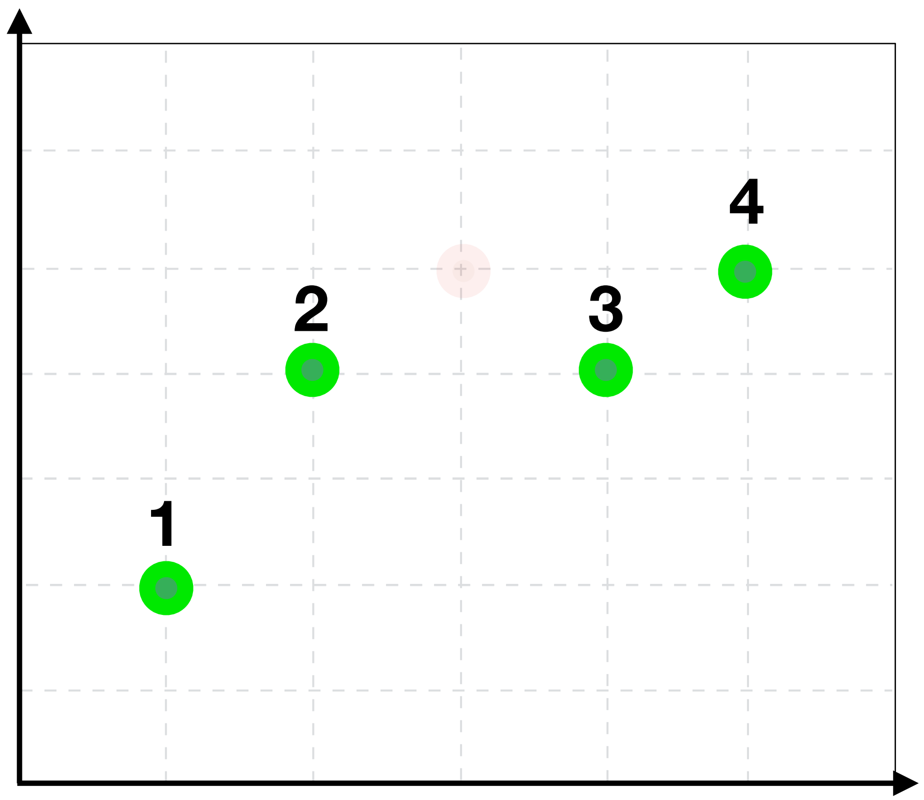
Complicated, but ideal

	Training	Val
I	0	4
II	0	1.25
III	0	1.25
IV	0	4
Average	0	2.65

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average	0.511	**0.625**



Complicated, but ideal

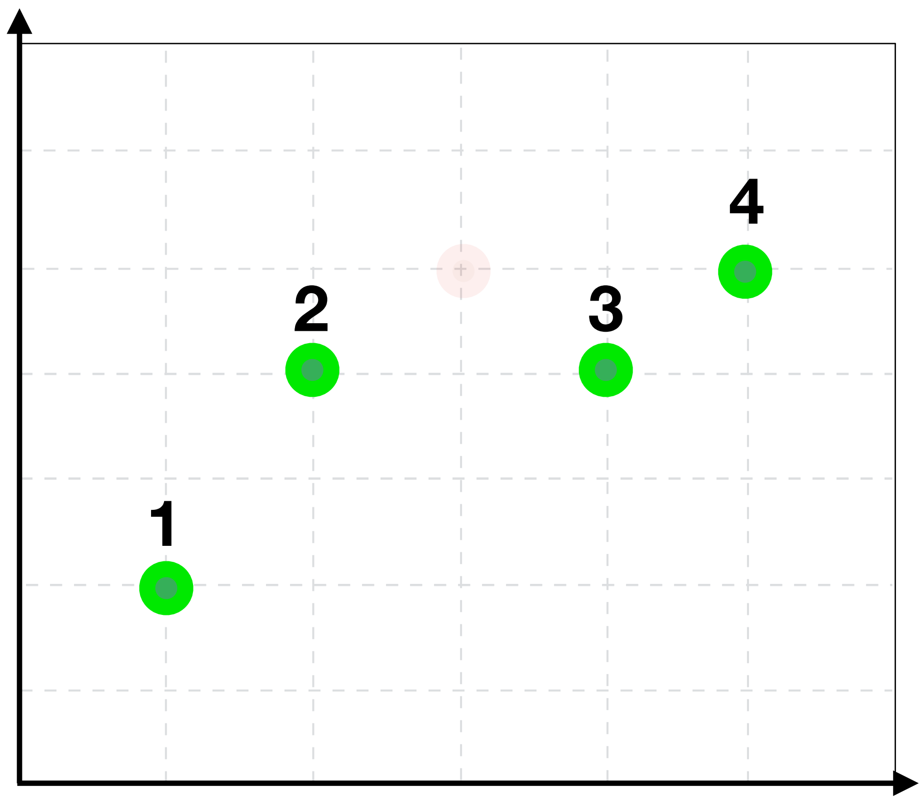
	Training	Val
I	0	4
II	0	1.25
III	0	1.25
IV	0	4
Average	0	**2.65**

Choose **the best** model based on **average validation** performance across CV folds

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average	0.511	**0.625**



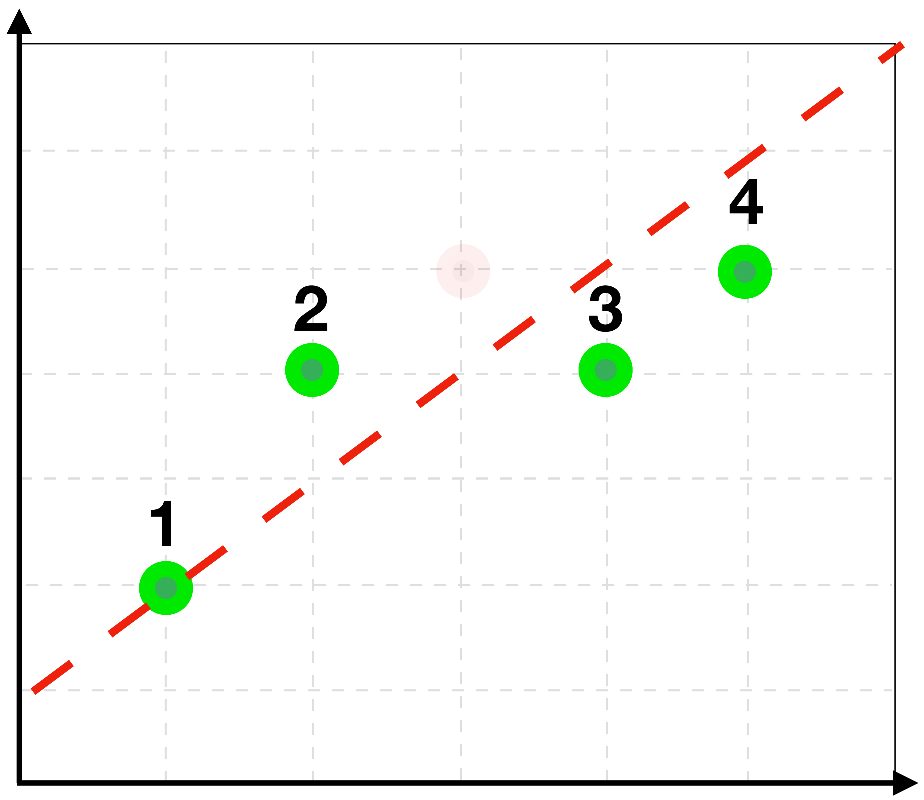
Choose **the best** model based on **average validation** performance across CV folds

Cross Validation (CV) Algorithm

Simple, but imperfect

The **final model** then should be trained on all training examples

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average	0.511	**0.625**



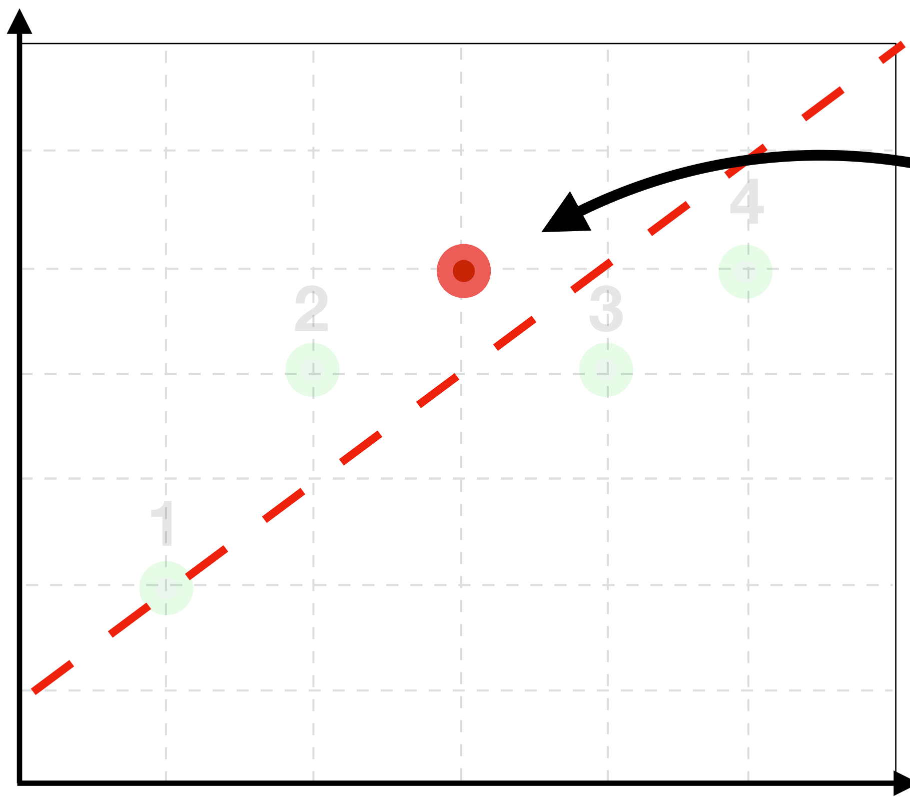
Choose **the best** model based on **average validation** performance across CV folds

Cross Validation (CV) Algorithm

Simple, but imperfect

	Training	Val
I	0.66	1
II	0.37	0.25
III	0.014	1.25
IV	1	0
Average	0.511	**0.625**

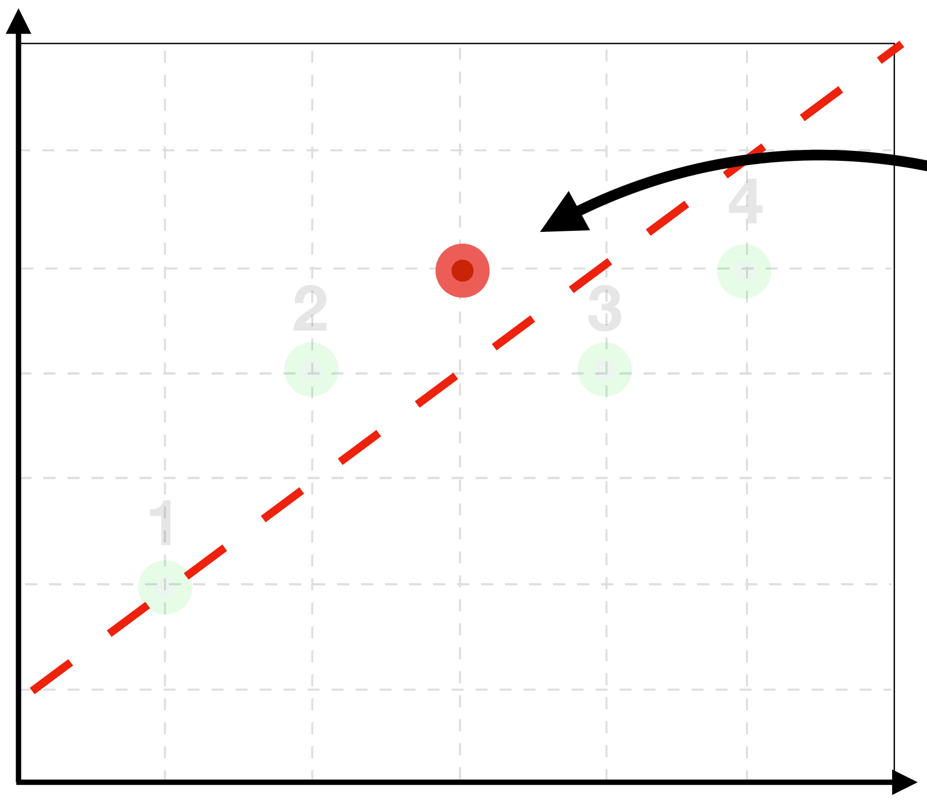
The **final model** then should be trained on all training examples



The **test set** is used for the final evaluation

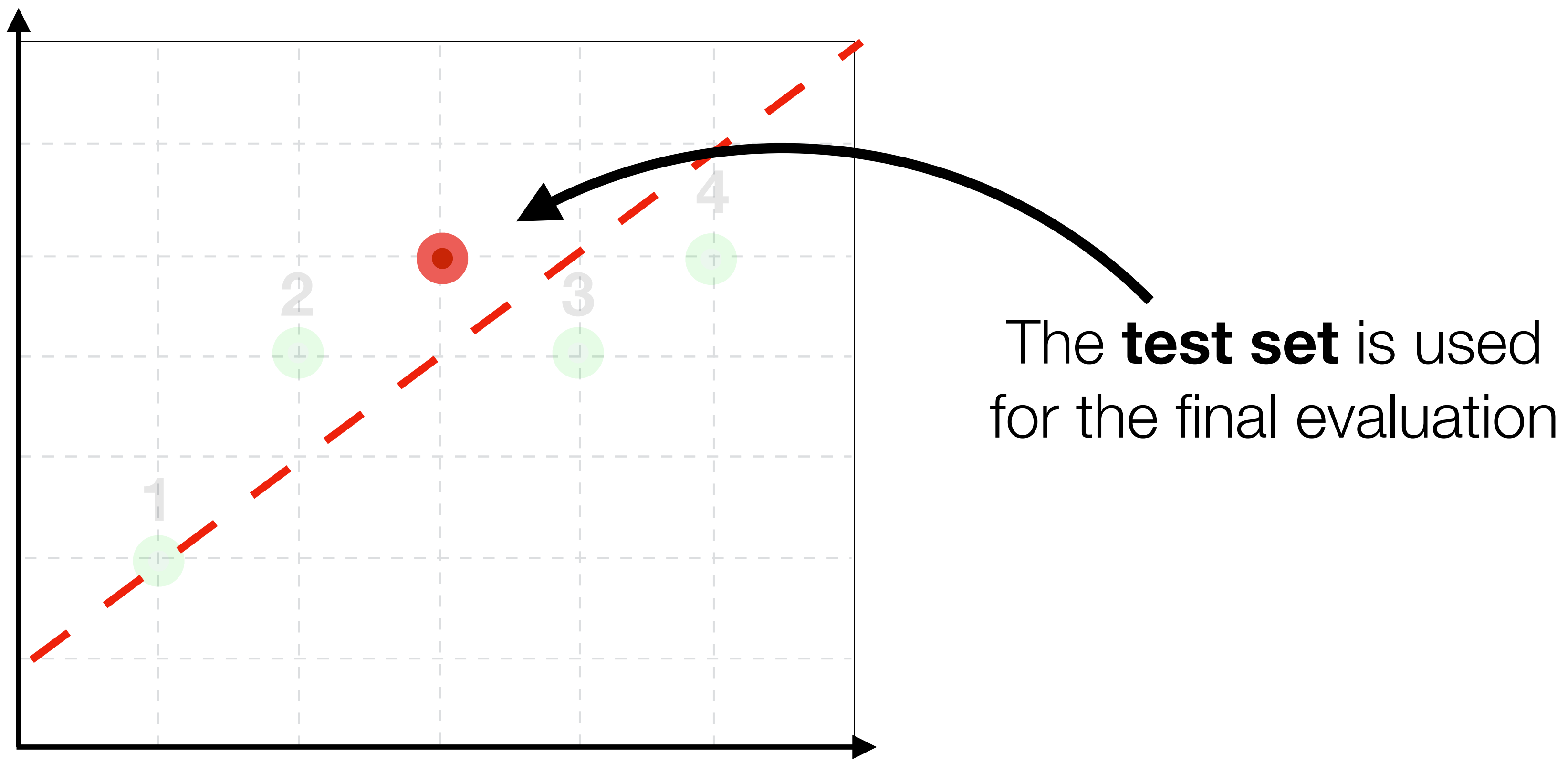
Choose **the best** model based on **average validation** performance across CV folds

Disclaimer: never evaluate models on 1 data point.
Here we show it only for the illustration purposes.



The **test set** is used for the final evaluation

Disclaimer: never evaluate models on 1 data point. Here we show it only for the illustration purposes.



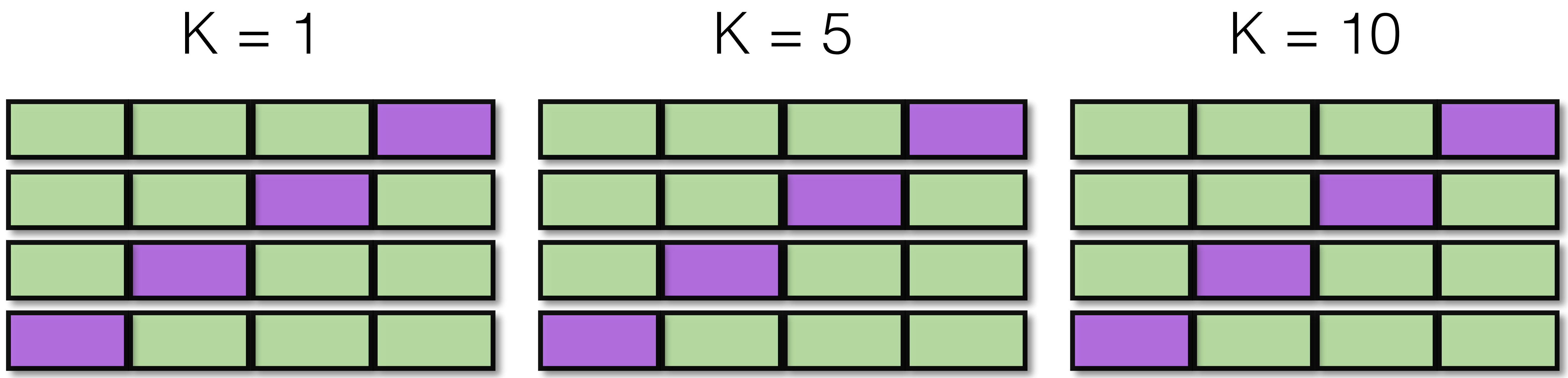
Disclaimer 2: simple models are not always better than complex. Just make sure you don't overfit your data

Cross Validation (CV) Algorithm

For each parameter (e.g. **K** in K-NN) choice - new loop of CV

Cross Validation (CV) Algorithm

For each parameter (e.g. **K** in K-NN) choice - new loop of CV



Average acc. = **0.87**

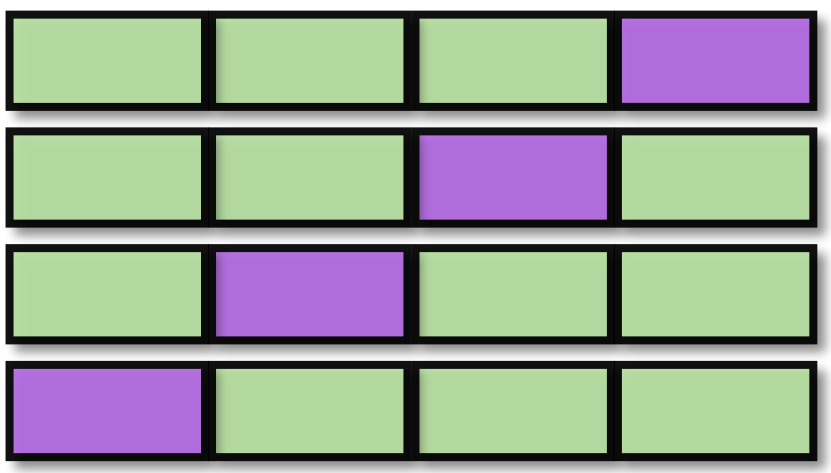
Average acc. = 0.81

Average acc. = 0.83

Cross Validation (CV) Algorithm

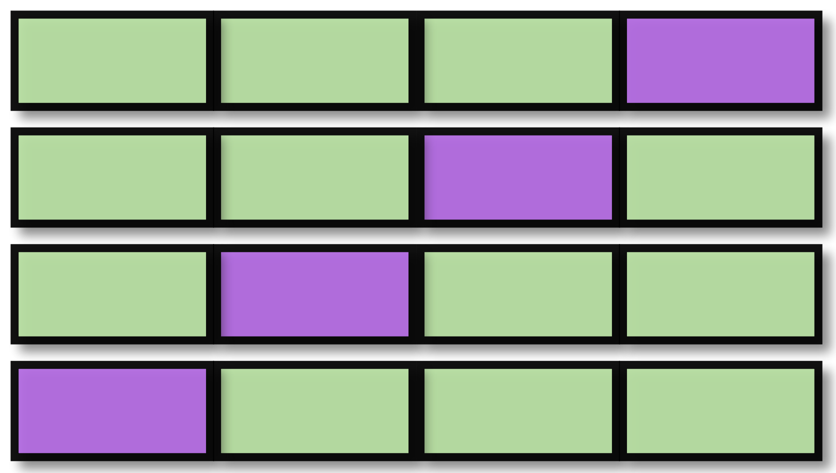
For each parameter (e.g. **K** in K-NN) choice - new loop of CV

K = 1



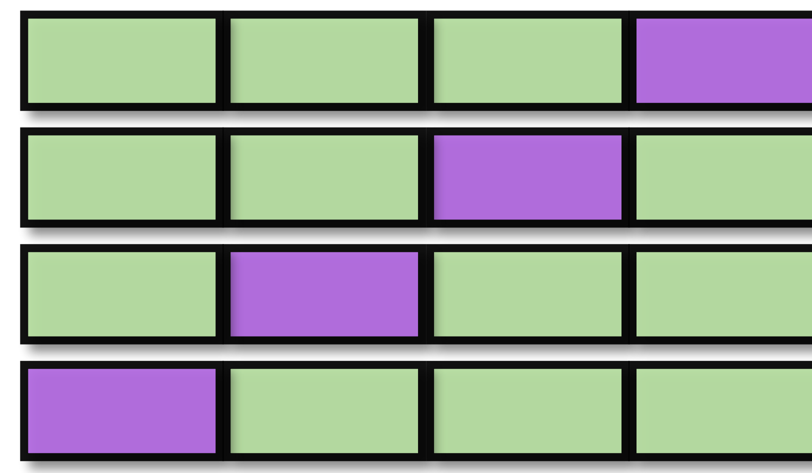
Average acc. = **0.87**

K = 5



Average acc. = 0.81

K = 10

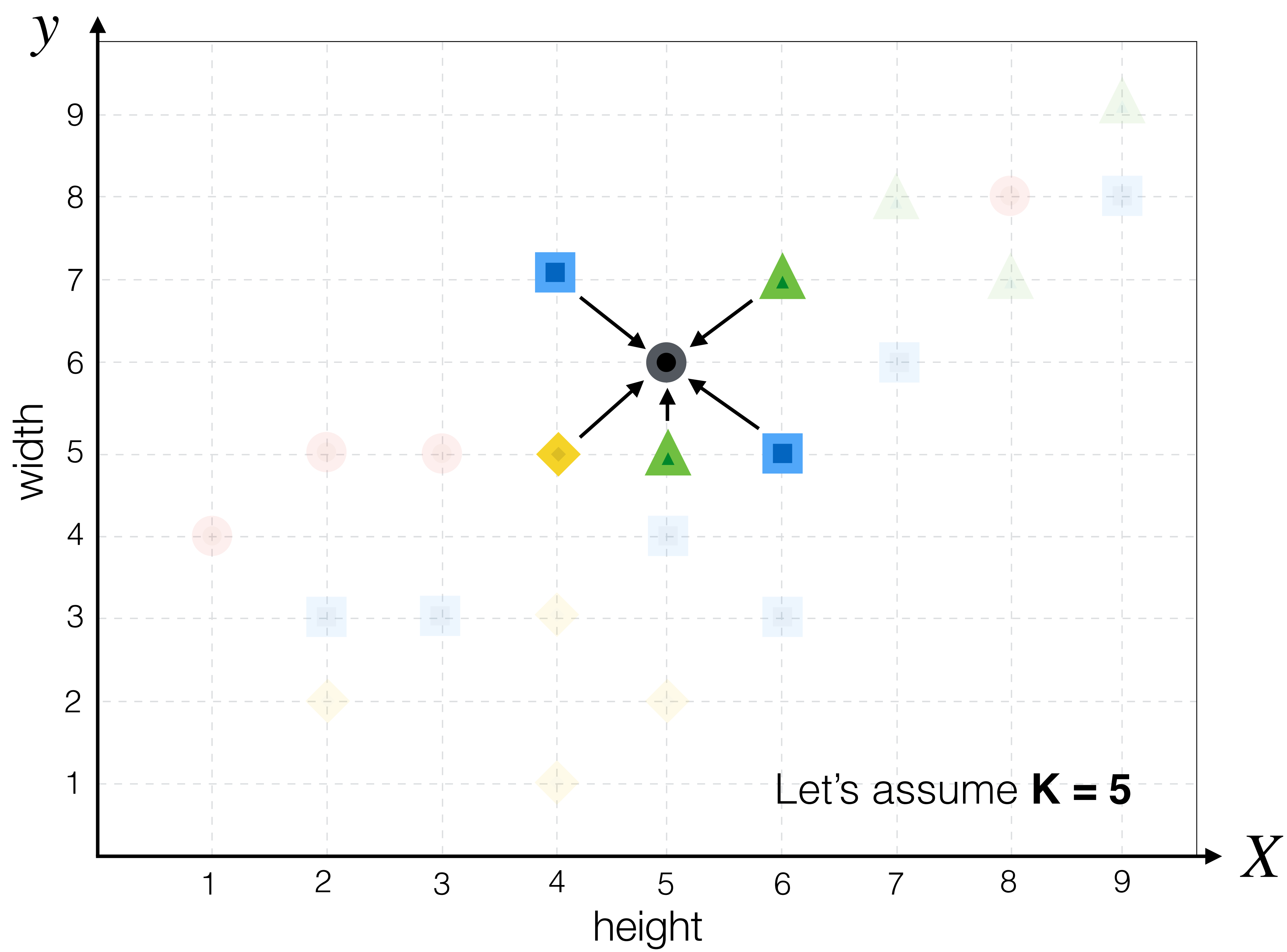


Average acc. = 0.83

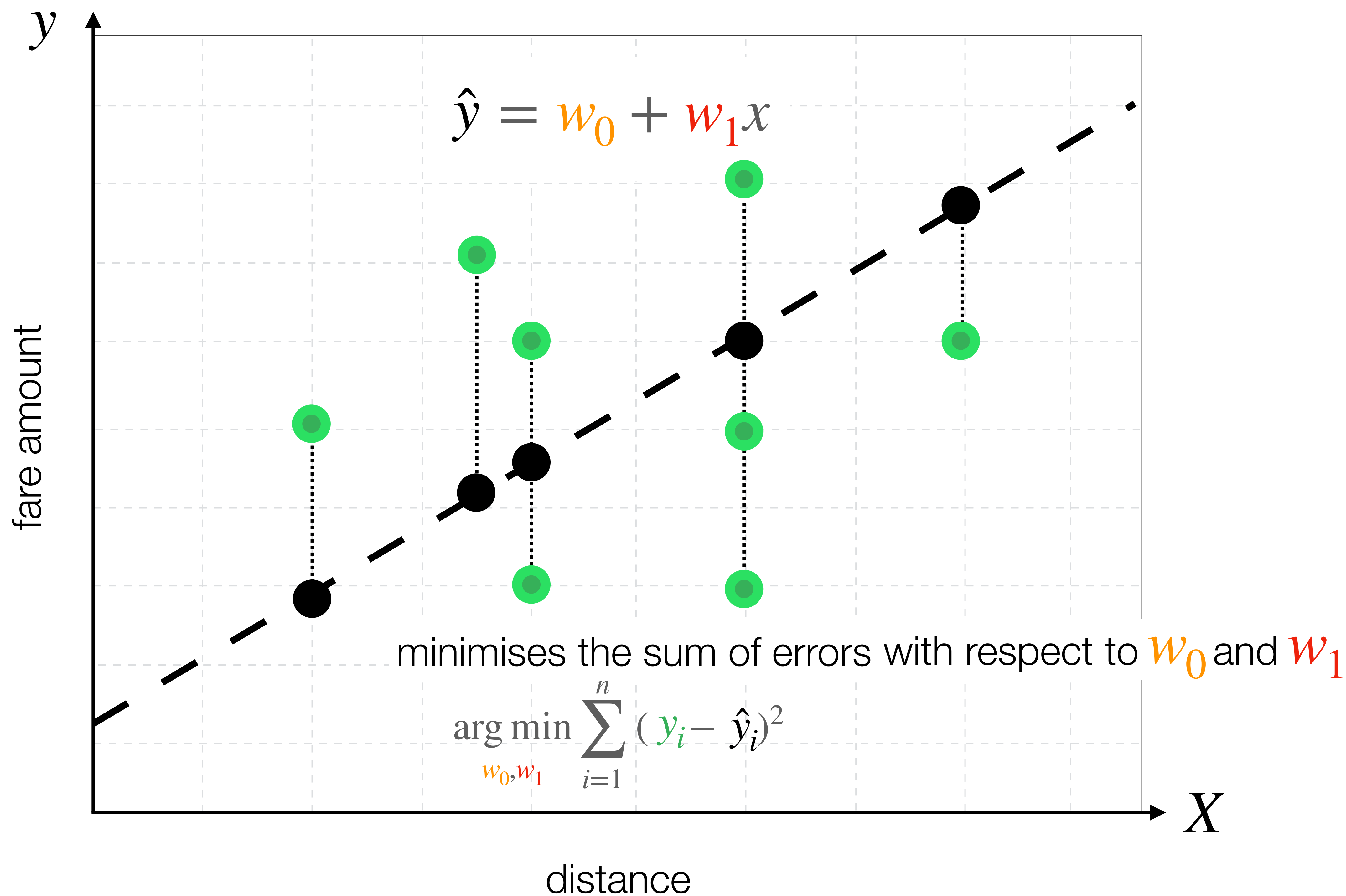
Before each CV loop we **shuffle** the data
to get new **random** folds

So far…

K-Nearest Neighbour Classifier

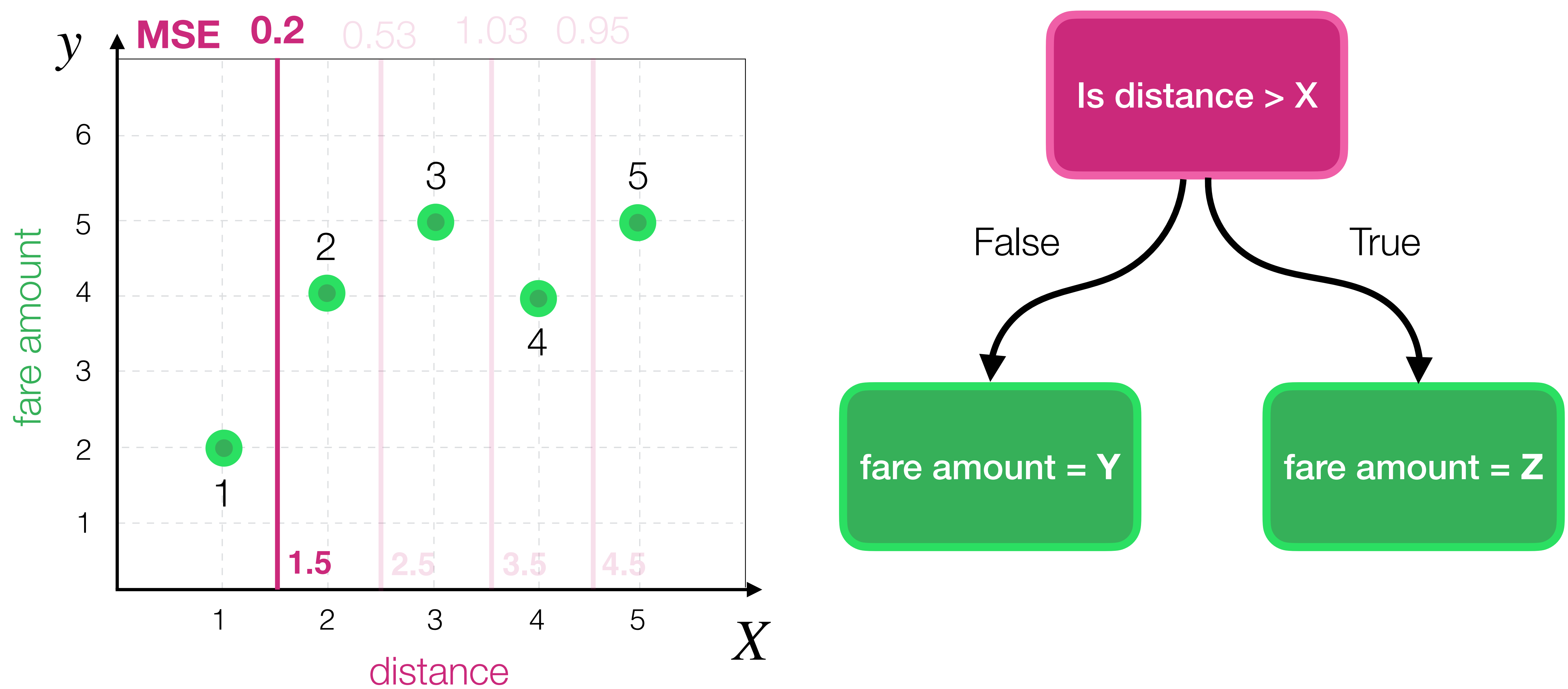


Linear Regression

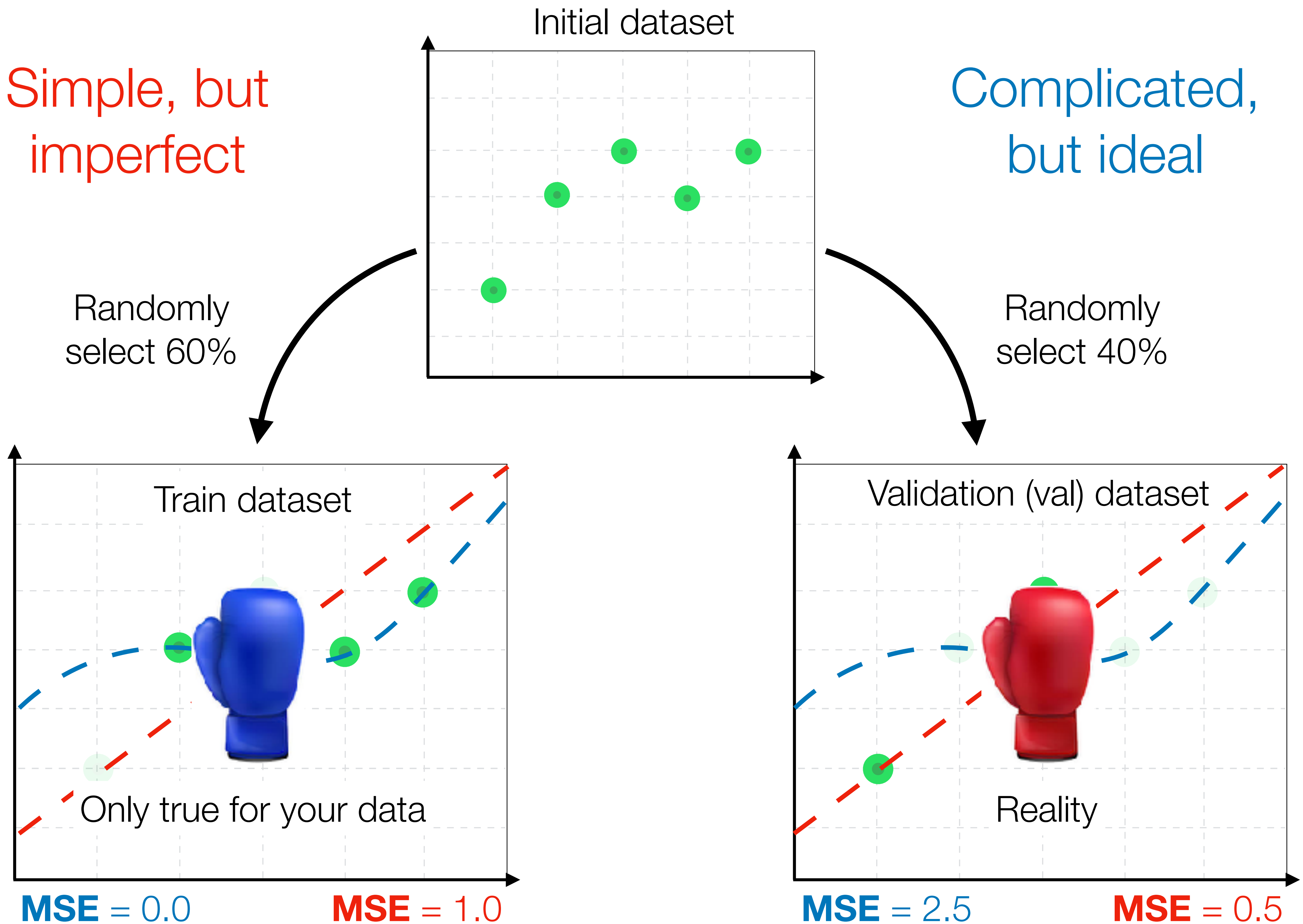


Decision Tree Algorithm

We choose the **split** that minimises total MSE



Train/val split



Cross Validation (CV) Algorithm

Simple, but imperfect

Average MSE = **0.65**

Complicated, but ideal

Average MSE = 1.4

Before each CV loop we **shuffle** the data to get new **random** folds

Supervised Learning pipeline

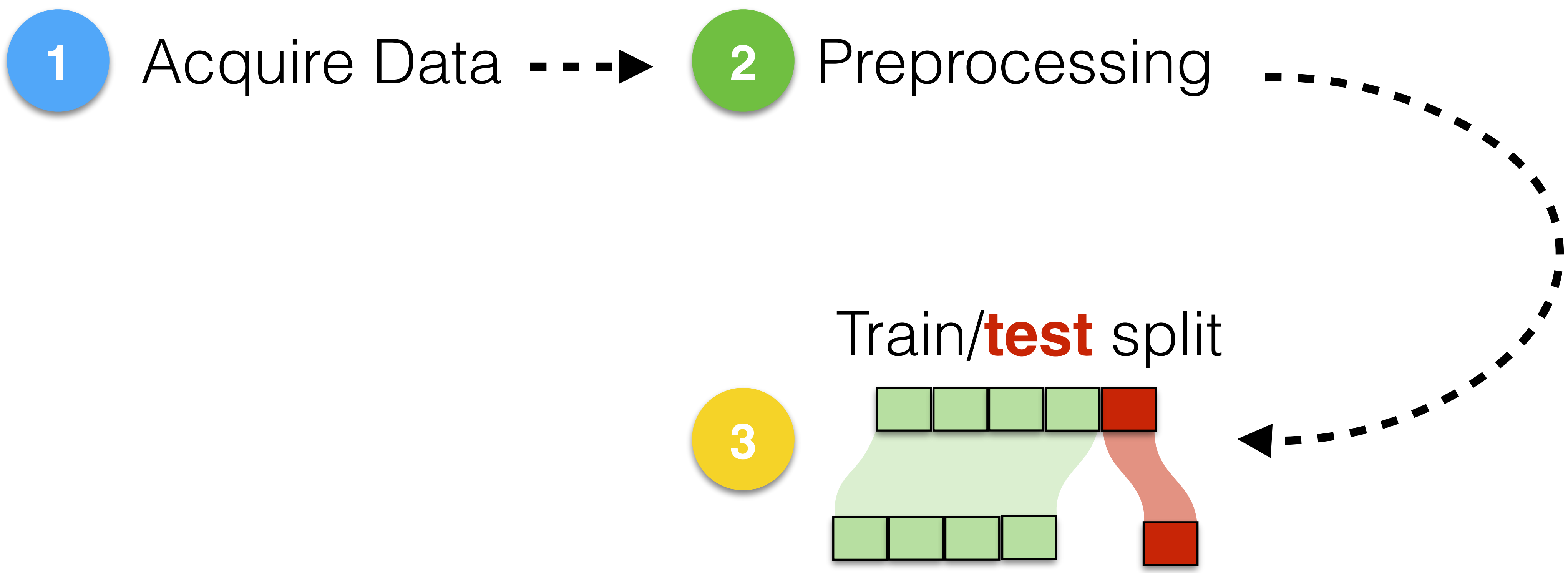
Supervised Learning pipeline

1. Acquire Data

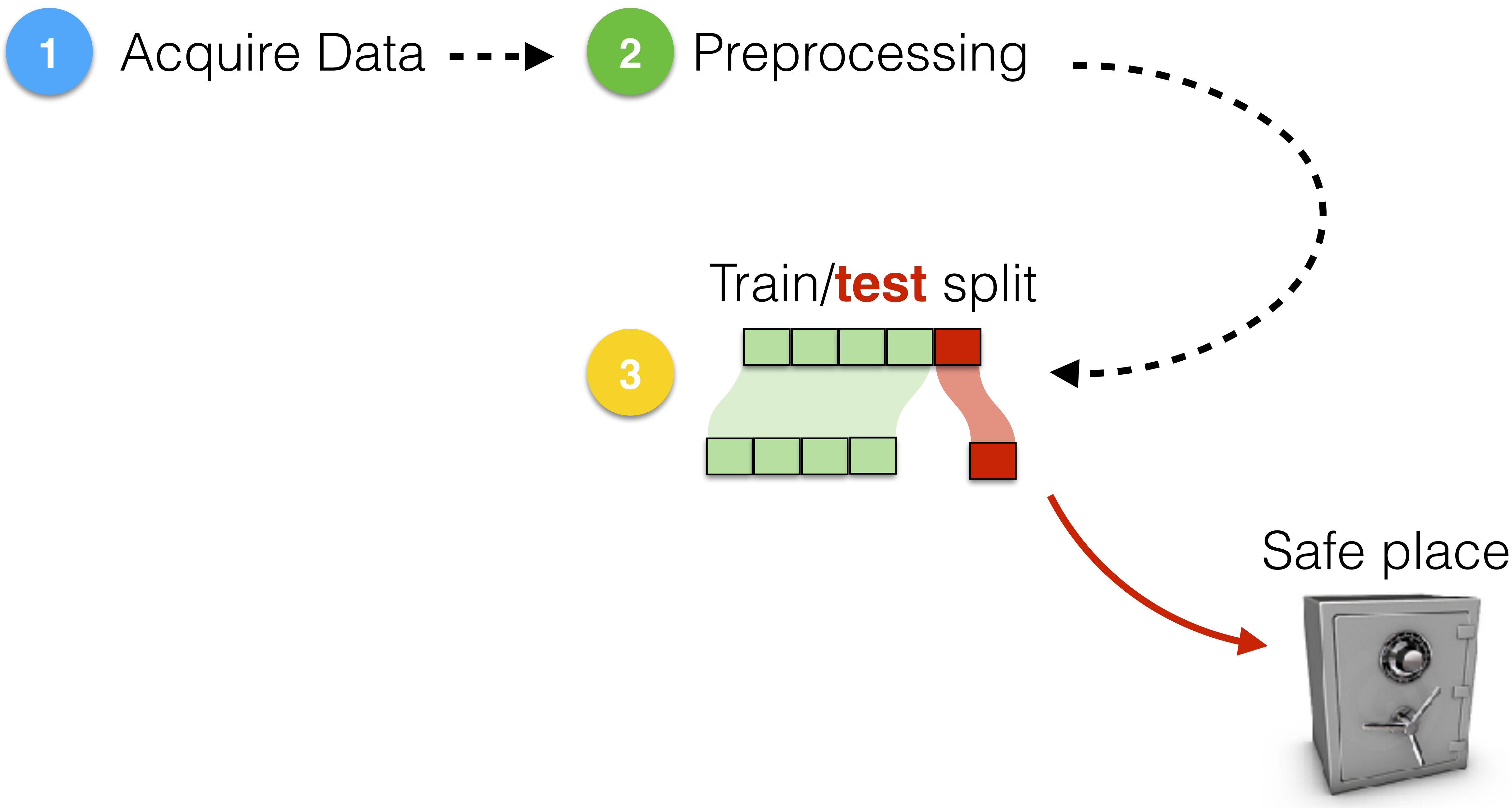
Supervised Learning pipeline

1 Acquire Data - - -▶ 2 Preprocessing

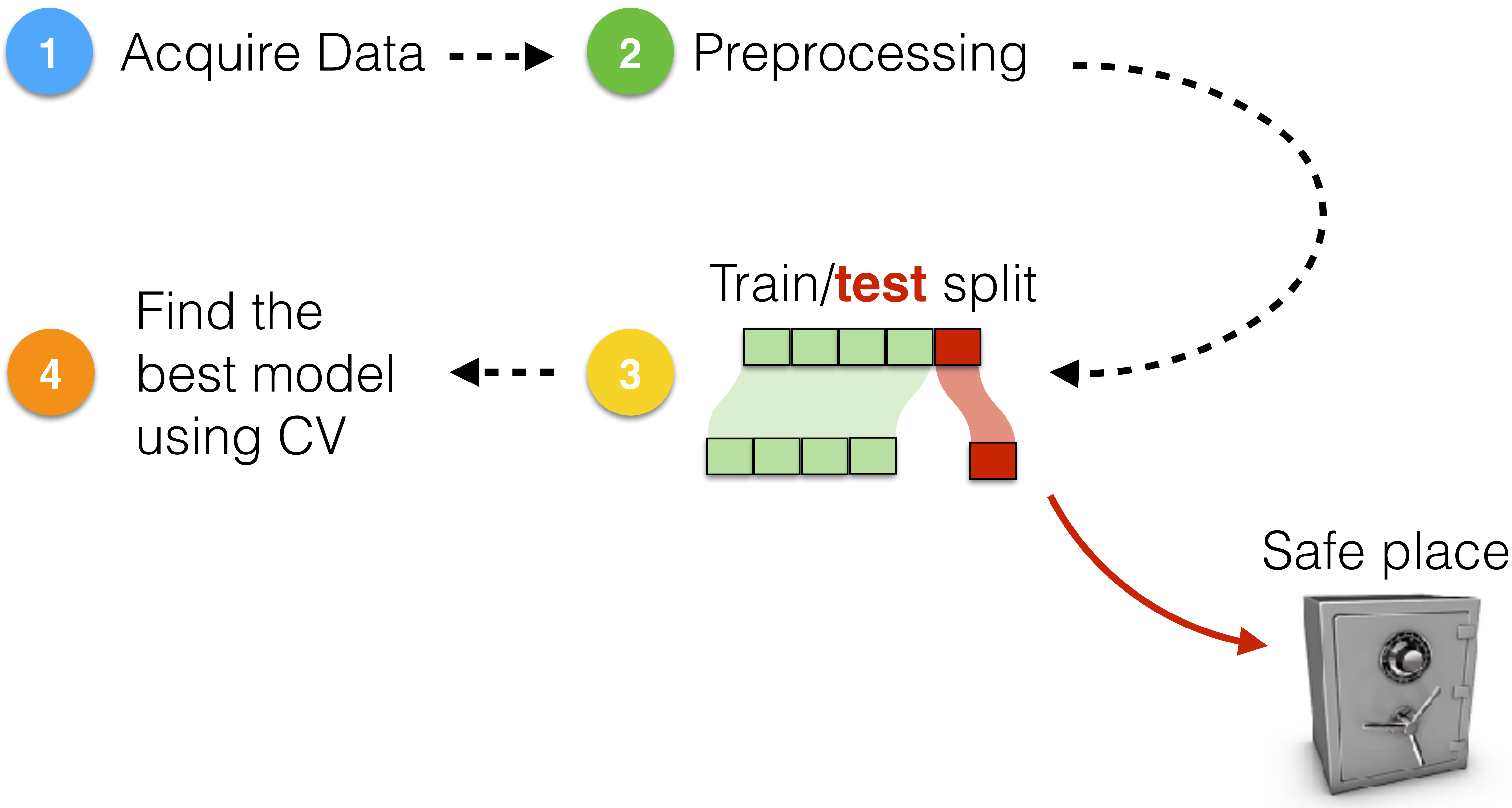
Supervised Learning pipeline



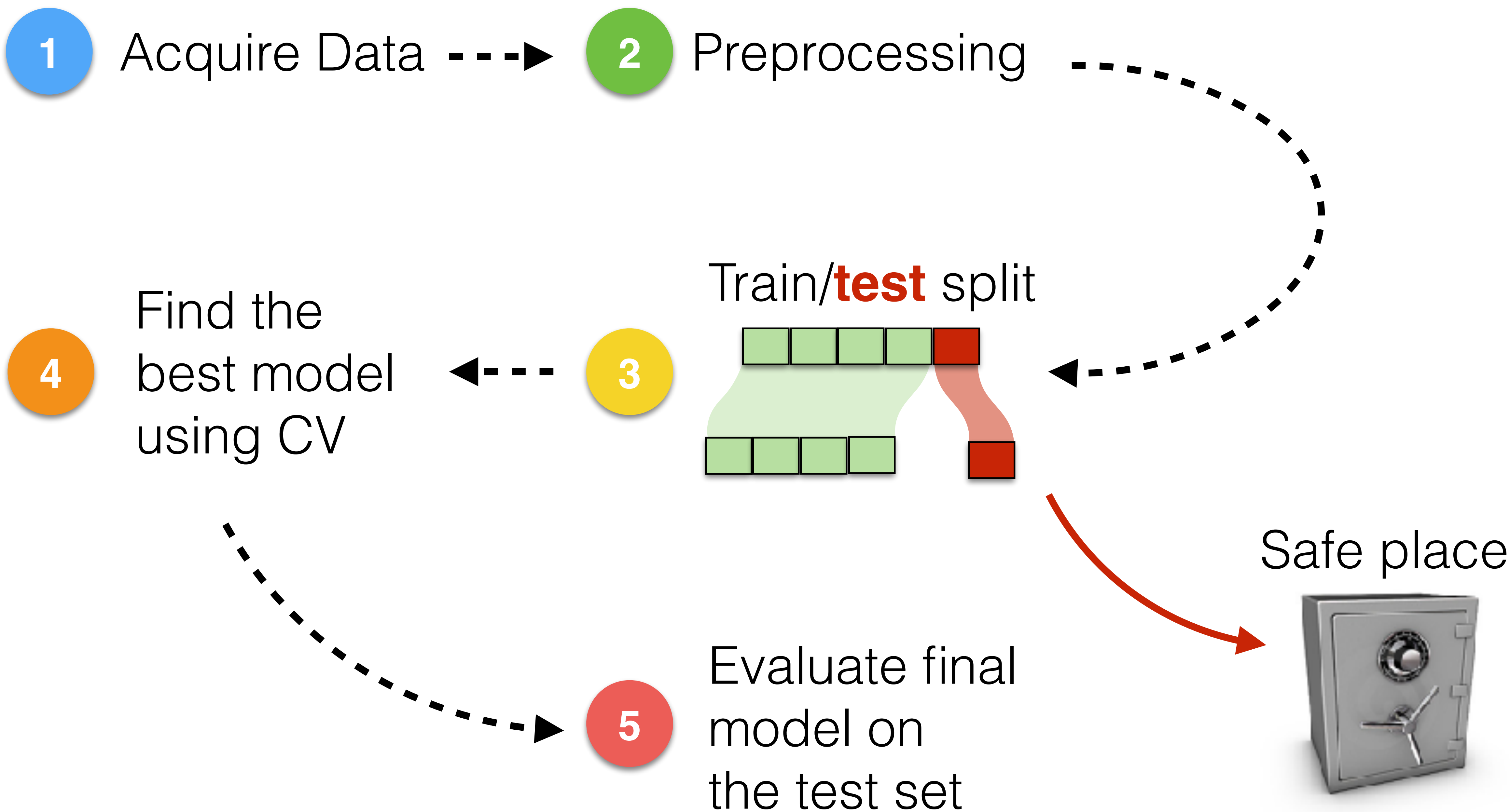
Supervised Learning pipeline



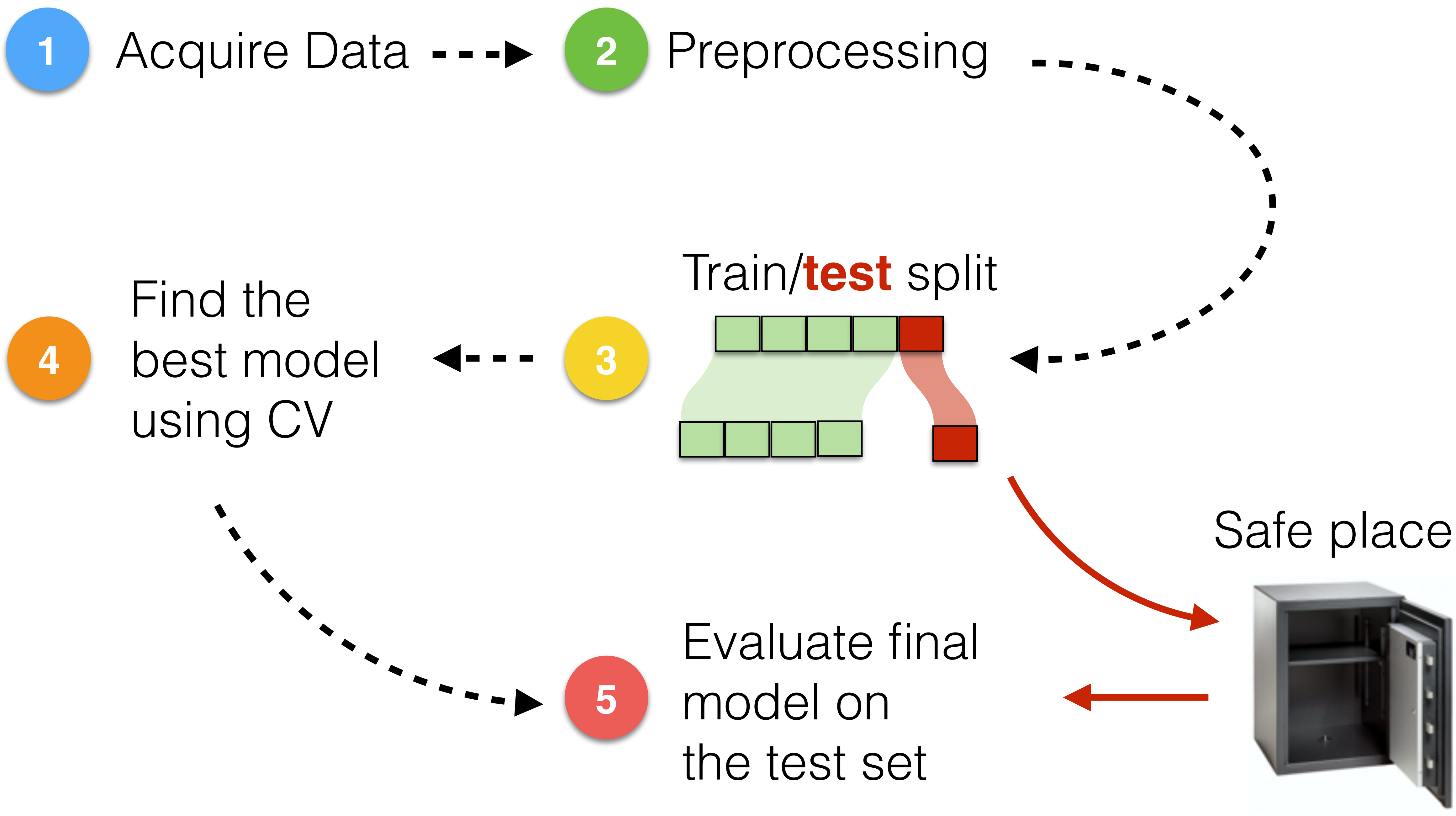
Supervised Learning pipeline



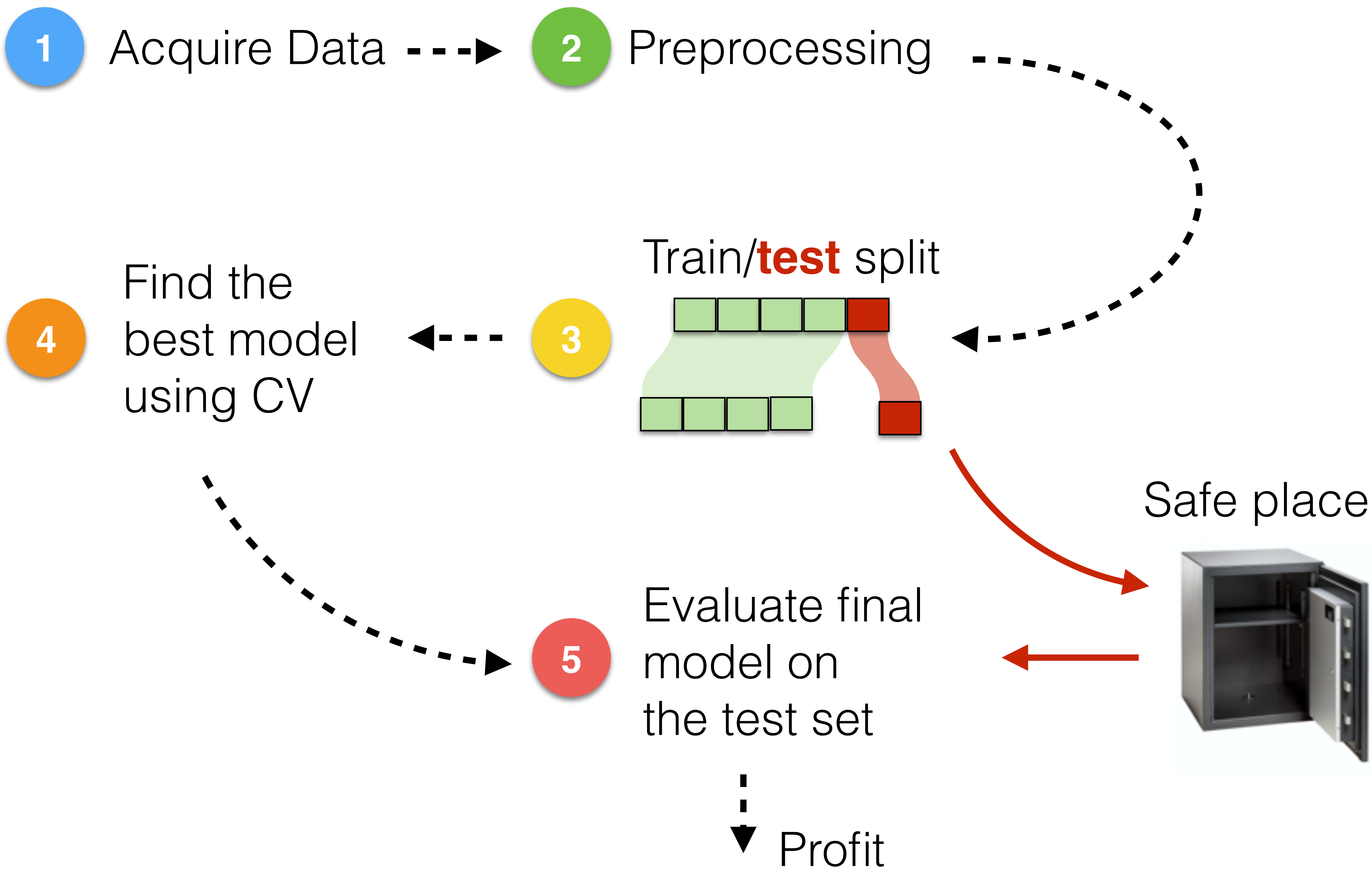
Supervised Learning pipeline



Supervised Learning pipeline



Supervised Learning pipeline



If I have seen further than others, it is by standing upon the shoulders of giants.

(Isaac Newton)

References

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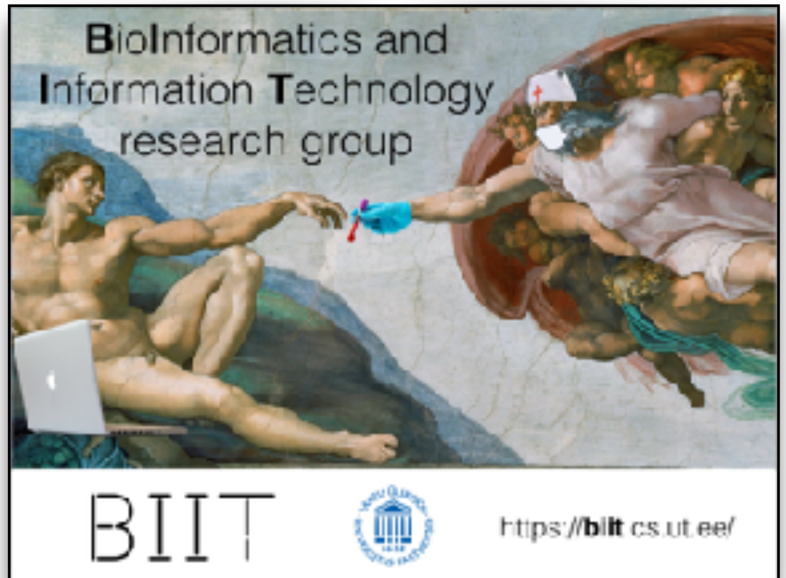

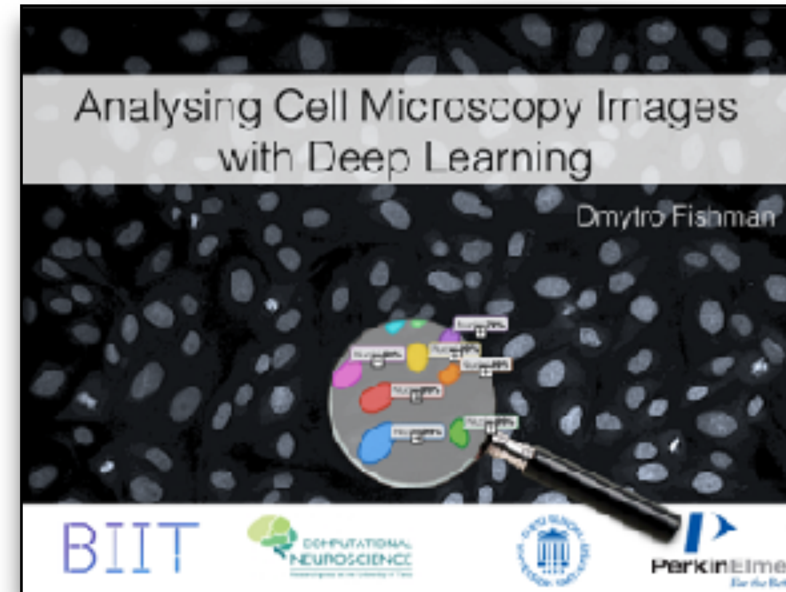

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